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COMPLETE FIVE-VOLUME MONOGRAPH · VOLUMES I–V

The Arithmetic Topos

Foundations of a Trace-Theoretic Approach to the Langlands Programme

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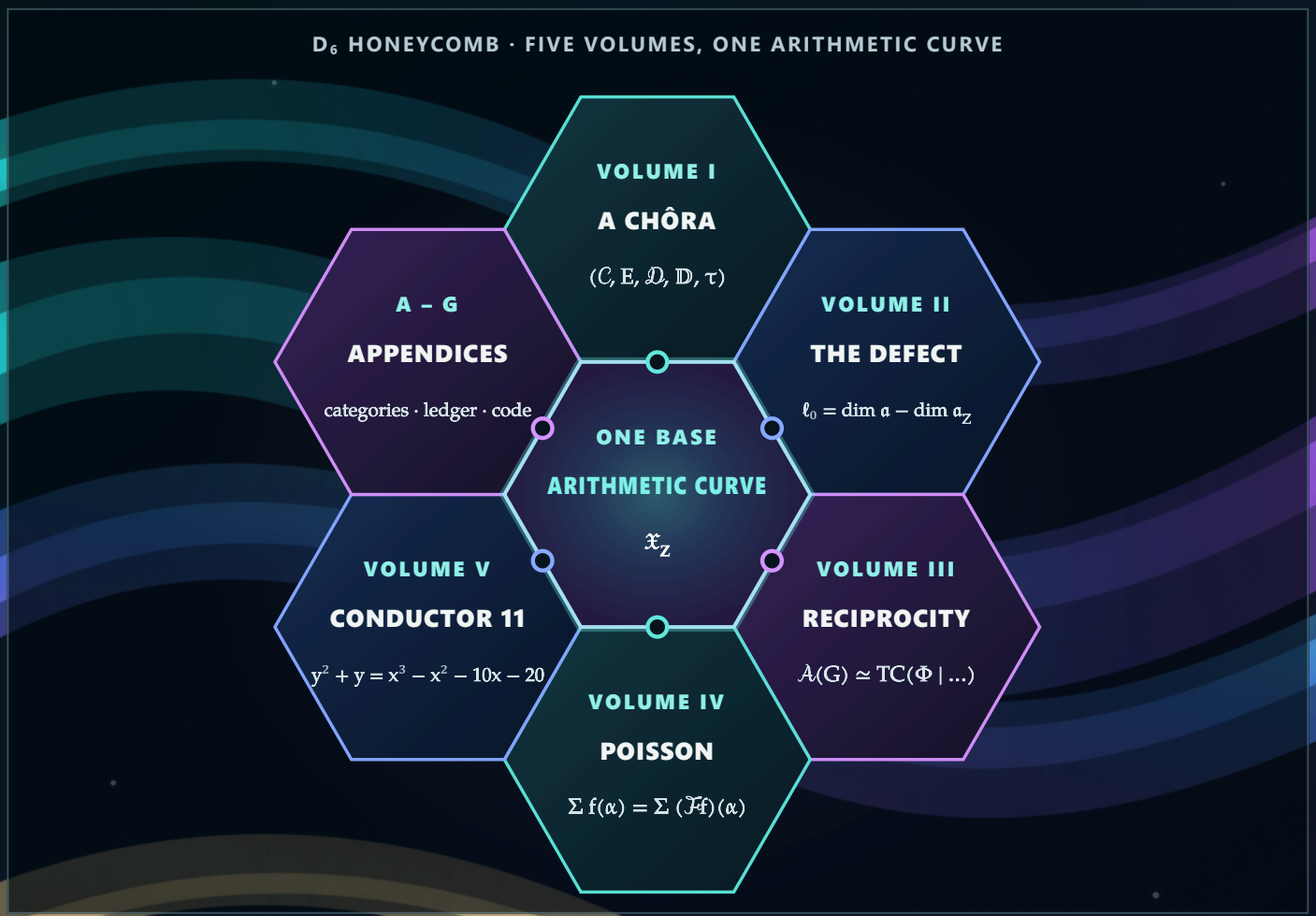

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The Arithmetic Topos

The Complete Monograph

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Claim boundary. *It is not a proof of the Langlands correspondence.* What is offered here is a *framework*: a foundational system, a construction of the objects that system requires, a precise formulation of the correspondence within it, an explicit inventory of what must be proved, and a computational and formal-verification programme by which the inventory might be discharged. Every numbered statement carries one of five epistemic status marks: [ESTABLISHED], [THEOREM], [CONJECTURE], [DESIGN], or [FORMAL TARGET]. Their meaning is set out in the Preface and in *How to Read This Book*, which the reader is asked to read before Chapter 1.

The author has taken care in the preparation of this book but makes no expressed or implied warranty of any kind and assumes no responsibility for errors or omissions. Results presented here that are explicitly labelled as conjectural are offered as programmatic proposals and not as established theorems; see *How to Read This Book* and the statement ledger of Appendix F.

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For those who believed the two halves of arithmetic were one.

Series Note

The Arithmetic Topos is the seventh work in the WiseMotif Monograph of Mathematical Intelligence and belongs to the Grothendieckian Programme. The first two works of the series are bound closely together. *The Grothendieckian Machine* set out how mathematical worlds are constructed, compared and carried from one setting to another, and how a machine may hold them as typed, versioned and verifiable objects. *GM-Omega-Infinity: Topos-Native Intelligence* stated what an intelligence must be, internally, if it is to reason inside such worlds. The third to sixth works, *Arithmetic Spectral Descent*, *Fractured Adelic Geometry*, *The Archimedean Prism* and *Strained Prismatic Geometry*, took the Riemann Hypothesis, the conjecture of Birch and Swinnerton-Dyer, the Hodge Conjecture and the *abc* conjecture as research subjects of the Programme. The present work is a self-contained system in the same sense, and it is the Programme's fifth research subject.

The Programme has a single research aim: machine intelligence that discovers mathematical structure on its own account. That aim asks for reasoning architectures held to the standard of proof, and for a discipline of recursive mathematical discovery, in which a proposed structure is tested, what survives the test is abstracted into a new setting, and the question is posed again in that setting. Its model is Grothendieck's rising sea. A hard problem is not attacked where it stands; a theory is raised around it until the problem becomes natural and gives way. A research subject of the Programme is a problem on which this method is carried through in full, with every step recorded in a form that a mathematician can check and a machine can execute.

This work carries it through on the Langlands programme. It begins from a contrast. Over a function field the programme has a base. The curve is a curve, its Frobenius is given by the arithmetic of the field, and the correspondence for GL_n, with the automorphic-to-Galois direction for every reductive group, is a theorem, proved by taking traces of Frobenius on moduli of shtukas; in the geometric setting the categorical correspondence itself has now been proved. Over $\mathbf{Z}$ there is no base: $\operatorname{Spec}\mathbf{Z}$ is not a curve over anything, it carries no Frobenius, and its archimedean place is invisible to sheaves. The recursion therefore changes the object rather than the statement. It asks for a global arithmetic curve $\mathfrak{X}_{\mathbf{Z}}$ — the syntomifications at the finite primes and an archimedean twistor chart, glued along an adelic period halo — carrying a six-functor formalism with a primitive trace, and an absolute Frobenius whose trace is to compute the automorphic spectrum. The Main Conjecture, Conjecture 19.5, is the resulting statement: three equivalences, an automorphic trace mechanism, a categorical correspondence and a categorical Lefschetz localisation, each a separate conjecture. Each step is then tested against what is known: the function-field case, the de Rham limit, the local fields, the abelian case of class field theory, the elliptic curve of conductor eleven, and the documented failures of the Ramanujan conjecture. What fails is recorded where it fails. The recursion ends in a statement ledger of one hundred and eleven items, twenty-eight axioms and eighty-three entries, fifty-three of them conjectural (Proposition 40.3); seventeen falsification tests, five of them fatal; four gaps of the first rank of difficulty and a fifth of a different kind; fourteen research problems in dependency order, with a fifteen-year critical path; and a

companion program, `aletheia`, with a Lean 4 statement skeleton, which implements the finite algorithms of the book and keeps its open comparisons open.

The work keeps one editorial principle throughout. No result is presented at a strength greater than its proof. Every numbered statement carries one of five status marks where it is made, printed as a label: [ESTABLISHED], [THEOREM], [CONJECTURE], [DESIGN] or [FORMAL TARGET]. Three of them answer to the labels of the whole series: an established result to a known one, a conjecture to an open one, and a theorem of the framework, proved here from its stated axioms, to a new result relative to those axioms. The other two are this work's own: a design is a definition, construction or principle, which carries no truth claim, and a formal target is a statement written so that a proof assistant could check it, which has not yet been machine-checked. Computation is reported as computation, at the level of certification it has, and is never promoted to a theorem. An obstruction the work fails to remove is stated as an obstruction, in the place where it stops the argument.

The Langlands programme remains open over number fields beyond the cases that the literature has proved. This work does not prove the Langlands correspondence, and no case of its Main Conjecture is established unconditionally in it (Proposition 40.4). What it offers is the state of the question after the method has been applied without concession: what the arithmetic curve would deliver, what it would need, and where each route to it stops.

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Abstract

This monograph asks why the Langlands correspondence can be proved over function fields — where it is a theorem for GL_n, and in the automorphic-to-Galois direction for every reductive group — and remains open over number fields in general; and what object would let the function-field proofs, which take traces of Frobenius over a curve, be imitated over $\mathbf{Z}$.

The diagnosis (Chapter 1) isolates four obstructions and argues that they are one: the absence of a base over which both sides of the correspondence live; the absence of a Frobenius, and hence of a trace; the absence of a uniform treatment of the archimedean place; and the cohomological defect ℓ_0, which halts the Taylor–Wiles method for all but a thin class of groups and fields.

[DESIGN] The foundations (Part I) define a geometry as a six-functor formalism with duality and a primitive trace theory — a chôra — by the twenty-eight axioms KH.1–KH.28, written in the modal directed type theory $\mathrm{TT}_{\mathrm{KH}}$ of Chapter 6; weight structures take the place of the missing motivic t-structure.

[CONJECTURE] The base (Part II) is specified, not built: the global arithmetic curve $\mathfrak{X}_{\mathbf{Z}}$, glued from the syntomifications at the finite primes, whose local theory is established, and an archimedean twistor chart, along an adelic period halo. Its existence (Conjecture 10.16) and effective halo descent (Conjecture 10.13) are conjectures, and so is its orientation of degree three (Conjecture 12.14), on which the shifted symplectic structure of the gauge and spectral stacks of Part III depends.

[CONJECTURE] The Main Conjecture (Conjecture 19.5) states that the automorphic spectrum of a reductive group over a number field is the twisted cyclic trace of the absolute Frobenius on nilpotent sheaves on the stack of G-bundles over $\mathfrak{X}_{\mathbf{Z}}$; that the same trace may be taken on ind-coherent sheaves on the stack of Langlands parameters; and that there it localises to the Frobenius-fixed locus as the global sections of a virtual dualising sheaf. The three equivalences are three separate conjectures. Eight corollaries are derived from them, among them reciprocity, functoriality as Fourier conjugation (Chapter 24), Ramanujan on the pure locus, Fontaine–Mazur as extension across the Hodge–Tate divisor and the trace formula as a derived index theorem; each corollary is stated with the hypotheses it consumes, and none of them is established by this route.

[THEOREM] What is proved is conditional on the axioms, and the book says so: twenty-four of the one hundred and eleven ledger items are theorems of the framework, each depending on at least one axiom or on a conjecture of the GC family (Proposition 40.3). Four results would survive the collapse of the framework around them: the categorical Lefschetz theorem in the form Theorem 21.28, the trace rigidity theorem of Chapter 20, the computation of the archimedean regularised determinant behind Theorem 9.27, and the local structure theory of matrix factorisations assembled in Chapter 17.

[ESTABLISHED] The established mathematics on which the book rests is reported as such and used as a control: the function-field theorems, the geometric Langlands equivalence, the Fargues–Scholze geometrisation, prismatic and syntomic cohomology, topological cyclic homology, modularity lifting and Arthur's classification. The four degenerations of the Main

Conjecture — function fields, the de Rham limit, local fields and the abelian case — are run in Chapter 19, and the curve of conductor eleven is followed from Chapter 1 to Chapter 36.

The computational part is executable, and it is reported as computation. The companion package `aletheia` implements the finite algorithms of Chapter 38, with twenty-one tests that pass; Appendix G records what it does and does not yet do. A Lean 4 statement skeleton, not yet compiled and with thirty-three open placeholders, fixes what a formal audit must check.

What remains is stated as a programme: seventeen falsification tests, five of them fatal (Chapter 39); four gaps of the first rank of difficulty and a fifth, the coherence data of the higher-categorical structures, of a different kind; fourteen research problems in dependency order and a fifteen-year critical path whose bottleneck is library-building (Chapter 40). The book does not prove the Langlands correspondence, and no case of its Main Conjecture is established unconditionally in it.

Preface

The unknown thing to be known appeared to me as some stretch of earth or hard marl, resisting penetration … the sea advances insensibly in silence, nothing seems to happen, nothing moves, the water is so far off you hardly hear it … yet it finally surrounds the resistant substance.

— A. Grothendieck, *Récoltes et Semailles*

What this book is

This is a book about a single question, and about the machinery that question demands.

The question is the one Robert Langlands asked in his letter to André Weil in January 1967: is there a correspondence, governed by the dual group, between the automorphic representations of a reductive group over a global field and the representations of the Galois group of that field? Sixty years of mathematics have converted that question from a speculation into the organising principle of number theory, and have proved, in an accumulation of deep and difficult special cases, that the answer is yes. They have not proved it in general. They have not, this book will argue, even produced a framework in which the general statement is *the kind of thing that could be proved*.

The thesis of this treatise is that the obstruction is not a missing lemma. It is a category error committed at the foundations, and inherited by every subsequent approach.

The classical Langlands correspondence is stated as a *bijection of sets*: a set of automorphic representations on one side, a set of Galois parameters on the other, matched place by place. A bijection of sets is a shadow. It records that two collections have the same cardinality; it does not record *why*, and it does not carry the structure that would let one propagate the statement from one group to another, from one field to another, or from one place to the next. Every technique that has succeeded — Taylor–Wiles patching, the stabilised trace formula, the Langlands–Kottwitz method, Vincent Lafforgue's excursion operators, Fargues–Scholze geometrisation — has succeeded exactly to the extent that it managed, locally and by force, to reconstruct some of the missing structure. Each such reconstruction has a range, and each has a boundary beyond which it demonstrably fails. Chapter 1 charts those boundaries.

The proposal of this book is to stop reconstructing the structure and start with it.

> **Key Idea.** The Langlands correspondence is not a bijection. It is the value, at π_0, of a **trace** taken on a self-dual six-functor geometry over a single global base. The bijection is what remains when one forgets everything that made it true.

Three descents organise the whole work, and the reader who carries them will never be lost in it.

The first descent is categorification. The bijection is replaced by an equivalence of stable ∞-categories, of which the bijection is the induced map on isomorphism classes of irreducible objects. This is not new: it is the geometric Langlands programme of Beilinson and Drinfeld, brought to completion in the de Rham setting by Gaitsgory, Raskin and their collaborators in

2024. What is new here is the insistence that the equivalence, not the bijection, is the *statement*, and that the arithmetic case must be formulated the same way from the outset.

The second descent is the search for a base. The geometric theory works because a curve is a curve: it has a Ran space, a factorisation structure, a category of sheaves with a six-functor formalism. Number theory has $\operatorname{Spec} \mathbf{Z}$, which is not a curve, and which has no archimedean point in any sense a sheaf can see. Part II constructs the object this book proposes as the substitute — the **global arithmetic curve** $\mathfrak{X}_{\mathbf{Z}}$ — by gluing the syntomifications at the finite primes to an archimedean chart built from Simpson's twistor line, the Connes–Consani scaling flow, and Deninger's Riemann flow, along an adelic period halo. Whether that object exists with all the properties demanded of it is the principal open problem of this treatise, and it is labelled as such wherever it occurs.

The third descent is the trace. In a 2-category of geometries whose objects are six-functor formalisms with duality, "take the trace of Frobenius" is a well-typed operation, and the cobordism hypothesis says what it computes. Reciprocity at a finite place becomes the Lefschetz trace; reciprocity at the archimedean place becomes a regularised determinant of the twistor flow; and functoriality, which has resisted every direct assault since 1967, becomes *naturality of the trace in the dual group*. It is not a separate conjecture. It is a corollary.

What this book is not

It is not a proof of the Langlands correspondence.

The reader is owed that sentence early and plainly, because the ambition of the surrounding pages could otherwise mislead. What is offered here is a *framework*: a foundational system, a construction of the objects that system requires, a precise formulation of the correspondence within it, an explicit inventory of what must be proved, and a computational and formal-verification programme by which the inventory might be discharged. Some items in the inventory are theorems, proved here or elsewhere. Most are conjectures. A few are, in truth, programmes for constructing objects that do not yet exist.

To keep this distinction visible on every page, every non-classical assertion in this book carries an **epistemic status mark**:

Mark	Meaning
[ESTABLISHED]	Established. A theorem of the existing literature, cited, or a result proved here in full from such theorems.
[THEOREM]	Theorem of the framework. Proved in this book from stated axioms.
[CONJECTURE]	Conjecture. Precisely stated; not proved here or elsewhere.
[DESIGN]	Design. A definition, construction or design principle: a choice of formulation, justified but not provable.
[FORMAL TARGET]	Formal-verification target. Stated so that a proof assistant could check it.

Appendix F tabulates every marked statement in the book together with its dependencies, so that a reader may see at a glance which conjectures carry which consequences, and what would collapse if any one of them failed. A framework that cannot be falsified is not mathematics, and the design of this one has been governed throughout by the requirement that its central claims be refutable. §1.6 lists the specific computations and comparisons that would refute them, and Chapter 39 sets them out as a protocol.

The place of this work in the Programme

This work belongs to the Grothendieckian Programme of the WiseMotif Monograph of Mathematical Intelligence, and it is the Programme's fifth research subject, after the Riemann Hypothesis, the conjecture of Birch and Swinnerton-Dyer, the Hodge Conjecture and the *abc* conjecture. The Programme has a single aim: machine intelligence that discovers mathematical structure on its own account, under reasoning architectures held to the standard of proof. A research subject is a problem on which that aim is tested in full. The test is not whether a machine can produce text about a famous problem; it is whether the method can raise a structure around the problem, check that structure against everything already known, and say precisely where it fails.

The method is recursive mathematical discovery. A structure is proposed; it is tested against the cases in which the answer is known; what survives is abstracted into a new setting, and the question is posed again there. What does not survive is not discarded. It becomes a conjecture, a gap or a falsification test, with a statement, a status and a place in the ledger, and it is carried forward as part of the result. This book is the record of one such recursion on the Langlands programme: the global arithmetic curve and the trace on it are the structure proposed; the function-field case, the de Rham limit, the local fields, the abelian case and the curve of conductor eleven are the cases it is tested against; and the ledger of Appendix F, the falsification protocol of Chapter 39 and the gaps and problems of Chapter 40 are what the recursion leaves behind.

Grothendieck described two ways of opening a hard nut: to strike it until it breaks, or to immerse it in a rising sea until it softens and opens of itself. Most of the theorems of six decades on the arithmetic side are of the first kind: Taylor–Wiles patching, the stabilised trace formula and the Langlands–Kottwitz method are each a precise blow at one point. The geometric programme over a curve is of the second kind, and it has succeeded there. This book tries the second way over $\mathbf{Z}$. It asks what the base would have to be over which the function-field proofs have somewhere to run, writes down the requirements such a base must meet, and follows their consequences. The sea has not risen far enough: the base is specified, its local charts are established mathematics, and its global assembly rests on conjectures of the first rank of difficulty. But the specification is exact, its degenerations reproduce the known cases, and it locates the remaining difficulty in a finite, named set of statements. That is the state the book reports.

What the Programme takes from this subject is a method rather than a result. A conditional architecture of this size is useful only if its conditions are finite, named and attackable, and if the boundary between what is proved and what is proposed survives the length of the argument. The ledger, the five status marks, the falsification protocol and the companion code are the instruments by which this book tries to keep that boundary fixed. They are offered to the reader in the same spirit as the mathematics: to be checked, and where they fail, to be corrected.

The Grothendieck standard

A book with these ambitions must say what it means by invoking Grothendieck, and must be measured against him fairly.

Grothendieck's method, reduced to its mechanism, was an inversion: *replace a space by the category of sheaves upon it, and replace properties of the space by universal properties of that category*. Schemes, topoi, the six operations and motives are four applications of that one move. Its power came from the fact that the new objects were more flexible than the old, and that theorems became easier as generality increased — the "rising sea."

The foundations proposed in Part I perform a second inversion, of the same kind and at the next level. Grothendieck kept the space and built the category over it. Here the category is kept and *the space is deleted*: a **geometry** is defined to be a six-functor formalism equipped with duality and — the new datum — a **primitive trace theory**. Spaces, where they exist, are recovered by a reconstruction theorem. The reason for this second inversion is specific and not merely aesthetic: reciprocity laws are trace identities, and in a foundation where trace is a derived notion, every reciprocity law must be built by hand. In a foundation where trace is primitive, reciprocity laws are type-correct by construction.

Whether this is a more radical break than Grothendieck's is not for the author to say, and the claim would in any case be tested only by what the framework proves. What can be said is that the *pattern* is his, that it is applied here to the one problem he did not reach, and that the standard of completeness, generality and systematic exposition set by the *Éléments* has been the standard aimed at on every page.

There is one respect in which this book deliberately departs from that model. The *Éléments* were written to be read by mathematicians and checked by mathematicians. The architecture proposed here has a coherence structure — the compatibilities among six functors, dualities, traces and modalities across infinitely many places — that exceeds what human refereeing can reliably audit. Chapter 6 and Part VII therefore specify the machine layer: a type theory in which the foundations are natively expressible (Chapter 6), a constitution that keeps definition, conjecture, demonstration and computation apart (Chapter 37), certified algorithms with independent checkers (Chapter 38), a formalisation ladder on the fifteen-year critical path (Chapter 40), and a discovery loop (§1.8). This is not decoration. In the author's judgement, a programme of this size that is not machine-checkable is not, at this point in the history of the subject, a serious proposal.

How to read this book

Reader's Guide. The book is long, and almost no one should read it linearly. Chapter 1 is a complete account of the whole architecture and should be read by everyone; §1.5 and §1.6 state the Main Conjecture informally and list what the book claims and what would falsify it, and a reader who wants only the claims may stop there. The Main Conjecture is stated in final form in Chapter 19. The part summaries and chapter synopses form a *Leitfaden* through the dependencies, and *How to Read This Book* gives four suggested routes: the arithmetician's, the categorist's, the sceptic's and the computational.

Part I builds the foundations. Part II constructs the global arithmetic curve. Part III builds the gauge and spectral stacks over it and explains the defect ℓ_0. Part IV develops the trace mechanism and states the Main Conjecture. Part V treats the analytic theory — Schwartz categories, Poisson summation, the trace formula recast as an index theorem, and the Langlands–Fourier transform. Part VI works out the local theory and the arithmetic consequences. Part VII is the computational and formal-verification programme, and it assesses what has and has not been achieved and sets out the research programme that follows.

Prerequisites are stated in *How to Read This Book*. The technical demands are real: algebraic number theory and class field theory throughout; the representation theory of reductive groups over local and global fields for Parts IV, V and VI; ∞-categories and derived algebraic geometry for Parts I, III and IV; p-adic Hodge theory for Parts II and VI. Appendices A–D supply compressed but complete accounts of the less standard prerequisites, and the reader is encouraged to consult them as needed rather than in advance; Appendix E collects the notation, Appendix F is the statement ledger and Appendix G describes the reference implementation.

The five volumes

The seven parts are bound as one monograph in five volumes, with the seven appendices as the sixth cell of the series' hexagonal cover; where the text says *this volume*, it means the whole monograph.

Volume I: The Kernel and the Base (Parts I–II, Chapters 1–12). Chapter 1 states the problem and the thesis. Part I defines a geometry as a six-functor formalism with duality and a primitive trace, by the axioms KH.1–KH.28, and writes them in the type theory $\mathrm{TT}_{\mathrm{KH}}$. Part II specifies the global arithmetic curve: the syntomic charts at the finite primes, the archimedean chart, the adelic halo that glues them, the absolute Frobenius and the orientation of degree three. The local charts rest on established mathematics; the gluing and the existence of the curve are conjectures.

Volume II: Gauge and Spectral Stacks (Part III, Chapters 13–18). The two sides of the correspondence as stacks over the curve, their shifted symplectic geometry, the defect ℓ_0 and its structural explanation, the derived Hecke algebra and Koszul duality, and singular support.

Volume III: The Trace Correspondence (Part IV, Chapters 19–24). The Main Conjecture with its three equivalences and eight corollaries, cyclotomic traces, the categorical Lefschetz theorem, the spectral action, the field-theoretic formulation, and functoriality as Fourier conjugation.

Volume IV: Analysis and the Trace Formula (Part V, Chapters 25–30). Schwartz categories, Poisson summation as a gluing axiom, the trace formula as a derived index theorem, Beyond Endoscopy and the deleted boundary, L-functions and Γ-factors, Eisenstein series and the continuous spectrum.

Volume V: Consequences and the Programme (Parts VI–VII, Chapters 31–40). The p-adic and archimedean local correspondences, modularity, Fontaine–Mazur, Ramanujan and Arthur, and special values; then the constitution of the treatise, the certified algorithms, the falsification protocol, and the assessment of what has and has not been achieved, with the research programme that follows.

The appendices (A–G). Higher categories, condensed and analytic mathematics, prismatic and syntomic cohomology and trace methods, in compressed but complete form; the notation; the statement ledger; and the reference implementation.

Acknowledgements

The mathematics on which this framework rests is the work of a large community over six decades, and the debt is recorded in the bibliography and, more importantly, in Part II, where the achievements of that community are set out in what the author hopes is proper detail. The specific intellectual debts are too numerous to list here; they are acknowledged at the point of use, which is the only proper place for them.

Any errors, and all speculative claims, are the author's own.

If you are willing to read it as pure speculation I would appreciate that; if not — I am sure you have a waste basket handy.

— R. P. Langlands, letter to A. Weil, 1967

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Acknowledgements

A work of this scope is never the product of a single mind, and it would be a species of dishonesty to present it as though it were. What follows is an attempt at a proper accounting.

The intellectual debt is, first and overwhelmingly, to Robert Langlands, whose 1967 letter to André Weil posed the question that this book is about and whose subsequent writings established that the question was not a curiosity but a programme. Whatever is of value here is a footnote to that letter.

The second debt is to Alexander Grothendieck. This treatise proposes to replace the foundations that *Éléments de géométrie algébrique* and the *Séminaire de géométrie algébrique* established, and it would be a poor sort of gratitude to attempt such a thing without saying plainly that the attempt is only possible because those foundations were built first, and built so well that their limits can be seen. The reader will find, in Chapter 2 and again in Chapter 40, an explicit accounting of what the chôral framework gains and what it surrenders relative to EGA. The surrenders are real. Grothendieck's theory proved its theorems; this one, for the most part, does not yet.

The mathematical substance of this book stands on work that is not mine. The prismatic and syntomic foundations of Chapter 8 are the work of Bhargav Bhatt, Peter Scholze, Jacob Lurie and Vladimir Drinfeld, and the reader should understand that where Chapter 8 is correct it is because it is reporting their theorems, and where it is speculative the speculation is mine alone. The cyclotomic formalism of Chapters 3 and 11 is due to Marcel Bökstedt, Lars Hesselholt, Ib Madsen, Thomas Nikolaus and Peter Scholze; the presentation of TC used throughout is theirs. The geometric Langlands theory that Chapters 13, 18, 21 and 32 take as their model is the work of Vladimir Drinfeld, Dennis Gaitsgory, Sam Raskin, Dima Arinkin, Alexander Beilinson and their collaborators, whose proof of the categorical de Rham conjecture is the single most important piece of evidence that anything of the kind attempted here can be made to work. The shifted symplectic geometry of Chapters 12 and 15 is due to Tony Pantev, Bertrand Toën, Michel Vaquié and Gabriele Vezzosi, and the local structure theory to Christopher Brav, Vittoria Bussi and Dominic Joyce. The derived Hecke theory of Chapters 16 and 17, which supplies what I believe to be the sharpest available test of the framework's central numerical claim, is due to Akshay Venkatesh, Frank Calegari, David Geraghty and Søren Galatius. The analytic architecture of Part V owes its shape to James Arthur, Ngô Bao Châu, Yiannis Sakellaridis and Akshay Venkatesh, and its condensed formulation to Dustin Clausen and Peter Scholze. Laurent Fargues and Peter Scholze's geometrisation of the local correspondence is the proof of concept for the entire enterprise, and Vincent Lafforgue's excursion operators are its direct ancestor.

I have tried throughout to be exact about attribution, and where an idea in this book has a recognisable ancestor I have said so in the text rather than in a footnote. Where I have failed to do so, the failure is an error and I should be grateful to be told of it. The Notes and references at the end of every chapter are written in a discursive manner precisely so that the lineage of each idea can be traced without ambiguity.

I am grateful to the mathematical community for the extraordinary openness of its literature. Essentially every result cited in this book was available to me without cost, in final or preprint form, at the moment I needed it. That is a collective achievement, and it is worth naming.

The computational apparatus described in Part VII — the `aletheia` reference implementation, the certificate formats, and the verification harness — was developed alongside the mathematics, and the discipline it imposed changed the mathematics more than once. Several claims that appear in this book as conjectures were, at earlier stages, stated as theorems; the verification layer is what caught them. I record this because it is the strongest argument I know for the constitutional separation described in Chapter 37.

Responsibility for every error, every overreach, and every claim that turns out to be false is mine alone. The epistemic status marks that appear beside every numbered statement in this book — [E], [T], [C], [D], [F] — exist so that a reader can see at a glance where that responsibility falls most heavily. I ask the reader to use them, and to be sceptical in exactly the proportion that they indicate.

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How to Read This Book

This book is long, and it is not uniform. Some of it reports established mathematics, some of it proves new results of a modest kind, and some of it proposes a programme whose central claims are conjectural. A reader who does not know which is which will be misled. This section explains the apparatus by which the three are kept apart, and offers four routes through the material.

The status marks

Every numbered statement in this book carries one of five marks, printed beside its number.

Mark	Meaning
[ESTABLISHED]	**Established.** A theorem of the existing literature, cited precisely, or a result proved here in full from such theorems.
[THEOREM]	**Theorem of the framework.** Proved in this book from stated axioms, completely, modulo results marked [ESTABLISHED].
[CONJECTURE]	**Conjecture.** Not proved, here or elsewhere. Stated with hypotheses precise enough to be false.
[DESIGN]	**Design.** A definition, construction or design principle: a choice of formulation, justified but not provable. Carries no truth claim; may nonetheless be the wrong choice.
[FORMAL TARGET]	**Formal-verification target.** Stated so that a proof assistant could check it. It is a target of the formalisation programme of Chapters 37 and 38 and Appendix G, not a machine-checked result: the Lean 4 statement skeleton that accompanies the book is not yet compiled and has thirty-three open placeholders.

The marks are load-bearing. A statement marked [C] is not a theorem whose proof has been omitted for reasons of space; it is a conjecture, and the author does not know whether it is true. The apex claim of the book — the Main Conjecture of Chapter 19 — is marked [C], and no case of it is proved unconditionally anywhere in these pages. A reader who wants that stated more bluntly should turn to Chapter 40.

Every chapter also contains at least one block marked **Caveat**, placed where the chapter is weakest rather than where it is strongest. These are the most useful paragraphs in the book for anyone trying to decide how much of it to believe.

The four routes

The arithmetician's route. Chapter 1; Chapter 7; Chapter 16; Chapter 19; Chapters 33 to 36; Chapter 40. This route assumes familiarity with automorphic forms and Galois representations and asks only what the framework says about problems the reader already cares about. It takes the foundational machinery on trust, which is legitimate provided one remembers that the machinery is the part that is not yet built.

The categorist's route. Chapters 2 to 6; Chapters 12, 15, 18; Chapters 19 to 23; Appendix A. This route follows the formal architecture and treats the arithmetic as an application. A reader taking it should nonetheless read Chapter 7, because the entire construction is motivated by an obstruction that the categorical language does not display.

The sceptic's route, which is recommended. Chapter 40 first — the assessment, including the four hardest gaps and what is likely to be wrong. Then Appendix F, the ledger of every claim in the book with its status and its dependencies. Then Chapter 39, the falsification protocol, which lists the explicit computations that would refute the framework. Only then Chapter 1, and then whatever the reader's scepticism has been directed towards. This route is recommended because the book's principal claim on a reader's attention is the precision of its accounting, and the accounting is at the back.

The computational route. Chapter 37; Chapter 38; Appendix G; then the `aletheia` source tree distributed with this work. Nothing in the software asserts a mathematical fact; it is designed to produce certificates to be checked by an independent kernel, and Appendix G records how far the present release does so. Chapter 37 explains why that separation is a mathematical requirement and not a software-engineering convenience.

Prerequisites

The book assumes: algebraic geometry at the level of *EGA* I–III or Hartshorne plus a working acquaintance with stacks; homological algebra and the language of derived and stable ∞-categories at the level of Lurie's *Higher Topos Theory* and *Higher Algebra*, of which Appendix A gives a self-contained summary sufficient for reading; algebraic number theory including class field theory; and the classical Langlands programme at the level of the Corvallis volumes. Familiarity with prismatic cohomology, topological cyclic homology, derived symplectic geometry and geometric Langlands is *not* assumed — Chapters 8, 3, 15 and 13 respectively develop what is needed, and Appendices A to D collect the background in expository form.

No part of the book assumes the reader has read the parts before it, except that: Part III presumes Part II's construction of the base; Part IV presumes Part III's two stacks; and Parts V and VI presume the Main Conjecture as stated in Chapter 19. Cross-references are given by number throughout and are exact.

A warning about the register

The book is written in the declarative style customary in this series, in which constructions are announced rather than negotiated. That style is efficient, but in a work whose central claims are conjectural it carries a risk: a sentence of the form "we now construct $\mathfrak{X}_{\mathbf{Z}}$" reads as though the construction were complete. It is not. The status marks and the caveats exist to counteract the register, and where the two conflict the marks are authoritative.

Notation and Conventions

The following conventions are in force throughout. Departures from them are announced locally and are always explicit.

Fields, rings, and places

F denotes a global field, $\mathcal{O}_F$ its ring of integers when F is a number field, and v a place of F. The completion at v is F_v, with ring of integers $\mathcal{O}_v$, uniformiser ϖ_v, residue field k_v of cardinality q_v, and normalised absolute value $|\cdot|_v$. The adele ring is $\mathbf{A} = \mathbf{A}_F$ and the ideles $\mathbf{A}^\times$. Finite adeles are $\mathbf{A}_f$; the archimedean component is $F_\infty = F \otimes_{\mathbf{Q}} \mathbf{R}$. A place is *finite* if it is non-archimedean and *infinite* otherwise; the treatise's persistent concern is to treat the two on the same footing, and the notation deliberately does not privilege either.

X denotes a smooth projective geometrically connected curve over a finite field $\mathbf{F}_q$, with function field $F = \mathbf{F}_q(X)$. Throughout the book $X/\mathbf{F}_q$ is the **control case**: whenever an assertion is made about the arithmetic setting, the corresponding assertion over X is stated, together with its status in the literature. This convention is the principal device by which the book is kept accurate, and the reader is encouraged to use it as such.

p is a prime, usually the residue characteristic; ℓ is a prime distinct from p, used for coefficients. $\mathbf{Z}_p$, $\mathbf{Q}_p$, $\mathbf{C}_p$, $\mathbf{Z}[1/N]$ have their usual meanings. $W(k)$ is the ring of Witt vectors of k and $W_n(k)$ its length-n truncation. $\mathbf{S}$ is the sphere spectrum. $\mathbf{F}_1$ appears only in Chapter 7 and always in quotation from other authors.

Groups

G is a connected reductive group over F, assumed split unless the contrary is stated. T is a maximal torus, B a Borel containing it, W the Weyl group, $X^*(T)$ and $X_*(T)$ the character and cocharacter lattices. The dual group is $\widehat{G}$, taken over $\mathbf{C}$ or over $\overline{\mathbf{Q}}_\ell$ as the context requires; the L-group is ${}^L G = \widehat{G} \rtimes \Gamma_F$. Where the distinction matters — and it does matter, in Chapter 14 and again in Chapter 35 — the C-group of Buzzard and Gee is written ${}^C G$ and is not conflated with ${}^L G$. The centre is Z and $K_\infty \subset G(F_\infty)$ is a maximal compact subgroup.

The **defect** is

$$\ell_0 \;=\; \operatorname{rank} G(F \otimes \mathbf{R}) \;-\; \operatorname{rank} K_\infty \;-\; \operatorname{rank} Z,$$

the invariant that governs Chapter 16 and that recurs, in a different guise, in Chapters 17, 27 and 39.

Categories

All categories are ∞-categories in the sense of Lurie unless explicitly called ordinary; all limits, colimits, and adjunctions are the ∞-categorical ones; all diagrams commute up to coherent homotopy, and the coherence data are suppressed in the notation but never in the hypotheses. Sp is spectra, Cat_∞ small ∞-categories, Pr^L presentable categories with left adjoints. For a stack Y we write $\mathrm{QCoh}(Y)$, $\mathrm{IndCoh}(Y)$, $\mathrm{Shv}(Y)$ for quasi-coherent sheaves, ind-coherent sheaves, and the relevant sheaf theory; $\mathrm{IndCoh}_{\mathcal{N}}(Y)$ and $\mathrm{Shv}_{\mathcal{N}}(Y)$ denote the full subcategories with singular support in the nilpotent cone $\mathcal{N}$, as defined in Chapter 18.

The six functors are written $f^*, f_*, f_!, f^!, \otimes, \underline{\mathrm{Hom}}$, with $f^* \dashv f_*$ and $f_! \dashv f^!$. $\mathrm{Corr}(\mathcal{C})$ is the $(\infty,2)$-category of correspondences of Chapter 5. A **chôra** is a tuple $(\mathcal{C}, E, D, \mathbb{D}, \tau)$ as in Definition 2.8, and $\mathrm{Geom}^{\mathrm{tr}}$ is the 2-category of chôrai.

Traces

$\mathrm{tr}(f \mid M)$ is the categorical trace of an endomorphism f of a dualisable object M; $\mathrm{Tr}(\mathcal{C})$ is the trace of a dualisable category, i.e. its Hochschild homology. THH and TC are topological Hochschild and topological cyclic homology, the latter always in the Nikolaus–Scholze formulation, with cyclotomic structure maps φ_p. The absolute Frobenius of Chapter 11 is Φ, and $\mathrm{TC}(\Phi \mid \mathcal{C})$ denotes the Φ-twisted cyclotomic trace of the category $\mathcal{C}$, defined in Chapter 20.

The arithmetic base

$\mathfrak{X}_{\mathbf{Z}}$ is the global arithmetic curve constructed in Part II. Its finite charts are the syntomifications $\mathrm{Spf}(\mathbf{Z}_p)^{\mathrm{syn}}$ of Chapter 8; its archimedean chart is the twistor object of Chapter 9, carrying the action of the Deligne torus $\mathbb{S}$ with its splitting $\mathbb{S}(\mathbf{R}) \cong \mathbf{R}_{>0} \times S^1$; the charts are glued by the period halo of Chapter 10. Prismatisation is written $(-)^{\Delta}$ and syntomification $(-)^{\mathrm{syn}}$. The dualising object is ω and the virtual dualising complex of Chapter 15 is ω^{vir}.

$\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}})$ is the arithmetic gauge stack of Chapter 13 and $\mathrm{LocSys}_{{}^L G}(\mathfrak{X}_{\mathbf{Z}})$ the arithmetic spectral stack of Chapter 14; LocSys^{Φ} denotes the Φ-fixed locus. $\mathcal{A}(G)$ is the automorphic category of G.

Numbering and cross-reference

Statements are numbered consecutively within each chapter by a single counter, so that a definition and the theorem immediately following it are consecutive objects and a reader need not know a statement's kind in advance. Equations are numbered within chapters. Figures are numbered as *Plate n.k*, where n is the chapter; each of the forty chapters carries three, and the List of Plates gives their titles. Axioms are numbered KH.1 through KH.28 globally and are stated in Chapters 2 to 6; Appendix F indexes every occurrence.

Citations are by author and year in the text and are resolved in the bibliography. A reference given in prose without a bracketed citation is a deliberate signal that the attribution is to a body of work rather than to a single document.

Conventions of language

Spelling is British. The first person singular, where it occurs, is the author's; the first person plural is the customary mathematical one and includes the reader. "Iff" does not appear; "if and only if" is written out. "Canonical" means *equipped with the coherence data making it canonical*, and where those data are not supplied the word is not used.

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Part I

The Foundational Kernel

Correspondential geometry: the proposal that a geometry is not a space but a six-functor formalism equipped with a primitive trace theory. The axioms KH.1-KH.28 are stated and their consequences developed; weight structures are installed in place of the missing motivic t-structure; and the classical Langlands correspondence is reformulated as a statement about categorical traces before any arithmetic base has been constructed.

Nothing in this part mentions a number field. That is deliberate. The foundational kernel is stated for an arbitrary chôra, and the reader who wishes to test it against a case where every statement is a theorem should keep a smooth projective curve over a finite field in view throughout. The arithmetic base does not appear until Chapter 7, and the correspondence itself not until Chapter 19.

Chapters 1–6.

Chapter 1

The Shape of the Problem

Synopsis. This chapter states the Langlands programme in its classical form, recounts what has actually been proved in the six decades since Langlands' letter to Weil, and then argues that the residual difficulty is not a sum of unrelated technical obstacles but a single structural absence. Four obstructions are isolated and stated as mathematical assertions: the absence of a base over which the two sides of the correspondence can be compared; the absence of a Frobenius, and hence of a trace; the absence of a uniform treatment of the archimedean place; and the cohomological defect measured by the invariant ℓ_0, which halts the Taylor–Wiles method for all but a thin class of groups and fields. The chapter then announces the thesis of the treatise — that these four are one — states the Main Conjecture informally and marks it as a conjecture, explains the system of epistemic status marks that every assertion in this book carries, gives a reader's guide to the seven parts, and compares the proposal carefully with Grothendieck's refoundation of algebraic geometry. Nothing in this chapter is proved. Its purpose is to make the reader see the same shape that the author sees.

1.1 Reciprocity in its classical form

A reciprocity law is a statement that an arithmetic question with an infinite and apparently unstructured answer has, in fact, a finite and structured one. The oldest example remains the clearest. Fix an odd prime p and ask, for each odd prime $q \neq p$, whether p is a square modulo q. The answer is a function on the primes q with no evident pattern. Gauss' quadratic reciprocity law asserts that this function is nevertheless determined by the residue class of q modulo $4p$: an infinite and irregular datum is governed by a character of a finite abelian group.

Every subsequent reciprocity law has had this form. Class field theory, in the hands of Takagi, Artin, Chevalley and Tate, generalised it as far as abelian extensions permit. To a number field F one attaches the idele class group $\mathbf{A}_F^\times / F^\times$, and the Artin reciprocity map identifies its profinite completion with the Galois group $\mathrm{Gal}(F^{\mathrm{ab}}/F)$ of the maximal abelian extension; the splitting of a prime $\mathfrak{p}$ in an abelian extension E/F is read off from the image of a uniformiser at $\mathfrak{p}$.

For non-abelian extensions the analogous statement was, before 1967, not even formulated. Artin had attached to every continuous representation $\rho\colon \mathrm{Gal}(\overline{F}/F) \to \mathrm{GL}_n(\mathbf{C})$ an L-function $L(s,\rho)$, defined as an Euler product over the primes of F whose local factor at an unramified $\mathfrak{p}$ is $\det(1 - \rho(\mathrm{Frob}_\mathfrak{p})\, N\mathfrak{p}^{-s})^{-1}$, and had conjectured that $L(s,\rho)$ extends to an entire function of s when ρ contains no trivial constituent. For abelian ρ this follows from class field theory, because $L(s,\rho)$ is then a Hecke L-function and Hecke, and later Tate, had proved the analytic continuation. For non-abelian ρ there was no mechanism at all. Analytic continuation is a statement about a function of a complex variable; Galois theory produces no functions of a complex variable. Something had to intervene.

Langlands' proposal, communicated in a seventeen-page handwritten letter to Weil in January 1967 (Langlands 1967) and developed in the 1970 Washington lecture "Problems in the theory of automorphic forms" (Langlands 1970), was that the intervening object is

the theory of automorphic representations, and that the dictionary between the two sides is mediated by a group — the dual group — constructed from the root datum of the group one starts with. The standard reference formulation, and the one this treatise takes as normative when it speaks of "the classical statement", is Borel's Corvallis article (Borel 1979), which appeared in the two volumes edited by Borel and Casselman (Borel and Casselman 1979a, 1979b).

1.1.1 The dual group and the L-group

The construction that makes the programme possible is the passage from a reductive group to its dual.

Definition 1.1 (Dual group and L-group). [ESTABLISHED]

Let F be a global or local field and let G be a connected reductive group over F. Fix a separable closure $\overline{F}$ and a maximal torus of $G_{\overline{F}}$ contained in a Borel subgroup. The based root datum of $G_{\overline{F}}$ is the quadruple $\Psi(G) = (X^*, \Delta, X_*, \Delta^\vee)$ consisting of the character lattice, a choice of simple roots, the cocharacter lattice, and the corresponding simple coroots. The *dual group* $\widehat{G}$ is the connected reductive group over $\mathbf{C}$ — unique up to inner automorphism — whose based root datum is $\Psi(\widehat{G}) = (X_*, \Delta^\vee, X^*, \Delta)$, obtained by interchanging characters with cocharacters and roots with coroots. The action of $\mathrm{Gal}(\overline{F}/F)$ on $\Psi(G)$ by outer automorphisms transports to an action on $\widehat{G}$ preserving a chosen splitting, and the *L-group* is the semidirect product

$$ {}^{L}G \;=\; \widehat{G} \rtimes \mathrm{Gal}(\overline{F}/F), $$

or the same with $\mathrm{Gal}(\overline{F}/F)$ replaced by the Weil group W_F, according to the context. The construction is due to Langlands (Langlands 1967); the interchange $\mathrm{SO}_{2n+1} \leftrightarrow \mathrm{Sp}_{2n}$, $\mathrm{GL}_n \leftrightarrow \mathrm{GL}_n$, $\mathrm{PGL}_n \leftrightarrow \mathrm{SL}_n$ is the standard table of examples.

Two features of Definition 1.1 deserve emphasis. First, the dual group is defined over $\mathbf{C}$ regardless of what F is; it is a combinatorial shadow of G and carries no arithmetic, all the arithmetic sitting in the Galois action. Second, the passage $G \mapsto \widehat{G}$ is an involution on root data, which is to say that it is a duality in the strict sense — and the entire programme is the assertion that this combinatorial duality lifts to a duality between two categories of analytic and arithmetic objects. This treatise takes the view that the lift exists because the combinatorial duality is the decategorified trace of a genuine duality in a six-functor formalism, and that the failure to see the trace has been the obstacle.

1.1.2 Global reciprocity

Conjecture 1.2 (Global reciprocity). [CONJECTURE]

Let F be a number field and G a connected reductive group over F. There should exist a map

$$ \pi \;\longmapsto\; \sigma_\pi $$

from the set of isomorphism classes of cuspidal automorphic representations $\pi = \bigotimes_v{}' \pi_v$ of $G(\mathbf{A}_F)$ — those occurring discretely in $L^2(G(F)Z(\mathbf{A}_F)\backslash G(\mathbf{A}_F))$ and orthogonal to the image of parabolic induction — to the set of $\widehat{G}(\mathbf{C})$-conjugacy classes of admissible homomorphisms

$$ \sigma\colon \mathcal{L}_F \;\longrightarrow\; {}^{L}G, $$

where $\mathcal{L}_F$ denotes the conjectural global Langlands group of F, subject to the following compatibility: for every place v of F the restriction $\sigma_\pi|_{\mathcal{L}_{F_v}}$ corresponds to π_v under the local correspondence of Theorem

1.4. The map should have finite fibres, and for $G = \mathrm{GL}_n$ it should be injective, so that a cuspidal π is determined by σ_π and conversely. Admissibility requires that σ commute with the projections to $\mathrm{Gal}(\overline{F}/F)$, that the image of a Frobenius lift be semisimple, and that σ be continuous for the relevant topology.

Three warnings must be attached to Conjecture 1.2, and this treatise attaches them once here and takes them as standing.

The first is that the global Langlands group $\mathcal{L}_F$ has no construction. It is a device introduced so that the conjecture may be stated in a form parallel to the local one; Langlands' original formulation avoided it, and Arthur's later formulation makes it a hypothetical topological group whose finite-dimensional complex representations should classify cuspidal automorphic representations of GL_n. For F a function field $\mathcal{L}_F$ may be taken to be the Weil group, and the conjecture becomes a theorem; for F a number field the group is not known to exist, and one of the reasons this treatise exists is the belief that $\mathcal{L}_F$ is not a group at all but the fundamental group of an object of higher-categorical nature, whose π_0 has been mistaken for the whole.

The second is that the quantifier structure is delicate. The conjecture as stated is a conjecture about a *map*, not about a bijection; even for GL_n the image is not all admissible σ but only those satisfying an irreducibility condition, and for general G the fibres are the L-packets, which are finite but whose internal structure is governed by the centraliser group $S_\sigma = \pi_0(Z_{\widehat{G}}(\mathrm{im}\,\sigma))$ and requires Arthur's refinement (Arthur 2013) to describe.

The third is that in practice one replaces $\mathcal{L}_F$ by $\mathrm{Gal}(\overline{F}/F)$ and demands ℓ-adic coefficients. The ℓ-adic form of the conjecture is the form in which almost all known cases are proved. It is weaker in one respect (it says nothing about the archimedean place except through Hodge–Tate weights) and stronger in another (it produces an actual Galois representation rather than a homomorphism from a group that may not exist).

1.1.3 Functoriality

Conjecture 1.3 (Functoriality). [CONJECTURE]

Let F be a global field, let H and G be connected reductive groups over F, and let

$$\eta\colon {}^{L}H \longrightarrow {}^{L}G$$

be a homomorphism of L-groups commuting with the projections to $\mathrm{Gal}(\overline{F}/F)$. Then to every automorphic representation π of $H(\mathbf{A}_F)$ there should be associated an automorphic representation $\Pi = \eta_*\pi$ of $G(\mathbf{A}_F)$, characterised by the condition that for almost all v the Satake parameter of Π_v is the image under η of the Satake parameter of π_v. Consequently $L(s,\Pi,r) = L(s,\pi,r\circ\eta)$ for every representation r of ${}^{L}G$.

Functoriality is the deeper of the two conjectures, and Langlands regarded it as such from the beginning (Langlands 1970). Reciprocity for GL_n follows from functoriality applied to the inclusion of the image of a Galois representation, but functoriality contains far more: base change, automorphic induction, the transfer of endoscopic data, symmetric power lifts, and, in the generality stated, the Ramanujan conjecture and the Artin conjecture together. It is also the conjecture with the least visible mechanism. Reciprocity at least resembles a classification statement. Functoriality asserts that an arbitrary map of complex algebraic groups induces a map of automorphic spectra, and no known method produces such a map from the datum of η alone; the known cases are proved one η at a time, by methods that do not generalise.

The reader should note the following, which is central to the argument of this treatise: *functoriality has the form of a naturality statement*. If the automorphic spectrum $\mathcal{A}(G)$ were the

value at G of a functor defined on a category whose morphisms include the η, functoriality would not be a conjecture but a definition unwinding. That it is instead a deep conjecture records the fact that no such functor is known. Constructing one is the substance of Part IV.

1.1.4 The local correspondence

Theorem 1.4 (Local Langlands for GL_n). [ESTABLISHED]
Let F be a non-archimedean local field. There is a bijection

$$\mathrm{Irr}(\mathrm{GL}_n(F)) \xrightarrow{\sim} \{\, n\text{-dimensional Frobenius-semisimple Weil–Deligne representations of } W_F \,\}$$

between isomorphism classes of irreducible admissible representations of $\mathrm{GL}_n(F)$ over $\mathbf{C}$ and isomorphism classes of Frobenius-semisimple representations of the Weil–Deligne group $W_F' = W_F \ltimes \mathbf{G}_a$, compatible with twisting, central characters, duality, and the L- and ε-factors of pairs. For F of positive characteristic this is a theorem of Laumon, Rapoport and Stuhler (Laumon et al. 1993); for F of characteristic zero it was proved by Harris and Taylor (Harris and Taylor 2001) by realising the correspondence in the vanishing cycles of Lubin–Tate space, and independently by Henniart (Henniart 2000) by a numerical characterisation using ε-factors.

The Weil–Deligne group deserves a word, since its role is often obscured. The Weil group W_F is the dense subgroup of $\mathrm{Gal}(\overline{F}/F)$ generated by the inertia subgroup and an integral power of Frobenius, given the topology in which inertia is open. Its finite-dimensional complex representations are too few: they are all semisimple with finite image on inertia, and hence cannot see the Steinberg representation. The Weil–Deligne group repairs this by adjoining a unipotent parameter, the *monodromy operator* N, satisfying $\mathrm{Ad}(\sigma(w))N = \|w\| N$. The pair (σ, N) is exactly the datum extracted from an ℓ-adic representation by Grothendieck's monodromy theorem. That the local correspondence requires N — that is, requires a *derived* or *filtered* enhancement of the naive Galois datum — is the first appearance in the classical theory of a phenomenon this treatise treats as fundamental: the correct spectral object is not a set of homomorphisms but a stack with a nilpotent direction. For general G the local correspondence is not a bijection but a partition of $\mathrm{Irr}(G(F))$ into finite L-packets indexed by parameters, with the members of a packet indexed by characters of a component group. It has been established in this form for many groups, and, in a formulation of a different nature, for all G by Fargues and Scholze (Fargues and Scholze 2026), to which §1.2 returns.

1.1.5 The L-function dictionary

Conjecture 1.5 (The L-function dictionary). [CONJECTURE]
Let F be a number field, G reductive over F, π a cuspidal automorphic representation of $G(\mathbf{A}_F)$, and $r\colon {}^L G \to \mathrm{GL}_N$ a finite-dimensional representation. Define

$$L(s, \pi, r) \;=\; \prod_{v} L(s, \pi_v, r),$$

the product over all places, with the local factor at a finite unramified v given by $\det(1 - r(c_{\pi_v})\, q_v^{-s})^{-1}$ in terms of the Satake parameter $c_{\pi_v} \in {}^L G(\mathbf{C})$, at a ramified finite place by the Weil–Deligne recipe applied to the local parameter, and at an archimedean place by a product of factors $\Gamma_{\mathbf{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ and $\Gamma_{\mathbf{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ determined by the archimedean parameter. Then $L(s, \pi, r)$ converges in a right half-plane, continues meromorphically to $\mathbf{C}$, is entire unless r contains the trivial representation, and satisfies a functional equation

$$L(s, \pi, r) \;=\; \varepsilon(s, \pi, r)\, L(1 - s, \widetilde{\pi}, r^{\vee}),$$

with ε a product of local ε-factors, all but finitely many equal to 1. Moreover if σ is the parameter attached to π by Conjecture 1.2, then $L(s, \pi, r) = L(s, r \circ \sigma)$, the Artin L-function of the composite.

The last clause of Conjecture 1.5 is the reciprocity law proper: it says that the analytic object built from harmonic analysis on $G(\mathbf{A}_F)$ coincides with the arithmetic object built from Galois theory. The Artin conjecture follows, because automorphic L-functions of cuspidal representations are entire by the theory of Godement–Jacquet and its descendants; the Ramanujan conjecture follows from the purity of the associated Galois representations.

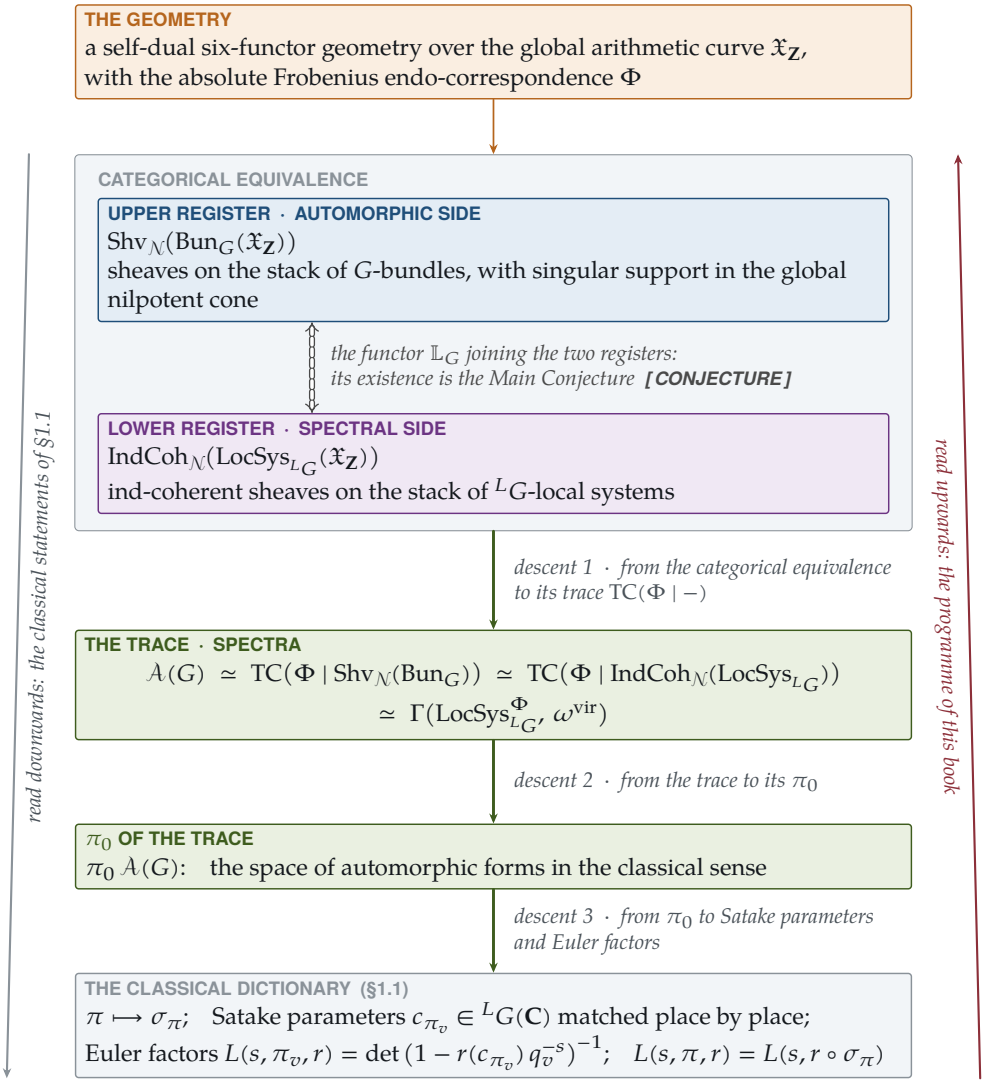


Plate 1.1. The three descents and the trace mechanism: the classical Langlands correspondence exhibited as the π_0-shadow of a categorical trace on a self-dual six-functor geometry over the global arithmetic curve. The upper register shows the automorphic side, sheaves on a stack of bundles; the lower register shows the spectral side, ind-coherent sheaves on a stack of local systems; the two are joined by a functor whose existence is the content of the Main Conjecture. The three vertical arrows are the descents: from the categorical equivalence to its trace, from the trace to its π_0, and from π_0 to the classical dictionary of Satake parameters and Euler factors. Reading the diagram downwards recovers the statements of §1.1; reading it upwards is the programme of this book.

What it does not show: a proof of the joining functor: the passage between the two registers is the Main Conjecture, marked [C], and no case of it is proved unconditionally in this book.

The plate reproduced as Plate 1.1 is the conceptual summary of what this treatise proposes, and the reader who wishes to know the destination before the journey may study it now. Three descents are indicated. The first descends from a six-functor formalism over a global arithmetic curve to a categorical trace; the second from the trace to its homotopy groups; the third from π_0 of the trace to the classical correspondence as stated in Conjecture 1.2. The assertion is that the classical statement is a shadow, and that its difficulty is the difficulty of any shadow: it retains the content but discards the mechanism.

1.2 Six decades of theorems

It would be dishonest to present the Langlands programme as an unsolved problem in the way that the Riemann hypothesis is an unsolved problem. A very great deal has been proved. This section records what, with attention to the exact content of each theorem, because the diagnosis in §1.4 depends on knowing precisely where each method stops.

1.2.1 The abelian case and the analytic foundation

Theorem 1.6 (Tate's thesis). [ESTABLISHED]
Let F be a global field and χ a unitary Hecke character of $\mathbf{A}_F^\times/F^\times$. For a Schwartz–Bruhat function Φ on $\mathbf{A}_F$ the zeta integral

$$Z(s,\Phi,\chi) \;=\; \int_{\mathbf{A}_F^\times} \Phi(x)\,\chi(x)\,|x|^{s}\,d^\times x$$

converges for $\mathrm{Re}(s)$ large, continues meromorphically to $\mathbf{C}$, and satisfies the functional equation $Z(s,\Phi,\chi) = Z(1-s,\widehat{\Phi},\chi^{-1})$, where $\widehat{\Phi}$ is the adelic Fourier transform. The Hecke L-function $L(s,\chi)$ is the ratio of Z to an explicit local normalisation, and inherits continuation and functional equation, including the archimedean Γ-factors (*Tate 1950*).

Tate's thesis is the case $G = \mathrm{GL}_1$ of the whole programme, and it is proved. It is worth pausing on *how*. The proof is Poisson summation on $\mathbf{A}_F$ for the discrete subgroup F, combined with the self-duality of $\mathbf{A}_F$. The functional equation is the Fourier involution; the Γ-factors emerge from the archimedean local zeta integrals on the same footing as the Euler factors emerge from the non-archimedean ones. This is the only place in the classical theory where the archimedean place is on the same footing as the finite ones, and the reason is that the mechanism — self-duality plus a summation formula — is available at all places at once. That observation is the seed of Problem 1.19 and of the programme of Part V.

1.2.2 Small rank, and the trace formula

Jacquet and Langlands (Jacquet and Langlands 1970) established the correspondence between automorphic representations of GL_2 and of the multiplicative group of a quaternion algebra, by comparing the two trace formulae term by term. This is the prototype of every subsequent comparison argument. Its content is that the geometric sides of the two trace formulae, which are sums over conjugacy classes, can be matched by a correspondence of conjugacy classes together with a matching of orbital integrals; the spectral sides then agree, and the spectral side is what one wants.

Base change for GL_2 over a cyclic extension of prime degree was proved by Langlands (Langlands 1980) and extended to GL_n by Arthur and Clozel (Arthur and Clozel 1989), again by the trace formula, here in twisted form. The role of base change in the proof of Fermat's last theorem, and in essentially every subsequent modularity lifting theorem, is difficult to

overstate: it is the mechanism by which one changes the field until the local conditions become tractable.

Arthur's classification (Arthur 2013) determines the discrete automorphic spectrum of quasi-split symplectic and special orthogonal groups in terms of self-dual cuspidal representations of general linear groups, by a sustained comparison of twisted and stable trace formulae. It is conditional in its published form on the stabilisation of the twisted trace formula, which was subsequently completed by Moeglin and Waldspurger. What it depends on is the fundamental lemma.

Theorem 1.7 (The fundamental lemma). [ESTABLISHED]
Let G be an unramified reductive group over a non-archimedean local field F of equal characteristic, and let H be an endoscopic group of G. Then the orbital integrals of the unit of the spherical Hecke algebra of G at a strongly regular semisimple element match, after multiplication by the Langlands–Shelstad transfer factors, the stable orbital integrals of the unit of the spherical Hecke algebra of H at a corresponding element (Ngô 2010). The transfer factors are those of Langlands and Shelstad (Langlands and Shelstad 1987); the passage from equal to mixed characteristic is due to Waldspurger (Waldspurger 2008).

Ngô's proof is the point at which the programme's dependence on geometry becomes explicit and unavoidable. The orbital integrals are interpreted as counting points on fibres of the Hitchin fibration for a curve over a finite field (Ngô 2006); the matching is deduced from a support theorem for the decomposition of the direct image. There is no known proof by local harmonic analysis alone. A statement about integrals over p-adic groups was proved by a cohomological argument over a curve, and the argument requires the curve.

1.2.3 Function fields

Over a global function field the programme is, for general linear groups, a theorem.

Theorem 1.8 (Global Langlands for GL_n over a function field). [ESTABLISHED]
Let X be a smooth projective geometrically connected curve over $\mathbf{F}_q$ with function field F, and let $\ell \nmid q$. There is a bijection between the set of cuspidal automorphic representations of $\mathrm{GL}_n(\mathbf{A}_F)$ with fixed central character and the set of irreducible rank-n $\overline{\mathbf{Q}}_\ell$-local systems on an open subset of X with the corresponding determinant, compatible with the local correspondence at every place, and matching L- and ε-factors. The case $n = 2$ is due to Drinfeld (Drinfeld 1983), building on his theory of elliptic modules (Drinfeld 1974); the general case is due to L. Lafforgue (Lafforgue 2002).

The proof proceeds by constructing moduli stacks of shtukas — vector bundles on X equipped with a modification relating them to their Frobenius twist — compactifying them, and computing the Grothendieck–Lefschetz trace of Frobenius on their ℓ-adic cohomology. The Ramanujan–Petersson conjecture for GL_n over function fields follows, by Deligne's purity theorem (Deligne 1974a, 1980). The entire argument is a trace computation on a cohomology theory attached to a geometric object over a base, and the base is the curve X.

V. Lafforgue's work (Lafforgue 2018) extends the construction of Galois parameters to arbitrary reductive groups.

Theorem 1.9 (Excursion operators and the automorphic-to-Galois direction). [ESTABLISHED]
Let X be a smooth projective geometrically connected curve over $\mathbf{F}_q$ with function field F, and let G be a connected reductive group over F. There is a canonical decomposition of the space of cuspidal automorphic forms on $G(\mathbf{A}_F)$ indexed by $\widehat{G}(\overline{\mathbf{Q}}_\ell)$-conjugacy classes of semisimple homomorphisms $\mathrm{Gal}(\overline{F}/F) \to {}^L G(\overline{\mathbf{Q}}_\ell)$ that are almost everywhere unramified, compatible with the Satake isomorphism

at unramified places. The decomposition is effected by a commutative algebra of excursion operators *constructed from the cohomology of moduli of shtukas with several legs* (*Lafforgue 2018*).

The excursion operators are, structurally, partial traces: one takes a cohomology class on a moduli space of shtukas with legs labelled by representations of ${}^{L}G$, and evaluates it against a conjugation-invariant function on $({}^{L}G)^{n}$, the resulting operator depending only on that function. §1.4 will argue that this is precisely the operation that cannot presently be performed over $\operatorname{Spec}\mathbf{Z}$.

1.2.4 Number fields: modularity

Theorem 1.10 (Modularity of elliptic curves over $\mathbf{Q}$). [ESTABLISHED]
Every elliptic curve E over $\mathbf{Q}$ is modular: there is a cuspidal newform f of weight 2 and level $N = \operatorname{cond}(E)$ with rational Hecke eigenvalues such that $L(E,s) = L(f,s)$, equivalently such that the ℓ-adic Tate module of E is isomorphic to the ℓ-adic representation attached to f for every ℓ. The semistable case is due to Wiles (*Wiles 1995*) *with the commutative-algebra criterion supplied jointly with Taylor* (*Taylor and Wiles 1995*)*; the general case is due to Breuil, Conrad, Diamond and Taylor* (*Breuil et al. 2001*).

The Taylor–Wiles method deserves a precise description, because Problem 1.20 is a statement about exactly where it fails. One wishes to prove that a deformation ring R, parametrising lifts of a residual Galois representation subject to local conditions, is isomorphic to a Hecke algebra $\mathbf{T}$ acting on a space of automorphic forms. There is a surjection $R \twoheadrightarrow \mathbf{T}$; one wants injectivity. The method proceeds by adjoining auxiliary sets Q_n of primes at which one allows extra ramification, in such a way that the enlarged deformation rings R_{Q_n} are generated by a bounded number of elements over a fixed power series ring, and the enlarged spaces of automorphic forms remain free of the same rank over the corresponding group rings. Patching produces a module over a power series ring in q variables, of depth q, and one concludes by a commutative-algebra argument that $R = \mathbf{T}$. The argument requires that the automorphic side contribute a module whose depth matches the dimension of the ring acting.

1.2.5 Locally, p-adically, and geometrically

Scholze's theory of perfectoid spaces (Scholze 2012) and the attendant p-adic Hodge theory (Scholze 2013) altered what is possible in the p-adic setting, by making tilting available as a systematic tool and by providing a framework in which the cohomology of infinite-level Shimura varieties can be analysed. Among the consequences is the construction of Galois representations attached to torsion classes in the cohomology of locally symmetric spaces for GL_n over a CM field — a result of Scholze's, published in the *Annals* in 2015, which this treatise cites in prose since the bibliography key has not been fixed.

Theorem 1.11 (Geometrisation of the local correspondence). [ESTABLISHED]
Let F be a non-archimedean local field and G a connected reductive group over F. There is a stack Bun_G of G-bundles on the Fargues–Fontaine curve (*Fargues and Fontaine 2018*)*, a six-functor formalism for ℓ-adic sheaves on it, and a construction attaching to every irreducible smooth representation of $G(F)$ a semisimple L-parameter, compatible with the classical correspondence where the latter is known, and characterised by the action of excursion operators built from Hecke correspondences on Bun_G* (*Fargues and Scholze 2026*).

The Fargues–Scholze theorem is the closest existing analogue, at the local level and for one place at a time, of what this treatise proposes globally. Its mechanism is that of Theorem

1.9: a geometric object playing the role of a curve, a six-functor formalism on a moduli stack of bundles over it, and excursion operators extracted from the cohomology. The Fargues–Fontaine curve supplies, for a single p, the "curve" that the number field lacks.

Theorem 1.12 (Geometric Langlands over a curve in characteristic zero). [ESTABLISHED]
Let X be a smooth projective curve over an algebraically closed field of characteristic zero and G a reductive group. There is an equivalence of DG *categories*

$$\mathrm{D\text{-}mod}(\mathrm{Bun}_G(X)) \;\simeq\; \mathrm{IndCoh}_{\mathcal{N}}(\mathrm{LocSys}_{L_G}(X))$$

compatible with Hecke actions and with Eisenstein series, where $\mathcal{N}$ is the global nilpotent cone and singular support is taken in the sense of Arinkin and Gaitsgory (Arinkin and Gaitsgory 2015). This is the theorem of Gaitsgory, Raskin and their collaborators, established in the five-paper series (Gaitsgory and Raskin 2024a, 2024b; Arinkin, Beraldo, Campbell, et al. 2024; Campbell et al. 2024; Arinkin, Beraldo, Chen, et al. 2024).

Warning 1.13 (What the geometric theorem does not say). [ESTABLISHED]
Theorem 1.12 is a theorem about curves over an algebraically closed field of characteristic zero. It is not a theorem about number fields, and it does not imply any case of Conjecture 1.2. The relationship between the geometric and the arithmetic statements is one of analogy mediated by the function-field case: over $\mathbf{F}_q$ one may hope to take the trace of Frobenius on a categorical equivalence and recover a numerical correspondence, and it is that operation — categorical trace — that this treatise proposes to import into the arithmetic setting. To describe the geometric theorem as "the Langlands correspondence, proved" is an error of exactly the kind this treatise's status marks exist to prevent.

1.2.6 The shape of the record

Observation 1.14 (The pattern in the record). [DESIGN]
Reviewing the theorems above, one observes the following. Every unconditional proof of a *global* case of Conjecture 1.2 in which the correspondence is established in both directions and for all n simultaneously — Theorem 1.8, Theorem 1.9 — takes place over a global function field and proceeds by computing a trace of Frobenius on the cohomology of a moduli space fibred over the curve. Every proof of a global case over a number field — Theorem 1.10 and its descendants — proceeds instead by deformation theory and patching, establishes only the Galois-to-automorphic direction for representations already known to be geometric, and is confined to situations in which a certain cohomological invariant vanishes. Every advance in the local theory of the last fifteen years — Theorem 1.11 — proceeds by manufacturing a curve where none was given. The record therefore suggests, though it does not prove, that the operative distinction is not between local and global, nor between characteristic p and characteristic zero, but between settings that possess a curve carrying a six-functor formalism with a trace, and settings that do not.

1.3 The curve of conductor eleven

Abstraction in this subject is cheap, and the reader is entitled to demand that any proposed framework be tested against an object that can be held in the hand. This section takes the simplest elliptic curve over $\mathbf{Q}$ of positive conductor, follows every piece of its arithmetic through the classical correspondence, and then states — with no proof, and with explicit acknowledgement of what is being assumed — which component of the framework would be responsible for recovering which fact.

1.3.1 The curve, its points, and its form

Example 1.15 (The elliptic curve of conductor 11). [ESTABLISHED]
Let E be the elliptic curve over $\mathbf{Q}$ given by the Weierstrass equation

$$E\colon \quad y^2 + y \;=\; x^3 - x^2 - 10x - 20,$$

the curve labelled $11a1$ in the tables of Cremona and in the L-functions and modular forms database. Its discriminant is -11^5 and its conductor is $N = 11$. It is the modular curve $X_0(11)$, the first curve of genus one to occur as a modular curve of prime level. Its Mordell–Weil group is $E(\mathbf{Q}) \cong \mathbf{Z}/5\mathbf{Z}$, generated by the point $(5, 5)$; the rank is zero. Its reduction at 11 is split multiplicative, with Tamagawa number $c_{11} = 5$, and it has good reduction at every other prime.

Counting points on the reduction $\overline{E}$ modulo small primes gives $\#\overline{E}(\mathbf{F}_2) = 5$ and $\#\overline{E}(\mathbf{F}_3) = 5$, hence $a_2 = 2 + 1 - 5 = -2$ and $a_3 = 3 + 1 - 5 = -1$. Continuing,

$$\begin{aligned} &a_2 = -2, \quad a_3 = -1, \quad a_5 = 1, \quad a_7 = -2, \quad a_{11} = 1,\\ &a_{13} = 4, \quad a_{17} = -2, \quad a_{19} = 0, \quad a_{23} = -1. \end{aligned}$$

The associated newform is

$$\begin{aligned} f \;&=\; q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2 \\ &=\; q - 2q^2 - q^3 + 2q^4 + q^5 + 2q^6 \\ &\qquad - 2q^7 - 2q^9 - 2q^{10} + q^{11} + \cdots, \end{aligned}$$

the unique normalised cuspidal newform of weight 2 and level $\Gamma_0(11)$, and $a_p(f) = a_p(E)$ for all p. The completed L-function is

$$\Lambda(E, s) \;=\; 11^{s/2}\,(2\pi)^{-s}\,\Gamma(s)\,L(E, s), \qquad \Lambda(E, s) = \Lambda(E, 2 - s),$$

with sign $+1$; the central value is $L(E, 1) \approx 0.2538418609$, non-zero, consistent with rank zero. The real period is $\Omega \approx 1.2692093$ and $L(E, 1)/\Omega = 1/5$, which is the value predicted by the conjecture of Birch and Swinnerton-Dyer with trivial Tate–Shafarevich group and $c_{11} = 5$, since $|\mathrm{Sha}(E)| \cdot c_{11} / |E(\mathbf{Q})_{\mathrm{tors}}|^2 = 5/25 = 1/5$. The residual representation $\overline{\rho}_{E,5}$ is reducible, being an extension of the trivial character by the mod-5 cyclotomic character on account of the rational 5-torsion; for every $\ell \neq 5$ the representation $\overline{\rho}_{E,\ell}$ is surjective onto $\mathrm{GL}_2(\mathbf{F}_\ell)$.

Every clause of Example 1.15 is classical and independently verifiable. The verification is worth carrying out in public, because the whole programme is an assertion that two lists of numbers, computed by procedures with nothing in common, coincide. On the left one counts points on a curve over finite fields; on the right one expands an infinite product in a formal variable. The table below records both computations for the first ten primes.

p	$\#\overline{E}(\mathbf{F}_p)$	$a_p = p + 1 - \#\overline{E}(\mathbf{F}_p)$	coefficient of q^p in f	agree
2	5	-2	-2	yes
3	5	-1	-1	yes
5	5	1	1	yes
7	10	-2	-2	yes
11	—	1	1	yes
13	10	4	4	yes
17	20	-2	-2	yes
19	20	0	0	yes
23	25	-1	-1	yes
29	30	0	0	yes

The left-hand column is obtained by enumerating the solutions of $y^2 + y = x^3 - x^2 - 10x - 20$ in $\mathbf{F}_p \times \mathbf{F}_p$ and adding one for the point at infinity; at $p = 11$ the reduction is singular and the entry is omitted, the value $a_{11} = 1$ being the local datum of split multiplicative reduction rather than a point count on a smooth curve. The right-hand column is obtained by expanding

$$f \;=\; q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2$$

as a power series. Expanding the two factors separately and multiplying, one finds

$$q - 2q^2 - q^3 + 2q^4 + q^5 + 2q^6 - 2q^7 - 2q^9 - 2q^{10} + q^{11} - 2q^{12} + 4q^{13} + 4q^{14} + \cdots,$$

and the multiplicativity relations $a_{mn} = a_m a_n$ for $(m, n) = 1$ and $a_{p^2} = a_p^2 - p$ for $p \nmid 11$ are visible in the expansion: for example $a_4 = a_2^2 - 2 = 4 - 2 = 2$, $a_6 = a_2 a_3 = (-2)(-1) = 2$, $a_9 = a_3^2 - 3 = 1 - 3 = -2$, $a_{10} = a_2 a_5 = -2$, and $a_{14} = a_2 a_7 = 4$. These relations are exactly the statement that the Hecke operators T_p generate a commutative algebra acting on a one-dimensional space; they are invisible on the point-counting side, where each a_p is an unrelated tally.

The ten agreements in the table are, individually, small facts. Collectively they are the content of Theorem 1.10 for this curve, and it should be emphasised that no proof of that theorem proceeds by comparing the two columns. The comparison is a consequence, not a method. What one wants — and what this treatise proposes to supply — is a single object of which both columns are computations of the same invariant, so that the agreement becomes a tautology rather than a coincidence to be established.

1.3.2 What the framework must explain

Consider now how the correspondence organises the data of Example 1.15. On the automorphic side one has $\pi = \bigotimes_v' \pi_v$, a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbf{A}_{\mathbf{Q}})$ generated by f: for $p \neq 11$ the local component π_p is the unramified principal series with Satake parameter the conjugacy class $\mathrm{diag}(\alpha_p, \beta_p)$ satisfying $\alpha_p + \beta_p = a_p p^{-1/2}$ and $\alpha_p \beta_p = 1$ after unitary normalisation; at $p = 11$ it is the Steinberg representation twisted by an unramified quadratic character; at the archimedean place π_∞ is the discrete series of weight 2, whose $(\mathfrak{g}, K)$-module has the infinitesimal character corresponding to Hodge–Tate weights $\{0, 1\}$. On the spectral side one has $\sigma = \rho_{E,\ell}$, the ℓ-adic Tate module, a two-dimensional representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ unramified outside 11ℓ, with $\det = \chi_\ell$ the cyclotomic character, crystalline at $\ell \neq 11$ and semistable non-crystalline at 11, where its Weil–Deligne parameter has $N \neq 0$. The correspondence identifies the two lists of local data place by place; Theorem 1.10 is the assertion that such a π exists.

Now the accurate statement of what the framework of this treatise is designed to do with this example. Every item below is a *design intention*, not a theorem of this book.

For each prime $p \neq 11$, the Euler factor $(1 - a_p p^{-s} + p^{1-2s})^{-1}$ is to be recovered as the local contribution at p of the endo-correspondence Φ to a categorical trace, computed on the syntomic chart of the global arithmetic curve $\mathfrak{X}_{\mathbf{Z}}$; the component φ_p of Φ plays the role that geometric Frobenius plays over $\mathbf{F}_q$, and the numerator $1 - a_p p^{-s} + p^{1-2s}$ is the characteristic polynomial of φ_p acting on a two-dimensional graded piece. At $p = 11$ the local factor degenerates to $(1 - 11^{-s})^{-1}$; in the framework this is the local term attached to a parameter with $N \neq 0$, that is, to a point of the spectral stack lying in the nilpotent cone rather than at a smooth point, and the drop in degree of the Euler factor is the drop in the rank of the φ_{11}-invariants. This is Part VI's business.

The Γ-factor $(2\pi)^{-s}\Gamma(s)$ and the conductor factor $11^{s/2}$ are to be recovered from the archimedean chart, as a regularised determinant of the scaling flow $\mathbf{R}_{>0}$ acting on the archimedean fibre — the mechanism stated as Corollary C3 of the framework (Appendix F) and developed in Part V. The functional equation $\Lambda(E, s) = \Lambda(E, 2 - s)$ is to be the self-duality of the six-functor formalism over $\mathfrak{X}_{\mathbf{Z}}$, expressed as Verdier duality exchanging s with $2 - s$; this is the structural analogue of Poisson summation in Theorem 1.6 and of Poincaré duality in the function-field case.

The modular form f itself is to appear not as an input but as an element of π_0 of the categorical trace $\mathrm{TC}(\Phi \mid \mathrm{Shv}_{\mathcal{N}}(\mathrm{Bun}_{\mathrm{GL}_2}(\mathfrak{X}_{\mathbf{Z}})))$; the statement that its Hecke eigenvalues agree with the traces of Frobenius on $\rho_{E,\ell}$ is to be the statement that the two computations of the same trace — one on the automorphic side, one on the spectral side — agree, because they are traces of the same endomorphism of the same dualisable object computed in two dual presentations.

The reducibility of $\overline{\rho}_{E,5}$ is to be visible as membership in the Eisenstein stratum of the spectral stack: the cofibre of the natural map $q_! \to q_*$ along the Eisenstein correspondence is precisely the obstruction to the residual representation being irreducible, and the failure of ambidexterity discussed in §1.4.5 is concentrated on that stratum. Finally, because $\ell_0 = 0$ for GL_2 over $\mathbf{Q}$ — see Problem 1.20 — the classical Taylor–Wiles patching applies without modification, and the framework must reproduce it rather than replace it; a framework that could not recover Theorem 1.10 in the case $\ell_0 = 0$ would be refuted.

Status and Caveat. It must be said plainly what the framework does *not* do for Example 1.15. It offers no independent access to the special value $L(E, 1)/\Omega = 1/5$, and no route to the Birch–Swinnerton-Dyer conjecture: the trace formalism developed here computes L-functions as characteristic polynomials of an endomorphism, and characteristic polynomials do not know about Tate–Shafarevich groups without a further input — a regulator, or a positivity — that this treatise does not supply. It offers no new proof of modularity for E: the modularity of $11a1$ was known to Eichler and Shimura, and the general theorem is Wiles'. It offers no mechanism for the Riemann hypothesis for $L(E, s)$, because it produces no positivity on the trace. What it offers is a single mechanism from which the Euler factors, the Γ-factor, the functional equation, the conductor and the local degeneration at 11 all descend, rather than five mechanisms that happen to agree. Whether that is worth having is for the reader to decide; whether it is true is the subject of Conjecture 1.24, which is not proved in this book.

1.3.3 The same computation over a function field

Set $q = 5$, let $X = \mathbf{P}^1_{\mathbf{F}_5}$ with coordinate t, and let $F = \mathbf{F}_5(t)$ be its function field. Consider the elliptic curve

$$\mathcal{E}\colon \quad y^2 \;=\; x^3 + x + t$$

over F. Its discriminant is proportional to $4 + 27t^2$, which reduces modulo 5 to $4 + 2t^2$; since $2t^2 = -4$ has no solution in $\mathbf{F}_5$ — the squares modulo 5 being $\{0, 1, 4\}$ and 3 not among them — the curve has good reduction at every degree-one place of the affine line, and the analogue of the table of §1.3.1 may be computed by the same enumeration. The places of degree one are $v_c = (t - c)$ for $c \in \mathbf{F}_5$, with residue field $\mathbf{F}_5$, and the reduction of $\mathcal{E}$ at v_c is $y^2 = x^3 + x + c$.

v	reduction	$\#\mathcal{E}_v(\mathbf{F}_5)$	$a_v = 5 + 1 - \#\mathcal{E}_v(\mathbf{F}_5)$
(t)	$y^2 = x^3 + x$	4	2
$(t-1)$	$y^2 = x^3 + x + 1$	9	-3
$(t-2)$	$y^2 = x^3 + x + 2$	4	2
$(t-3)$	$y^2 = x^3 + x + 3$	4	2
$(t-4)$	$y^2 = x^3 + x + 4$	9	-3

Each a_v satisfies the Hasse bound $|a_v| \le 2\sqrt{5} < 4.48$. The sum of the five entries is zero, and the affine surface $S\colon y^2 = x^3 + x + t$ over $\mathbf{F}_5$ therefore has

$$\#S(\mathbf{F}_5) \;=\; \sum_{c \in \mathbf{F}_5} (5 - a_{v_c}) \;=\; 25 \;=\; q^2.$$

So far the two computations — this one and the one in §1.3.1 — are formally identical: a list of local point counts indexed by the places of a global field. The difference appears at the next step, and it is total.

Over F the numbers a_v are not merely a list. The Grothendieck–Lefschetz trace formula asserts that

$$\#S(\mathbf{F}_{q^n}) \;=\; \sum_i (-1)^i \operatorname{tr}\left(\operatorname{Frob}_q^n \mid H^i_c(S_{\overline{\mathbf{F}}_q}, \overline{\mathbf{Q}}_\ell)\right),$$

so that every entry of the table, and every entry of every table one could compute over every finite extension, is a trace of a power of a *single* endomorphism acting on a *single* finite-dimensional graded vector space. The L-function

$$L(\mathcal{E}, T) \;=\; \prod_v \det\left(1 - \operatorname{Frob}_v \, T^{\deg v} \mid \mathcal{F}\right)^{-1}, \qquad \mathcal{F} = R^1\pi_*\overline{\mathbf{Q}}_\ell,$$

is a polynomial in $T = q^{-s}$, by Grothendieck's rationality theorem, of a degree computable from the conductor and the genus, and its inverse roots are the eigenvalues of that endomorphism on H^1. The functional equation is Poincaré duality for X; the Riemann hypothesis for $L(\mathcal{E}, T)$ is Deligne's purity theorem (Deligne 1974a). The rationality, the functional equation and the bound are not three theorems; they are three properties of one linear operator.

The shtuka picture makes this structural rather than incidental. A shtuka of rank n with legs at $x_1, \dots, x_k$ on X is a vector bundle $\mathcal{V}$ of rank n on X together with a modification

$$\mathcal{V} \dashrightarrow {}^{\tau}\mathcal{V} \;=\; (\operatorname{Frob}_X \times \mathrm{id})^*\mathcal{V}$$

which is an isomorphism away from the legs and has prescribed relative position at each leg. The moduli of such objects, $\mathrm{Sht}_{n,k}$, fibres over X^k, and its ℓ-adic cohomology carries

commuting actions of the Hecke algebra and of $\mathrm{Gal}(\overline{F}/F)^k$. Both Theorem 1.8 and Theorem 1.9 are obtained by taking traces on this cohomology; the excursion operators of the latter are constructed by letting the legs collide and separate, an operation that requires X^k — that is, requires the base to have a self-product with more than one factor.

1.3.4 The two columns

The comparison may now be set out. Every entry in the left column is a theorem or a construction in the published literature; every entry in the right column is either absent or replaced by an ad hoc substitute, and the third column names what this treatise proposes in its place.

Function field $F = \mathbf{F}_q(X)$	Number field F	Proposed replacement
the curve X, a scheme of dimension 1 over $\mathbf{F}_q$	$\operatorname{Spec} O_F$, no base beneath it	$\mathfrak{X}_{\mathbf{Z}}$, Part II
the base field $\mathbf{F}_q$	absent; $\mathbf{Z}$ is initial	the sphere spectrum $\mathbf{S}$
the self-product $X \times_{\mathbf{F}_q} X$	$\operatorname{Spec}\mathbf{Z} \times \operatorname{Spec}\mathbf{Z} = \operatorname{Spec}\mathbf{Z}$	$\mathbf{Z} \otimes_{\mathbf{S}} \mathbf{Z}$, Proposition 1.17
geometric Frobenius Frob_q, a global endomorphism	local φ_p only, no global endomorphism	the endo-correspondence Φ
$\mathrm{Sht}_{n,k} \to X^k$, moduli of shtukas	no moduli of shtukas	mapping stacks over $\mathfrak{X}_{\mathbf{Z}}$, Part III
excursion operators from X^k	no X^k, hence none	trace of Φ on a dualisable category
Poincaré–Verdier duality on X	Artin–Verdier duality, but with the archimedean places inserted	self-duality of the six-functor formalism
Deligne purity, hence Ramanujan	no purity theorem available	weight structure, Part IV
all places alike; no archimedean place	one or more archimedean places, treated separately	the archimedean chart, Part V
L-function a polynomial in q^{-s}	L-function transcendental, entire at best	characteristic series of Φ

The right-hand column of that table is not a list of ten problems. It is one problem listed ten times, and the argument for that claim is the business of §1.4.

1.3.5 What fails under transport

Before making the argument it is worth being exact about the failure, because the failure has been studied by several authors from several directions and the partial successes are instructive. Suppose one attempts to transport the proof of Theorem 1.8 to $\operatorname{Spec}\mathbf{Z}$, step by step. The following six things go wrong, in this order.

1. *There is no base field.* The curve X is a scheme over $\mathbf{F}_q$, and almost every construction in the proof is relative to that structure morphism. $\operatorname{Spec}\mathbf{Z}$ has no structure morphism to anything smaller. The search for a base — for "$\mathbf{F}_1$" — is old; Borger's proposal that descent to $\mathbf{F}_1$ be encoded by a λ-ring structure, that is, by commuting lifts of all the Frobenii (Borger 2009), is the most structurally disciplined version, and it does supply a candidate base; the "below $\operatorname{Spec}\mathbf{Z}$" constructions of Toën and Vaquié (Toën and Vaquié 2009) supply another, by working with monoidal categories in place of rings. What neither has supplied is a six-functor formalism over that base with a duality; the λ-structure organises the Frobenii but does not by itself make anything dualisable.
2. *There is no product $X \times X$.* Even granted a base, one needs the self-product, and by Proposition 1.17 the scheme-theoretic self-product of $\operatorname{Spec}\mathbf{Z}$ is $\operatorname{Spec}\mathbf{Z}$ itself. Deninger's programme (Deninger 1998, 2002) proposed instead a dynamical system — a foliated

three-manifold with a flow whose closed orbits are the primes — on which the missing cohomology would be an infinite-dimensional space with a regularised determinant. The arithmetic predictions of that picture are strikingly accurate, and the expected shape of the archimedean factor is one of them; what has not been constructed is the space itself, nor a formalism in which its cohomology is defined rather than postulated.

3. *There is no global Frobenius.* Each prime supplies φ_p on its own p-adic chart, and these do not glue to an endomorphism of anything. Connes' spectral realisation of the zeros of ζ (Connes 1999) and the later scaling site of Connes and Consani (Connes and Consani 2016) construct a space carrying a genuine flow whose trace reproduces the Weil explicit formula, with the archimedean term included. That is a real achievement and this treatise is indebted to it. Its limitation, for present purposes, is that the construction is analytic and one-dimensional — it sees GL_1 — and no version of it is known that carries a six-functor formalism in which non-abelian G could be accommodated.
4. *There is no perfect obstruction theory of the right dimension.* On the spectral side one wants $\mathrm{LocSys}_{{}^L G}$ to be a derived stack with a (-1)-shifted symplectic structure, since it is that structure which produces a virtual dualising sheaf and hence a trace. By the Alexandrov–Kontsevich–Schwarz–Zaboronsky construction, as developed by Pantev, Toën, Vaquié and Vezzosi (Pantev et al. 2013), such a structure on a mapping stack $\mathrm{Map}(\mathfrak{X}, B^L G)$ requires the source to be oriented of degree three. Derived algebraic geometry, in the form given it by Toën and Vezzosi (Toën and Vezzosi 2005, 2008), supplies the machinery, and their later work on Chern characters and traces (Toën and Vezzosi 2015) supplies the trace formalism in the derived setting; what none of it supplies is the oriented source. The number-theoretic candidate — the compactified $\mathrm{Spec}\, O_F$ — is oriented of degree three in the étale-cohomological sense of Artin–Verdier duality (Artin and Verdier 1973), but that duality holds for the arithmetic site with archimedean places adjoined by hand and has never been extended to a six-functor formalism in which the sheaves one needs are objects.
5. *There is no motivic t-structure.* The function-field argument uses purity and the weight filtration, which come from Deligne's theory (Deligne 1980) and ultimately from the Frobenius eigenvalues. Over $\mathbf{Z}$ the corresponding statements would follow from a motivic t-structure on the triangulated category of mixed motives, whose existence is equivalent to Grothendieck's standard conjectures (Grothendieck 1969) and is open. This treatise does not prove the standard conjectures and does not assume them; it replaces the t-structure by a weight structure in Bondarko's sense (Bondarko 2010), which is a weaker datum and is part of the axiom system rather than a theorem.
6. *There is no compact object at the archimedean place.* Every trace computation requires compact or dualisable objects. At a finite place the compact objects of the relevant category are supplied by the smooth representations of a p-adic group with their finitely generated modules; at the archimedean place the natural category of $(\mathfrak{g}, K)$-modules has no comparable compactness, and the constructions of Scholze's programme — perfectoid spaces (Scholze 2012), diamonds (Scholze 2017), the Fargues–Fontaine curve (Fargues and Fontaine 2018) and the geometrisation (Fargues and Scholze 2026) — are, by construction, p-adic. They manufacture, for a single prime, exactly the curve that Problem 1.16 says is missing; they do not manufacture one at ∞, and Scholze has been explicit that the archimedean analogue is not available. Recent work on analytic stacks (Clausen and Scholze 2024) narrows the gap by putting real and p-adic analytic geometry in one framework, and Part V is written on the assumption that this is the right ambient setting; but the compact objects at ∞ remain to be produced.

Each of these six programmes obtained something real, and none obtained a global object carrying all three inputs of Principle 1.22 at once. The diagnosis of §1.4 is that this is not six failures but one.

1.4 Four obstructions, and one

This is the central section of the chapter. Four difficulties are isolated. Each is stated as a mathematical assertion, not as a mood; each is accompanied by references to the literature that documents it; and each is marked with a status indicating whether it is a citable theorem or a diagnosis offered by the author. The reader who accepts the four statements but rejects the synthesis in §1.4.5 will still, I hope, find the four worth having isolated.

1.4.1 The missing base

Problem 1.16 (The missing base). [DESIGN]

Over a global function field $F = \mathbf{F}_q(X)$ the automorphic side and the spectral side are both categories of sheaves on stacks living over the single scheme X: automorphic forms are functions on $\mathrm{Bun}_G(X)(\mathbf{F}_q)$, obtained as the trace of Frobenius on $\mathrm{Shv}(\mathrm{Bun}_G(X))$, and Galois parameters are points of $\mathrm{LocSys}_{^L G}(X)$. The comparison between them is effected by correspondences over X — Hecke correspondences, shtuka moduli — and by the sheaf-theoretic operations of pullback, pushforward and duality relative to X. Over a number field F there is no such X. The scheme $\mathrm{Spec}\, O_F$ is one-dimensional and has no non-trivial morphism to a base other than $\mathrm{Spec}\,\mathbf{Z}$; and $\mathrm{Spec}\,\mathbf{Z}$ is terminal in schemes, so that

$$\mathrm{Spec}\,\mathbf{Z} \times_{?} \mathrm{Spec}\,\mathbf{Z} \;=\; \mathrm{Spec}(\mathbf{Z} \otimes_{\mathbf{Z}} \mathbf{Z}) \;=\; \mathrm{Spec}\,\mathbf{Z},$$

the diagonal being an isomorphism. Consequently there is no self-product over which to form a Hecke correspondence with two distinct legs, no notion of a graph of Frobenius meeting the diagonal transversally, and no relative duality. The assertion is that every known proof of a global reciprocity statement in the strong form of Theorem 1.8 uses the base essentially, and that no substitute for it exists in the category of schemes.

The observation is old, and §1.3.5 has already recorded the principal responses to it; the analogy between primes and knots developed by Mazur, Morishita and others (Mazur 1973; Morishita 2012) should be added to that list. The following proposition records the precise sense in which the diagonal fails to be a divisor.

Proposition 1.17 (Failure of the diagonal to be a divisor). [ESTABLISHED]

In the category of schemes the diagonal $\Delta\colon \mathrm{Spec}\,\mathbf{Z} \to \mathrm{Spec}\,\mathbf{Z} \times_{\mathrm{Spec}\,\mathbf{Z}} \mathrm{Spec}\,\mathbf{Z}$ *is an isomorphism, so its conormal sheaf and its self-intersection vanish. In the category of* $\mathbf{E}_\infty$*-rings, by contrast, the absolute tensor square is* $\mathbf{Z} \otimes_{\mathbf{S}} \mathbf{Z}$*, where* $\mathbf{S}$ *is the sphere spectrum, and the derived self-intersection of the diagonal is*

$$\Delta^* \Delta_* O \;\simeq\; \mathrm{THH}(\mathbf{Z}),$$

topological Hochschild homology, whose homotopy groups were computed by Bökstedt: $\pi_0\,\mathrm{THH}(\mathbf{Z}) = \mathbf{Z}$*,* $\pi_{2i-1}\,\mathrm{THH}(\mathbf{Z}) = \mathbf{Z}/i$ *for* $i \geq 1$*, and* $\pi_{2i}\,\mathrm{THH}(\mathbf{Z}) = 0$ *for* $i > 0$ *(Bökstedt et al. 1993). The self-intersection is therefore non-trivial once one leaves the category of ordinary rings.*

Proposition 1.17 is the first indication of the direction this treatise takes. The base is missing in schemes; it is not missing in spectra, and the price of admitting it is a homotopical foundation. Part II is devoted to constructing, conjecturally, an object $\mathfrak{X}_{\mathbf{Z}}$ that plays over $\mathbf{Z}$ the

role that X plays over $\mathbf{F}_q$, and to identifying the three candidate constructions — the Arakelov compactification with an archimedean fibre, the absolute prismatisation of Bhatt and Lurie (Bhatt and Lurie 2022b, 2022a), and the $\mathbf{F}_1$-theoretic or λ-ring approach of Borger (Borger 2009) — as three charts on a single object.

1.4.2 The missing trace

Problem 1.18 (The missing trace). [DESIGN]

Every unconditional proof of a global reciprocity statement over a function field is, at its core, the computation of a trace. In Theorem 1.8 it is the Grothendieck–Lefschetz trace of the geometric Frobenius on the compactly supported ℓ-adic cohomology of a moduli space of shtukas; in Theorem 1.9 it is a family of partial traces, the excursion operators, indexed by invariant functions on powers of $\widehat{G}$; in Theorem 1.7 it is a trace on the cohomology of the fibres of the Hitchin map. In each case the trace is well-defined because there is an endomorphism — Frobenius — of a dualisable object in a symmetric monoidal category, namely the cohomology of a proper or suitably compactified space. Over $\operatorname{Spec}\mathbf{Z}$ there is no endomorphism playing the role of Frobenius: the primes p furnish local Frobenii φ_p acting on the p-adic charts, but there is no single endomorphism of a global object restricting to all of them, and no category in which the global object is known to be dualisable. The assertion is that the absence of a *global* trace, rather than the absence of local Frobenii, is what prevents the transfer of the function-field arguments.

Two remarks sharpen this. First, the local Frobenii do assemble in one known context: topological cyclic homology. The cyclotomic structure on THH is exactly the datum of compatible Frobenius maps $\varphi_p\colon \mathrm{THH}(R) \to \mathrm{THH}(R)^{tC_p}$ for all p simultaneously, and TC is the equaliser that extracts the invariants of this system (Nikolaus and Scholze 2018). Historically this structure arose from the cyclotomic trace of Bökstedt, Hsiang and Madsen and from the Witt-vector computations of Hesselholt and Madsen (Bökstedt et al. 1993; Hesselholt and Madsen 1997). It is not an accident that TC appears on the right-hand side of the Main Conjecture; it appears because it is the only known construction that takes all the φ_p at once and produces a global invariant.

Second, dualisability is a real condition and not a formality. The categories appearing on both sides of the correspondence are large — presentable $\mathbf{E}_\infty$-categories with no finiteness — and the trace of an endomorphism of a large category is not defined without further structure. The theory of continuous K-theory and traces on dualisable presentable categories, due to Efimov (Efimov 2024), and the theory of nuclear modules of Clausen and Scholze (Clausen and Scholze 2019, 2024), are what make the right-hand side of Conjecture 1.24 even a candidate for being well-formed. The point to retain is that "take the trace" is a demand with a precise cost, and the cost is dualisability.

1.4.3 The missing archimedean place

Problem 1.19 (The missing archimedean place). [DESIGN]

In the classical formulation the local factor of an automorphic L-function at a finite place v is determined by the local representation π_v through the Satake or Weil–Deligne recipe, which is uniform in v and is a statement about the same kind of object at every v. At an archimedean place the local factor is a product of $\Gamma_{\mathbf{R}}$ and $\Gamma_{\mathbf{C}}$ factors, determined by the archimedean parameter through the archimedean local Langlands correspondence of Langlands, which classifies $(\mathfrak{g}, K)$-modules by homomorphisms $W_{\mathbf{R}} \to {}^L G$. The machinery on the two sides has

no common foundation: on the finite side one has smooth representations of a totally disconnected group and, geometrically, ℓ-adic sheaves on a stack of bundles; on the archimedean side one has Harish-Chandra modules, asymptotic expansions and the analytic theory of the discrete series. There is no known six-functor formalism whose values at the archimedean place are $(\mathfrak{g}, K)$-modules and whose values at a finite place are smooth representations, and consequently there is no known geometric object of which the completed L-function is the characteristic polynomial. The assertion is that the Γ-factor is, in every existing treatment except Tate's thesis, an insertion rather than a deduction.

The exception is instructive and is the reason Theorem 1.6 was stated above with its mechanism attached. For GL_1 the archimedean factor is produced by the same zeta integral as the finite factors, and the functional equation is a single Fourier involution on $\mathbf{A}_F$; the uniformity is available precisely because the adele ring is self-dual, and the self-duality does not distinguish the places. For GL_n, $n \geq 2$, the Godement–Jacquet integral achieves the same uniformity for the standard L-function, and the Braverman–Kazhdan proposal (Braverman and Kazhdan 2000) attempts to extend it to a general representation r of ${}^L G$ by constructing a γ-function as the Fourier transform of a suitable kernel. That proposal remains open in general; its difficulty is exactly the difficulty of Problem 1.19. The treatment offered here, in Part V, is to construct an archimedean chart on which the group $\mathbb{S} = \mathbf{R}_{>0} \times S^1$ acts, and to obtain the Γ-factor as a regularised determinant of the generator of the scaling flow, so that φ_p and the archimedean flow become two components of a single endo-correspondence Φ.

1.4.4 The defect

The fourth obstruction is different in kind from the first three: it is not a diagnosis but a theorem, and it says exactly where the most successful method in the number-field case stops.

Problem 1.20 (The defect ℓ_0). [ESTABLISHED]

Let F be a number field, G a connected reductive group over F, and $X = G(F)\backslash G(\mathbf{A}_F)/K_\infty K_f$ the associated locally symmetric space. Set

$$q_0 \;=\; \tfrac{1}{2}\bigl(\dim(G(F\otimes\mathbf{R})/K_\infty) - \operatorname{rank}\bigr), \qquad \ell_0 \;=\; \operatorname{rank}_{\mathbf{R}} G(F\otimes\mathbf{R}) - \operatorname{rank}_{\mathbf{R}} K_\infty Z_\infty,$$

so that tempered cuspidal cohomology is concentrated in the ℓ_0+1 degrees $q_0, q_0+1, \dots, q_0+\ell_0$. Then $\ell_0 = 0$ if and only if X admits a complex structure making it a Shimura variety in the relevant cases, and the Taylor–Wiles method in its classical form — in which a patched module is free of finite rank over a power series ring — applies only when $\ell_0 = 0$. For $\ell_0 > 0$ the patched module has depth ℓ_0 less than the dimension of the ring acting on it, and the numerical criterion of (Taylor and Wiles 1995) fails. Examples with $\ell_0 > 0$: GL_2 over an imaginary quadratic field ($\ell_0 = 1$); GL_n over $\mathbf{Q}$ for $n \geq 3$ ($\ell_0 = \lfloor (n-1)/2 \rfloor$); GL_2 over a field with a real and a complex place. The derived modification of the method, in which the patched object is a complex and the conclusion is an isomorphism of derived deformation rings with derived Hecke algebras, is due to Calegari and Geraghty (Calegari and Geraghty 2018), and remains conditional on conjectures about the existence and local properties of Galois representations attached to torsion classes.

Proposition 1.21 (The defect as a homotopy-theoretic invariant). [ESTABLISHED]

With notation as in Problem 1.20, Venkatesh and his collaborators have shown that the cohomology of the locally symmetric space, localised at a suitable non-Eisenstein maximal ideal, carries an action of a graded algebra $\Lambda^\bullet(\mathbf{Z}^{\ell_0})$ — an exterior algebra on ℓ_0 generators in degree one — under which it becomes free, so that the multiplicity $\ell_0 + 1$ of the tempered spectrum is accounted for by an exterior

algebra rather than by an accident of the archimedean component. The algebra is realised both by derived Hecke operators (Venkatesh 2019) and, on the spectral side, by the tangent complex of a derived deformation ring (Galatius and Venkatesh 2018); the compatibility of the two realisations is the motivic conjecture of Prasanna and Venkatesh (Prasanna and Venkatesh 2019).

The reader should now see why Problem 1.20 is called a cohomological rather than a technical difficulty. The Taylor–Wiles method does not fail because the patching is hard to arrange; it fails because the automorphic side supplies a module of the wrong depth, and the depth deficit equals ℓ_0, which is determined by the real Lie group $G(F \otimes \mathbf{R})$ and its maximal compact subgroup. It is an archimedean invariant. That the obstruction to a p-adic method should be measured by an invariant of the real place is a hint, and this treatise takes it as such: Problem 1.20 and Problem 1.19 are, on the thesis advanced below, the same obstruction seen from two sides.

1.4.5 One obstruction

Key Idea. The four problems of §1.4 are not four difficulties but one. The single difficulty is the absence, over $\operatorname{Spec}\mathbf{Z}$, of a *dualisable trace-theoretic geometry*: a six-functor formalism on a global object $\mathfrak{X}_{\mathbf{Z}}$, self-dual in the sense that Verdier duality is an equivalence with the expected shift, carrying a distinguished endo-correspondence Φ whose components are the Frobenii φ_p at the finite places and the scaling flow $\mathbb{S} = \mathbf{R}_{>0} \times S^1$ at the archimedean place, and on which the relevant categories of sheaves are dualisable so that $\operatorname{Tr}(\Phi \mid -)$ is defined. Supplying such a geometry dissolves the four simultaneously: the base is $\mathfrak{X}_{\mathbf{Z}}$ itself; the trace is $\operatorname{Tr}(\Phi \mid -)$; the archimedean place is a chart of $\mathfrak{X}_{\mathbf{Z}}$ on the same footing as the p-adic charts, so that Γ-factors are computed rather than inserted; and the defect ℓ_0 is the rank of the compact torus in the archimedean gauge group, so that the exterior algebra of Proposition 1.21 is the cohomology of that torus and the derived structure that Calegari and Geraghty must impose by hand is present from the start. The four obstructions are one because they are four symptoms of the absence of one object.

The claim in the box is the thesis of this treatise.

1.4.6 The unifying principle

Principle 1.22 (Single-obstruction principle). [DESIGN]
Let $\mathcal{C}$ be a symmetric monoidal ∞-category and Φ an endomorphism of a dualisable object M of $\mathcal{C}$. The trace $\operatorname{Tr}(\Phi \mid M)$ is defined by the composite

$$\mathbf{1} \xrightarrow{\text{coev}} M \otimes M^\vee \xrightarrow{\Phi \otimes \mathrm{id}} M \otimes M^\vee \xrightarrow{\tau} M^\vee \otimes M \xrightarrow{\text{ev}} \mathbf{1},$$

and requires exactly three inputs: a monoidal structure, a duality $M \simeq M^\vee$ up to twist, and an endomorphism. In the function-field setting these are supplied respectively by the tensor structure on ℓ-adic sheaves over X, by Poincaré–Verdier duality for the proper morphism $\operatorname{Bun}_G(X) \to \operatorname{Spec}\mathbf{F}_q$ after truncation, and by geometric Frobenius. The principle asserted here is that Problem 1.16 is the failure of the first, Problem 1.18 the failure of the third, Problem 1.19 the failure of the second at one place, and Problem 1.20 the numerical measure of the failure of the second — so that all four are failures of the trace to be definable, and any construction supplying the three inputs supplies all four repairs at once.

The reader will object that Principle 1.22 is a reformulation rather than an argument, and the objection is just: nothing has been proved, and the status mark *D* records this. Its value, if it has any, is that it converts four research programmes into one, and that the one is of a kind for which techniques have recently become available.

1.4.7 Ambidexterity as the technical form

The duality input in Principle 1.22 admits a sharper formulation in terms of a condition familiar from higher algebra: ambidexterity, the coincidence of left and right adjoints to pullback.

Conjecture 1.23 (Ambidexterity–Compactness Principle). [CONJECTURE]
Let $\mathfrak{X}_{\mathbf{Z}}$ be the global arithmetic curve of Part II, G a reductive group scheme over it, and $q\colon \mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}}) \to \mathfrak{X}_{\mathbf{Z}}$ the structure morphism. After Whittaker renormalisation and restriction of singular support to the global nilpotent cone $\mathcal{N}$, the natural transformation

$$q_! \longrightarrow q_*$$

is an equivalence on $\mathrm{Shv}_{\mathcal{N}}(\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}}))$. Equivalently, the category $\mathrm{Shv}_{\mathcal{N}}(\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}}))$ is dualisable in the sense of Efimov and self-dual under the resulting pairing, and Φ acts on it by a trace-class endomorphism. Each of the four problems of §1.4 is an instance of the failure of this equivalence in a setting where the object $\mathfrak{X}_{\mathbf{Z}}$ has not been constructed.

The formulation via $q_! \to q_*$ is the one that makes contact with existing technology. In the geometric setting the corresponding statement is a theorem, and is one of the principal inputs to Theorem 1.12. In the p-adic setting the six-functor formalisms of Mann (Mann 2022), of Heyer and Mann (Heyer and Mann 2024), and Scholze's abstract axiomatisation (Scholze 2025), together with the general framework of Gallauer (Gallauer 2021), supply the ambient language in which Conjecture 1.23 can even be posed. What is missing is the global object over which to pose it.

1.4.8 What would follow

Suppose Conjecture 1.23 and the construction of $\mathfrak{X}_{\mathbf{Z}}$ were in hand. The following would then be available, and the reader may treat the list as a specification of what Parts III to VI attempt to deliver. Reciprocity becomes a computation of π_0 of a trace, because the two presentations — automorphic and spectral — of the same dualisable object give the same trace. Functoriality becomes naturality: an L-homomorphism $\eta\colon {}^L H \to {}^L G$ induces a map of spectral stacks, hence a map of trace spectra, and the lift is $\mathrm{Lift}_\eta = \mathfrak{F}_G^{-1} \circ \eta_* \circ \mathfrak{F}_H$, where $\mathfrak{F}$ denotes the correspondence functor. The Γ-factor becomes a regularised determinant; the Ramanujan conjecture a purity statement for the weight filtration on the trace; the Fontaine–Mazur conjecture the statement that a parameter defined on a punctured neighbourhood extends across a divisor; and the Arthur–Selberg trace formula an index theorem, equating a spectral decomposition of a trace with a geometric localisation of the same trace. Each of these is developed as a corollary of the Main Conjecture in Part IV, and each is marked *C*.

1.5 The Main Conjecture, stated informally

The following is the central assertion of the treatise. It is stated here informally, with the symbols explained in prose; it is stated properly, with all hypotheses, in Chapter 19. No case of it is proved unconditionally in this book.

Conjecture 1.24 (Main Conjecture (informal form)). [CONJECTURE]

Let G be a connected reductive group scheme over the global arithmetic curve $\mathfrak{X}_{\mathbf{Z}}$ of Part II, with dual group $\widehat{G}$ and L-group ${}^{L}G$, and let Φ denote the absolute Frobenius endo-correspondence of $\mathfrak{X}_{\mathbf{Z}}$. Then there are canonical equivalences

$$\begin{aligned} \mathcal{A}(G) &\simeq \mathrm{TC}(\Phi \mid \mathrm{Shv}_{\mathcal{N}}(\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}}))) \\ &\simeq \mathrm{TC}(\Phi \mid \mathrm{IndCoh}_{\mathcal{N}}(\mathrm{LocSys}_{{}^{L}G}(\mathfrak{X}_{\mathbf{Z}}))) \\ &\simeq \Gamma(\mathrm{LocSys}_{{}^{L}G}^{\Phi}, \omega^{\mathrm{vir}}) \end{aligned}$$

of spectra, compatible with the Hecke action on the first two terms, with the action of excursion operators on the third, and with the local correspondences at every place of $\mathfrak{X}_{\mathbf{Z}}$, including the archimedean place.

Each symbol requires explanation, and the explanations are the content of the conjecture.

$\mathcal{A}(G)$ denotes the *automorphic spectrum* of G: a spectrum, not a vector space, whose homotopy groups in degree zero recover the space of automorphic forms in the classical sense and whose higher homotopy groups encode the derived structure that Venkatesh's work (Venkatesh 2019) and Proposition 1.21 show to be present. The passage from a vector space to a spectrum is what permits $\ell_0 > 0$ to be a statement about homotopy groups rather than a failure of a numerical criterion.

$\mathfrak{X}_{\mathbf{Z}}$ is the global arithmetic curve, the object whose construction Part II attempts, conjecturally a stack in a category of analytic or spectral geometry, with one chart for each place of $\mathbf{Q}$ — prismatic or syntomic at each finite p, and an archimedean chart carrying the action of $\mathbb{S} = \mathbf{R}_{>0} \times S^1$ — glued along an adelic overlap. It is expected to have geometric dimension two but orientation degree three, the extra unit coming from the circle factor; that mismatch is the source of the (-1)-shifted symplectic structure on the spectral side.

Φ is the *absolute Frobenius endo-correspondence*: not an endomorphism of $\mathfrak{X}_{\mathbf{Z}}$ but a correspondence $\mathfrak{X}_{\mathbf{Z}} \leftarrow \mathfrak{Z} \rightarrow \mathfrak{X}_{\mathbf{Z}}$, restricting on the chart at p to the cyclotomic Frobenius φ_p of (Nikolaus and Scholze 2018) and on the archimedean chart to the scaling flow. That Φ is a correspondence rather than a map is essential and is the reason TC rather than a naive fixed-point construction appears.

$\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}})$ is the moduli stack of G-bundles on $\mathfrak{X}_{\mathbf{Z}}$; its $\mathbf{F}_1$-points, so to speak, are the classical adelic double cosets $G(\mathbf{Q})\backslash G(\mathbf{A})/K$. $\mathrm{Shv}_{\mathcal{N}}$ denotes sheaves with singular support in the global nilpotent cone $\mathcal{N}$, in the sense of Arinkin and Gaitsgory (Arinkin and Gaitsgory 2015); the restriction is what makes the category dualisable and is not optional. $\mathrm{LocSys}_{{}^{L}G}(\mathfrak{X}_{\mathbf{Z}})$ is the derived stack of ${}^{L}G$-local systems on $\mathfrak{X}_{\mathbf{Z}}$, which by the dimension count above carries a (-1)-shifted symplectic structure in the sense of Pantev, Toën, Vaquié and Vezzosi (Pantev et al. 2013), and $\mathrm{IndCoh}_{\mathcal{N}}$ denotes ind-coherent sheaves with the corresponding singular-support condition.

TC is topological cyclic homology in the formulation of Nikolaus and Scholze (Nikolaus and Scholze 2018), applied not to a ring but to the pair consisting of a dualisable category and the endo-correspondence Φ; it is the trace construction that takes all the φ_p at once. Finally $\mathrm{LocSys}_{{}^{L}G}^{\Phi}$ is the derived fixed locus of Φ on the spectral stack — the arithmetic analogue of the set of $\mathbf{F}_q$-rational points — and ω^{vir} is the virtual dualising sheaf on it, equipped with orientation data in the sense of Behrend and of Brav, Bussi and Joyce.

The first equivalence is a construction of the automorphic spectrum as a trace, the second is the correspondence proper, and the third is a localisation — an arithmetic Lefschetz–Verdier or Atiyah–Bott statement identifying a global trace with an integral over the fixed locus.

Remark 1.25 (What is and is not proved). [DESIGN]

No case of Conjecture 1.24 is proved unconditionally in this treatise. The object $\mathfrak{X}_{\mathbf{Z}}$ is not constructed; it is characterised by axioms and shown to be consistent with three partial constructions. Conjecture 1.23 is not proved. What is proved here consists of: formal consequences of the axioms, marked *T*; the degeneration of the framework to the known function-field and geometric statements, which serves as a consistency check; the reconstruction of classical local and archimedean factors from the axioms in low rank; and a number of computations, in Part VII, that could have refuted the framework and did not. A reader who wishes to know at once what would falsify the whole should turn to Chapter 40 and to Appendix F.

The relation of Conjecture 1.24 to the classical statements of §1.1 is as follows. Conjecture 1.2 is expected to follow by taking π_0 of the composite equivalence and matching Satake parameters place by place; Conjecture 1.3 by naturality in G, as described in §1.4.8; Conjecture 1.5 by computing the characteristic polynomial of Φ on the trace, place by place, with the archimedean factor supplied by the archimedean chart. All three implications are themselves conjectural, since they depend on compatibilities not established here.

1.6 What this book claims, and what would falsify it

The preceding section stated a conjecture of very wide scope, and a reader is entitled to a precise account of the epistemic position the treatise occupies. I give it here rather than in an appendix, because a reader who forms a false impression in Chapter 1 will carry it through forty chapters. The account is written in the first person because it concerns what I, as author, claim; the work is issued under my name and that of WiseMotif.ai, and the distinction between what the literature has established and what is proposed here is my responsibility to police.

1.6.1 The status of the Main Conjecture

Conjecture 1.24 is a conjecture. It is not a theorem with a gap, nor a theorem modulo a technical lemma, nor a theorem whose proof is deferred to a sequel. No case of it is proved unconditionally in this book, and I do not know how to prove any case of it. The strongest positive evidence I can offer is of four kinds: that the conjecture degenerates, under the substitutions described in Part IV, to statements known to be true over function fields and over complex curves; that it predicts the correct archimedean factor in the one case where that prediction has been carried out in full, namely the case of Criterion 1.26; that it accounts, by a single mechanism, for the invariant ℓ_0 whose independent discovery on the automorphic and spectral sides is recorded in Proposition 1.21; and that it survives the numerical tests of Part VII. None of these is a proof, and the fourth is the weakest kind of evidence there is.

1.6.2 The status of the axioms

The framework rests on twenty-eight axioms, designated KH.1 to KH.28 and stated in full in Part I and Part II. They are grouped as follows: KH.1–KH.5 fix the notion of a geometry as a six-functor formalism and the resulting notion of duality; KH.6–KH.11 impose the trace and weight structure and define the height of such a geometry; KH.12–KH.18 describe the global arithmetic base, its charts at each place, the gluing along an adelic halo, the absolute diagonal and the orientation of degree three; KH.19–KH.22 give the gauge and spectral stacks and their shifted symplectic structure; KH.23–KH.26 govern the analytic side, nuclearity, the Fourier kernel and hyperspherical duality; KH.27–KH.28 fix the internal language and the verification discipline.

These are hypotheses. They are not self-evident, they are not derived from anything, and the great majority carry the mark [D], which is to say that they are design choices for which I have argued but which I have not proved to be forced. Two of them, KH.18 and KH.5, carry the mark [C] and are genuine conjectures with content that could be false. One, KH.20, carries [T] because it is derived from KH.18 together with the construction of Pantev, Toën, Vaquié and Vezzosi (Pantev et al. 2013). The reader should treat every theorem of this book as a theorem about that specification rather than about the arithmetic of number fields.

1.6.3 What is unconditional

A small amount of the book is unconditional, and it should be identified so that it is not lost among the conjectures. The formal results of Part I — the rigidity of the trace on a dualisable object, the behaviour of traces under composition of correspondences, the comparison of the trace with topological cyclic homology in the cases where both are defined — are theorems of higher algebra and depend on no arithmetic hypothesis. The degeneration statements are unconditional in the direction that matters: given the framework's axioms, the recovery of the function-field and geometric statements is proved. The computations of Part VII are unconditional, being computations. The identification in Part III of ℓ_0 with the rank of a compact torus is conditional on the construction of the archimedean chart.

Everything else — the existence of $\mathfrak{X}_{\mathbf{Z}}$, the ambidexterity of Conjecture 1.23, the Main Conjecture and each of its six corollaries, the arithmetic applications of Part VI — is conditional. In particular this book proves no new case of modularity, no new case of functoriality, no instance of the Ramanujan conjecture, and nothing whatever about the location of the zeros of any L-function. A reader who takes away the impression that the Langlands programme has been resolved here has read the book incorrectly, and the fault will be partly mine for having written a long book about a conjecture.

1.6.4 What would falsify the framework

A specification that cannot fail is worthless, and I have tried to write one that can. Four kinds of failure would be decisive.

First, a computational refutation. If the trace construction applied to the arithmetic Picard stack yields anything other than the completed Riemann zeta function — see Criterion 1.26 — the framework is wrong at the level of its central mechanism, and no repair in the upper strata could save it. The same applies to the eleven further tests of Chapter 40: each is a case where the framework makes a prediction that can be checked against an independently known answer, and a single failure is fatal to the stratum in which it occurs.

Second, an inconsistency of the axiom system. If KH.1–KH.28 can be shown to have no model — for instance if the orientation degree demanded by KH.18 can be shown incompatible with the ambidexterity demanded by KH.11 — then the specification describes nothing. Appendix F records which fragments have been formalised in Lean 4 (Moura and Ullrich 2021) against *mathlib* (Community 2020).

Third, a degeneration failure. If the framework, specialised to a global function field, fails to reproduce Theorem 1.8 or Theorem 1.9, or specialised to a complex curve fails to reproduce Theorem 1.12, then it is not a generalisation of the known theory.

Fourth, a demonstration that the four problems of §1.4 are independent — for example a construction supplying a dualisable trace-theoretic geometry over $\operatorname{Spec}\mathbf{Z}$ in which the defect ℓ_0 still obstructs patching. That would refute Principle 1.22, which is the organising claim of the whole work.

Status and Caveat. There is a fifth possibility, which is neither refutation nor confirmation and which I regard as the most likely outcome: that the specification is consistent, that an object satisfying it exists, and that the object is not related to the automorphic spectrum in the manner Conjecture 1.24 asserts — that the equivalences hold with the wrong normalisations, or on the wrong subcategory, or only after a modification that destroys the corollaries. In that case the framework would be neither true nor false but useless, and there is no test in Chapter 40 that would detect it. The only defence against this outcome is to carry the construction far enough that the arithmetic consequences become checkable, which is what Part VI attempts and does not complete.

1.7 The architecture, and how to read this book

A conjecture of the scope of Conjecture 1.24 cannot be presented as a single argument; it must be presented as a structure with declared interfaces, so that a reader may locate which layer a given claim inhabits, and so that the failure of one layer does not silently contaminate another.

1.7.1 The strata

Plate 1.2 displays the eight strata and the objects each exports. The foundational kernel exports the two-category of geometric objects, correspondences and two-morphisms, with its symmetric monoidal structure and notion of dualisability. The global arithmetic curve stratum exports a self-dual site with a degree map; this export is marked C, and everything above it inherits that conditionality. The spectral geometry stratum exports a quasi-coherent stack with a (-1)-shifted symplectic form, together with the invariant ℓ_0; the automorphic geometry stratum exports a module category over the Hecke algebra; the correspondence stratum exports the functor $\mathfrak{F}$ with its unit and counit; and the trace stratum exports $\mathrm{Tr}(\Phi \mid -)$ and the local factors.

The discipline governing the strata is that each consumes only the interface exported by the stratum beneath it, and never the internal constructions of that stratum; a replacement construction satisfying the same interface leaves the strata above unchanged. The three candidate constructions of $\mathfrak{X}_{\mathbf{Z}}$ in Part II are offered in this spirit: they are alternative implementations of one interface, and the arguments of Parts III to VI are written against the interface alone.

The division of labour among the top three strata is deliberate. The discovery stratum proposes and asserts nothing; the construction strata build, and assert only relative to declared axioms; the verification stratum certifies, and is the sole source of factual status. The alternative is a work of this length in which a proposed analogy and a verified theorem are typeset identically.

1.7.2 The status marks

Every non-classical assertion in this treatise carries one of five marks. The system is explained here once and used everywhere.

Mark	Meaning	Obligation on the author
[E]	Established: a theorem in the published literature	A citation must accompany it
[T]	Theorem of this framework: proved here from stated axioms	A proof must be given or referenced by number
[C]	Conjecture: believed, not proved	The evidence must be stated, and the consequences marked [C] in turn
[D]	Design principle: a definitional or architectural choice	The alternatives must be acknowledged

Mark	Meaning	Obligation on the author
[F]	Formal-verification target: intended for machine checking	The current status of the formalisation must be recorded

The marks are not decorative and they are not a disclaimer. They are the mechanism by which a reader may compute, for any assertion in the book, the set of unproved statements on which it depends. A treatise that proposes a resolution of a problem of this standing and does not carry such marks is asking the reader to take on trust a distinction — between what is proved and what is hoped — that the reader has no means to reconstruct. Presenting a [C] as a [T] is the one error this book cannot survive, and the author has preferred, in every doubtful case, the weaker mark.

A word on [D]. Many of the assertions in §1.4 carry it. A statement such as Problem 1.16 is partly mathematical and partly a claim about what the literature does: the mathematical content is proved (as in Proposition 1.17) and marked [E], while the claim about method is a reading of the record, offered as the organising choice on which the architecture rests. Marking it [D] says: this is why the book is shaped as it is, and if you reject it you should reject the shape.

1.7.3 A reader's guide

Reader's Guide. The treatise is in seven parts.
Part I, The Foundational Kernel (Chapters 1–6), fixes the language: correspondential geometry, six-functor formalisms, dualisability, and the abstract theory of categorical traces. It assumes familiarity with ∞-categories at the level of a first course.
Part II, The Global Arithmetic Curve (Chapters 7–12), constructs, as far as present technique allows, the object $\mathfrak{X}_{\mathbf{Z}}$, presenting the prismatic, Arakelov and λ-ring charts and stating the gluing axioms.
Part III, Gauge and Spectral Stacks (Chapters 13–18), constructs $\mathrm{Bun}_G(\mathfrak{X}_{\mathbf{Z}})$ and $\mathrm{LocSys}_{^L G}(\mathfrak{X}_{\mathbf{Z}})$, establishes the (-1)-shifted symplectic structure, and identifies ℓ_0 with the rank of the archimedean gauge torus.
Part IV, The Trace Correspondence (Chapters 19–24), states and analyses Conjecture 1.24, derives the corollaries listed in §1.4.8, and treats functoriality as naturality.
Part V, Analysis, Duality, and the Trace Formula (Chapters 25–30), builds the archimedean chart, obtains Γ-factors as regularised determinants, and reformulates the Arthur–Selberg trace formula as an index theorem.
Part VI, Local Theory and Arithmetic Consequences (Chapters 31–36), works out the local theory place by place and derives the arithmetic consequences — modularity lifting with $\ell_0 > 0$, purity, and the local correspondence at ramified places.
Part VII, Computation, Verification, and the Programme (Chapters 37–40), gives the computational tests, the formalisation status, the falsification protocol and the research programme.
Three paths through the book are suggested.
The *arithmetician's path*: Chapter 1, then Part II for the object, then Chapter 19 for the statement, then Part VI for the consequences, returning to Part I only for definitions actually needed.
The *categorist's path*: Part I in full, then Part III, then Part IV, then Part V. This path takes the arithmetic on trust and asks whether the categorical constructions are coherent.
The *sceptic's path*: begin with Chapter 40, which lists twelve computations and structural tests any of which would refute the framework, and Appendix F, which records exactly what has been formalised and what has not; then read §1.4 of this chapter for the diagnosis, then Chapter 19 for the statement, and only then, if at all, the constructions. The author regards this as the most rational path and would take it himself.

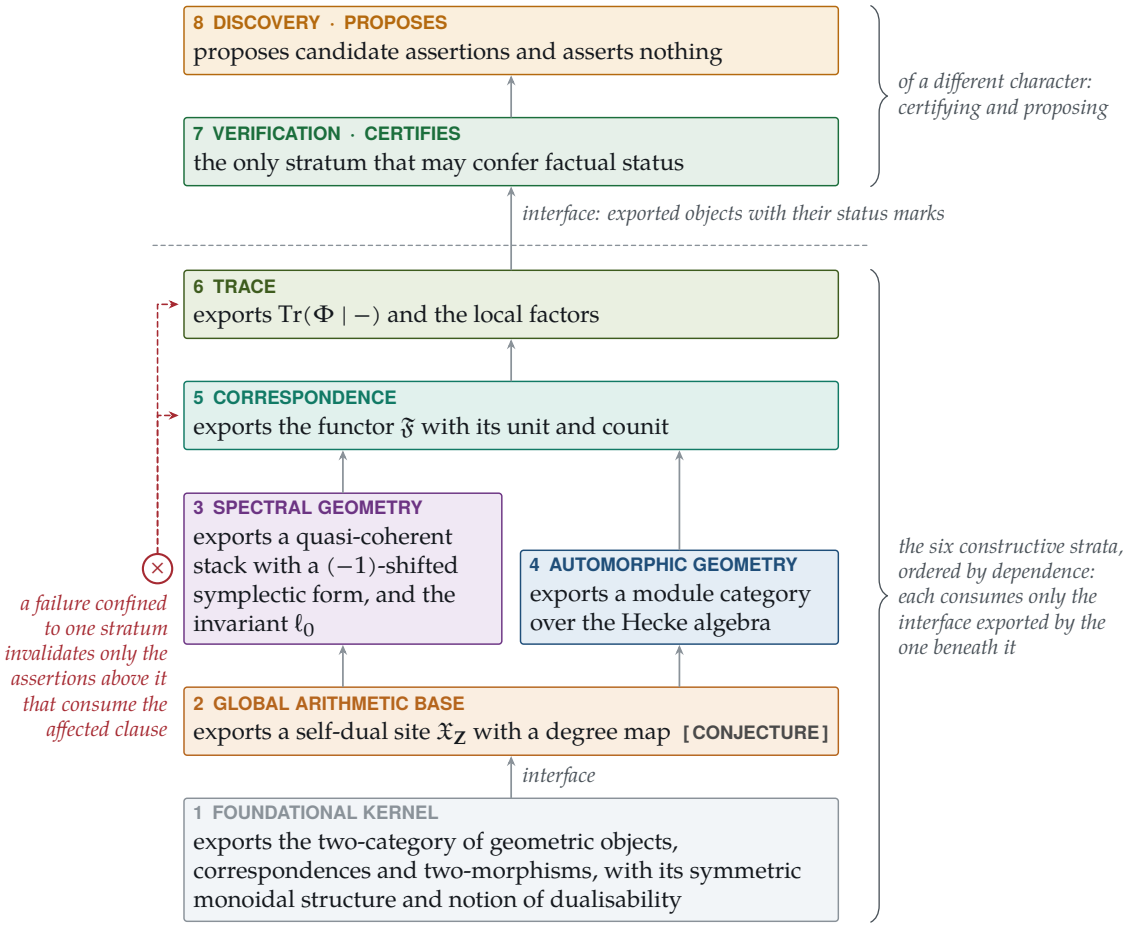


Plate 1.2. The eight strata of the framework and the interfaces between them. Each stratum consumes only the interface exported by the one beneath it; only the verification stratum may confer factual status. The lower six strata are constructive and are ordered by dependence: the foundational kernel, the global arithmetic base, the spectral and automorphic geometries, the correspondence and the trace. The upper two are of a different character, the verification stratum certifying and the discovery stratum proposing. An interface is a declared list of exported objects with their status marks, and a failure confined to one stratum invalidates only those assertions above it that consume the affected clause.

What it does not show: the construction of the global arithmetic base: its export, a self-dual site with a degree map, is marked [C]*, and every stratum above it inherits that conditionality.*

1.8 Certification and falsification

A framework of this size will contain errors. The question is not whether but whether they can be found, and how quickly. This section describes the two mechanisms the treatise carries for that purpose: a certification discipline for how assertions acquire status, and a falsification protocol listing what would refute the framework.

Plate 1.3 displays that loop. The sixth stage is the kernel gate: an assertion acquires the mark [E] or [T] only by passing it, which means either a located citation to the published literature or a proof whose every step is either such a citation or a previously certified assertion.

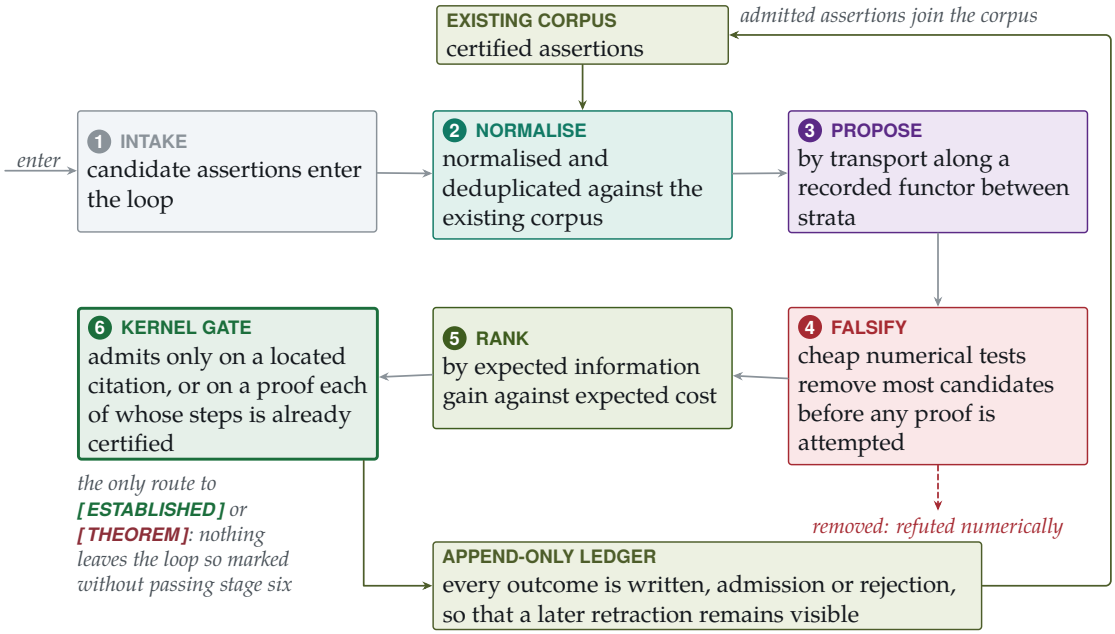


Plate 1.3. The Aletheia discovery loop. Nothing leaves the loop marked [E] or [T] that has not passed the kernel gate at stage six. Candidate assertions enter at the left, are normalised and deduplicated against the existing corpus, and are then proposed by transport along a recorded functor between strata. Cheap numerical falsification removes most of them before any proof is attempted, and what survives is ranked by expected information gain against expected cost. The gate admits an assertion only on a located citation or on a proof each of whose steps is itself already certified, and the outcome, whether admission or rejection, is written to an append-only ledger so that a later retraction remains visible.

What it does not show: a certificate issued by the loop itself: stages one to five propose, filter and rank, and only the kernel gate confers [E] or [T].

Criterion 1.26 (The multiplicative-group gate). [FORMAL TARGET]

The following is the primary go/no-go test for the framework, and it is stated here so that a reader may check it before investing in the rest. Let $\mathrm{Bun}_{\mathbb{G}_m}^{\mathrm{ar}}$ *denote the arithmetic Picard stack of* $\mathfrak{X}_{\mathbf{Z}}$ *and* Gauge *the associated category of gauge sheaves. Then the zeta function of the trace spectrum must satisfy*

$$\zeta(\,\mathrm{TC}(\Phi \mid \mathrm{Gauge}(\mathrm{Bun}_{\mathbb{G}_m}^{\mathrm{ar}})), s) \;=\; \pi^{-s/2}\,\Gamma(s/2)\,\zeta(s),$$

the completed Riemann zeta function, with the archimedean factor produced by the archimedean chart and not inserted. If the framework yields anything else — in particular, if it yields $\zeta(s)$ *without the* Γ*-factor, or with a* Γ*-factor of the wrong argument — then it is wrong, and no amount of repair at higher strata will save it. This criterion is designated for formal verification; its current status is recorded in Appendix F.*

Eleven further tests are given in Chapter 40, of three kinds: degeneration tests, which require the framework to reproduce a known theorem when a parameter is specialised; consistency tests, which require two computations of the same quantity by different routes to agree; and numerical tests, which require a computed invariant to match a table. The framework has passed those that have been carried out. That is a weak statement and is not offered as more.

It remains to record a dependence. The framework is parasitic, in a precise and unembarrassed sense, on several bodies of recent work that it does not reprove: the geometrisation

of the local correspondence (Fargues and Scholze 2026); prismatic cohomology (Bhatt and Scholze 2022, 2023); condensed and analytic geometry (Clausen and Scholze 2019, 2024); dualisable presentable categories (Efimov 2024); the geometric Langlands equivalence (Gaitsgory and Raskin 2024a); and the fundamental lemma (Ngô 2010). If any of these were to fail, the corresponding stratum would fail with it, and this dependence is recorded at each point of use rather than absorbed into a general acknowledgement.

1.9 Grothendieck's programme and this one

The comparison is unavoidable, and it is better made by the author, plainly, than left to be made by others as an accusation. It is proper to say what is analogous, what is more radical, and what the earlier programme had that this one does not.

1.9.1 What EGA and SGA did

Grothendieck's programme, executed in the *Éléments* (Grothendieck and Dieudonné 1960) and the *Séminaires* (Artin et al. 1972a, 1972b, 1973; Grothendieck and Illusie 1977), made five moves. It replaced varieties by schemes, so that arithmetic and geometry became instances of one theory and nilpotents became legitimate. It replaced points by functors of points, so that a space became determined by what maps into it. It replaced open coverings by sites, so that cohomology could be defined where the Zariski topology was too coarse. It replaced cohomology theories by the six operations, so that duality became a formal consequence of adjunctions rather than a theorem proved anew for each theory. And it replaced theorems about objects by theorems about morphisms, so that every absolute statement acquired a relative form and the relative form was the useful one.

The purpose of these moves was, in every case, the same: to make a hard theorem into a formal consequence of a well-chosen definition. The Weil conjectures were the target. Rationality and the functional equation followed from the Lefschetz trace formula in étale cohomology, which followed from the six operations and proper base change, which followed from the definitions. Only the Riemann hypothesis resisted this treatment, and the standard conjectures (Grothendieck 1969) were Grothendieck's proposal for making it too a formal consequence. They remain open; Deligne proved the Riemann hypothesis by other means (Deligne 1974a), and the standard conjectures are the one part of the programme that did not close.

1.9.2 What is analogous

This treatise makes a move of the same type. Where Grothendieck replaced spaces by functors and coverings by sites, the proposal here is to replace *spaces* by *six-functor formalisms carrying a primitive trace*: the fundamental object is not a geometric object from which a cohomology theory is derived, but a symmetric monoidal ∞-category with the six operations and a distinguished trace, from which a geometry may or may not be extracted. The motivation is identical to Grothendieck's — to make a hard theorem the formal consequence of a definition — and the target is the same kind of theorem: a trace formula, from which a reciprocity law follows.

The analogy extends to the treatment of the base. Grothendieck's relative point of view made every statement a statement over a base, and the base was permitted to be arbitrary. The obstruction identified in Problem 1.16 is that $\operatorname{Spec}\mathbf{Z}$ is terminal *in schemes*, and the response — to work over the sphere spectrum, where the absolute tensor square is non-trivial by Proposition 1.17 — is the natural continuation of the relative point of view into a category

where more bases exist. Grothendieck enlarged the category of spaces until Spec $\mathbf{Z}$ was an object; the proposal here is to enlarge it until Spec $\mathbf{Z}$ has a self-product.

1.9.3 What is more radical

Three things go beyond anything in EGA. First, the primitive is a trace, not a cohomology theory. In Grothendieck's foundations the trace formula is a theorem about a cohomology theory; here the trace is part of the data, and cohomology is what one computes from it. This inverts the logical order and is the reason the treatise can contemplate a setting in which no underlying space is available.

Second, the foundations are homotopical rather than abelian. Grothendieck's derived categories were a technical device for handling complexes; here the ∞-categorical structure is essential, because the objects that must be dualisable are large categories, and dualisability in that sense (Efimov 2024) has no abelian shadow. The derived structure that Venkatesh's work exhibits on the automorphic side (Venkatesh 2019; Galatius and Venkatesh 2018) is not a refinement of a classical statement; it is the statement.

Third, and most radically, the archimedean place is to be an object of the same geometry as the finite places. Grothendieck's foundations have nothing to say about $\mathbf{R}$; Arakelov theory attaches archimedean data by hand, and the success of that theory is a success of bookkeeping rather than of unification. The proposal that a six-functor formalism can have $(\mathfrak{g}, K)$-modules as its values at one place and smooth representations at another is, of the four principal conjectures of this book, the one for which the least evidence exists.

1.9.4 What EGA had that this does not

EGA proved its theorems. This must be said without qualification, because everything else in the comparison is worth nothing beside it. Grothendieck's foundations were not a proposal but a machine delivered in working order, and within a decade they had produced the rationality of zeta functions, ℓ-adic cohomology, the theory of the fundamental group, Deligne's proof of the Weil conjectures, and the modern theory of moduli.

This treatise has no comparable yield. It proves no new modularity theorem. It proves no case of functoriality that was not known. It supplies no new information about the zeros of any L-function. Its central object is not constructed but axiomatised, and its central conjecture is not proved in any case. What it offers is a reorganisation: a claim that four difficulties are one difficulty, a specification of what an object would have to be in order to dissolve them, and a set of tests by which the claim can be refuted. Grothendieck's own assessment of his work, recorded in *Récoltes et Semailles* (Grothendieck 1986), was that the essential act was the patient construction of the right notions rather than the solution of any particular problem; but the right notions in his case were vindicated by the problems they then solved, and no such vindication is available here.

The proposal is therefore offered as a proposal. Whether it deserves the comparison it has invited is not for its author to say.

Summary. The Langlands programme asks for a correspondence between automorphic representations of a reductive group over a global field and homomorphisms from a Galois-theoretic object to the dual L-group, matching L-functions place by place; functoriality asks in addition that a map of L-groups induce a map of automorphic spectra. Over function fields the programme is, for general linear groups, a theorem, proved by computing traces of Frobenius on the cohomology of moduli of shtukas; over number fields the known results are of a different character, proved by deformation theory and confined to situations where an invariant ℓ_0 vanishes. Four structural

> obstructions separate the two settings: the absence of a base over which to compare the two sides; the absence of a global Frobenius and hence of a trace; the absence of a uniform treatment of the archimedean place; and the cohomological defect ℓ_0. This treatise asserts that the four are one, namely the absence of a dualisable trace-theoretic geometry over $\operatorname{Spec}\mathbf{Z}$, and proposes to supply one. The Main Conjecture identifies the automorphic spectrum with a topological cyclic homology of an absolute Frobenius endo-correspondence acting on a category of sheaves on a stack of bundles over a global arithmetic curve, and with the corresponding object on the spectral side, and with the cohomology of a derived fixed locus. No case of it is proved here. Every assertion in the book carries a status mark recording exactly what it rests on.

1.10 Notes and references

The analogy on which the whole subject rests is older than the programme. In a letter written from prison at Rouen in March 1940 to his sister Simone, Weil described three parallel columns — number fields, function fields over finite fields, and Riemann surfaces — and compared the mathematician's situation to that of a decipherer possessing a text in three scripts of which only one is legible. Weil's point was not that the three columns say similar things; it was that the middle column is the one in which one can work, and that the intended target is the first. His own converse theorem (Weil 1967), characterising the Dirichlet series attached to modular forms by the functional equations of their twists, is the technical bridge that made the automorphic side of Langlands' proposal formulable, and it is the reason Langlands wrote to Weil rather than to anyone else. The present treatise may be read as an attempt to supply, for the first column, the object that the middle column has and the first lacks; the diagnosis of §1.3.4 is Weil's comparison carried out with modern vocabulary and with the missing entries named.

The primary sources for the programme itself are Langlands' letter to Weil of January 1967 (Langlands 1967), which is reproduced in facsimile and transcription in the digital archive of his collected works, and the Washington lecture (Langlands 1970), which is where the conjectures first appear in print. The paper on Euler products (Langlands 1971) contains the analytic technique that underlies the constant term computation and hence the whole theory of Eisenstein series. Borel's Corvallis article (Borel 1979) remains the standard formulation and is the one adopted here; the two Corvallis volumes (Borel and Casselman 1979a, 1979b) are still the best single point of entry to the classical theory. For the later shape of the programme, Langlands' own reflections on beyond-endoscopy (Langlands 2004) should be read alongside the Corvallis formulation; it is there that the limits of the trace-formula method as then understood are first stated by their principal architect.

Tate's thesis (Tate 1950) is the model of exposition for the GL_1 theory and, as argued in §1.2.1, the only classical treatment in which the archimedean place is handled by the same mechanism as the finite places. Jacquet and Langlands (Jacquet and Langlands 1970) is the origin of the local-global method for GL_2 and of the comparison of trace formulae; Langlands' base change monograph (Langlands 1980) and its extension by Arthur and Clozel (Arthur and Clozel 1989) are the technical foundation of nearly every subsequent modularity argument. Selberg's original trace formula (Selberg 1956) and Arthur's development of it (Arthur 1978), together with Kottwitz's stabilisation (Kottwitz 1986) and Waldspurger's work on endoscopy (Waldspurger 1997), constitute the analytic backbone of the subject and are treated in Part V from the index-theoretic point of view.

For the function-field case the reader should begin with Drinfeld's elliptic modules (Drinfeld 1974) and his proof for GL_2 (Drinfeld 1983), then L. Lafforgue (Lafforgue 2002), and then

V. Lafforgue (Lafforgue 2018), whose excursion operators are the direct ancestor of the trace construction in Part IV. The dependence of all of these on Deligne's purity theorems (Deligne 1974a, 1980) is absolute, and the reader who has not read *Weil II* has not seen the mechanism by which a trace computation becomes an arithmetic theorem. Grothendieck's foundations are set out in the *Éléments* (Grothendieck and Dieudonné 1960) and in the seminars (Artin et al. 1972a; Deligne 1977; Grothendieck and Illusie 1977); the standard conjectures (Grothendieck 1969) are the unclosed part of that programme, and the comparison in §1.9 should be read with them in mind.

The number-field results begin with Wiles (Wiles 1995) and Taylor–Wiles (Taylor and Wiles 1995), reach their classical culmination in (Breuil et al. 2001) and in Khare and Wintenberger's proof of Serre's conjecture (Khare and Wintenberger 2009a, 2009b), and enter their derived phase with Calegari and Geraghty (Calegari and Geraghty 2018). The identification of the defect ℓ_0 as a homotopy-theoretic invariant, rather than an obstruction to be circumvented, is due to Venkatesh and his collaborators (Venkatesh 2019; Galatius and Venkatesh 2018; Prasanna and Venkatesh 2019); of all recent work in the arithmetic direction this is the closest in spirit to the present treatise, and Part III's identification of ℓ_0 with the rank of an archimedean torus is an attempt to explain what Venkatesh's exterior algebra is the cohomology of. Scholze's construction of Galois representations attached to torsion classes in the cohomology of locally symmetric spaces for GL_n over CM fields — published in the *Annals of Mathematics* in 2015, and cited here in prose because the bibliography key for it has not been fixed — rests on the perfectoid methods of (Scholze 2012) and (Scholze 2013).

The local theory is treated in (Harris and Taylor 2001) and (Henniart 2000) classically, and geometrically in (Fargues and Scholze 2026), which rests on the curve of (Fargues and Fontaine 2018) and on the theory of diamonds (Scholze 2017). The six-functor formalism used throughout this book is that of (Scholze 2025), with the p-adic instances of (Mann 2022) and (Heyer and Mann 2024) and the general framework of (Gallauer 2021); the analytic and condensed foundations are (Clausen and Scholze 2019) and (Clausen and Scholze 2024), and the theory of traces on large categories is (Efimov 2024). On the trace side, the formulation of topological cyclic homology adopted here is that of Nikolaus and Scholze (Nikolaus and Scholze 2018), whose account makes the cyclotomic structure — the simultaneous presence of all the φ_p — transparent; the historical route is through (Bökstedt et al. 1993) and (Hesselholt and Madsen 1997). The reader who wishes to understand why a homotopy-theoretic invariant should have anything to say about arithmetic should read the integral p-adic Hodge theory of Bhatt, Morrow and Scholze (Bhatt et al. 2018) together with its topological companion (Bhatt et al. 2019): it is there that topological Hochschild homology of a p-adic ring is shown to compute, after suitable completions and Frobenius twists, the crystalline and de Rham cohomologies that arithmetic geometry had been using for fifty years. That identification is the single strongest piece of evidence that the trace-theoretic language of this treatise is not a formal analogy but a description of the same objects in different coordinates; and the prismatic theory (Bhatt and Scholze 2022, 2023; Bhatt and Lurie 2022b, 2022a) supplies the finite charts of $\mathfrak{X}_{\mathbf{Z}}$, with Borger's λ-ring approach (Borger 2009) supplying a third and the derived foundations of Toën and Vezzosi (Toën and Vezzosi 2005, 2008) the ambient geometry. The analytic L-function theory used in §1.1.5 is that of Godement and Jacquet (Godement and Jacquet 1972); Laumon's work on the Fourier transform and the local correspondence (Laumon 1987b, 1987a) is the geometric ancestor of the kernel constructions in Part V, and the local-global compatibility results of Emerton (Emerton 2006a, 2006b) are used throughout Part VI. The theory of perverse sheaves (Beilinson et al. 1982) is assumed without further comment.

On the geometric side, (Beilinson and Drinfeld 2004) is the origin of the categorical formulation; (Arinkin and Gaitsgory 2015) introduces the singular-support condition without which no dualisability statement is available; (Mirković and Vilonen 2007) and (Satake 1963) supply the geometric and classical Satake isomorphisms; and the equivalence itself is (Gaitsgory and Raskin 2024a) and its four companions (Arinkin, Beraldo, Campbell, et al. 2024; Campbell et al. 2024; Arinkin, Beraldo, Chen, et al. 2024; Gaitsgory and Raskin 2024b). The physical interpretation (Kapustin and Witten 2007) and Kapranov's earlier proposal (Kapranov 1995) are the source of the topological-field-theory reading used in Part IV, with Lurie's classification (Lurie 2009b) supplying the formal statement and (Lurie 2017) the ambient algebra. The shifted symplectic structures are those of (Pantev et al. 2013).

The traditions that have addressed Problem 1.16 directly are discussed in §1.3.5; to the works cited there should be added Kim's arithmetic Chern–Simons theory (Kim 2015, 2018) and the duality of Artin and Verdier (Artin and Verdier 1973), which is the closest thing in the classical literature to the self-duality demanded of $\mathfrak{X}_{\mathbf{Z}}$. Simpson's non-abelian Hodge theory (Simpson 1992, 1997a) is the source of the twistor picture used in Part V for the archimedean chart. The period programme of Sakellaridis and Venkatesh (Sakellaridis and Venkatesh 2017) and the γ-function proposal of Braverman and Kazhdan (Braverman and Kazhdan 2000) are the two existing programmes with which the present one has the most overlap, and Part V is written with both in view. Finally, the formalisation reported in Part VII and Appendix F is carried out in Lean 4 (Moura and Ullrich 2021) against *mathlib* (Community 2020); the numerical data on elliptic curves used in Example 1.15 are drawn from Cremona's tables and the L-functions and modular forms database, cited in prose since no bibliography key has been assigned to them.

1.11 Exercises

1. Verify by direct point counting that the curve of Example 1.15 satisfies $a_5 = 1$ and $a_7 = -2$, and check these against the coefficients of the eta product $q\prod_{n\geq 1}(1-q^n)^2(1-q^{11n})^2$.
2. Show that for $G = \mathrm{GL}_n$ the dual group is again GL_n, that for $G = \mathrm{SO}_{2n+1}$ it is Sp_{2n}, and that $\widehat{\widehat{G}} \cong G$ for any connected reductive G. Deduce that $G \mapsto \widehat{G}$ is an involution on the set of based root data.
3. Compute ℓ_0 and q_0 for GL_2 over $\mathbf{Q}$, over an imaginary quadratic field, over a real quadratic field, and for GL_3 over $\mathbf{Q}$. In which of these cases does the classical Taylor–Wiles method apply without modification, and why?
4. Let F be a number field. Show that $\operatorname{Spec} \mathcal{O}_F$ admits no morphism to a scheme S making it a relative curve with $\dim S = 1$, and state precisely which step of the argument of Theorem 1.8 fails as a consequence.
5. Using the computation of $\pi_*\,\mathrm{THH}(\mathbf{Z})$ recorded in Proposition 1.17, compute the first three homotopy groups of $\mathrm{THH}(\mathbf{Z})$ and interpret the appearance of $\mathbf{Z}/i$ in degree $2i-1$ as a statement about the conormal complex of the diagonal.
6. Verify that the Weil–Deligne parameter of the local component at 11 of the automorphic representation attached to the curve of Example 1.15 has $N \neq 0$, and that the resulting Euler factor is $(1-11^{-s})^{-1}$ rather than a factor of degree two. Explain the drop in degree in terms of the rank of the invariants of Frobenius.
7. Show that the completed zeta function $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the only Dirichlet series with Euler product over the primes of $\mathbf{Q}$ and a functional equation $s \mapsto 1-s$ of the stated shape whose archimedean factor is a single $\Gamma_{\mathbf{R}}$. Use this to explain why Criterion 1.26 is a genuine test rather than a normalisation.
8. Let $\eta\colon {}^{L}H \to {}^{L}G$ be an L-homomorphism. Assuming Conjecture 1.3, show that the composite of two such lifts is the lift of the composite, and state precisely the naturality diagram that §1.4.8 proposes to make this a formal consequence.
9. Read the statement of Theorem 1.12 and write down, in not more than one page, exactly which hypotheses prevent it from implying any case of Conjecture 1.2. Identify which of the four problems of §1.4 each hypothesis corresponds to.
10. Suppose an object $\mathfrak{X}_{\mathbf{Z}}$ satisfying the axioms of Part II exists, and suppose its archimedean chart carries an action of $\mathbb{S} = \mathbf{R}_{>0} \times S^1$ with the circle acting with finite-dimensional weight spaces. Show that the rank of the induced compact torus in the gauge group is an invariant of the real group $G(F \otimes \mathbf{R})$, and compare with the definition of ℓ_0 in Problem 1.20.
11. Assign a status mark to each of the following, justifying your choice by the criteria of §1.7.2: the modularity of every elliptic curve over a real quadratic field; the existence of the global Langlands group $\mathcal{L}_{\mathbf{Q}}$; the assertion that $\mathrm{Shv}_{\mathcal{N}}(\mathrm{Bun}_G(X))$ is dualisable for X a smooth projective complex curve; and the assertion of Criterion 1.26.
12. Formulate, in the style of §1.4, a fifth obstruction concerning the non-algebraic automorphic spectrum — Maass forms of eigenvalue not corresponding to any motive — and say why the thesis of §1.4.5 does not obviously dissolve it. This is not an exercise with a known answer; it is the gap the author regards as the most serious unaddressed one.
13. Repeat the computation of §1.3.3 for the curve $y^2 = x^3 + x + t$ over $\mathbf{F}_7(t)$: determine the degree-one places of good reduction, tabulate a_v at each, verify the Hasse bound, and compute the number of points of the affine surface over $\mathbf{F}_7$. Compare with the value q^2 obtained for $q = 5$ and decide whether the coincidence there is structural or accidental.

This exercise tests a claim of the book, namely that the entries of such a table are traces of one endomorphism; state what would have to be exhibited to verify that claim directly.

14. The two-column table of §1.3.4 asserts that each entry in its right-hand column is absent rather than merely unknown. Select any two rows and either construct the missing object using only techniques available in the published literature, or give a precise reason why the techniques of §1.3.5 do not suffice. A successful construction in a single row would falsify the corresponding clause of Principle 1.22 and should be reported as such.

Chapter 2

Chôral Geometry: Geometry Without Space

Synopsis. A geometry, in this treatise, is not a space. It is a six-functor formalism together with a primitive trace. The present chapter turns that sentence into a definition. We recall the presentation of the six operations as functoriality on a category of correspondences; we define a chôra to be a tuple consisting of a category of figures, a class of admissible legs, a sheaf theory, a duality involution and a trace; and we explain why the trace must be axiomatised rather than derived. The five foundational axioms KH.1–KH.5 are stated in full, each is commented upon, and for each we record what observation would falsify it. We then construct the 2-category of chôrai with trace, giving objects, 1-morphisms, 2-morphisms, composition and coherence data explicitly rather than gesturing at them. The central theorem of the chapter identifies the chôral trace, for the sheaf theory of ℓ-adic complexes on separated schemes of finite type over $\mathbf{F}_q$, with the Grothendieck–Lefschetz trace, hence with point counts. Four realisations are worked out in detail: schemes, condensed analytic stacks, spectra with topological Hochschild homology, and a deliberate non-example which satisfies the first three axioms and violates the fourth. The chapter closes with a comparison with the foundations of the *Éléments de géométrie algébrique*, stating plainly what the chôra buys and what is surrendered.

2.1 The inversion

Grothendieck's reconstruction of algebraic geometry can be described, without much violence, as five successive moves. The first was *relativisation*: the object of study is not a variety but a morphism, and every property worth naming is a property of morphisms, stable under base change. The second was the *functor of points*: a scheme is determined by, and may as well be replaced by, the functor it represents on commutative rings; geometry becomes a question about representability. The third was the *topos*: a space is determined by its category of sheaves, so one may discard the underlying point-set and retain only a site, or indeed only the topos. The fourth was the *six operations*: cohomology is not a sequence of groups attached to a space but a system of functors attached to morphisms, and the content of a cohomology theory lies in the adjunctions and base-change compatibilities among $f^*, f_*, f_!, f^!$, $\otimes$ and $\underline{\mathrm{Hom}}$. The fifth was *motives*: all the cohomology theories one meets are shadows of a single universal one, and the arithmetic content of a variety lies in that universal object.

Each move enlarges the class of objects by weakening what an object is required to be. A scheme has fewer obligations than a variety; a topos has fewer than a scheme; a six-functor formalism, as an object in its own right, has fewer still. The moves also share a direction: at each stage a structure that had been *primitive* is demoted to a *consequence*. Points cease to be primitive and become a functor; opens cease to be primitive and become a site; cohomology ceases to be primitive and becomes functoriality.

The programme was not completed, and the places where it stopped are the places from which this treatise starts. We name five; they will recur throughout Part I.

F1. No motivic t-structure. The category of mixed motives exists as a triangulated or stable ∞-category, but the abelian category of mixed motives whose derived category it should be does not exist unconditionally; its existence is equivalent to a package containing Grothendieck's standard conjectures (Grothendieck 1969; Cisinski and Déglise 2019). Consequently the motivic picture supplies a formalism but not the abelian heart that would make its objects behave like sheaves in the classical sense.

F2. The archimedean place is not uniform. For a finite place one has étale cohomology of a scheme over a p-adic field; for the archimedean place one has Hodge theory, and the two are related by analogy rather than by a common construction. Arakelov theory patches the gap with hermitian metrics but does not exhibit $\mathbf{R}$ and $\mathbf{Q}_p$ as objects of one geometry.

F3. The six operations are rebuilt for every geometry. Étale, ℓ-adic, de Rham, rigid analytic, p-adic, motivic, condensed: in each case the operations are constructed afresh, and the labour of verifying base change and the projection formula is repeated. The formalism is a template that is instantiated by hand, not a structure that is transported.

F4. Trace formulas are proved one at a time. Grothendieck–Lefschetz for schemes over finite fields, Atiyah–Bott for elliptic complexes, Arthur–Selberg for automorphic representations, Bismut's formulas in index theory: each is a theorem with its own proof, and the statement that they are instances of one identity is at present a slogan. There is no ambient structure in which a reciprocity law is *type-correct* by construction.

F5. There is no absolute base. $\operatorname{Spec}\mathbf{Z}$ is a final object in schemes, but it is not a base over which $\operatorname{Spec}\mathbf{Z}$ itself is a curve. The analogy between number fields and function fields fails precisely where it would be most useful, and the search for the missing base — the field with one element, Λ-rings, the various sites of Connes and Consani, the prismatisation of $\mathbf{Z}_p$ — is a search for a geometry rather than for an object within one (Toën and Vaquié 2009; Borger 2009; Connes and Consani 2016; Bhatt and Lurie 2022b).

The inversion proposed here is the sixth move, and it is of the same kind as the previous five: something primitive is demoted, and something derived is promoted. What is demoted is the locally ringed space, together with the whole apparatus of underlying topological spaces, points, opens and stalks. What is promoted is the six-functor formalism itself, which ceases to be an invariant *of* a geometry and becomes the *definition* of one.

Principle 2.1 (The chôral inversion). [DESIGN]
A geometry is not a space equipped with a sheaf theory. A geometry is *a sheaf theory: a lax symmetric monoidal functor from a category of correspondences to presentable stable categories, together with a duality involution and a trace. Spaces, when they exist, are reconstructed from this datum; when they do not exist, nothing is lost that was being used.*

The word *chôra* is taken from the *Timaeus*, where it names the receptacle in which figures appear without itself being a figure. The usage is not decorative, and Plate 2.1 contrasts the two pictures. The objects of C below are called *figures* rather than spaces precisely because they need not have points, and the chôra is the ambient structure in which they are related, not a thing of which they are parts. Whether a chôra nevertheless has enough points to be reconstructed as a space is the content of the fifth axiom, and it is a conjecture, not a theorem.

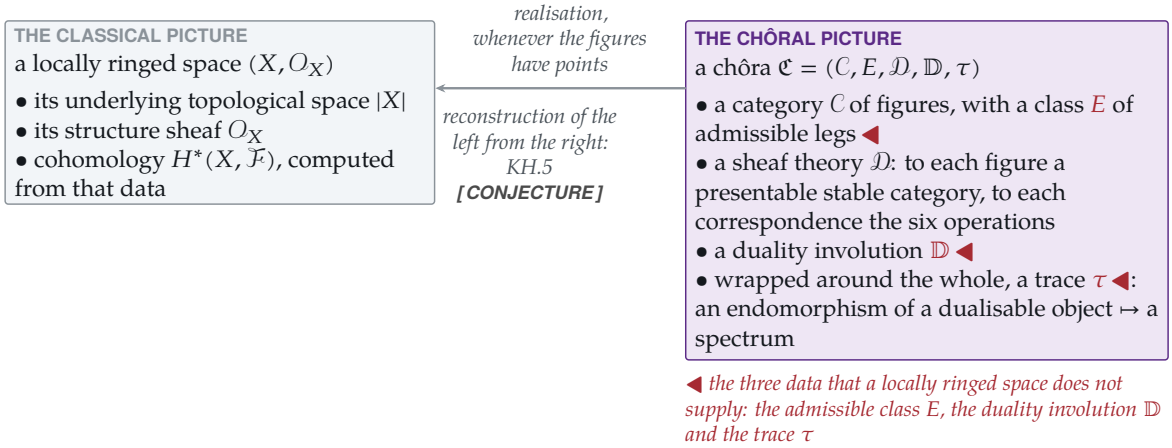


Plate 2.1. The chôra as receptacle. On the left, the classical picture: a locally ringed space, with its underlying topological space, its structure sheaf, and cohomology computed from that data. On the right, the chôral picture: a category of figures with a class of admissible legs, a sheaf theory assigning a presentable stable category to each figure and the six operations to each correspondence, and, wrapped around the whole, a trace which sends an endomorphism of a dualisable object to a spectrum. The arrow between the two pictures runs from right to left: the classical picture is a realisation of the chôral one whenever the figures happen to have points, and the reconstruction of the left-hand picture from the right-hand one is the content of KH.5, which is conjectural. The three annotations on the right-hand panel mark the three data that a locally ringed space does not supply: the admissible class E, the duality involution $\mathbb{D}$, and the trace τ.
What it does not show: a theorem of reconstruction: recovering the classical picture from the chôral one is the content of KH.5, which is a conjecture, and a chôra need not have points at all.

The claim of this chapter is therefore precise and limited. It is not claimed that the chôra proves anything that schemes do not; it is claimed that the chôra is a definition of *geometry* under which the four realisations of §2.8 are objects of one category, under which the six operations are transported rather than rebuilt, and under which a trace formula is a statement about a morphism of chôrai rather than a coincidence between two theorems. Whether that is worth the price is argued, as carefully as we are able, in §2.9.

2.2 Correspondences and the six operations

The technical device that makes the inversion possible is the observation that the six operations are not six independent functors but the components of a single functor defined on a category whose morphisms are correspondences. This observation is due in its modern form to Liu and Zheng, whose work on gluing pseudofunctors made the ∞-categorical bookkeeping tractable, and it was systematised by Gaitsgory and Rozenblyum (Gaitsgory and Rozenblyum 2017a, 2017b); the axiomatic form we use is that of Mann and of Heyer and Mann (Mann 2022; Heyer and Mann 2024), with Scholze's lectures as the standard reference (Scholze 2025). Earlier axiomatisations in the motivic setting are Ayoub's (Ayoub 2007a, 2007b) and Cisinski–Déglise's (Cisinski and Déglise 2019); a comparison of the several notions of six-functor formalism now in use is Gallauer's (Gallauer 2021).

Definition 2.2 (The category of correspondences). [DESIGN]

Let $\mathcal{C}$ be an ∞-category admitting finite limits and let E be a class of morphisms of $\mathcal{C}$, called the *admissible* morphisms, which contains all equivalences and is stable under composition and under base change. The *category of correspondences* $\mathrm{Corr}(\mathcal{C}, E)$ is the ∞-category with

- objects those of $\mathcal{C}$;
- 1-morphisms $X \to Y$ the correspondences $X \xleftarrow{f} Z \xrightarrow{g} Y$ with $g \in E$, the left leg f being arbitrary and the right leg g admissible;
- composition of $X \leftarrow Z \to Y$ and $Y \leftarrow W \to V$ given by the pullback $Z \times_Y W$, which exists because $\mathcal{C}$ has finite limits and whose right leg to V lies in E because E is stable under base change and composition;
- identity on X the correspondence $X \leftarrow X \to X$ with both legs the identity.

Cartesian product in $\mathcal{C}$, when it exists, makes $\mathrm{Corr}(\mathcal{C}, E)$ symmetric monoidal with unit the final object; we write $\mathrm{Corr}(\mathcal{C}, E)^{\otimes}$ for this symmetric monoidal ∞-category. Composition is associative only up to the canonical equivalences between iterated pullbacks, so that the construction is ∞-categorical: the coherence data are the higher pullback comparisons.

Two remarks on the definition are needed at once. First, the asymmetry between the legs is the whole point. The left leg records a pullback f^* and the right leg records a pushforward $g_!$, and it is because g is required to be admissible that $g_!$ exists at all; in the scheme case E is the class of separated morphisms of finite type, and $g_!$ is defined by compactification. Second, the composition rule *is* proper base change: the assertion that composing two correspondences means forming a pullback is exactly the assertion that $g'^* g_! \simeq g'_! g^*$ for a cartesian square. Base change is not an axiom in this presentation; it is the statement that Corr is a category.

Definition 2.3 (Six-functor formalism). [DESIGN]

A *six-functor formalism* on $(\mathcal{C}, E)$ is a lax symmetric monoidal functor

$$\mathcal{D}\colon \mathrm{Corr}(\mathcal{C}, E)^{\otimes} \longrightarrow \mathrm{Pr}^{L,\otimes}_{\mathrm{st}}$$

into the symmetric monoidal ∞-category of presentable stable ∞-categories and colimit-preserving functors, with Lurie's tensor product (Lurie 2017). We write $\mathcal{D}(X)$ for the value on a figure X, and for a morphism $f\colon X \to Y$ we write

$$f^* := \mathcal{D}(X \xleftarrow{f} X \xrightarrow{\mathrm{id}} X), \qquad f_! := \mathcal{D}(X \xleftarrow{\mathrm{id}} X \xrightarrow{f} Y)$$

the second being defined when $f \in E$.

Proposition 2.4 (The operations and their formal consequences). [ESTABLISHED]
Let $\mathcal{D}$ be a six-functor formalism on (C, E). Then

1. *each $\mathcal{D}(X)$ is a presentable stable symmetric monoidal ∞-category, the monoidal structure $\otimes$ and its right adjoint $\underline{\mathrm{Hom}}$ coming from the lax monoidality of $\mathcal{D}$ applied to the diagonal;*
2. *f^* is symmetric monoidal and, being colimit-preserving between presentable categories, admits a right adjoint f_*;*
3. *$f_!$ is colimit-preserving and admits a right adjoint $f^!$;*
4. *for every cartesian square with $g \in E$ one has a canonical equivalence $f^* g_! \simeq g'_! f'^*$;*
5. *the projection formula $g_!(A \otimes g^* B) \simeq g_! A \otimes B$ holds canonically.*

Statements (4) and (5) are consequences of the functoriality and lax monoidality of $\mathcal{D}$, not additional hypotheses.

Proof. This is the content of the cited axiomatisations; see (Mann 2022, sect. A.5), (Heyer and Mann 2024) and (Scholze 2025). The adjoints in (2) and (3) exist by the adjoint functor theorem for presentable categories (Lurie 2009a, sect. 5.5.2). Item (4) is the composition law of Definition 2.2 applied to the two ways of factoring a correspondence, and (5) is lax monoidality applied to the correspondence $X \leftarrow X \rightarrow Y$ tensored with the identity of Y. □

One point in Definition 2.3 is easy to pass over and should not be. The functor $\mathcal{D}$ is required to be *lax* symmetric monoidal, not symmetric monoidal. Laxity means that one is given comparison maps

$$\mathcal{D}(X) \otimes \mathcal{D}(Y) \longrightarrow \mathcal{D}(X \times Y)$$

which need not be equivalences. When they are equivalences one says that the formalism satisfies *Künneth*, and this is a genuine restriction: it holds for ℓ-adic sheaves on separated finite-type schemes over a field, by the Künneth formula together with the generation of constructible complexes on a product by exterior products, and it fails in settings where the sheaf theory has room for objects on $X \times Y$ that are not assembled from objects on the factors. Almost every duality statement in this chapter — Criterion 2.22, and hence KH.4 in the form we use it — requires Künneth in the two instances $Y = X$ and $Y = X \times X$, since ev and coev are otherwise maps between the wrong objects. Laxity is retained in the definition, rather than strengthened to strong monoidality, because the interesting examples are lax and one wants Künneth to be a hypothesis checked on the figures one cares about, not a condition built into the notion of a geometry.

A second point concerns the direction of the adjunctions. The functor $\mathcal{D}$ is covariant on Corr, and a correspondence has a backwards leg; so $\mathcal{D}$ is contravariant in the left leg and covariant in the right. The four functors $f^*, f_!, f_*, f^!$ therefore split into two which are values of $\mathcal{D}$ and two which are adjoints, and only the first two are functorial in a way visible inside Corr. This asymmetry is the source of most of the technical care in §2.5: a morphism of chôrai is a transformation of the values, its compatibility with the adjoints is then a formal consequence, whereas its compatibility with the duality involution and with the trace has to be imposed as extra data.

There is a further formal feature of Corr which is so useful, and so misleading, that it must be isolated.

Observation 2.5 (Formal self-duality of figures). [ESTABLISHED]
Every object X of $\mathrm{Corr}(C, E)$ is self-dual, provided the diagonal $\Delta_X \colon X \to X \times X$ and the structure map $p_X \colon X \to *$ lie in E. The evaluation and coevaluation are

$$\mathrm{ev} \colon X \times X \xleftarrow{\Delta_X} X \xrightarrow{p_X} *, \qquad \mathrm{coev} \colon * \xleftarrow{p_X} X \xrightarrow{\Delta_X} X \times X,$$

and the two triangle identities hold, so that the trace, *in* Corr, of a correspondence $X \leftarrow Z \rightarrow X$ is the figure $Z \times_{X\times X} X$ of its fixed points.

Proof. Both legs of ev and coev are admissible, so the two correspondences are 1-morphisms of $\mathrm{Corr}(\mathcal{C}, E)$. We verify the first triangle identity, that the composite

$$X \;\simeq\; X \times * \xrightarrow{\mathrm{id}_X \times \mathrm{coev}} X \times X \times X \xrightarrow{\mathrm{ev} \times \mathrm{id}_X} * \times X \;\simeq\; X$$

is the identity; the second is obtained from the first by exchanging the two factors, since ev and coev are interchanged by the symmetry of Corr. correspondences out. The first is

$$X \times * \xleftarrow{\mathrm{id}\times p_X} X \times X \xrightarrow{\mathrm{id}\times \Delta_X} X \times X \times X,$$

and the second is

$$X \times X \times X \xleftarrow{\Delta_X \times \mathrm{id}} X \times X \xrightarrow{p_X \times \mathrm{id}} * \times X.$$

By Definition 2.2 their composite is computed by the pullback

$$P \;=\; (X \times X) \times_{X\times X\times X} (X \times X),$$

the maps being $\mathrm{id} \times \Delta_X$ and $\Delta_X \times \mathrm{id}$. Writing points formally, the first sends (x, y) to (x, y, y) and the second sends (y', z) to (y', y', z); the fibre product therefore imposes $x = y'$, $y = y'$ and $y = z$, so that $P \simeq X$ with both projections the identity. This needs no points to justify it: P is the limit of the two glued squares

$$X \times_{X\times X} (X \times X) \quad \text{and} \quad (X \times X) \times_{X\times X} X,$$

each of which is a pullback along a diagonal, and a pullback along a diagonal of a map that factors through that diagonal returns the source. The composite correspondence is therefore

$$X \times * \xleftarrow{\mathrm{id}} X \xrightarrow{\mathrm{id}} * \times X,$$

which under $X \times * \simeq X \simeq * \times X$ is the identity 1-morphism. Applying $\mathcal{D}$ is not involved anywhere in the argument, which is why the conclusion is available for every figure. □

The final sentence of the observation follows by the same computation: the trace of $X \leftarrow Z \rightarrow X$ is the composite $\mathrm{ev} \circ (c \times \mathrm{id}) \circ \mathrm{coev}$ from $*$ to $*$, and unwinding the three pullbacks leaves the single fibre product $Z \times_{X\times X} X$ formed along $(f, g) \colon Z \to X \times X$ and Δ_X.

Warning 2.6 (Formal duality is not categorical duality). [DESIGN]

The self-duality of Observation 2.5 lives in $\mathrm{Corr}(\mathcal{C}, E)$ and is available for every figure, with no hypotheses beyond admissibility of Δ_X and p_X. It does *not* follow that $\mathcal{D}(X)$ is a dualisable object of $\mathrm{Pr}^L_{\mathrm{st}}$, nor that the trace computed through the image of ev and coev agrees with the categorical trace of an endomorphism of $\mathcal{D}(X)$. A lax monoidal functor carries duality data to duality data only when the lax structure maps involved are equivalences, and the relevant structure map is the natural transformation $p_{X!} \to p_{X*}$. The two traces agree when that norm map is an equivalence, and in general they do not agree at all. This is the exact reason why duality is imposed as an axiom in §2.4 rather than proved, and the non-example of §2.8 is built to make the discrepancy visible.

2.3 The chôra and the primitive trace

We can now write down the object of study.

Definition 2.7 (Chôra). [DESIGN]
A *chôra* is a tuple $\mathfrak{C} = (\mathcal{C}, E, \mathcal{D}, \mathbb{D}, \tau)$ in which

- $\mathcal{C}$ is an ∞-category with finite limits, whose objects are called *figures*;
- E is a class of morphisms of $\mathcal{C}$ containing the equivalences and stable under composition and base change, whose members are called *admissible legs*;
- $\mathcal{D}\colon \mathrm{Corr}(\mathcal{C}, E)^{\otimes} \to \mathrm{Pr}^{L,\otimes}_{\mathrm{st}}$ is a six-functor formalism in the sense of Definition 2.3;
- $\mathbb{D}$ is a *duality involution*: a natural family of equivalences $\mathbb{D}_X\colon \mathcal{D}(X)^{\mathrm{op}} \xrightarrow{\sim} \mathcal{D}(X)$ with $\mathbb{D}_X^2 \simeq \mathrm{id}$, exchanging $f_!$ with f_* and f^* with $f^!$ up to coherent equivalence;
- τ is a *trace theory* in the sense of Definition 2.8.

The figures are not assumed to have points, underlying topological spaces, or structure sheaves, and no site is postulated.

The first four entries of the tuple are, up to packaging, familiar. The fifth is the novelty, and the substance of this section is to say what it is and why it cannot be omitted.

Definition 2.8 (Trace theory). [DESIGN]
Let $(\mathcal{C}, E, \mathcal{D})$ be as above and let $\mathcal{D}^{\mathrm{dbl}} \subseteq \mathcal{D}$ denote the sub-formalism spanned by the figures X for which $\mathcal{D}(X)$ is a dualisable object of $\mathrm{Pr}^L_{\mathrm{st}}$. A *trace theory* on $\mathcal{D}$ consists of

1. for each such X and each endomorphism F of $\mathcal{D}(X)$ in $\mathrm{Pr}^L_{\mathrm{st}}$, a spectrum $\tau(X, F)$, functorial in F and invariant under conjugation;
2. a *localisation datum*: a figure $\mathrm{Fix}(F)$ in $\mathcal{C}$ together with a class $\mathrm{loc}(F) \in \tau(\mathrm{Fix}(F))$ whose image under the structure map computes $\tau(X, F)$;
3. an *additivity datum*: for every exact sequence of endomorphisms, an equivalence exhibiting τ as additive in the sense of May (May 2001), compatibly with (2);
4. a *multiplicativity datum*: equivalences $\tau(X \times Y, F \boxtimes G) \simeq \tau(X, F) \otimes \tau(Y, G)$, coherently associative and unital;
5. a *cyclic datum*: an action of the circle on $\tau(X, \mathrm{id})$, together with the coherences making $\tau(-, \mathrm{id})$ a functor to spectra with circle action.

A trace theory is *primitive* when it is part of the data of the chôra, that is, when it is not required to be induced by any construction from $(\mathcal{C}, E, \mathcal{D}, \mathbb{D})$.

Definition 2.9 (Primitive trace theory). [DESIGN]
A chôra $\mathfrak{C}$ has a *primitive trace* if τ is specified as the fifth entry of the tuple and morphisms of chôrai are required to respect it, in the sense of Definition 2.18. Equivalently: the trace is a structure on a six-functor formalism, not a property of it, and two chôrai may have the same $(\mathcal{C}, E, \mathcal{D}, \mathbb{D})$ and different τ.

Why insist on this? There is a familiar construction which appears to make the insistence unnecessary. If $\mathcal{M}$ is a dualisable object of a symmetric monoidal ∞-category and F is an endomorphism of $\mathcal{M}$, one has the categorical trace

$$\mathrm{tr}(F)\colon \mathbf{1} \xrightarrow{\mathrm{coev}} \mathcal{M} \otimes \mathcal{M}^{\vee} \xrightarrow{F \otimes \mathrm{id}} \mathcal{M} \otimes \mathcal{M}^{\vee} \xrightarrow{\sigma} \mathcal{M}^{\vee} \otimes \mathcal{M} \xrightarrow{\mathrm{ev}} \mathbf{1},$$

and this construction is entirely determined by the monoidal structure (Toën and Vezzosi 2015; Hoyois et al. 2017; Ponto and Shulman 2012). Since a six-functor formalism lands in a symmetric monoidal category, one might expect the trace to come for free. Three separate obstructions stand in the way, and each is a reason for primitivity.

First, the bare categorical trace is a value, not a theory. What $\mathrm{tr}(F)$ produces is a single endomorphism of the unit. What a trace formula requires is a decomposition of that endomorphism into contributions indexed by a figure of fixed points, and that decomposition is item (2) of Definition 2.8. In the classical setting this is the theory of Lefschetz–Verdier local terms (SGA 5, Exposé III), whose refinement and computation is the subject of Varshavsky's work on local terms and of the subsequent analysis by Gaitsgory, Kazhdan, Rozenblyum, Varshavsky and Yom Din. That the local terms exist and are computable is not formal; the categorical trace knows the sum but not the summands.

Second, dualisability is not decidable from the operations. In the analytic settings of §2.8 the categories $\mathcal{D}(X)$ are typically not compactly generated, and whether they are dualisable is a genuine question whose answer requires the theory of continuous K-theory for dualisable categories developed by Efimov (Efimov 2024). A structure which is only defined on an undecidable subclass is not a structure one can axiomatise as derived.

Third, and decisively, the trace is not determined by the six operations. We prove this.

Proposition 2.10 (Non-uniqueness of the trace). [THEOREM]
There exist two chôrai with identical underlying data $(C, E, \mathcal{D}, \mathbb{D})$ and non-equivalent trace theories τ, τ'. Hence no construction can produce τ from $(C, E, \mathcal{D}, \mathbb{D})$ alone, and τ must be primitive.

Proof. Take C to be the one-object category $\{*\}$, E all morphisms, and $\mathcal{D}(*) = \mathrm{Mod}_k$ for $k = \mathbf{F}_p$, with $\mathbb{D}$ the standard k-linear duality on perfect complexes extended by ind-completion. Since C has a single object and a single morphism, the six-functor formalism $\mathcal{D}$ consists of exactly one symmetric monoidal presentable stable category and the identity operations; there is no further data to distinguish.

Now consider two trace theories on this formalism. The first, τ, assigns to the identity endomorphism the topological Hochschild homology of k relative to the sphere,

$$\tau(*, \mathrm{id}) = \mathrm{THH}(\mathbf{F}_p/\mathbb{S}) = \mathrm{THH}(\mathbf{F}_p),$$

whose homotopy is, by Bökstedt's computation, the polynomial algebra $\pi_* \mathrm{THH}(\mathbf{F}_p) = \mathbf{F}_p[x]$ on a class x in degree 2. The circle action of item (5) is the standard cyclotomic one, and the additivity and multiplicativity data are those of (Nikolaus and Scholze 2018). The second, τ', assigns the Hochschild homology of k relative to k itself,

$$\tau'(*, \mathrm{id}) = \mathrm{HH}(\mathbf{F}_p/\mathbf{F}_p) = \mathbf{F}_p,$$

concentrated in degree 0, with the trivial circle action; additivity and multiplicativity are again standard. Both satisfy all five items of Definition 2.8. But $\pi_2 \tau(*, \mathrm{id}) = \mathbf{F}_p \neq 0 = \pi_2 \tau'(*, \mathrm{id})$, so the two trace theories are not equivalent. Since the underlying $(C, E, \mathcal{D}, \mathbb{D})$ is literally the same tuple in both cases, no functorial construction can recover τ from it. □

The proof is short, and it is worth saying exactly what it shows. It shows that the trace records the *base of linearity*: whether one regards $\mathbf{F}_p$ as an algebra over the sphere spectrum or over itself. That choice is invisible to the six operations, which see only the category $\mathrm{Mod}_{\mathbf{F}_p}$ and its monoidal structure, and it is exactly the choice that distinguishes topological from ordinary Hochschild homology. Since the arithmetic content of the theory — p-adic Hodge theory, the cyclotomic trace, the entire apparatus of TC — is carried by that choice and not by the operations, one cannot have it for free.

Among the five items of Definition 2.8 the second is the one which does the arithmetic work, and a word about it is in order. Without a localisation datum, a trace theory assigns a spectrum to a pair (X, F) and nothing more; with one, it assigns a *distribution* on the figure of

fixed points, of which the spectrum is the total mass. Every trace formula in the literature is a statement about the distribution rather than the mass: Grothendieck–Lefschetz says that the local terms at Frobenius fixed points are all equal to one; Atiyah–Bott says that the local term at a non-degenerate fixed point is the reciprocal of a determinant; the Arthur–Selberg formula organises its geometric side by conjugacy classes, which are the fixed points of a correspondence on a quotient. To require only the mass would be to require nothing usable. On the other hand the localisation datum cannot be extracted from the mass, since a spectrum does not remember a decomposition, and it is not determined by the operations, by the same argument as in Proposition 2.10. This is a second, independent reason for primitivity, and it is the reason the localisation datum was written into the definition of a trace theory rather than added later as a property.

Remark 2.11 (Primitivity is not arbitrariness). [CONJECTURE]
Primitivity does not mean that τ is unconstrained. The counterpart of Proposition 2.10 is a rigidity statement: once one demands that the trace be linear over the sphere spectrum and compatible with the cyclotomic structure, the trace theory is expected to be unique up to equivalence, and to be topological Hochschild homology. That statement is a conjecture of the treatise and is taken up in Chapter 3; it is stated there, not here, and nothing in this chapter depends on it.

2.4 The axioms

We now state the five foundational axioms of chôral geometry. They are numbered KH.1–KH.5 throughout the treatise; the remaining axioms of the system, which concern the trace and its rigidity, dualisability and ambidexterity, and the absolute base, are stated in Chapter 3, Chapter 4 and Chapter 7. Four of the five are design principles: they carry status D, meaning that they are definitional commitments whose justification is the coherence of what follows, not propositions awaiting proof. The fifth is a conjecture, and it is the weakest link in the group. After each axiom we give a commentary and, separately, a falsification condition: a statement of what one would have to exhibit in order to refute the axiom, or to show that adopting it was a mistake. A design principle cannot be false in the way a theorem can, but it can be shown to be empty, or incoherent, or to exclude the examples it was invented to include, and those are the failures we record.

2.4.1 KH.1: Geometry is a six-functor formalism

Axiom 2.12 (KH.1 (Geometry is a six-functor formalism)). [DESIGN]
A geometry is a pair $(\mathcal{C}, E)$ as in Definition 2.2 together with a lax symmetric monoidal functor

$$\mathcal{D}\colon \operatorname{Corr}(\mathcal{C}, E)^{\otimes} \longrightarrow \mathrm{Pr}^{L,\otimes}_{\mathrm{st}}.$$

No underlying topological space, locale, site, or set of points is postulated, and no structure sheaf is part of the data. The objects of $\mathcal{C}$ are figures; the geometry is the functor.

Commentary. The axiom is the inversion of Principle 2.1 in technical form. Its content is negative as much as positive: it forbids the use of points, opens, stalks, generic points and local rings in the definition of a geometry, and thereby forces every geometric notion to be expressed in terms of the operations. The gain is that the class of geometries is enlarged to include structures which have no space underlying them — condensed analytic stacks, categories of modules over structured ring spectra, the conjectural absolute base of Chapter 7

— while the loss is the entire toolkit of local verification. One does not, under KH.1, check a statement on an open cover; one checks it by exhibiting an adjunction.

It is worth being clear that KH.1 does not assert that spaces are useless, nor that the six-functor formalisms arising in practice fail to come from spaces. Almost all of them do. The assertion is that the space is not part of the definition, so that theorems proved from the axioms apply equally to the formalisms which have no space.

Falsification. KH.1 fails if it is empty or if it is too generous. It is empty if no theorem of substance can be proved from it: if every result requires, in its proof, a return to points. It is too generous if the class of six-functor formalisms admits pathologies that make the subsequent axioms vacuous — for instance, if there is a six-functor formalism satisfying KH.2–KH.4 in which the trace of the identity is zero on every figure, so that the trace carries no information. The practical falsification is the first: a proof of the comparison theorem of §2.7 that could not be carried out without reintroducing the underlying scheme would show that the axiom is doing no work. We claim in §2.7 that it can be carried out; the reader should judge.

2.4.2 KH.2: The operations are functoriality on correspondences

Axiom 2.13 (KH.2 (Six operations as functoriality)). [DESIGN]

The six operations of a geometry are the components of $\mathscr{D}$ in the sense of Definition 2.3: f^* and $f_!$ are the values of $\mathscr{D}$ on the two elementary correspondences, f_* and $f^!$ are their right adjoints, and $\otimes$ and $\underline{\mathrm{Hom}}$ *come from lax monoidality. Proper base change and the projection formula are therefore theorems (Proposition 2.4), obtained from the composition law in* $\mathrm{Corr}(C, E)$ *and the monoidal structure, and are not additional axioms.*

Commentary. The mathematical content of KH.2 is the reduction of a long list of compatibilities to a single functor. In the classical development, the construction of $f_!$ for a separated morphism of finite type requires a compactification, the verification that the result is independent of the compactification, and then a further verification of base change and the projection formula; each of these is a substantial argument, and each must be redone in every new setting. Presented as functoriality on correspondences, the compatibilities are consequences of associativity of pullback. This is the technical justification for the claim in §2.9 that the chôra addresses failure F3.

The reduction is not free: it displaces the labour into showing that the assignment $\mathscr{D}$ is a functor of ∞-categories at all, which is where the coherence problems live. The gluing techniques of Liu and Zheng, and the machinery of Gaitsgory and Rozenblyum, exist precisely to discharge that burden (Gaitsgory and Rozenblyum 2017a). We record in §2.5 that the status of some of this machinery has been questioned in the literature, and we do not conceal that our construction depends on it.

Falsification. KH.2 fails if there is a geometry of interest whose operations demonstrably do not assemble into a functor on correspondences — for example, a setting in which $f_!$ exists and satisfies base change but for which the coherence data required to make $\mathscr{D}$ a functor of ∞-categories provably do not exist. It also fails, in a weaker but still fatal sense, if the passage to Corr loses information that the separate operations retain, so that two inequivalent geometries have equivalent $\mathscr{D}$.

2.4.3 KH.3: Geometricity by adjointability

Axiom 2.14 (KH.3 (Geometricity by adjointability)). [DESIGN]

The geometric adjectives are defined *by adjointability properties of the operations, not characterised by them. For $f\colon X \to Y$ in E:*

- *f is* proper *if the canonical norm map* $\mathrm{Nm}_f \colon f_! \to f_*$ *is an equivalence;*
- *f is* cohomologically smooth *if* $f^! \simeq f^*(-) \otimes \omega_f$ *for an invertible object* $\omega_f \in \mathcal{D}(X)$*, the dualising object of f, compatibly with base change;*
- *f is* étale *if it is cohomologically smooth with* $\omega_f \simeq \mathbf{1}$*, equivalently if* $f^! \simeq f^*$*;*
- *f is* cohomologically proper *if $f_!$ preserves compact objects.*

These are definitions in every chôra; in the chôra of schemes they agree with the classical notions.

Commentary. This axiom is where the inversion becomes visible in daily practice. Classically, properness is a condition on the underlying topological spaces (universally closed, separated, of finite type) and the equivalence $f_! \simeq f_*$ is the *proper base change theorem*, a hard result. Here the implication runs the other way: the adjointability is the definition, and the agreement with the classical notion in the scheme case is a comparison statement to be proved once, in that realisation, rather than a theorem to be reproved in each new geometry.

The word "cohomologically" is not decoration. In the analytic and spectral realisations there are morphisms which satisfy $f_! \simeq f_*$ without being proper in any topological sense, and morphisms which are topologically proper without satisfying it. The axiom commits us to the cohomological side of every such divergence. The separation of *proper* from *cohomologically proper* in the list above is deliberate: the first is about the norm map, the second about preservation of compact objects, and the two are independent. In the non-example of §2.8 every morphism is proper by fiat, since $f_!$ is *defined* to be f_*, and yet duality fails; conversely there are chôrai in which $f_!$ preserves compact objects while the norm map is very far from an equivalence, the failure being measured by a Tate construction. That second phenomenon is ambidexterity, and it is the subject of Chapter 4.

Falsification. KH.3 fails if the adjointability definitions disagree with the classical ones in the scheme realisation, since the scheme case is the calibration. It fails in a subtler way if the adjectives so defined are not stable under the operations one expects: if, say, the composite of two morphisms with $f_! \simeq f_*$ can fail to have $f_! \simeq f_*$, or if the class of cohomologically smooth morphisms is not stable under base change in some chôra of interest. Stability under base change is automatic from the definitions given; stability under composition is not, and a counterexample in a chôra satisfying KH.1 and KH.2 would require the definitions to be amended.

2.4.4 KH.4: Duality

Axiom 2.15 (KH.4 (Duality)). [DESIGN]

A chôra carries a duality involution $\mathbb{D}$: a natural family of equivalences $\mathbb{D}_X \colon \mathcal{D}(X)^{\mathrm{op}} \xrightarrow{\sim} \mathcal{D}(X)$ *with* $\mathbb{D}_X^2 \simeq \mathrm{id}$*, exchanging* $f_! \leftrightarrow f_*$ *and* $f^* \leftrightarrow f^!$ *up to coherent equivalence. On the full subcategory of dualisable figures — those X for which $\mathcal{D}(X)$ is dualisable in* $\mathrm{Pr}^L_{\mathrm{st}}$ *— the involution restricts to a perfect pairing, and the image under $\mathcal{D}$ of the formal self-duality of Observation 2.5 is a genuine self-duality of $\mathcal{D}(X)$.*

Commentary. This is the axiom that does the most work and is the most easily overlooked, because in the scheme case it is a theorem — Verdier duality — and one is tempted to expect it always. It is not always available. As Warning 2.6 records, the self-duality of a figure in Corr is formal and universal, but a lax monoidal functor carries it forward only when the relevant structure maps are equivalences. KH.4 postulates that the chôrai we admit are those for which it does.

Two consequences of adopting KH.4 as an axiom deserve to be stated. First, chôrai without self-duality exist, are perfectly reasonable six-functor formalisms, and are *excluded from the theory by fiat*. The non-example of §2.8 is exactly such an object, and we present it in detail so

that the cost of the exclusion is visible. Second, since dualisability is the hypothesis under which the categorical trace is defined at all, KH.4 is what makes the trace of Definition 2.8 well posed. The axiom and the fifth entry of the tuple are therefore not independent: KH.4 delimits the domain on which τ is defined.

Falsification. KH.4 is falsified as a *useful* axiom if the class of chôrai satisfying it is too small to contain the intended examples. The sharp test is the four realisations of §2.8: if condensed analytic stacks, or the spectral realisation, failed KH.4, the axiom would have to be weakened to a relative or partial duality. As matters stand they satisfy it, the analytic case by the dualisability results for nuclear modules and the theory of (Efimov 2024). The axiom is falsified as a *correct* axiom if some construction the treatise depends upon — the trace formula of Chapter 27, the categorical Lefschetz theorem of Chapter 21 — can be shown to require a chôra in which $\mathbb{D}^2 \not\simeq \mathrm{id}$.

2.4.5 KH.5: Reconstruction

Axiom 2.16 (KH.5 (Reconstruction, or Tannakian adequacy)). [CONJECTURE]
For a chôra $\mathfrak{C}$ in the intended class, the category of figures is recovered from the sheaf theory: the canonical functor

$$C \longrightarrow \mathrm{Spec}(\mathrm{CAlg}(\mathcal{D}))$$

sending a figure to the spectrum of the commutative algebra objects of its sheaf theory is an equivalence onto its essential image, and that image is all of the target. Equivalently: a chôra has enough points, in the Tannakian sense, to be reconstructed from its sheaf theory.

Commentary. KH.5 is a conjecture and it is marked as such. It is the weakest link in the group, and we say plainly why it matters. Without KH.5 the framework is still a framework: the operations still compose, the trace is still defined, and the comparison theorem of §2.7 still holds, since its proof takes place inside a chôra which is known to come from schemes. What is lost without KH.5 is the *claim to have replaced geometry*. If the sheaf theory does not determine the figures, then the figures are additional data and the chôra has not absorbed them; one is then studying sheaf theories over spaces, which is what one was doing before.

The Tannakian precedents are encouraging but not conclusive (Rivano 1972; Deligne and Milne 1982; Lurie 2018). Reconstruction is known for quasi-compact quasi-separated schemes from their symmetric monoidal categories of quasi-coherent sheaves, by results in the tradition of Balmer's tensor-triangular reconstruction and Lurie's spectral Tannaka duality; it is known in the derived and spectral settings under hypotheses; and it is open in the analytic and condensed settings, where the relevant categories are not compactly generated. The accurate statement is that we expect reconstruction where it has been proved and hope for it elsewhere.

It is worth distinguishing two things that "having enough points" might mean, because they behave differently. The first is the topos-theoretic sense: a topos has enough points when a map of sheaves is an equivalence as soon as it is one on every stalk. In that sense the étale topos of a scheme has enough points, by a theorem of Deligne, and the exodromy theorems in the tradition of Barwick, Glasman and Haine — which we attribute in prose — recover the étale topos from the profinite stratified shape, that is, from a category of points together with specialisation data. The second sense is the Tannakian one used in KH.5: the *figure* is recovered, as an object of C, from the sheaf theory as an algebraic structure. The first sense is a statement about a fixed topos; the second is a statement about the functor $C \to \mathrm{CAlg}(\mathcal{D})^{\mathrm{op}}$ being an equivalence onto its image. KH.5 asks for the second, and the first does not imply it.

This matters because several of the realisations of §2.8 have enough points in the first sense and are not known to satisfy the second; realisation (c), where the figures are formal duals of ring spectra, satisfies the second essentially by construction and has no points at all in the first sense. The two notions are independent, and only the second is the axiom.

Conjecture 2.17 (Reconstruction in the quasi-compact case). [CONJECTURE]
Let $\mathfrak{C}$ be a chôra whose figures form a full subcategory of quasi-compact quasi-separated derived schemes and whose sheaf theory is quasi-coherent, with E the class of morphisms of finite presentation. Then the functor of KH.5 is fully faithful, and its essential image consists of those objects of $\mathrm{Spec}(\mathrm{CAlg}(\mathcal{D}))$ *whose associated sheaf theory is compactly generated by a single object.*

Falsification. KH.5 is falsified by exhibiting two non-equivalent chôrai in the intended class with equivalent sheaf theories, duality and trace. Such an example would not damage the internal coherence of the framework, but it would demote the framework from a replacement for geometry to a technique within it. A weaker falsification, which we regard as more likely, is that reconstruction holds only after restricting to a subclass so narrow that the interesting analytic and absolute examples fall outside it. In that case KH.5 survives as a theorem about a small class and fails as an axiom about the intended one.

2.5 The 2-category of chôrai

A definition of geometry is of no use unless geometries can be compared, and the comparison of two chôrai is not a single map but a structure with its own morphisms. We therefore give the 2-category $\mathrm{Geom}^{\mathrm{tr}}$ in full. The reader who wants only the statement of the comparison theorem may pass to §2.7; the definition is given here in detail because the comparison with EGA in §2.9 turns on whether it is a definition or a gesture.

That the structure is a 2-category and not a category is forced, not chosen. A 1-morphism of chôrai includes a natural transformation α of sheaf theories, and natural transformations of functors between ∞-categories are themselves the objects of a category, not the elements of a set; two comparisons of the same pair of geometries may differ by a homotopy, and the homotopy is data. In the scheme realisation this is visible already in the simplest example: the identity comparison of $\mathfrak{C}_{\mathrm{Sch}}$ with itself admits the Frobenius twist as a non-identity self-comparison, and the 2-morphisms record how the twist interacts with base change. A framework which flattened these into equalities would be unable to state the comparison theorems that Part IV needs, all of which are equivalences of categories rather than identities of objects.

Definition 2.18 (The 2-category of chôrai with trace). [DESIGN]
The $(\infty,2)$-category $\mathrm{Geom}^{\mathrm{tr}}$ has the following data.
Objects. Chôrai $\mathfrak{C} = (C, E, \mathcal{D}, \mathbb{D}, \tau)$ in the sense of Definition 2.7, satisfying KH.1–KH.4.
1-morphisms. A 1-morphism $\mathfrak{C} \to \mathfrak{C}'$ is a quadruple $(u, \alpha, \delta, \beta)$ where

- $u\colon C \to C'$ preserves finite limits and satisfies $u(E) \subseteq E'$. These two conditions are exactly what is needed for u to induce a symmetric monoidal functor $\mathrm{Corr}(u)\colon \mathrm{Corr}(C,E)^{\otimes} \to \mathrm{Corr}(C',E')^{\otimes}$, since preservation of finite limits makes u compatible with the composition law and $u(E) \subseteq E'$ makes it land in admissible legs;
- $\alpha\colon \mathcal{D} \Rightarrow \mathcal{D}' \circ \mathrm{Corr}(u)$ is a lax symmetric monoidal natural transformation whose components $\alpha_X\colon \mathcal{D}(X) \to \mathcal{D}'(uX)$ preserve colimits;
- δ is a duality compatibility: a family of equivalences $\delta_X\colon \alpha_X \circ \mathbb{D}_X \simeq \mathbb{D}'_{uX} \circ \alpha_X^{\mathrm{op}}$, coherent with the involutivity data $\mathbb{D}^2 \simeq \mathrm{id}$ and $\mathbb{D}'^2 \simeq \mathrm{id}$;
- β is a trace comparison: for every dualisable figure X of $\mathfrak{C}$ such that uX is dualisable in $\mathfrak{C}'$, and every endomorphism F of $\mathcal{D}(X)$, a map of spectra $\beta_{X,F}\colon \tau(X,F) \to \tau'(uX, \alpha_* F)$,

natural in F, compatible with the localisation, additivity, multiplicativity and cyclic data of Definition 2.8.

A 1-morphism is *trace-exact* if every $\beta_{X,F}$ is an equivalence, and *dualisability-preserving* if uX is dualisable whenever X is.

2-morphisms. A 2-morphism $(u,\alpha,\delta,\beta) \Rightarrow (u',\alpha',\delta',\beta')$ is a pair (θ,Θ) where

- $\theta\colon u \Rightarrow u'$ is a *cartesian* natural transformation, meaning that every naturality square

$$\begin{array}{ccc} uX & \longrightarrow & u'X \\ \downarrow & & \downarrow \\ uY & \longrightarrow & u'Y \end{array}$$

  is cartesian in $\mathcal{C}'$, and whose components lie in E';
- Θ is a modification $\alpha' \Rightarrow \mathcal{D}'(\mathrm{Corr}(\theta)) \circ \alpha$, compatible with δ, δ' and with β, β'.

Composition of 1-morphisms. Given $(u,\alpha,\delta,\beta)\colon \mathfrak{C} \to \mathfrak{C}'$ and $(v,\alpha',\delta',\beta')\colon \mathfrak{C}' \to \mathfrak{C}''$, the composite is

$$(v \circ u,\ (\alpha' * \mathrm{Corr}(u)) \circ \alpha,\ \delta'',\ \beta'')$$

where $\alpha' * \mathrm{Corr}(u)$ is the whiskering of α' along $\mathrm{Corr}(u)$, and δ'', β'' are the evident pastings. The identity 1-morphism on $\mathfrak{C}$ is $(\mathrm{id}_{\mathcal{C}}, \mathrm{id}_{\mathcal{D}}, \mathrm{id}, \mathrm{id})$.

Composition of 2-morphisms. Vertical composition of (θ,Θ) and (θ_1,Θ_1) is $(\theta_1 \circ \theta, \Theta_1 * \Theta)$; horizontal composition is the usual pasting of whiskerings. Both are defined up to the coherences of Proposition 2.19.

Three points in this definition are not routine and should be emphasised, since they are the places where an incomplete definition would break.

The first is the cartesianness requirement on θ. A natural transformation between functors $\mathcal{C} \to \mathcal{C}'$ does not in general induce a natural transformation between the induced functors on correspondence categories, because a correspondence has a *backwards* leg and naturality supplies only a forward square. What is needed is that the naturality square be cartesian, for then it can be read in either direction and supplies the required comparison of pullbacks. Requiring in addition that the components lie in E' makes θ itself a correspondence, so that $\mathrm{Corr}(\theta)$ is a natural transformation valued in 1-morphisms of $\mathrm{Corr}(\mathcal{C}', E')$. Without both conditions the 2-morphisms of $\mathrm{Geom}^{\mathrm{tr}}$ are not composable.

The second is that β is a *map*, not an equivalence. This is a deliberate weakening. If one demanded that every 1-morphism be trace-exact, then the comparison map from the chôra of schemes over $\mathbf{F}_q$ to the spectral chôra, which is the vehicle for the cyclotomic trace, would fail to be a 1-morphism at all, since the cyclotomic trace $K \to \mathrm{TC}$ is not an equivalence (Bökstedt et al. 1993; Dundas et al. 2013). The interesting comparisons in this treatise are lax, and a definition that excluded them would be useless. Trace-exactness is then a property one proves in the cases where it holds.

The third is the restriction of β to dualisable figures. Since τ is only defined there, a 1-morphism which does not preserve dualisability carries no trace information; this is why dualisability-preserving 1-morphisms are singled out, and it is the reason for the certificate pipeline of §2.6.

Proposition 2.19 (Coherence). [THEOREM]
The data of Definition 2.18 assemble into an $(\infty,2)$-category. Composition of 1-morphisms is associative up to a canonical equivalence, and the associativity equivalences satisfy the pentagon condition; identities are units up to canonical equivalence, and the unit and associativity coherences are compatible.

Proof. Associativity of composition reduces to associativity of whiskering, since

$$(\alpha'' * \mathrm{Corr}(v \circ u)) \circ (\alpha' * \mathrm{Corr}(u)) \circ \alpha$$

may be bracketed in either way and the two bracketings are identified by the canonical equivalence $\mathrm{Corr}(v \circ u) \simeq \mathrm{Corr}(v) \circ \mathrm{Corr}(u)$, which itself comes from the fact that u and v preserve finite limits. The pentagon condition follows because the equivalences in question are all instances of the comparison of iterated pullbacks, for which the pentagon is the standard coherence of finite limits (Lurie 2009a, sect. 4.4). Let us make that reduction explicit. Given composable $u\colon C \to C'$, $v\colon C' \to C''$ and $w\colon C'' \to C'''$, the five vertices of the pentagon are the five bracketings of $\mathrm{Corr}(w) \circ \mathrm{Corr}(v) \circ \mathrm{Corr}(u)$ applied to a 1-morphism, and the five edges are the comparison equivalences. Each comparison equivalence is, at the level of underlying spans, the statement that for a finite-limit-preserving functor u and a cospan $Y \to Z \leftarrow Y'$ in C the canonical map

$$u(Y \times_Z Y') \longrightarrow u(Y) \times_{u(Z)} u(Y')$$

is an equivalence. The pentagon therefore asserts precisely the coherence of the comparison maps for a *doubly* iterated fibre product, namely that the two ways of identifying

$$wvu((Y \times_Z Y') \times_{Z'} Y'')$$

with the corresponding triple pullback in C''' agree. This is the coherence of finite limits under limit-preserving functors, and it holds because the comparison map is natural in the index diagram and the two bracketings correspond to two cofinal subdiagrams of the same cospan-of-cospans, whose comparison is unique up to contractible choice.

The unit coherence is the case $u = \mathrm{id}$ of the same argument: $\mathrm{Corr}(\mathrm{id}) = \mathrm{id}$ on the nose, and the left and right unit equivalences are the canonical identifications $Y \times_Y Y' \simeq Y'$ and $Y \times_{Y'} Y' \simeq Y$, whose compatibility with associativity is again an instance of the coherence of finite limits. The compatibility of δ and β with these equivalences is imposed as part of the data of a 1-morphism, so nothing further is required. The 2-categorical structure — that is, the enrichment of the mapping objects in ∞-categories — follows from the fact that the 2-morphisms of Definition 2.18 are themselves the objects of a mapping space in the ∞-category of lax symmetric monoidal transformations. □

Proposition 2.20 (Symmetric monoidal structure). [THEOREM]
$\mathrm{Geom}^{\mathrm{tr}}$ *is symmetric monoidal. On objects,*

$$\mathfrak{C} \boxtimes \mathfrak{C}' = (C \times C',\ E \times E',\ \mathcal{D} \otimes \mathcal{D}',\ \mathbb{D} \boxtimes \mathbb{D}',\ \tau \otimes \tau'),$$

where $\mathcal{D} \otimes \mathcal{D}'$ is defined using the canonical equivalence

$$\mathrm{Corr}(C \times C', E \times E') \simeq \mathrm{Corr}(C,E) \times \mathrm{Corr}(C',E')$$

followed by the tensor product of $\mathrm{Pr}^L_{\mathrm{st}}$, and $\tau \otimes \tau'$ is defined by the multiplicativity datum. The unit is the trivial chôra $1 = (*, \mathit{all}, X \mapsto \mathrm{Sp}, \mathrm{id}, \mathrm{THH})$.

Proof. The displayed equivalence of correspondence categories holds because limits in a product are computed componentwise, so the pullback defining composition is the pair of pullbacks. The tensor product of $\mathrm{Pr}^L_{\mathrm{st}}$ is symmetric monoidal with unit Sp (Lurie 2017, sect. 4.8), which gives the symmetric monoidal structure on the $\mathcal{D}$-entry; the duality involutions tensor because $\mathbb{D}$ is required to be compatible with $\otimes$; the trace theories tensor by item (4) of Definition 2.8. That 1 is a unit is the observation that $\mathrm{Corr}(*, \mathrm{all})$ is the trivial symmetric monoidal category and $\mathrm{Sp} \otimes \mathcal{E} \simeq \mathcal{E}$. □

The three points just made are recorded schematically in Plate 2.2, which also arranges the realisations of §2.8 by the axioms they satisfy.

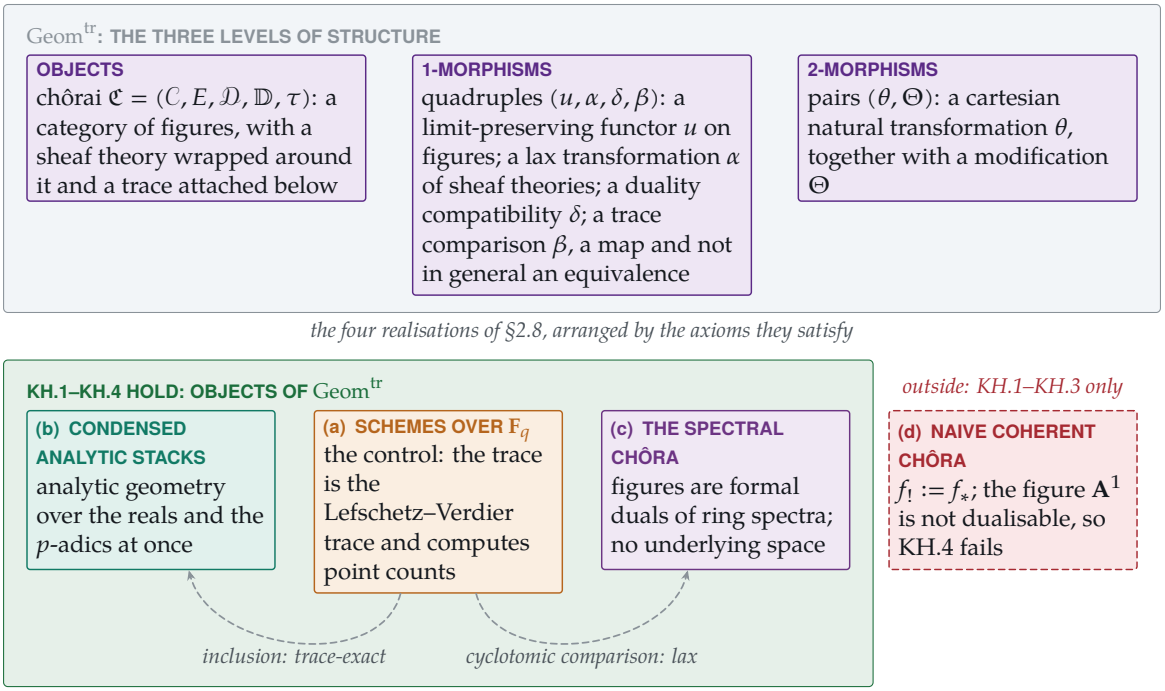


Plate 2.2. The 2-category of chôrai and its realisations. The upper band shows the three levels of structure: objects are chôrai, drawn as a category of figures with a sheaf theory wrapped around it and a trace attached below; 1-morphisms are quadruples consisting of a limit-preserving functor on figures, a lax transformation of sheaf theories, a duality compatibility, and a trace comparison which is a map and not in general an equivalence; 2-morphisms are cartesian natural transformations together with a modification. The lower band shows the four realisations of the final section arranged by which axioms they satisfy: schemes over a finite field and condensed analytic stacks and the spectral chôra all satisfy KH.1 through KH.4, while the fourth, the naive coherent chôra in which $f_!$ is declared to be f_*, sits outside the shaded region because its figure $\mathbf{A}^1$ is not a dualisable figure. The dashed arrows indicate the 1-morphisms discussed in the text: the cyclotomic comparison from schemes to the spectral chôra is lax, and the inclusion of schemes into analytic stacks is trace-exact.

What it does not show: the coherence of the construction: the symmetric monoidal $(\infty, 2)$-category of correspondences on which $\mathrm{Geom}^{\mathrm{tr}}$ *rests is borrowed from the literature, whose coherence arguments have been questioned, and the $(\infty, 3)$-category needed for the cobordism hypothesis is not constructed.*

Status and Caveat. The construction of $\mathrm{Corr}(C, E)^{\otimes}$ as a symmetric monoidal $(\infty, 2)$-category, on which Definition 2.18 rests, uses the machinery of Gaitsgory and Rozenblyum (Gaitsgory and Rozenblyum 2017a, 2017b). The completeness of the coherence arguments in that machinery has been questioned in the literature, and the questions have not been fully resolved in print. Alternative treatments exist — the gluing method of Liu and Zheng, and the more recent axiomatisations of Mann and of Heyer and Mann (Mann 2022; Heyer and Mann 2024) — and we

have written Definition 2.18 so that it can be read in any of them, since it uses only the symmetric monoidal $(\infty,1)$-category of correspondences together with mapping-object enrichment. But the reader should understand that the foundations of this chapter are, at the level of coherence, borrowed rather than owned. A further and larger debt is deferred: the $(\infty,3)$-category of chôrai with dualisable figures, which would be needed to apply the cobordism hypothesis (Lurie 2009b) in the form used in Chapter 23, is not constructed here and may not be constructible with present technology.

2.6 Dualisable figures and how to certify them

KH.4 quantifies over dualisable figures, so the axiom is only as sharp as that notion. Two notions must be kept apart, and conflating them is the commonest error in this subject.

Definition 2.21 (Categorical and chôral dualisability). [DESIGN]
Let $\mathfrak{C}$ be a chôra and X a figure with p_X and Δ_X in E.

- X is *categorically dualisable* if $\mathcal{D}(X)$ is a dualisable object of $\mathrm{Pr}^L_{\mathrm{st}}$.
- X is a *dualisable figure*, or *chôrally dualisable*, if the images under $\mathcal{D}$ of the evaluation and coevaluation of Observation 2.5, namely

$$\mathrm{ev} = p_{X!}\,\Delta_X^*, \qquad \mathrm{coev} = \Delta_{X!}\,p_X^!,$$

satisfy the triangle identities and exhibit $\mathcal{D}(X)$ as self-dual over $\mathcal{D}(*)$.

Chôral dualisability implies categorical dualisability. The converse is false, and KH.4 refers to the former.

The distinction matters because categorical dualisability is cheap: any compactly generated presentable stable category is dualisable, with dual the ind-completion of the opposite of its compact objects. Chôral dualisability asks something quite different — that the *geometric* pairing, built from the operations, be the duality — and that is a finiteness condition on the geometry, not on the category.

Criterion 2.22 (Recognising dualisable figures). [THEOREM]
Let $\mathfrak{C}$ satisfy KH.1–KH.3, let X be a figure with $p_X, \Delta_X \in E$, and assume the Künneth condition that the lax structure maps $\mathcal{D}(X)\otimes\mathcal{D}(Y)\to\mathcal{D}(X\times Y)$ are equivalences for $Y = X$ and $Y = X\times X$. Write $\mathcal{A} = \mathcal{D}(X)^{\omega}$ for the compact objects.

1. (Sufficiency.) *Suppose $\mathcal{D}(X)$ is compactly generated, $p_{X!}$ carries $\mathcal{A}$ into the compact objects of $\mathcal{D}(*)$, and the pairing $\langle M,N\rangle = p_{X!}(M\otimes N)$ induces an equivalence $\mathcal{A}^{\mathrm{op}}\to\mathcal{A}$. Then X is a dualisable figure.*
2. (Necessity.) *Suppose $\mathcal{D}(X)$ is compactly generated and its monoidal unit $\mathbf{1}_X$ is compact. If X is a dualisable figure then $p_{X!}$ carries compact objects to compact objects; that is, p_X is cohomologically proper.*
3. (Beyond compact generation.) *If $\mathcal{D}(X)$ is not compactly generated, categorical dualisability is equivalent to $\mathcal{D}(X)$ being a retract in $\mathrm{Pr}^L_{\mathrm{st}}$ of a compactly generated category, and the relevant invariants are those of Efimov's continuous K-theory (Efimov 2024).*

Proof. (1) Under the Künneth condition, $\mathcal{D}(X)\otimes\mathcal{D}(X)\simeq\mathcal{D}(X\times X)$, so ev and coev are morphisms of the expected shape. Since $\mathcal{D}(X) = \mathrm{Ind}\,\mathcal{A}$, a self-duality is determined by its restriction to $\mathcal{A}$, and that restriction is the pairing $\langle -,-\rangle$, which by hypothesis takes values in compact objects and induces $\mathcal{A}^{\mathrm{op}}\simeq\mathcal{A}$. The triangle identities for the resulting duality are then the assertions

$$p_{13!}(\Delta_{12}^*(M\boxtimes\Delta_{X!}p_X^!\mathbf{1}))\simeq M,$$

which follow from proper base change and the projection formula of Proposition 2.4 applied to the two cartesian squares expressing the self-intersection of the diagonal, together with $\Delta_X^* \Delta_{X!} \simeq \mathrm{id}$ on the relevant objects.

(2) A duality between compactly generated categories is an equivalence $\mathrm{Ind}(\mathcal{A}^{\mathrm{op}}) \simeq \mathrm{Ind}(\mathcal{A})$; being an equivalence it preserves compact objects, so it restricts to $\mathcal{A}^{\mathrm{op}} \simeq \mathcal{A}$ and the pairing ev takes pairs of compact objects to compact objects. Taking $N = \mathbf{1}_X$, which is compact by hypothesis, gives $\mathrm{ev}(M \boxtimes \mathbf{1}_X) = p_{X!}(M \otimes \mathbf{1}_X) = p_{X!}M$ compact for every compact M.

(3) This is Efimov's theorem, restated. □

Part (2) is the working form of the criterion, and it is worth stating its contrapositive in words: *if the unit is compact and pushforward with compact supports does not preserve compactness, the figure is not dualisable and KH.4 excludes the chôra.* We shall use exactly this in §2.8 to dispose of the non-example in a single line.

The criterion is finitary enough to be mechanised, and the treatise's formalisation programme takes it as one of its first targets.

Algorithm 2.23 (The dualisability certificate pipeline). [FORMAL TARGET]

Input: a finite presentation of a chôra $\mathfrak{C}$ — generators and relations for $\mathcal{C}$, a decidable predicate for E, a presentation of $\mathcal{D}$ by generators of $\mathcal{D}(X)$ and structure maps — together with a figure X.

1. *Admissibility.* Verify $p_X \in E$ and $\Delta_X \in E$. If either fails, emit `NOT-APPLICABLE` with the failing leg.
2. *Künneth.* Verify that the structure maps $\mathcal{D}(X) \otimes \mathcal{D}(Y) \to \mathcal{D}(X \times Y)$ are equivalences for $Y = X, X \times X$, by checking on generators. On failure emit `KUNNETH-FAIL`.
3. *Generation.* Produce a set G of compact generators of $\mathcal{D}(X)$ and a certificate that G generates, that is, that an object right-orthogonal to G vanishes.
4. *Finiteness.* For each $g \in G$, verify that $p_{X!}g$ is compact in $\mathcal{D}(*)$. On failure emit `NOT-DUALISABLE` together with the witness g; by Criterion 2.22 this is a proof, not a heuristic, provided $\mathbf{1}_X$ is compact.
5. *Duality.* Construct the candidate involution $\mathbb{D}_X = \underline{\mathrm{Hom}}(-, \omega_X)$ with $\omega_X = p_X^! \mathbf{1}$, verify $\mathbb{D}_X^2 \simeq \mathrm{id}$ on G, and verify that $\langle -, - \rangle$ induces $\mathcal{A}^{\mathrm{op}} \simeq \mathcal{A}$.
6. *Triangles.* Verify the two triangle identities on G.
7. *Fallback.* If step 3 fails because $\mathcal{D}(X)$ is not compactly generated, attempt instead to exhibit $\mathcal{D}(X)$ as a retract of a compactly generated category and emit an Efimov-style certificate.

Output: either `DUALISABLE` with the tuple $(G, \mathbb{D}_X, \omega_X)$ and the verification transcript, or `NOT-DUALISABLE` with a witness, or `UNKNOWN` with the step at which the pipeline stopped.

The status mark on Algorithm 2.23 is F: the pipeline is specified here as a target for machine formalisation, and it has not been implemented. Steps 1, 2 and 4 are decidable in the presentations that arise in §2.8; steps 5 and 6 involve coherence data and are the part for which a proof assistant is needed. The architecture of the formalisation, and the reason for insisting that every such claim in this treatise carry a certificate rather than an assurance, is the subject of Chapter 38. The control flow, together with the two ways the pipeline can fail, is drawn in Plate 2.3.

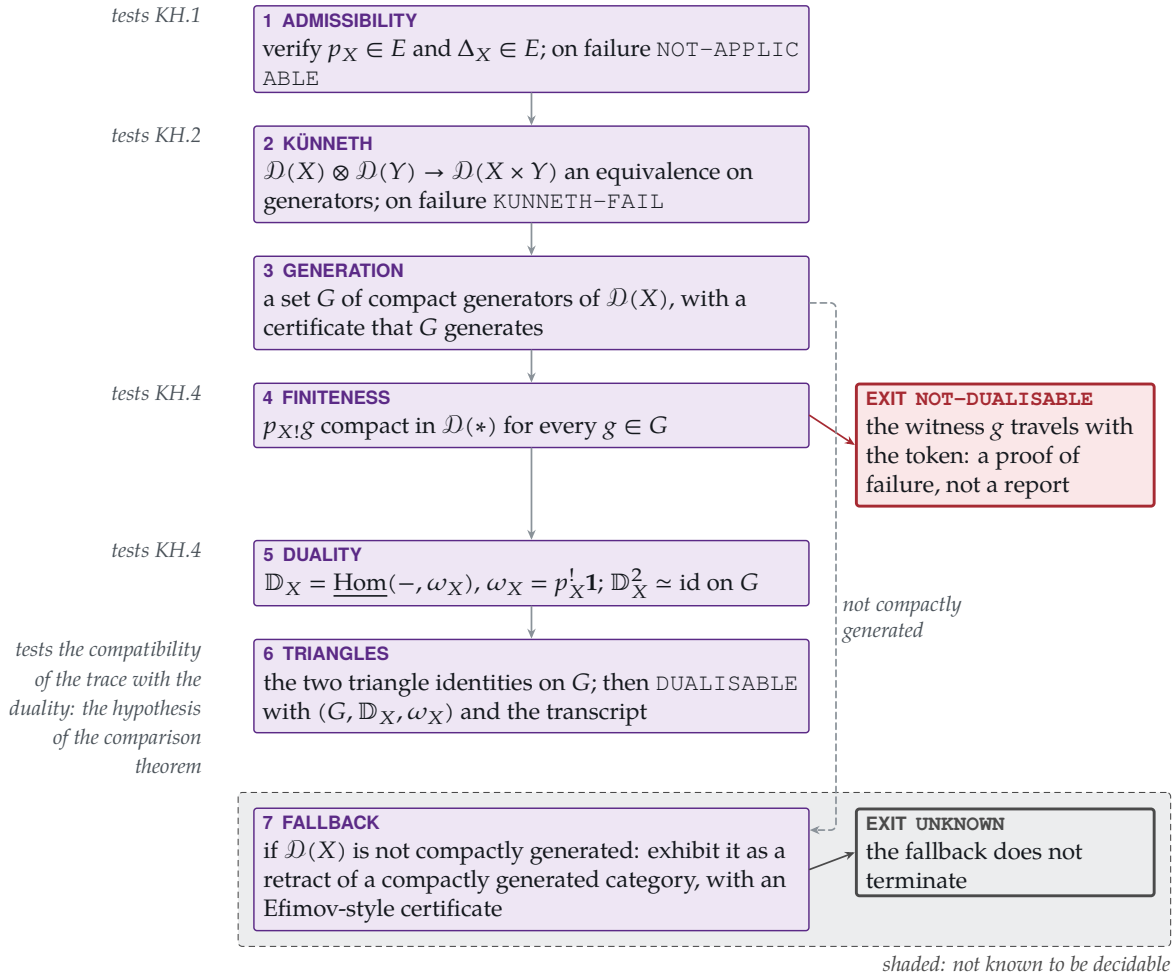


Plate 2.3. The dualisability certificate pipeline. The diagram is a flow chart with seven boxes corresponding to the steps of the algorithm in the text, read downwards, with two exit points on the right. The first exit is NOT-DUALISABLE, reached from the finiteness step when some compact generator has non-compact pushforward with compact supports; the witness travels with the exit token, so that the output is a proof of failure rather than a report of it. The second exit is UNKNOWN, reached when compact generation fails and the Efimov fallback does not terminate; the shaded region on the lower right marks the part of the pipeline which is not known to be decidable. The left-hand column annotates each step with the axiom it tests: steps one and two test KH.1 and KH.2, steps four and five test KH.4, and step six tests the compatibility of the trace with the duality, which is the hypothesis of the comparison theorem.

What it does not show: an implementation: the pipeline is a target for machine formalisation and has not been implemented; steps 5 and 6 carry coherence data for which a proof assistant is needed.

2.7 The comparison with Grothendieck–Lefschetz

Everything so far has been definition. This section contains the theorem which the definitions must earn: that in the geometry of schemes over a finite field the chôral trace is the trace of Frobenius on ℓ-adic cohomology with compact supports, hence the number of rational points. The function-field case is the control throughout this treatise, and it is the calibration for every subsequent claim.

Notation 2.24 (The scheme chôra over a finite field). [DESIGN]
Fix a prime power $q = p^r$, a prime $\ell \neq p$, and $\Lambda = \overline{\mathbf{Q}}_\ell$. Let $\mathcal{C}_{\mathrm{Sch}}$ be the ∞-category of separated schemes of finite type over $k = \mathbf{F}_q$, let E be all morphisms of $\mathcal{C}_{\mathrm{Sch}}$, and let

$$\mathcal{D}_{\mathrm{Sch}}(X) = \mathrm{Ind}\, \mathcal{D}_c^b(X, \Lambda)$$

be the ind-completion of the bounded constructible derived category, with the six operations of SGA 4 and SGA 5 (Artin et al. 1972a, 1973; Grothendieck and Illusie 1977; Deligne 1977; Milne 1980). Let $\mathbb{D}_X = \underline{\mathrm{Hom}}(-, \omega_X)$ with $\omega_X = p_X^! \Lambda$ be Verdier duality, and let τ be the trace theory whose value on (X, F) is the categorical trace of F with its Lefschetz–Verdier localisation. We write $\mathfrak{C}_{\mathrm{Sch}}$ for the resulting chôra.

Theorem 2.25 (Comparison of the chôral and Grothendieck–Lefschetz traces). [THEOREM]
Let X be a separated scheme of finite type over $\mathbf{F}_q$ and let Frob_q denote the q-power geometric Frobenius, regarded as the correspondence $X \xleftarrow{\mathrm{Frob}_q} X \xrightarrow{\mathrm{id}} X$ in $\mathrm{Corr}(\mathcal{C}_{\mathrm{Sch}}, E)$. Then X is a dualisable figure of $\mathfrak{C}_{\mathrm{Sch}}$, and the chôral trace of the induced endomorphism of $\mathcal{D}_{\mathrm{Sch}}(X)$ satisfies

$$\tau(X, \mathrm{Frob}_q) \;=\; \sum_i (-1)^i \,\mathrm{tr}(\mathrm{Frob}_q^* \mid H_c^i(X_{\bar{k}}, \Lambda)) \;=\; \#X(\mathbf{F}_q).$$

More generally, for a correspondence $c\colon X \leftarrow Z \to X$ with $Z \to X \times X$ of finite type, the chôral trace equals the Lefschetz–Verdier trace, and the localisation datum of Definition 2.8 is the classical decomposition into local terms supported on $\mathrm{Fix}(c)$.

Proof. The argument has four steps; each is a citation to the classical theory, and the content of the theorem is that the four fit together to identify a categorical construction with a classical one.

Step 1: X is a dualisable figure. We apply Criterion 2.22. The Künneth condition $\mathcal{D}_{\mathrm{Sch}}(X) \otimes \mathcal{D}_{\mathrm{Sch}}(Y) \simeq \mathcal{D}_{\mathrm{Sch}}(X \times_k Y)$ holds for separated finite-type schemes over a field, by the Künneth formula for ℓ-adic cohomology together with the fact that constructible complexes on a product are generated by exterior products. The category $\mathcal{D}_{\mathrm{Sch}}(X)$ is compactly generated with compact objects the constructible complexes. That $p_{X!}$ carries constructible complexes to constructible, that is, perfect, complexes over Λ is Deligne's finiteness theorem (Deligne 1977). Finally the pairing $\langle M, N\rangle = p_{X!}(M \otimes N)$ induces an equivalence $\mathcal{D}_c^b(X)^{\mathrm{op}} \simeq \mathcal{D}_c^b(X)$, this being Verdier duality (Artin et al. 1973). Hence X is dualisable and $\mathcal{D}_{\mathrm{Sch}}(X)$ is self-dual over $\mathcal{D}_{\mathrm{Sch}}(\mathrm{Spec}\, k)$ with evaluation $p_{X!}\Delta_X^*$ and coevaluation $\Delta_{X!}p_X^!$. As a sanity check, for $X = \mathrm{Spec}\, k$ the two maps are the identity, since Δ and p_X are both the identity.

Step 2: the categorical trace is the Lefschetz–Verdier trace. For a dualisable figure and an endomorphism given by a correspondence c, unwinding the definition of tr through the evaluation and coevaluation of Step 1 gives

$$\mathrm{tr}(c) = p_{\mathrm{Fix}(c)!}(\mathrm{loc}(c)), \qquad \mathrm{Fix}(c) = Z \times_{X \times X} X,$$

where $\mathrm{loc}(c) \in H^0(\mathrm{Fix}(c), \omega_{\mathrm{Fix}(c)})$ is the class constructed in SGA 5, Exposé III (Grothendieck and Illusie 1977). That the categorical trace, formed in the symmetric monoidal ∞-category $\mathrm{Pr}^L_{\mathrm{st}}$, agrees with the classical Lefschetz–Verdier construction is the content of the modern trace formalism (Toën and Vezzosi 2015; Hoyois et al. 2017; Ponto and Shulman 2012), applied to the sheaf-theoretic setting; the identification of the local terms and their behaviour under the operations is the subject of Varshavsky's work on local terms and of the categorical trace formalism of Gaitsgory, Kazhdan, Rozenblyum, Varshavsky and Yom Din.

Step 3: the fixed points of Frobenius, and transversality. For $c = \mathrm{Frob}_q$ the fixed-point scheme is

$$\mathrm{Fix}(\mathrm{Frob}_q) = X \times_{X\times X,(\mathrm{Frob}_q,\mathrm{id})} X = X^{\mathrm{Frob}_q} = X(\mathbf{F}_q),$$

a finite discrete scheme, by Lang's description of the Frobenius-fixed points. The intersection is transversal: the differential of Frob_q vanishes identically, because q-th powering has derivative $q \cdot x^{q-1} = 0$ in characteristic p, so the graph of Frobenius meets the diagonal with multiplicity one at every fixed point. Consequently every local term equals 1 (SGA 5, Exposé III), and $\mathrm{tr}(\mathrm{Frob}_q) = \#X(\mathbf{F}_q)$.

Step 4: identification with cohomology. Applying $p_{X!}$ to the duality datum and using $\mathrm{tr}(F \mid \mathcal{D}(X)) = \sum_i (-1)^i \mathrm{tr}(F \mid H^i_c)$, which is the additivity datum of Definition 2.8 applied to the finite filtration of $p_{X!}\Lambda$ by cohomological degree, gives the first equality of the theorem. Combining with Step 3 gives the second, which is the Grothendieck–Lefschetz trace formula. □

The theorem is marked T: it is proved here from the axioms together with the classical results cited, all of which are established literature. Nothing in the proof is new; what is new, if anything is, is that the statement is now an identification of two constructions inside one framework rather than a coincidence between two theorems.

Corollary 2.26 (Point counts as chôral traces). [THEOREM]
For every separated finite-type X over $\mathbf{F}_q$ and every $n \geq 1$,

$$\tau(X, \mathrm{Frob}_q^n) = \#X(\mathbf{F}_{q^n}),$$

and the zeta function $Z(X,t) = \exp(\sum_{n\geq 1} \tau(X, \mathrm{Frob}_q^n)\, t^n/n)$ is the generating function of the chôral traces of the powers of Frobenius.

Proof. Apply Theorem 2.25 over $\mathbf{F}_{q^n}$, noting that $\mathrm{Frob}_q^n = \mathrm{Frob}_{q^n}$ and that base change along $\mathbf{F}_q \to \mathbf{F}_{q^n}$ is a trace-exact 1-morphism of chôrai in the sense of Definition 2.18. □

Example 2.27 (Projective spaces and finite étale figures). [ESTABLISHED]
Take $X = \mathbf{P}^1_{\mathbf{F}_q}$. Then $H^0_c = \Lambda$ with Frobenius acting trivially and $H^2_c = \Lambda(-1)$ with Frobenius acting by q, all other compactly supported cohomology vanishing. Hence

$$\tau(\mathbf{P}^1, \mathrm{Frob}_q) = 1 + q,$$

which is $\#\mathbf{P}^1(\mathbf{F}_q)$, as it must be. Two features of the general theory are already visible. First, the trace is a sum over a *filtration*, not over points: the summands 1 and q are the two weight-graded pieces, and the localisation datum redistributes them over the $q+1$ points. Second, the eigenvalue q is not a root of unity, so the trace is not the trace of a finite permutation representation; the weight structure is essential, and it is the subject of Chapter 3.

The neighbouring cases are worth computing, because they show how little of the answer is carried by the geometry and how much by the weights. For $X = \mathbf{A}^n_{\mathbf{F}_q}$ the only non-vanishing compactly supported cohomology is $H^{2n}_c = \Lambda(-n)$, on which Frobenius acts by q^n, so $\tau = q^n = \#\mathbf{A}^n(\mathbf{F}_q)$. For $X = \mathbf{P}^n_{\mathbf{F}_q}$ one has $H^{2i}_c = \Lambda(-i)$ for $0 \leq i \leq n$ and nothing else, so

$$\tau(\mathbf{P}^n, \mathrm{Frob}_q) = 1 + q + \cdots + q^n = \frac{q^{n+1}-1}{q-1},$$

which is the number of lines through the origin in $\mathbf{F}_q^{n+1}$. For $X = \mathbf{G}_{m,\mathbf{F}_q}$ the excision sequence for $\mathbf{G}_m \subset \mathbf{A}^1 \supset \{0\}$ gives $H^1_c = \Lambda$ with trivial Frobenius action and $H^2_c = \Lambda(-1)$, so $\tau =$

$-1+q=q-1$, the order of $\mathbf{F}_q^\times$. Here the Euler characteristic with compact supports is 0 while the chôral trace is $q-1$; the two invariants are different, and only the second is a trace.

A case of a different character, and one which will matter when we come to the absolute base, is a figure with no rational points at all. Let $X=\operatorname{Spec}\mathbf{F}_{q^n}$ over $\mathbf{F}_q$ with $n>1$. It is finite étale of degree n, so $H^0(X_{\bar{\mathbf{F}}_q},\Lambda)\simeq\Lambda^n$, with basis the n embeddings $\mathbf{F}_{q^n}\hookrightarrow\bar{\mathbf{F}}_q$, and all other cohomology vanishes. Frobenius permutes the embeddings cyclically, so with respect to that basis it acts by the $n\times n$ permutation matrix

$$P=\begin{pmatrix}0&0&\cdots&0&1\\1&0&\cdots&0&0\\0&1&\cdots&0&0\\\vdots&&\ddots&&\vdots\\0&0&\cdots&1&0\end{pmatrix}.$$

Since P has no fixed basis vector for $n>1$, and more precisely since $\operatorname{tr}(P^k)=n$ when $n\mid k$ and 0 otherwise, we obtain

$$\tau(X,\mathrm{Frob}_q^k)=\begin{cases}n, & n\mid k,\\ 0, & \text{otherwise},\end{cases}$$

in agreement with $\#X(\mathbf{F}_{q^k})$, which is n when $\mathbf{F}_{q^n}\subseteq\mathbf{F}_{q^k}$ and empty otherwise. The zeta function is accordingly

$$Z(X,t)=\exp\Big(\sum_{k\geq1}\#X(\mathbf{F}_{q^k})\frac{t^k}{k}\Big)=\exp\Big(\sum_{m\geq1}\frac{t^{nm}}{m}\Big)=\frac{1}{1-t^n}.$$

The point to retain is that the trace vanishes for k not divisible by n even though X is not the empty figure. A chôral trace can be zero without the figure being trivial; this is the elementary form of the phenomenon that makes vanishing theorems in Chapter 3 delicate.

Example 2.28 (Elliptic curves: ordinary, supersingular, and a non-Frobenius correspondence). [ESTABLISHED]

Let E be the elliptic curve of conductor 11 given by

$$y^2+y=x^3-x^2-10x-20,$$

whose L-function is that of the weight-two newform $\eta(z)^2\eta(11z)^2$. Reduction modulo a prime $p\neq11$ gives a smooth projective curve over $\mathbf{F}_p$, hence a figure of $\mathfrak{C}_{\mathrm{Sch}}$, and Theorem 2.25 gives

$$\tau(E_{\mathbf{F}_p},\mathrm{Frob}_p)=\#E(\mathbf{F}_p)=p+1-a_p,$$

where $a_p=\operatorname{tr}(\mathrm{Frob}_p\mid H^1(E_{\bar{\mathbf{F}}_p},\Lambda))$ is the p-th coefficient of the newform. The first values are

p	2	3	5	7	13	17	19	23
a_p	-2	-1	1	-2	4	-2	0	-1
τ	5	5	5	10	10	20	20	25

Two entries can be checked by hand, and it is worth doing so, because the treatise rests on the claim that the abstract trace is this concrete count. Modulo 2 the equation reduces to $y^2+y=x^3+x^2$. Substituting $x=0$ gives $y^2+y=0$, so $y\in\{0,1\}$; substituting $x=1$ gives $y^2+y=0$ again, so $y\in\{0,1\}$. That is four affine points, and with the point at infinity

$\#E(\mathbf{F}_2) = 5 = 2 + 1 - (-2)$. Modulo 3 the equation reduces to $y^2 + y = x^3 + 2x^2 + 2x + 1$. The right-hand side takes the values $1, 0, 0$ at $x = 0, 1, 2$, while $y^2 + y$ takes the value 0 at $y = 0, 2$ and the value 2 at $y = 1$. So $x = 0$ contributes nothing, and each of $x = 1, 2$ contributes two, giving four affine points and, with infinity, $\#E(\mathbf{F}_3) = 5 = 3 + 1 - (-1)$.

Passing to $\mathbf{F}_{p^2}$ costs nothing, and it exhibits the multiplicativity of the trace under composition of correspondences. Write α, β for the eigenvalues of Frob_p on H^1, so that $\alpha + \beta = a_p$ and $\alpha\beta = p$ by Poincaré duality. Then $\mathrm{tr}(\mathrm{Frob}_p^2) = \alpha^2 + \beta^2 = a_p^2 - 2p$, whence

$$\tau(E_{\mathbf{F}_{p^2}}, \mathrm{Frob}_{p^2}) = p^2 + 1 - (a_p^2 - 2p).$$

For $p = 2$ this is $4 + 1 - (4 - 4) = 5$; for $p = 3$ it is $9 + 1 - (1 - 6) = 15$; for $p = 5$ it is $25 + 1 - (1 - 10) = 35$. The first is a useful check, since $E(\mathbf{F}_2)$ has five points and must inject into $E(\mathbf{F}_4)$.

Now take a curve of a different arithmetic type. Let

$$E' : \quad y^2 = x^3 - x$$

over $\mathbf{F}_p$ with $p \equiv 3 \pmod 4$. This curve has complex multiplication by $\mathbf{Z}[i]$, and at such p the prime is inert, so the reduction is supersingular and $a_p = 0$. The count can be done by hand and should be. Over $\mathbf{F}_3$ the cube map is a bijection, so $x^3 - x$ takes the values $0, 0, 0$ at $x = 0, 1, 2$; each gives the single solution $y = 0$, so there are three affine points and $\#E'(\mathbf{F}_3) = 4 = 3 + 1$. Over $\mathbf{F}_7$ the values of $x^3 - x$ at $x = 0, \dots, 6$ are $0, 0, 6, 3, 4, 1, 0$, while the squares in $\mathbf{F}_7^\times$ are $\{1, 2, 4\}$; so $x = 0, 1, 6$ each contribute one point, $x = 4, 5$ each contribute two, and $x = 2, 3$ contribute none, giving seven affine points and $\#E'(\mathbf{F}_7) = 8 = 7 + 1$. Over $\mathbf{F}_{11}$ the values are $0, 0, 6, 2, 5, 10, 1, 6, 9, 5, 0$ and the squares are $\{1, 3, 4, 5, 9\}$; the three zeroes contribute one point each and $x = 4, 6, 8, 9$ contribute two each, giving eleven affine points and $\#E'(\mathbf{F}_{11}) = 12 = 11 + 1$. In all three cases $\tau = p + 1$ and $a_p = 0$, as the supersingularity predicts. Over $\mathbf{F}_{p^2}$ the eigenvalues are $\pm i\sqrt{p}$, so $a_{p^2} = -2p$ and $\tau = p^2 + 1 + 2p = (p + 1)^2$: for $p = 3$ this is 16, for $p = 7$ it is 64. The contrast with the ordinary curve E is instructive. For E the two Frobenius eigenvalues are distinct elements of a real quadratic field and the trace varies erratically with p; for E' they are purely imaginary and the trace is constant. The difference lies entirely in the weight-graded piece H^1 and not in the formalism.

Finally, a correspondence which is *not* a Frobenius, which is where the localisation datum begins to do real work. Fix $n \geq 2$ and let c be the correspondence $E \xleftarrow{\mathrm{id}} E \xrightarrow{[n]} E$ given by the graph of multiplication by n. On cohomology, $[n]^*$ acts by 1 on H^0, by n on H^1 (because $[n]$ induces multiplication by n on the Tate module), and by $\deg[n] = n^2$ on H^2. Hence

$$\tau(E, [n]) \;=\; 1 - 2n + n^2 \;=\; (n - 1)^2.$$

The fixed locus is the group scheme $E[n - 1]$ of $(n - 1)$-torsion, whose degree is $(n - 1)^2$, so the count matches — but only as a *degree*, not as a number of points. Transversality of $\Gamma_{[n]}$ with the diagonal requires $1 - d[n] = 1 - n$ to be invertible on the tangent space, that is, $p \nmid n - 1$. When $p \mid n - 1$ the intersection is not transversal and $E[n - 1]$ is not reduced. Take $p = 5$ and $n = 6$, so that $(n - 1)^2 = 25$. If E is ordinary at 5 then $E[5] \simeq \mu_5 \times \mathbf{Z}/5$ as a group scheme, so $\#E[5](\bar{\mathbf{F}}_5) = 5$: the trace 25 is distributed over five points, each carrying local term 5. If E is supersingular at 5 then $E[5]$ is a local-local group scheme with a single $\bar{\mathbf{F}}_5$-point, and the whole of the trace 25 sits at the origin. The chôral trace is 25 in both cases and does not know the difference; the localisation datum does. This is the phenomenon that step (3) of Theorem 2.25 avoids by the computation $d(\mathrm{Frob}_q) = 0$, and it is the reason that theorem concerns Frobenius and not arbitrary endomorphisms.

The interest of the example is not the arithmetic, which is classical, but the placement. The left-hand side of the displayed equality is a categorical trace formed in $\mathrm{Pr}^{L}_{\mathrm{st}}$; the right-hand side is a coefficient of a modular form. The Main Conjecture of this treatise, stated in Chapter 19, asserts an equality of the same shape in which the left-hand side is a chôral trace over the absolute base and the right-hand side is an automorphic period. The present example is the case in which both sides are already known, and it is therefore the case against which the general statement is calibrated.

The reader should be clear about why the function-field case is treated throughout this treatise as the control, and about the exact respect in which it is one. Over $\mathbf{F}_q$ there is a distinguished endomorphism of every figure, the Frobenius, and it is available because the base has positive characteristic; it is a morphism of schemes over $\mathbf{F}_q$, so it defines a correspondence, so it defines an endomorphism of $\mathcal{D}(X)$, and its trace is computable by Theorem 2.25. Everything in the function-field Langlands correspondence — Drinfeld's modules and shtukas (Drinfeld 1974), Lafforgue's proof for GL_n (Lafforgue 2002), Vincent Lafforgue's construction of the excursion operators for a general reductive group (Lafforgue 2018) — takes place inside the chôra $\mathfrak{C}_{\mathrm{Sch}}$ of Notation 2.24, enlarged to admit the stacks Bun_G and LocSys. That enlargement is not innocent, since Bun_G is not of finite type and one must work with a suitable class of sheaves, but it is by now standard, and the geometric Langlands equivalence in the recent series of papers (Gaitsgory and Raskin 2024a, 2024b; Arinkin, Beraldo, Campbell, et al. 2024; Campbell et al. 2024; Arinkin, Beraldo, Chen, et al. 2024) is a statement about a 1-morphism of chôrai in the sense of Definition 2.18, though it is not stated that way by its authors.

Over $\mathbf{Z}$ there is no Frobenius. This is the whole difficulty, and it is worth naming it here so that no reader mistakes the comparison theorem for more than it is. $\operatorname{Spec}\mathbf{Z}$ has no endomorphism playing the part of Frob_q; there is a Frobenius at each finite place, but the places do not assemble into a global endomorphism, and there is nothing at all at the archimedean place. Consequently Theorem 2.25 has no direct arithmetic analogue: the left-hand side of its displayed equality can be written down only when a Frobenius exists. The programme of Part II is to manufacture the missing base, over which such an endomorphism does exist; the programme of Part IV is to compute its trace. The present chapter supplies the language in which both can be stated and neither can be proved.

2.8 Four realisations

A definition of geometry is tested by its instances. We give three, and then one object which satisfies the first three axioms and fails the fourth, so that the boundary of the definition is visible. The three positive cases are chosen to be as far apart as possible: one is classical algebraic geometry over a finite field, one is analytic geometry over the reals and the p-adics simultaneously, and one has no underlying space of any kind.

Examples 2.29 (Three chôrai). [ESTABLISHED]

(a) Schemes over a finite field. This is $\mathfrak{C}_{\mathrm{Sch}}$ of Notation 2.24. KH.1 and KH.2 hold by the construction of the six operations in SGA 4 and SGA 5. KH.3 holds and agrees with the classical adjectives: $f_! \simeq f_*$ for proper f is proper base change, $f^! \simeq f^*(d)[2d]$ for smooth f of relative dimension d is the smooth base change and purity package, and $\omega_f \simeq \Lambda$ exactly for étale f. KH.4 holds by Verdier duality and Deligne's finiteness theorem, as verified in Step 1 of the proof of Theorem 2.25. KH.5 holds in the restricted form of Conjecture 2.17 for the quasi-coherent variant, by Tannakian reconstruction. The trace is the Lefschetz–Verdier trace, and by Corollary 2.26 it computes point counts, hence zeta functions. This realisation is

the *control*: every structural claim of this treatise is required to specialise here to something already known, and where it does not, the claim is withdrawn rather than the control.

(b) Condensed analytic stacks. Let $\mathcal{C}$ be the ∞-category of analytic stacks in the sense of Clausen and Scholze (Clausen and Scholze 2019, 2024; Scholze 2020), whose affine objects are analytic rings — a condensed ring together with a choice of which condensed modules count as complete — and let $\mathcal{D}(X)$ be the derived category of quasi-coherent sheaves in that sense. The class E consists of the morphisms which are !-able, that is, admit a factorisation into an open immersion followed by a proper morphism in the analytic sense. That this yields a six-functor formalism is the substance of the p-adic case treated by Mann (Mann 2022) and of the general framework of Heyer and Mann (Heyer and Mann 2024) and Scholze's lectures (Scholze 2025); KH.1 and KH.2 hold by construction, and KH.3 is the definition of the analytic adjectives, which is exactly the point of the theory. What is new here, relative to (a), is that $\mathbf{R}$ and $\mathbf{Q}_p$ are objects of a single category: $\mathbf{Z}_{\blacksquare}$-modules give the solid theory and $\mathbf{R}_{\mathrm{liq}}$ the liquid theory (Commelin and Scholze 2023; Clausen and Scholze 2022), and the archimedean place ceases to be an exception. This is the realisation which addresses failure F2, and it is the reason the absolute base of Chapter 7 is built condensed rather than schematic.

KH.4 is the delicate axiom here, because the categories $\mathcal{D}(X)$ are typically *not* compactly generated, so Criterion 2.22 does not apply and one is thrown onto part (3) of the criterion. The relevant finiteness notion is nuclearity: the trace of an endomorphism is defined when the endomorphism is trace-class, and the categories of nuclear modules are dualisable without being compactly generated. Efimov's continuous K-theory (Efimov 2024) is the invariant that makes this tractable, and Andreychev's work on pseudocoherent and nuclear modules (Andreychev 2021) supplies the finiteness statements. The trace theory τ is therefore only defined on the trace-class part, and the localisation datum is correspondingly weaker than in (a). We regard this as the least secure of the three positive realisations, and say so again in the assessment of Chapter 40.

(c) Spectra with topological Hochschild homology. Let $\mathcal{C} = \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{op}}$, the opposite of $\mathbf{E}_\infty$-ring spectra, whose objects we write $\operatorname{Spec} R$, and let $\mathcal{D}(\operatorname{Spec} R) = \mathrm{Mod}_R$. For $f\colon \operatorname{Spec} S \to \operatorname{Spec} R$ corresponding to $R \to S$, set $f^* = S \otimes_R -$ and $f_! = f_* =$ restriction of scalars. Take E to be the class of those f for which S is a *perfect* R-module, so that f_* preserves perfect modules. This restriction of E is not a technicality: it is precisely what makes the realisation satisfy KH.4, as the non-example of Example 2.30 shows. With this E, KH.1 and KH.2 hold — base change is the computation $R' \otimes_R S \simeq S'$ and the projection formula is associativity of the tensor product — and KH.3 holds with every admissible morphism proper by construction. KH.4 holds by Criterion 2.22: Mod_R is compactly generated by R, the unit is compact, $f_!$ preserves perfect modules by the definition of E, and the pairing $(M, N) \mapsto M \otimes_R N$ is perfect on perfect modules.

The trace is $\tau(\operatorname{Spec} R, \mathrm{id}) = \mathrm{THH}(R)$, the topological Hochschild homology, that is, the Hochschild homology of R relative to the sphere spectrum. Its localisation datum is the *cyclotomic structure*: the family of maps $\varphi_p\colon \mathrm{THH}(R) \to \mathrm{THH}(R)^{tC_p}$ for each prime p, in the sense of Nikolaus and Scholze (Nikolaus and Scholze 2018), from which topological cyclic homology is recovered as the equaliser

$$\mathrm{TC}(R) = \mathrm{eq}\big(\mathrm{TC}^-(R) \rightrightarrows \mathrm{TP}(R)\big), \qquad \mathrm{TC}^-(R) = \mathrm{THH}(R)^{hS^1}, \quad \mathrm{TP}(R) = \mathrm{THH}(R)^{tS^1}.$$

Two computations fix the scale. Bökstedt's calculation, the founding result of the subject, gives

$$\pi_* \mathrm{THH}(\mathbf{F}_p) = \mathbf{F}_p[x], \qquad |x| = 2,$$

in contrast with $\mathrm{HH}(\mathbf{F}_p/\mathbf{F}_p) = \mathbf{F}_p$; this is the computation that drove Proposition 2.10. For the integers, Bökstedt's calculation gives

$$\pi_0 \mathrm{THH}(\mathbf{Z}) = \mathbf{Z}, \qquad \pi_{2i-1} \mathrm{THH}(\mathbf{Z}) = \mathbf{Z}/i \ (i \geq 1), \qquad \pi_{2i} \mathrm{THH}(\mathbf{Z}) = 0 \ (i \geq 1).$$

The appearance of the divisor i in odd degree is the first sign that THH over the sphere sees arithmetic that HH over $\mathbf{Z}$ does not; the same phenomenon, in its p-adic form, is what makes TC compute p-adic étale cohomology in the range of the Beilinson–Lichtenbaum comparison (Bhatt et al. 2019, 2018; Hesselholt 2018).

It is worth carrying the computation one step further, because the periodic and cyclotomic refinements are what make the trace a genuine arithmetic invariant rather than a formal one. Applying the S^1-Tate construction to $\mathrm{THH}(\mathbf{F}_p)$ gives

$$\pi_* \mathrm{TP}(\mathbf{F}_p) = \mathbf{Z}_p[\sigma^{\pm 1}], \qquad |\sigma| = 2,$$

so that $\pi_0 \mathrm{TP}(\mathbf{F}_p) = \mathbf{Z}_p$: the Tate construction has manufactured the p-adic integers out of a theory defined over a field of characteristic p. Nothing in $\mathrm{HH}(\mathbf{F}_p/\mathbf{F}_p) = \mathbf{F}_p$ can do this, and it is the cleanest available illustration of the assertion of Principle 2.1 that the trace carries information the underlying category does not. The equaliser defining TC then gives

$$\mathrm{TC}(\mathbf{F}_p; \mathbf{Z}_p) \ \simeq \ \mathbf{Z}_p \oplus \Sigma^{-1}\mathbf{Z}_p,$$

which by the theorem of Quillen on the K-theory of finite fields and the cyclotomic trace matches $K(\mathbf{F}_p; \mathbf{Z}_p)$ in non-negative degrees. For $\mathbf{F}_q$ with $q = p^n$ one has $\pi_* \mathrm{THH}(\mathbf{F}_q) = \mathbf{F}_q[x]$, $|x| = 2$, so the arithmetic of q enters only through the cyclotomic structure maps and not through the homotopy groups.

This realisation has no underlying space at all: the figures are formal duals of ring spectra, there is no topology, and the trace is nevertheless richer than in (a).

We now come to the non-example, which is included because a definition which excludes nothing defines nothing.

Example 2.30 (The naive coherent chôra: a non-example). [THEOREM]

Let k be a field, let C be the opposite of the category of connective commutative k-algebras, let E be *all* morphisms, let $\mathcal{D}(\operatorname{Spec} A) = \mathrm{Mod}_A$, and set

$$f^* = B \otimes_A -, \qquad f_! := f_* = \text{restriction of scalars along } A \to B.$$

This is the realisation (c) above with the restriction on E removed. We record what survives and what does not.

KH.1 and KH.2 hold. The assignment is a lax symmetric monoidal functor on correspondences. Base change is the computation, for a pushout square of rings with $B' = B \otimes_A A'$,

$$g^* f_* M = A' \otimes_A M \ \simeq \ B' \otimes_B M = f'_* g'^* M,$$

and the projection formula is $f_*(M \otimes_B f^* N) = M \otimes_B (B \otimes_A N) \simeq M \otimes_A N$. Both are equivalences of derived tensor products, and no hypotheses are needed.

KH.3 holds, vacuously. Every morphism is proper, since $f_! = f_*$ by fiat, and the smooth and étale classes are as usual.

KH.4 fails. Take $X = \mathbf{A}^1_k = \operatorname{Spec} k[t]$. The unit $\mathbf{1}_X = k[t]$ is a compact object of $\mathrm{Mod}_{k[t]}$, and $\mathrm{Mod}_{k[t]}$ is compactly generated by it, so the hypotheses of Criterion 2.22 are met. But

$$p_{X!}\mathbf{1}_X = p_{X*}k[t] = k[t]|_k = \bigoplus_{n \geq 0} k \cdot t^n,$$

an infinite-dimensional k-vector space, which is not a compact object of Mod_k. By the necessity part of the criterion, $\mathbf{A}^1$ is not a dualisable figure. *The violated axiom is KH.4, and only KH.4.*

The failure can be seen directly, without the criterion, and it is instructive to do so. The coevaluation is $\mathrm{coev} = \Delta_{X!} p_X^!$, and $p_X^!$ is the right adjoint of restriction of scalars $\mathrm{Mod}_{k[t]} \to \mathrm{Mod}_k$, namely $V \mapsto \mathrm{Hom}_k(k[t], V)$. Hence

$$\mathrm{coev}(k) = \Delta_{X*}\,\mathrm{Hom}_k(k[t], k) = k[t]^\vee = \prod_{n \geq 0} k \cdot t^{-n},$$

the full linear dual, which is not isomorphic to $k[t]$. Feeding this into the first triangle identity, the composite $(\mathrm{ev} \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \mathrm{coev})$ evaluated on $M = k[t]$ returns $k[t] \otimes_{k[t]} k[t]^\vee = k[t]^\vee$ rather than $k[t]$. The triangle identity fails, and it fails by exactly the discrepancy between a module and its dual which properness would have removed.

Diagnosis and repair. The failure is not a failure of the figure. The very same figure $\mathbf{A}^1$, taken in $\mathfrak{C}_{\mathrm{Sch}}$ of Notation 2.24 over $k = \mathbf{F}_q$, *is* a dualisable figure, and Theorem 2.25 gives $\tau(\mathbf{A}^1, \mathrm{Frob}_q) = q$. What differs is the sheaf theory: in $\mathfrak{C}_{\mathrm{Sch}}$ the operation $f_!$ is defined by compactification and is *compactly supported*, whereas here it was declared to be f_*. There are exactly two ways to repair the non-example, and both occur in this treatise. One may shrink E until f_* preserves compact objects, which produces realisation (c); or one may keep E large and construct a genuine $f_!$ by compactification, which produces realisation (a). The lesson, which Warning 2.6 anticipated in the abstract, is that $f_!$ is not f_*, that the difference is measured by the norm map, and that KH.4 is the axiom which insists the difference be under control.

The four cases also illustrate the point of §2.5. Realisations (a) and (b) are related by a 1-morphism of chôrai — the functor sending a scheme of finite type over a complete discretely valued field to its associated analytic stack — which is trace-exact where both traces are defined. Realisations (a) and (c) are related by a 1-morphism whose trace comparison β is the cyclotomic trace and is emphatically *not* an equivalence, which is why Definition 2.18 was written to allow lax β. The non-example is not an object of $\mathrm{Geom}^{\mathrm{tr}}$ at all, since the objects were required to satisfy KH.4; it lives in the larger 2-category obtained by dropping that requirement, and the inclusion of $\mathrm{Geom}^{\mathrm{tr}}$ into that larger 2-category is the formal expression of what KH.4 costs.

2.8.1 The control case, assertion by assertion

Because the whole method of this treatise is to state over $\mathbf{Z}$ what is a theorem over a curve, it is worth going through the principal assertions of this chapter one at a time and recording the status of each in the control case. Let X be a smooth projective geometrically connected curve over $\mathbf{F}_q$, and let $\mathfrak{C}_X$ be the chôra of separated finite-type X-schemes with ℓ-adic coefficients, $\ell \nmid q$.

Assertion	Over $X/\mathbf{F}_q$	Due to
KH.1	theorem	Grothendieck, Artin, Verdier, 1963–1973
KH.2	theorem	Deligne; ∞-categorical form, Liu–Zheng, 2012–2017
KH.3	theorem	Grothendieck (proper base change), 1963
KH.4	theorem	Poincaré–Verdier duality; Deligne, 1977
KH.5	partial	Uchida, Tamagawa, anabelian geometry
Criterion 2.22	theorem	Deligne (finiteness), 1977
Theorem 2.25	theorem	Grothendieck, 1964; Verdier–Illusie
Corollary 2.26	theorem	Grothendieck, 1964
Conjecture 2.17	open in general	—
Remark 2.11	open	—

Each line deserves a sentence. KH.1 asserts that a geometry carries a six-functor formalism; over X this is the content of SGA 4 and SGA 5 (Artin et al. 1972a, 1972b, 1973; Grothendieck and Illusie 1977), with the finiteness statements that make $f_!$ well behaved supplied by Deligne's *Théorèmes de finitude* (Deligne 1977). Its transposition to the absolute base is a definition, not a claim.

KH.2 asserts that the six operations are the functoriality of a single functor on correspondences. Over X the classical statements — base change, projection formula, and the compatibilities between them — are all in SGA 4, but the assembly into a functor $\mathrm{Corr}(C, E) \to \mathrm{Pr}^L_{\mathrm{st}}$, with all higher coherences, is a much later achievement, carried out in the ℓ-adic setting by Liu and Zheng in a series of preprints beginning around 2012 and in a general axiomatic setting by Gaitsgory and Rozenblyum (Gaitsgory and Rozenblyum 2017a, 2017b) and, more recently, by Mann (Mann 2022) and by Heyer and Mann (Heyer and Mann 2024). The gap between the classical compatibilities and the coherent functor is the gap between a 1-categorical and an ∞-categorical formulation, and it took forty years to close.

KH.3 asserts that geometricity is adjointability. Over X the proper base change theorem (SGA 4, Exp. XII) says that for f proper $f_! = f_*$ and base change holds; the smooth base change theorem says that for f smooth $f^! = f^*(d)[2d]$. Neither is formal: proper base change is proved by dévissage to curves and rests on the computation of the cohomology of curves with torsion coefficients. This is the axiom whose function-field status is most secure and whose absolute status is least clear, because the absolute base has no proper morphisms in the naive sense.

KH.4 asserts duality. Over X the relevant theorem is Poincaré–Verdier duality for separated finite-type schemes over a field, together with the constructibility and finiteness theorems that make the dualising complex behave. In the language of Criterion 2.22, the hypotheses are verified by: compact generation of the ℓ-adic derived category by constructible objects; preservation of compactness by $p_{X!}$, which is Deligne's finiteness theorem; and Künneth, which is again SGA 4. It is precisely this theorem that has no absolute analogue at present.

KH.5 asserts reconstruction. Over X the answer is *partial*. A scheme is not determined by its étale topos in general, but for hyperbolic curves over finite fields the arithmetic fundamental group does determine the curve, by a theorem of Tamagawa from the mid-nineteen-nineties, building on Uchida's reconstruction of a global function field from its absolute Galois group in the nineteen-seventies. On the categorical side, the symmetric monoidal derived category of constructible sheaves determines the étale homotopy type by the exodromy theorem of Barwick, Glasman and Haine, and tensor-triangular methods in the style of Balmer recover the underlying space of a scheme from its perfect derived category. None of these is KH.5 as stated, which is why KH.5 carries status C and the others do not.

Criterion 2.22 is, in the control case, a repackaging of Deligne's finiteness theorem, and it is a theorem there. What is new is not the content but the shape: the criterion is stated so that its hypotheses can be checked in a chôra where no finiteness theorem is available.

Theorem 2.25 and Corollary 2.26 are theorems over X, and they are due to Grothendieck; the Lefschetz–Verdier form with local terms is Exposé III of SGA 5 (Grothendieck and Illusie 1977), and the fine analysis of the local terms in the non-transversal case is due to Varshavsky in work of the two-thousands. The Weil conjectures that motivated the whole construction were proved by Deligne (Deligne 1974a, 1980).

Conjecture 2.17 and Remark 2.11 are open even over X, and their appearance in this chapter should be read as a statement of what would need to be true, not of what is.

Two further assertions have no control case. The definition of $\mathrm{Geom}^{\mathrm{tr}}$ (Definition 2.18) is a definition, so the question does not arise; and realisations (b) and (c) lie outside the analogy,

the first because analytic stacks were constructed only in the present decade (Clausen and Scholze 2024), the second because THH is not a function-field construction.

2.9 A comparison with EGA

The *Éléments de géométrie algébrique* is the standard against which any proposal to refound algebraic geometry must be measured, and the measurement should be made in public. This section states what the chôra buys and what it gives up. We have tried to make the second list longer and more specific than the first, because that is the true proportion.

2.9.1 What the chôra buys

The operations are transported rather than rebuilt. In EGA and its successors, each new cohomology theory requires the construction of $f_!$ by compactification, the proof of its independence of the compactification, and then proper base change and the projection formula. Étale, ℓ-adic, de Rham, rigid, p-adic, motivic, condensed: the labour has been performed at least seven times. Under KH.2 the six operations are the components of a single functor on correspondences, and the compatibilities are the composition law. A new geometry requires the construction of one functor, not the verification of a list. This is the response to failure F3, and it is the most secure of the gains, because it is a theorem about the formalism rather than a hope about its applications.

Geometric adjectives become definitions. Under KH.3, proper, smooth and étale are adjointability conditions. In EGA they are conditions on the underlying topological spaces and on finiteness of the algebra, and the equivalence with the cohomological conditions is a sequence of hard theorems. The chôral convention makes the theorems into definitions and demotes the classical statements to comparison lemmas in the scheme realisation. The gain is that the adjectives are available in geometries which have no topological space; the loss is discussed below.

The trace becomes type-correct. Under Definition 2.9 a trace formula is a statement about a 1-morphism of chôrai and its comparison map β. Grothendieck–Lefschetz, the categorical Lefschetz theorem of Chapter 21, and the trace formula of Chapter 27 then have, at least, the same *shape*. This is the response to F4. It is a response at the level of type discipline, not of proof: making two statements instances of one schema does not prove either, and we do not claim otherwise.

The archimedean place stops being an exception. Realisation (b) places $\mathbf{R}$ and $\mathbf{Q}_p$ in one category of analytic rings, so the six operations, KH.3 and the trace are available uniformly. This is the response to F2, and it is a genuine structural gain, though it belongs to Clausen and Scholze rather than to us.

Weights replace the heart. A chôra is not required to have a t-structure, and none of KH.1–KH.4 mentions one. The role played classically by the abelian category of perverse or mixed sheaves is played here by the weight structure of Chapter 3 and the trace itself. This sidesteps F1 rather than solving it: the motivic t-structure is still not constructed, but nothing in the framework needs it.

The absolute base becomes expressible. Since a chôra need not have points, the object $\mathfrak{X}_{\mathbf{Z}}$ of Chapter 7 can be specified as a chôra without first being constructed as a space. This is the response to F5, and it is the most speculative of the gains: an object which can be specified need not exist.

2.9.2 What is given up

EGA's objects have points; chôrai in general do not. This is the central cost and it should not be softened. A scheme has an underlying topological space, and with it: points, including generic points; a sober topology in which irreducible closed subsets have generic points; local rings, residue fields, and the entire technique of passing to a stalk; Krull dimension, and with it dimension theory, systems of parameters, and the local criteria that depend on them; Noetherian induction, which proves statements by reducing to a proper closed subset; the valuative criteria for separatedness and properness; Chevalley's theorem that the image of a constructible set is constructible; and the flatness and fibre-dimension technology of EGA IV, which is the working mathematician's daily instrument (Grothendieck and Dieudonné 1964, 1965, 1966, 1967).

None of these is available in a general chôra, and KH.5, which would restore points in a Tannakian form, is a conjecture. The practical effect is severe: a proof by Noetherian induction, or by checking on stalks, or by choosing a system of parameters, simply cannot be transcribed. It must be replaced by an argument in terms of adjunctions, and such arguments are longer, less intuitive, and much harder to find. We have found no substitute for the local intuition that a scheme's points supply.

Local gluing is not available as an intuition. In EGA one verifies a statement locally and glues. In a chôra without a site there is no local, and descent must be imposed as extra structure rather than inherited from a topology. Where descent is available it is because the particular realisation has a site; the axioms do not supply one.

The 2-category is large and its coherence exceeds human audit. The data of Definition 2.18 involve lax symmetric monoidal transformations between functors of $(\infty,2)$-categories, with modifications between them. We have stated the coherence conditions, and Proposition 2.19 asserts they are satisfiable; we do not claim to have checked every one by hand, and we doubt that anyone could. The response of this treatise is the formalisation programme of Chapter 38, which is a promissory note.

The primitivity of the trace is a cost as well as a gain. Making τ part of the data means that a chôra is not determined by its geometry in the older sense: one must say which trace one intends, and Proposition 2.10 shows that the choice is not forced. Where EGA had a single cohomology theory attached to a scheme and proved its properties, we have a datum that must be supplied and whose supply is not canonical. The treatise's answer is the rigidity conjecture of Chapter 3, which asserts that linearity over the sphere spectrum together with the cyclotomic structure pins the trace down; but until that is proved, primitivity means an extra choice at every step, and an extra hypothesis in every comparison. A framework that requires more input to say the same thing is, other things being equal, worse.

And, decisively: EGA proved its theorems. The *Éléments* is not a specification. It constructs schemes, proves descent, constructs Hilbert and Picard schemes, establishes cohomology and base change, proves the finiteness and comparison theorems, and does so in complete detail. The present framework constructs nothing comparable. It does not construct the absolute base $\mathfrak{X}_{\mathbf{Z}}$; it proves no new case of modularity; it does not prove the Ramanujan conjecture, or Fontaine–Mazur, or any statement in the classical literature that was open before. Theorem 2.25, the one theorem of this chapter with real content, is a reformulation of results proved by Grothendieck, Deligne and Verdier fifty years ago. The correct comparison is therefore not between two bodies of theorems but between a body of theorems and a proposed language, and the burden of proof lies entirely on the language.

The following table records the comparison in the form we shall refer back to.

Aspect	EGA	Chôral geometry
Primitive notion	locally ringed space	six-functor formalism with trace
Points	intrinsic	conjectural, KH.5
Six operations	constructed per theory	one functor on correspondences
Properness	universally closed	$f_! \simeq f_*$
Duality	Verdier's theorem	KH.4, an axiom
Trace	a theorem	primitive datum τ
Archimedean place	absent	realisation (b)
Local methods	central	unavailable
Status	proved	specified

2.9.3 The register of the claim

We state, finally, what is and is not being claimed, since the foregoing could be read either too strongly or too weakly.

It is *not* claimed that schemes should be abandoned, that EGA is superseded, or that the chôra proves anything EGA does not. It is not claimed that KH.1–KH.4 are true, since design principles are not the sort of thing that is true; they are adopted, and their adoption is justified by what can be built with them. It is not claimed that KH.5 holds.

It *is* claimed that the definition of Definition 2.7 is a definition, that Definition 2.18 is a 2-category, that the four realisations of §2.8 are as described, that the non-example fails exactly KH.4 and for the reason given, and that Theorem 2.25 holds as stated. Those claims are falsifiable in the ordinary mathematical sense, and the protocol by which the treatise proposes to have them tested is set out in Chapter 39.

Summary. A geometry, in the sense of this treatise, is a chôra: a category of figures, a class of admissible legs, a six-functor formalism on the resulting category of correspondences, a duality involution, and a primitive trace. The five foundational axioms are KH.1, that a geometry is such a formalism; KH.2, that the six operations are functoriality on correspondences, so that base change and the projection formula are theorems; KH.3, that the geometric adjectives are adjointability conditions; KH.4, that duality holds and the formal self-duality of a figure is carried by the sheaf theory to a genuine self-duality; and KH.5, conjecturally, that the figures are reconstructible from the sheaf theory. The trace must be primitive rather than derived, because two chôrai can share their six operations and differ in their traces; the witness is $\mathrm{THH}(\mathbf{F}_p)$ against $\mathrm{HH}(\mathbf{F}_p/\mathbf{F}_p)$. Chôrai form a symmetric monoidal $(\infty,2)$-category $\mathrm{Geom}^{\mathrm{tr}}$ whose 1-morphisms carry a trace comparison that is a map and not in general an equivalence, which is what allows the cyclotomic trace to be a morphism. For separated finite-type schemes over $\mathbf{F}_q$ with ℓ-adic coefficients the chôral trace of Frobenius is the Grothendieck–Lefschetz trace and equals the number of rational points. The framework has three realisations of very different character and at least one instructive non-example, the naive coherent chôra, which satisfies KH.1–KH.3 and fails KH.4 because pushforward does not preserve compact objects. Against EGA, the chôra gains transportability of the operations, a type-correct trace, and the archimedean place; it gives up points, local methods, and — most importantly — the fact that EGA proved its theorems and this does not.

2.10 Notes and references

The image of the receptacle is from the *Timaeus*, and the choice of the word *chôra* for the ambient structure of a geometry without points is meant to record that the receptacle is not itself a figure. The mathematical content owes nothing to Plato and everything to Grothendieck, whose five moves are described in the introductions to the *Éléments* (Grothendieck and Dieudonné 1960) and, retrospectively and at much greater length, in *Récoltes et semailles* (Grothendieck 1986) and *Pursuing Stacks* (Grothendieck 1983). The functor-of-points move is already fully explicit in the Bourbaki seminars collected as *FGA* (Grothendieck 1962); the topos is SGA 4 (Artin et al. 1972a, 1972b, 1973); the six operations are SGA 4 and SGA 5 (Grothendieck and Illusie 1977), with the essential streamlining of finiteness in Deligne's SGA 4½ (Deligne 1977); and the motivic programme is the *Standard conjectures* note (Grothendieck 1969), now best approached through Cisinski and Déglise (Cisinski and Déglise 2019).

The presentation of the six operations as functoriality on correspondences has several independent sources. The technique of gluing pseudofunctors along correspondence categories, which is what makes the ∞-categorical bookkeeping feasible, is due to Yifeng Liu and Weizhe Zheng, whose long preprint on enhanced six operations we have cited in prose rather than by key. The systematic treatment is Gaitsgory and Rozenblyum's two volumes (Gaitsgory and Rozenblyum 2017a, 2017b), whose coherence arguments have been questioned; readers who wish to avoid the dependency should use the axiomatisation of Mann (Mann 2022), of Heyer and Mann (Heyer and Mann 2024), or the lecture notes of Scholze (Scholze 2025), and Gallauer's comparison of the several definitions now in circulation (Gallauer 2021). Earlier axiomatisations in the motivic setting are Ayoub's (Ayoub 2007a, 2007b). The categorical infrastructure — presentability, the tensor product of presentable categories, adjoint functor theorems — is Lurie's (Lurie 2009a, 2017), with Cisinski's book as an alternative foundation (Cisinski 2019) and Toën and Vezzosi's homotopical algebraic geometry as the geometric antecedent (Toën and Vezzosi 2005, 2008).

The theory of traces in symmetric monoidal ∞-categories, which supplies Step 2 of Theorem 2.25, is developed by Toën and Vezzosi (Toën and Vezzosi 2015) and by Hoyois, Scherotzke and Sibilla (Hoyois et al. 2017), with the bicategorical version of Ponto and Shulman (Ponto and Shulman 2012) and the additivity theorem of May (May 2001). The refinement of Lefschetz–Verdier local terms which we have used and attributed in prose is Varshavsky's, and the categorical trace formalism which places it in the present language is that of Gaitsgory, Kazhdan, Rozenblyum, Varshavsky and Yom Din; we have not cited these by key because we were not confident of the entries. The character-theory perspective, in which traces of endofunctors are functions on loop spaces, is Ben-Zvi and Nadler's (Ben-Zvi and Nadler 2015), and the integral-transform formalism which underlies it is (Ben-Zvi et al. 2010).

The history of the comparison theorem itself is worth setting down, because it is the one result of this chapter with a genuine pedigree. Weil formulated in 1949 the conjectures which make point counts into traces, and observed that a suitable cohomology theory with a Lefschetz fixed-point formula would prove them; his announcement rests on the case of curves, which he had settled in the nineteen-forties. The cohomology theory was constructed by Grothendieck and Artin between 1958 and 1964, and Grothendieck announced the Lefschetz trace formula and the rationality of L-functions in a Bourbaki seminar of 1964; the detailed treatment, with the local terms and the Lefschetz–Verdier formalism due to Verdier and written up by Illusie, is SGA 5 (Grothendieck and Illusie 1977), which appeared only in 1977. The finiteness theorems that make the whole apparatus usable are Deligne's *Théorèmes de finitude*, of the same year (Deligne 1977). The Riemann hypothesis component of the Weil

conjectures was proved by Deligne in 1974 (Deligne 1974a) and given its definitive form, with the theory of weights that Chapter 3 takes up, in 1980 (Deligne 1980). Grothendieck's own preferred route, through the standard conjectures announced in 1968 (Grothendieck 1969), remains open. The analysis of the local terms in the non-transversal case, which is what the treatment of the $[n]$-correspondence in Example 2.28 gestures at, was carried out by Varshavsky in the two-thousands and is attributed here in prose.

Topological Hochschild homology and its cyclotomic structure are treated in the form we use by Nikolaus and Scholze (Nikolaus and Scholze 2018); the original construction of the cyclotomic trace is Bökstedt, Hsiang and Madsen's (Bökstedt et al. 1993), and the standard textbook is Dundas, Goodwillie and McCarthy (Dundas et al. 2013). Bökstedt's computation of $\mathrm{THH}(\mathbf{F}_p)$ and $\mathrm{THH}(\mathbf{Z})$ circulated as a preprint and we have attributed it in prose. The arithmetic applications which motivate realisation (c) are Bhatt, Morrow and Scholze's (Bhatt et al. 2018, 2019), Hesselholt's (Hesselholt 2018; Hesselholt and Madsen 1997), and the prismatic theory (Bhatt and Scholze 2022; Bhatt and Lurie 2022b), which is the subject of Chapter 8. The universal characterisation of K-theory that makes the cyclotomic trace canonical is (Blumberg et al. 2013).

Condensed mathematics is Clausen and Scholze's (Clausen and Scholze 2019; Scholze 2020), with the liquid theory (Commelin and Scholze 2023), the complex analytic application (Clausen and Scholze 2022) and the analytic stacks course (Clausen and Scholze 2024). The finiteness theory we invoke for KH.4 in that setting is Andreychev's (Andreychev 2021) together with Efimov's theory of dualisable categories and continuous K-theory (Efimov 2024), which is what makes realisation (b) tractable at all.

For KH.5 the Tannakian tradition begins with Saavedra Rivano (Rivano 1972) and Deligne and Milne (Deligne and Milne 1982), and its derived and spectral forms are in Lurie's *Spectral Algebraic Geometry* (Lurie 2018). The tensor-triangular reconstruction theorems in the tradition of Balmer are attributed in prose. The question of what a geometry is when it has no points has been raised from several directions — Toën and Vaquié's schemes below $\operatorname{Spec}\mathbf{Z}$ (Toën and Vaquié 2009), Borger's Λ-rings (Borger 2009), Connes and Consani's scaling site (Connes and Consani 2016) — and Chapter 7 returns to it.

Finally, the ambidexterity phenomenon mentioned in the commentary on KH.3, in which the norm map $f_! \to f_*$ fails to be an equivalence by a Tate construction, is the subject of Hopkins and Lurie's work on ambidexterity in chromatic homotopy theory and of the subsequent higher-semiadditivity theory of Carmeli, Schlank and Yanovski; both are attributed in prose here and treated in Chapter 4. The functorial field theory perspective which motivates the $(\infty,3)$-categorical extension deferred in the caveat of §2.5 is Lurie's account of the cobordism hypothesis (Lurie 2009b), with Kapranov's early proposal that Langlands duality should be a field theory (Kapranov 1995) and the Kapustin–Witten picture (Kapustin and Witten 2007) as the physical antecedents; these are taken up in Chapter 23.

2.11 Exercises

2.1. Let C have finite limits and let E be stable under composition and base change and contain the equivalences. Show directly that the composition rule of Definition 2.2 is well defined, that is, that the right leg of the composite lies in E, and identify the associativity constraint with the canonical comparison of the two iterated pullbacks.

2.2. Show that if E consists only of the equivalences, then $\mathrm{Corr}(C,E) \simeq C^{\mathrm{op}}$, and deduce that a six-functor formalism on (C,E) is then just a symmetric monoidal presheaf of presentable stable categories. Explain why KH.4 is vacuous in this case, and why the resulting chôra carries no interesting trace.

2.3. Verify the two triangle identities for the formal self-duality of Observation 2.5 by writing out the two relevant pullbacks, and identify precisely which pullback square is used at each step.

2.4. In the naive coherent chôra of Example 2.30, compute $p^!_X$ for $X = \operatorname{Spec} A$ with A a finite-dimensional k-algebra, and show that in this case $\mathbf{A} = \operatorname{Spec} A$ *is* a dualisable figure. Conclude that the failure in the example is a failure of finiteness, not of the construction.

2.5. Let G be a finite group and k a field. Compute the norm map $(-)_{hG} \to (-)^{hG}$ for the sheaf theory $\mathcal{D}(X) = \operatorname{Fun}(X, D(k))$ on π-finite spaces, and show that its cofibre is the Tate construction. Determine for which k and G the norm map is an equivalence, and relate your answer to the commentary on KH.3.

2.6. Show that a 1-morphism of chôrai which is trace-exact and dualisability-preserving carries dualisable figures to dualisable figures and identifies the two trace theories on them. Give an example of a 1-morphism which preserves dualisability but is not trace-exact.

2.7. Prove that the cartesianness hypothesis on the natural transformation θ in Definition 2.18 cannot be dropped, by exhibiting a natural transformation between limit-preserving functors which does not induce a transformation on correspondence categories.

2.8. Compute $\tau(X, \mathrm{Frob}_q)$ for $X = \mathbf{P}^1 \times \mathbf{P}^1$, for a smooth conic in $\mathbf{P}^2_{\bar{\mathbf{F}}_q}$ with a rational point, and for X a smooth projective curve of genus g expressed in terms of the Weil numbers $\alpha_1, \dots, \alpha_{2g}$; in each case obtain the answer both from Theorem 2.25 and by counting points directly, and check that the Weil bound $|\#X(\mathbf{F}_q) - q - 1| \le 2g\sqrt{q}$ is respected.

2.9. Using the table in Example 2.28, compute $\#E(\mathbf{F}_{p^2})$ and $\#E(\mathbf{F}_{p^3})$ for $p = 7$ and $p = 13$ from the Weil numbers, using Newton's identities to pass from a_p to $\mathrm{tr}(\mathrm{Frob}_p^k)$, and verify your answers against Corollary 2.26 applied to Frob_p^k.

2.10. Show that the trivial chôra $\mathbf{1}$ of Proposition 2.20 is a unit for $\boxtimes$, and determine its endomorphism ∞-category in $\mathrm{Geom}^{\mathrm{tr}}$.

2.11. Formulate precisely the statement that KH.5 fails, in the form "there exist chôrai $\mathfrak{C} \not\simeq \mathfrak{C}'$ with ...", and explain which parts of the framework of this chapter would survive such a failure and which would not.

2.12. Run steps 1 to 4 of Algorithm 2.23 by hand on the figure $\operatorname{Spec} k[t]/(t^2)$ in the naive coherent chôra and on $\mathbf{A}^1$ in $\mathfrak{C}_{\mathrm{Sch}}$, and compare the transcripts.

2.13. The curve $E' : y^2 = x^3 - x$ of Example 2.28 is supersingular at $p \equiv 3 \pmod 4$ and ordinary at $p \equiv 1 \pmod 4$. Count $E'(\mathbf{F}_p)$ by hand for $p = 5$ and $p = 13$ and read off a_5 and a_{13}. Then write $p = a^2 + b^2$ with a odd and check in both cases that $a_p = \pm 2a$; determine the sign convention that makes the statement uniform, and explain why no such formula can exist at $p \equiv 3 \pmod 4$. Finally compute $\tau(E'_{\mathbf{F}_{p^2}}, \mathrm{Frob}_{p^2})$ in both regimes and account for the difference in terms of the endomorphism ring.

2.14. Two conjectures of this chapter can be tested on very small figures. First, test Conjecture 2.17 on the finite étale figures $\operatorname{Spec} \mathbf{F}_{q^n}$ of Example 2.27: decide whether the symmetric monoidal category $\mathcal{D}(X)$ together with the values $\tau(X, \mathrm{Frob}_q^k)$ for all k determines n, and whether it determines X as a figure. Second, test Remark 2.11 on the trivial chôra: classify the trace theories which agree with THH on the unit and are compatible with the multiplicativity datum, and decide whether rigidity holds there. State clearly, in each case, whether your answer supports the conjecture, refutes it, or is neutral.

Chapter 3

Trace, Weight, and Purity

Synopsis. This chapter constructs the trace theory required by the arithmetic chôra. It begins with the categorical trace of an endomorphism of a dualisable object and proves cyclicity by identifying the trace with a labelled circle, rather than postulating cyclicity as an algebraic rule. It then develops the circle action, topological Hochschild homology, cyclotomic structure and the Nikolaus–Scholze definition of topological cyclic homology, with $\mathrm{THH}(\mathbf{Z})$ and $\mathrm{THH}(\mathbf{F}_p)$ as persistent tests. Finally it installs Bondarko's weight structures, proves additivity of trace along a weight tower, and defines chôral purity so that Deligne's purity theorem over a finite field is recovered exactly, while the corresponding assertion over $\mathfrak{X}_{\mathbf{Z}}$ remains explicitly conjectural.

3.1 The circle behind the trace

A trace is often introduced by a formula and then furnished with a list of properties. That order conceals the source of the properties. The categorical trace is not fundamentally a scalar attached to a square matrix; it is the value of a closed one-dimensional bordism labelled by an endomorphism. The closed bordism has no preferred point at which it should be cut. Cyclicity is the assertion that changing the cut does not change the circle.

Let $(\mathcal{V}, \otimes, \mathbf{1})$ be a symmetric monoidal ∞-category. All equivalences below are coherent equivalences in $\mathcal{V}$; after passage to the homotopy category they become equalities of morphisms. This distinction matters because the circle action encountered later is precisely the coherent family of these equivalences.

Definition 3.1 (Dualisability and categorical trace). [ESTABLISHED]
An object $X \in \mathcal{V}$ is **dualisable** if there exist an object $X^\vee$ and morphisms

$$\mathrm{coev}_X : \mathbf{1} \longrightarrow X \otimes X^\vee, \qquad \mathrm{ev}_X : X^\vee \otimes X \longrightarrow \mathbf{1}$$

such that the two composites obtained by inserting coevaluation and then evaluation are equivalent to id_X and $\mathrm{id}_{X^\vee}$. For an endomorphism $f : X \to X$ of a dualisable object, its **categorical trace** is

$$\mathrm{tr}_X(f) = \mathrm{ev}_X \circ \beta_{X,X^\vee} \circ (f \otimes \mathrm{id}_{X^\vee}) \circ \mathrm{coev}_X \in \mathrm{End}_{\mathcal{V}}(\mathbf{1}),$$

where β is the symmetry. The Euler characteristic of X is $\chi(X) = \mathrm{tr}_X(\mathrm{id}_X)$.

The definition is independent, up to a contractible space of choices, of the chosen duality datum. In spectra the target is a spectrum rather than merely a number. A numerical trace appears only after applying a homotopy group, a character, an Euler characteristic, or another realisation functor. This is why the primitive trace in a chôra must be spectrum-valued.

Theorem 3.2 (Cyclicity is rotation of the circle). [ESTABLISHED]

Let X and Y be dualisable objects of a symmetric monoidal ∞-category, and let $f : X \to Y$ and $g : Y \to X$. Then there is a canonical equivalence

$$\mathrm{tr}_X(gf) \simeq \mathrm{tr}_Y(fg).$$

These equivalences are coherently cyclic for any finite cyclic word of composable morphisms. They arise from rotation of a labelled circle and are therefore consequences of dualisability and symmetric monoidal coherence, not additional axioms on a trace function.

Proof of Theorem 3.2.

Consider the free rigid symmetric monoidal category generated by two objects X, Y, their chosen duals, and coupons $f : X \to Y$, $g : Y \to X$. The graphical coherence theorem for rigid symmetric monoidal categories identifies morphisms in this free category with isotopy classes of framed string diagrams, subject only to the two triangle relations and the symmetric braiding relations.

The composite defining $\mathrm{tr}_X(gf)$ is represented by a closed oriented strand. Travelling around it, one encounters first the coupon f and then the coupon g; the apparent order gf records only that the diagram was cut at the segment labelled X. The composite defining $\mathrm{tr}_Y(fg)$ is the same closed strand with the cut moved across the coupon f, so that the exposed segment is labelled Y. The two diagrams are related by an orientation-preserving rotation of the annulus and are therefore isotopic. Graphical coherence makes the represented morphisms $\mathbf{1} \to \mathbf{1}$ equal in the free category. Every rigid symmetric monoidal category with the displayed data receives a symmetric monoidal functor from the free one, so the equality holds in every such category.

For a cyclic word $X_0 \xrightarrow{f_1} X_1 \to \cdots \xrightarrow{f_n} X_0$, the identical argument moves the cut through any coupon. The space of orientation-preserving rotations of the unmarked circle is S^1; retaining all higher isotopies gives the coherent cyclic structure rather than merely a collection of pairwise equalities. This proves both assertions.

□

3.1.1 From cyclic operators to the circle action

The coherence in the proof can be written without pictures. For an $\mathbb{E}_1$-algebra A, the cyclic bar object has

$$B^{\mathrm{cy}}_n(A) = A^{\otimes(n+1)}.$$

Its face maps multiply adjacent factors, with the last face multiplying the last factor into the first; its degeneracies insert the unit. In addition it has the cyclic operator

$$t_n(a_0 \otimes a_1 \otimes \cdots \otimes a_n) = a_n \otimes a_0 \otimes \cdots \otimes a_{n-1}.$$

In a symmetric monoidal ∞-category this formula denotes the composite of the symmetry isomorphisms required to move the last factor past the preceding ones. The relations

$$t_n^{n+1} = \mathrm{id}, \qquad d_i t_n = t_{n-1} d_{i-1} \quad (1 \le i \le n), \qquad d_0 t_n = d_n,$$

together with the analogous degeneracy relations, say exactly that the simplicial bar object extends to Connes' cyclic category.

The geometric realisation of a cyclic object has a canonical S^1-action. One way to see it is to realise an n-simplex as an oriented circle divided into $n+1$ arcs. The barycentric

coordinates record the arc lengths, and t_n moves the chosen initial vertex to the next one. As the subdivision is refined, these finite rotations assemble into all rotations of the circle. The identity $t_n^{n+1} = \mathrm{id}$ closes the rotation after one full turn, while the face and degeneracy relations ensure that collapsing or inserting an arc commutes with rotation.

For a cyclic word of composable maps

$$X_0 \xrightarrow{f_1} X_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} X_0,$$

choose a marked point on the circle immediately before f_1. Cutting there and using duality gives $\mathrm{tr}(f_n \cdots f_1)$. Moving the mark past f_1 gives

$$\mathrm{tr}(f_1 f_n \cdots f_2).$$

The path of marked points between the two cuts is the required homotopy. Moving the mark through every coupon produces a loop of homotopies, and the cyclic identities above identify that loop with one rotation in S^1. Thus the equality of two traces is the degree-zero shadow of an action, not an isolated algebraic coincidence.

For two maps $f : X \to Y$ and $g : Y \to X$, this gives the requested equality without assuming either map invertible. The labelled circle has one arc labelled X, one labelled Y, and two coupons. A cut in the X-arc produces $gf : X \to X$; a cut in the Y-arc produces $fg : Y \to Y$. The rotation path crossing the coupon f supplies the canonical equivalence

$$\mathrm{tr}_X(gf) \simeq \mathrm{tr}_Y(fg).$$

Any attempted proof which first writes this equality as an axiom discards the path and therefore cannot recover its compatibility with a third cut, a tensor product, or a family of endomorphisms.

The same circle appears in the function-field control. The categorical trace of Frobenius is represented by a mapping circle with Frobenius monodromy. Rotating its cut is formal; identifying the resulting trace with functions on the Frobenius-fixed stack is the geometric fixed-point theorem. The former is higher-categorical coherence, established by trace theory; the latter is the Lefschetz theorem of Grothendieck and its categorical refinements. Over the arithmetic base the rotation remains formal wherever the trace exists, but the fixed-point identification and the assembly of all local Frobenii remain conjectural.

The proof explains a point which will recur. If one writes $\mathrm{tr}(fg) = \mathrm{tr}(gf)$ as an axiom in a formal language, one records the shadow but loses the homotopy which produced it. The lost homotopy is the beginning of the S^1-action on Hochschild homology. Connes' cyclic category, the cyclic bar construction and the rotation action on factorisation homology are three presentations of that same circle.

Over a smooth proper curve $X/\mathbf{F}_q$, the endomorphism of interest is geometric Frobenius. For a constructible complex $\mathcal{F} \in D_c^b(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell)$, the numerical shadow of the categorical trace is the Grothendieck–Lefschetz formula

$$\sum_i (-1)^i \,\mathrm{Tr}\left(\mathrm{Frob}_q \mid H_c^i(X_{\overline{\mathbf{F}}_q}, \mathcal{F})\right) = \sum_{x \in X(\mathbf{F}_q)} \mathrm{Tr}\left(\mathrm{Frob}_x \mid \mathcal{F}_{\bar{x}}\right),$$

with the usual local refinements at closed points (Grothendieck and Illusie 1977). Thus, in the function-field control case, the passage from a global trace to fixed-point contributions is a theorem. Over $\mathfrak{X}_{\mathbf{Z}}$ there is no constructed global Frobenius endomorphism and no established Lefschetz theorem for the proposed analytic stack. The chôral localisation datum records exactly the theorem that is missing.

3.2 Primitive trace and the chôral axioms

The trace used here is not reconstructed from a previously chosen cohomology theory. A chôra is already a six-functor formalism with duality, and its trace is part of the geometric datum. The consolidated axiom system of the framework (KH.1–KH.28, Appendix F) is reproduced precisely because later statements depend on the differences among design, theorem and conjecture.

Axiom 3.3 (KH.6 — Trace is primitive). [DESIGN]
A chôra carries a symmetric monoidal trace theory τ on pairs (M, f) consisting of a dualisable object and an endomorphism. Its values are spectra. It is natural under equivalences, multiplicative under tensor products, additive for invariant exact triangles, and localising for exact sequences of stable categories. Cohomological traces are realisations of τ; they do not define it.

Axiom 3.4 (KH.7 — Localisation datum). [DESIGN]
For every admissible endo-correspondence f of a figure X, the trace theory comes with a functorial comparison from the global trace to local contributions supported on the derived fixed-point figure

$$X^f = X \times_{X\times X,(\mathrm{id},f)} X.$$

Whenever a fixed-point figure decomposes into admissible components, the comparison identifies the global trace with the corresponding sum or integral of local terms.

Axiom 3.5 (KH.8 — Universality of topological Hochschild homology). [CONJECTURE]
For the sphere-linear, dualisable and compactly generated part of a chôra, the primitive cyclic trace is represented by topological Hochschild homology. More precisely, a localising symmetric monoidal trace theory equipped with coherent circle invariance receives the canonical comparison from THH *described in Theorem 3.9. Equivalence with* THH *requires the non-degeneracy and finiteness hypotheses of Axiom 3.6. The comparison map is established; the assertion that these hypotheses force it to be an equivalence in the stated chôral generality is conjectural, and it is not asserted for arbitrary large analytic categories.*

Axiom 3.6 (KH.9 — Trace non-degeneracy). [DESIGN]
On the chosen dualisable subcategory, the pairing induced by evaluation and the trace has trivial radical after the coefficient realisation under consideration. This is a genuine restriction on the chôra. Without it, THH *can map to a quotient trace theory without being recovered from that quotient.*

Axiom 3.7 (KH.10 — Weights replace the heart). [DESIGN]
The stable category of constructible objects in an arithmetic chôra is equipped with a weight structure in Bondarko's sense, compatible with duality and with the six operations on their stated domains. No motivic t-structure is assumed. The weight tower, its pure factors, and the trace of Frobenius or scaling on those factors are the invariants used in place of an abelian heart.

Axiom 3.8 (KH.11 — Ambidexterity and height). [DESIGN]
The height of a chôra is the least n for which the stipulated class of n-truncated maps is ambidextrous, in the sense that the norm comparison $f_! \to f_*$ is an equivalence. The proposed arithmetic chôra is required to have height equal to its orientation degree, conjecturally three. The identification of chromatic height with arithmetic dimension is not an established theorem and no result of this chapter depends on it.

The trace of the identity of a dualisable category is Hochschild homology by established higher-categorical trace theory (Hoyois et al. 2017; Ponto and Shulman 2012). What is not

established is the non-degenerate equivalence claimed by Axiom 3.5, that the automorphic categories over the unconstructed base are dualisable in the required category, or that their analytic traces satisfy the continuity needed for cyclotomic constructions. Efimov's extension of localising invariants to large dualisable categories is the relevant current technology, but it does not by itself construct the arithmetic chôra.

The governing mathematical relations are collected in Plate 3.1.

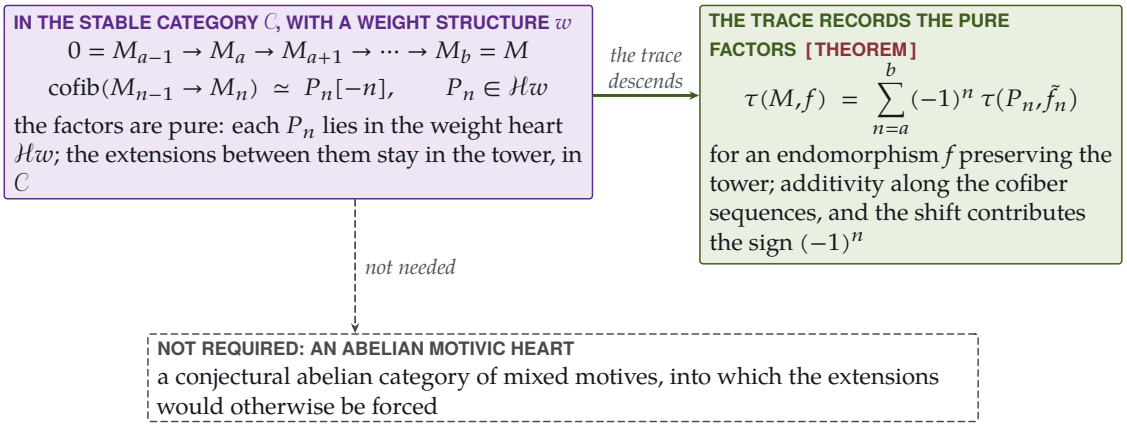


Plate 3.1. A weight structure supplies a tower whose factors are pure without requiring an abelian motivic heart. The trace descends through the tower and records the pure factors, while the missing extensions remain in the stable category rather than being forced into a conjectural category of mixed motives.

What it does not show: a category of mixed motives: the extensions stay in the stable category, and no abelian heart, motivic or otherwise, is constructed or assumed.

3.3 Topological Hochschild homology and the rotating trace

The trace of an identity endofunctor is already a familiar object. For an ordinary algebra it is Hochschild homology; for a ring spectrum it is topological Hochschild homology. The adjective *topological* does not indicate an optional enrichment of an algebraic construction. It records that the relative tensor products are formed over the sphere spectrum and therefore retain stable homotopy which ordinary Hochschild homology discards.

Definition (topological Hochschild homology). Let A be an $\mathbb{E}_1$-algebra in a presentable symmetric monoidal stable ∞-category whose tensor product preserves colimits separately. Its topological Hochschild homology is

$$\mathrm{THH}(A) = A \otimes_{A \otimes A^{\mathrm{op}}} A.$$

Equivalently, it is the geometric realisation of the cyclic bar object with n-simplices $A^{\otimes(n+1)}$. For a small idempotent-complete stable ∞-category C, $\mathrm{THH}(C)$ is the Morita-invariant extension of this construction from categories of perfect modules. It is the categorical trace $\mathrm{Tr}(\mathrm{id}_C)$.

The cyclic bar object is more than a simplicial object: it extends across Connes' cyclic category. Geometric realisation therefore carries a canonical S^1-action. At the level of the circle picture, a point of the simplicial bar construction chooses cuts and subdivisions of the circle, while realisation forgets the subdivision. Rotation moves the cut continuously.

Theorem 3.9 (The circle action and the universal cyclic trace). [ESTABLISHED]
For every $\mathbb{E}_1$-algebra A, $\mathrm{THH}(A)$ has a canonical S^1-action. If M is an object with an S^1-action, maps $\mathrm{THH}(A) \to M$ in the S^1-equivariant category are naturally equivalent to coherently cyclic trace maps from A to M: maps which identify the images of ab and ba together with all higher compatibilities. The same statement holds Morita-invariantly for small stable categories. Consequently THH *is the universal receptacle for a coherent cyclic trace in this precise sense; it is not the universal localising invariant without the word* trace.

Proof of Theorem 3.9.
The cyclic bar construction $B^{\mathrm{cy}}_\bullet(A)$ is generated by cyclic words $a_0|\cdots|a_n$. A map from its realisation to M is the same as a compatible family of maps from these tensor powers. Compatibility with the simplicial face maps imposes multiplication, compatibility with degeneracies imposes the unit, and compatibility with the cyclic operator imposes the homotopy carrying the image of ab to that of ba. The relations in the cyclic category impose exactly the iterated coherence conditions. Passing to geometric realisations identifies the automorphism group of the unmarked oriented circle with S^1, giving the action.

This proves the algebra statement by the universal property of a colimit. A small stable category is replaced by its spectral category of compact objects; the same cyclic nerve is taken with mapping spectra in place of elements. Morita invariance identifies the result for $\mathrm{Perf}(A)$ with the algebraic construction. The higher-categorical trace theorem identifies that cyclic nerve with $\mathrm{Tr}(\mathrm{id})$ (Hoyois et al. 2017; Lurie 2017).

□

This formulation avoids an overstatement sometimes made about universality. The universal additive or localising invariant of stable categories is the noncommutative motive constructed by Blumberg, Gepner and Tabuada; it is not THH (Blumberg et al. 2013). What THH universally receives is cyclic trace data. Axiom Axiom 3.5 therefore requires the trace structure, not merely additivity.

3.4 The arithmetic diagonal, corrected

The distinction between an absolute tensor square and a self-intersection is load-bearing. The framework corrects the tempting identity

$$\mathbf{Z} \otimes_{\mathbf{S}} \mathbf{Z} \simeq \mathrm{THH}(\mathbf{Z}).$$

It is false. On the left is an enveloping algebra over the sphere; on the right is the derived tensor product over that enveloping algebra.

Proposition 3.10 (The absolute diagonal correction). [ESTABLISHED]
Write $H\mathbf{Z}$ for the Eilenberg–Mac Lane $\mathbb{E}_\infty$-ring. Then

$$H\mathbf{Z} \otimes_{\mathbf{S}} H\mathbf{Z} \not\simeq \mathrm{THH}(H\mathbf{Z}),$$

and the correct formula is

$$\mathrm{THH}(H\mathbf{Z}) \simeq H\mathbf{Z} \otimes_{H\mathbf{Z}\otimes_{\mathbf{S}} H\mathbf{Z}} H\mathbf{Z}.$$

Geometrically, if $\Delta : \operatorname{Spec} H\mathbf{Z} \to \operatorname{Spec} H\mathbf{Z} \times_{\operatorname{Spec}\mathbf{S}} \operatorname{Spec} H\mathbf{Z}$ is the diagonal, then the algebra of its derived self-intersection is

$$\Delta^*\Delta_* \mathcal{O} \simeq \mathrm{THH}(H\mathbf{Z}).$$

Thus the ambient absolute product supplies two potentially distinct legs, whereas THH *measures their derived collision along the diagonal. Shtuka constructions need both statements in that order.*

Proof of Proposition 3.10.

For any commutative algebra object A over a symmetric monoidal base R, the diagonal is affine with enveloping algebra $A^e = A \otimes_R A$. Pulling its pushforward of the structure sheaf back along the diagonal is extension of scalars along the multiplication $A^e \to A$. Hence

$$\Delta^* \Delta_* \mathcal{O} \simeq A \otimes_{A^e} A.$$

By the definition above, this is $\mathrm{THH}_R(A)$. Taking $R = \mathbf{S}$ and $A = H\mathbf{Z}$ gives the displayed formula.

To see that the tensor square cannot be identified with the self-intersection, observe that the first is the algebra over which the second relative tensor product is formed. An equivalence between them would in particular force the multiplication resolution of $H\mathbf{Z}$ as an A^e-module to collapse. The homotopy groups computed in Theorem 3.11 exhibit the nontrivial derived torsion created by that resolution.

□

There is a concrete base-change test which separates the two spectra. Regard both as left $H\mathbf{Z}$-modules and extend scalars to $H\mathbf{F}_2$. For the tensor square one obtains

$$H\mathbf{F}_2 \otimes_{H\mathbf{Z}} (H\mathbf{Z} \otimes_{\mathbf{S}} H\mathbf{Z}) \simeq H\mathbf{F}_2 \otimes_{\mathbf{S}} H\mathbf{Z}.$$

Its homotopy is the mod-two homology of $H\mathbf{Z}$. The standard calculation gives

$$H_*(H\mathbf{Z}; \mathbf{F}_2) \cong \mathbf{F}_2[\xi_1^2, \xi_2, \xi_3, \dots], \qquad |\xi_i| = 2^i - 1,$$

so the degree-two group is generated by ξ_1^2 and is nonzero.

For the self-intersection, derived extension of scalars gives a universal coefficient exact sequence

$$0 \longrightarrow \pi_n \mathrm{THH}(\mathbf{Z}) \otimes \mathbf{F}_2 \longrightarrow \pi_n (H\mathbf{F}_2 \otimes_{H\mathbf{Z}} \mathrm{THH}(\mathbf{Z})) \longrightarrow \mathrm{Tor}_1^{\mathbf{Z}} (\pi_{n-1} \mathrm{THH}(\mathbf{Z}), \mathbf{F}_2) \longrightarrow 0.$$

Bökstedt's answer has $\pi_1 \mathrm{THH}(\mathbf{Z}) = \pi_2 \mathrm{THH}(\mathbf{Z}) = 0$. Hence the degree-two group after this base change is zero. We have therefore distinguished the objects by an explicit invariant:

$$\pi_2 (H\mathbf{F}_2 \otimes_{H\mathbf{Z}} (H\mathbf{Z} \otimes_{\mathbf{S}} H\mathbf{Z})) \cong \mathbf{F}_2, \qquad \pi_2 (H\mathbf{F}_2 \otimes_{H\mathbf{Z}} \mathrm{THH}(\mathbf{Z})) = 0.$$

The calculation also identifies their different jobs. The algebra $H\mathbf{Z} \otimes_{\mathbf{S}} H\mathbf{Z}$ is the coordinate algebra of the ambient absolute product. It carries two structure maps from $H\mathbf{Z}$, one for each projection, and these maps are what permit two legs to move independently. The spectrum $\mathrm{THH}(\mathbf{Z})$ is obtained only after both copies of $H\mathbf{Z}$ are pulled back to the diagonal. It records the derived excess intersection of the two legs when they coincide.

This order is required by the shtuka construction developed later in Chapter 22. A shtuka first needs an ambient product in which distinct legs and partial Frobenius modifications can be stated. Its fusion law then restricts to diagonals, where derived self-intersections control collisions. Replacing the ambient product by $\mathrm{THH}(\mathbf{Z})$ at the outset would retain collision data but erase the distinction between the two projections. Replacing $\mathrm{THH}(\mathbf{Z})$ by the tensor square would retain two projections but discard the derived collision. The false identification therefore collapses two successive constructions which that chapter must keep separate.

The correction strengthens the geometric interpretation. If the diagonal were an equivalence, two legs would collapse. Its derived self-intersection records all ways in which they meet non-transversely. What remains conjectural is the existence of an arithmetic Ran construction in which this spectral diagonal is compatible with the syntomic charts, the halo gluing and the archimedean chart. The identity above is established spectral algebra; its use as the arithmetic base is a design proposal.

3.5 Bökstedt's two calculations

Theorem 3.11 (Bökstedt's calculations for the integers and a finite field). [ESTABLISHED]
There are graded-ring and graded-group calculations

$$\pi_* \operatorname{THH}(\mathbf{F}_p) \cong \mathbf{F}_p[\sigma], \qquad |\sigma| = 2,$$

and

$$\pi_0 \operatorname{THH}(\mathbf{Z}) \cong \mathbf{Z}, \qquad \pi_{2i-1} \operatorname{THH}(\mathbf{Z}) \cong \mathbf{Z}/i \quad (i > 0), \qquad \pi_{2i} \operatorname{THH}(\mathbf{Z}) = 0 \quad (i > 0).$$

The second formula is an assertion about the additive homotopy groups; it does not identify the integral ring with a polynomial algebra.

Bökstedt proved these results by the spectral sequence now bearing his name. Modern accounts and refinements occur throughout the literature on the cyclotomic trace and integral p-adic Hodge theory (Bökstedt et al. 1993; Bhatt et al. 2019). The formula for $i = 1$ gives $\mathbf{Z}/1 = 0$, so the first positive nonzero integral group is $\pi_3 \cong \mathbf{Z}/2$.

3.5.1 Worked computation: the first ten homotopy groups

Bökstedt's formula gives the following concrete comparison.

Degree	$\pi_n \operatorname{THH}(\mathbf{Z})$	$\pi_n \operatorname{THH}(\mathbf{F}_p)$
0	$\mathbf{Z}$	$\mathbf{F}_p$
1	0	0
2	0	$\mathbf{F}_p\{\sigma\}$
3	$\mathbf{Z}/2$	0
4	0	$\mathbf{F}_p\{\sigma^2\}$
5	$\mathbf{Z}/3$	0
6	0	$\mathbf{F}_p\{\sigma^3\}$
7	$\mathbf{Z}/4$	0
8	0	$\mathbf{F}_p\{\sigma^4\}$
9	$\mathbf{Z}/5$	0

The contrast is structural. Over $\mathbf{F}_p$ one generator produces an unbounded even polynomial tower. Over $\mathbf{Z}$ positive even homotopy vanishes and the derived diagonal records torsion of order i in degree $2i - 1$. Rationally this torsion disappears, so $\operatorname{THH}(\mathbf{Z}) \otimes \mathbf{Q}$ sees only degree zero. Any arithmetic trace theory which is rational from the outset therefore discards the most visible evidence that the absolute diagonal is derived.

There is also a warning for weights. Topological degree is not automatically a motivic weight. In $\mathbf{F}_p[\sigma]$ it is natural, after choosing a Rees normalisation, to assign scaling weight one to σ and weight i to σ^i. No canonical global $\mathbf{G}_m$-action on $\operatorname{THH}(\mathbf{Z})$ with the required arithmetic interpretation has been constructed. The odd torsion groups are therefore a test for a prospective weight formalism, not evidence that such a formalism already exists.

3.5.2 Polynomial and perfect characteristic-p algebras

The calculation over $\mathbf{F}_p$ extends in two instructive directions. If A is a smooth $\mathbf{F}_p$-algebra, the Bökstedt spectral sequence and the Hochschild–Kostant–Rosenberg calculation give, as a graded A-algebra,

$$\pi_* \operatorname{THH}(A) \cong \Omega^*_{A/\mathbf{F}_p} \otimes_{\mathbf{F}_p} \mathbf{F}_p[\sigma], \qquad |\sigma| = 2,$$

where a differential form of degree j is placed in homotopy degree j. For the polynomial ring $A = \mathbf{F}_p[x]$ this becomes

$$\pi_* \operatorname{THH}(\mathbf{F}_p[x]) \cong \mathbf{F}_p[x] \otimes \Lambda_{\mathbf{F}_p}(dx) \otimes \mathbf{F}_p[\sigma], \qquad |dx| = 1, \quad |\sigma| = 2.$$

Thus the first six groups are

Degree	Homotopy group
0	$\mathbf{F}_p[x]$
1	$\mathbf{F}_p[x]\,dx$
2	$\mathbf{F}_p[x]\,\sigma$
3	$\mathbf{F}_p[x]\,dx\,\sigma$
4	$\mathbf{F}_p[x]\,\sigma^2$
5	$\mathbf{F}_p[x]\,dx\,\sigma^2$

Multiplication is determined by $(dx)^2 = 0$ and by the polynomial powers of σ. The odd groups are not torsion anomalies of the integral kind; they are the ordinary differential direction of the affine line. The even Bökstedt direction remains after that geometric differential direction has been removed.

Now let R be a perfect $\mathbf{F}_p$-algebra. Frobenius is an isomorphism on R. Its action on the cotangent complex is simultaneously an equivalence and zero, because $d(r^p) = 0$ in characteristic p; hence $L_{R/\mathbf{F}_p} \simeq 0$. The topological Hochschild calculation reduces to

$$\pi_* \operatorname{THH}(R) \cong R[\sigma], \qquad |\sigma| = 2.$$

In particular

$$\pi_{2j} \operatorname{THH}(R) \cong R \cdot \sigma^j, \qquad \pi_{2j+1} \operatorname{THH}(R) = 0.$$

Taking $R = \mathbf{F}_{p^r}$ gives one copy of $\mathbf{F}_{p^r}$ in every nonnegative even degree and zero in every odd degree.

These examples separate geometric imperfection from cyclotomic periodicity. The polynomial ring has a nonzero cotangent complex and therefore the exterior class dx. Passing to its perfection

$$\mathbf{F}_p[x^{1/p^\infty}] = \operatorname{colim}\left(\mathbf{F}_p[x] \xrightarrow{x \mapsto x^p} \mathbf{F}_p[x] \xrightarrow{x \mapsto x^p} \cdots\right)$$

kills the differential direction but not σ. Over a perfect function-field base, the surviving polynomial tower is therefore not a Kähler differential of the curve. It is the topological trace direction which feeds the cyclotomic Frobenius and the de Rham–Witt constructions.

For the arithmetic comparison, the lesson is that $\operatorname{THH}(\mathbf{F}_p) = \mathbf{F}_p[\sigma]$ is stable under perfect extension, whereas $\operatorname{THH}(\mathbf{Z})$ is assembled from finite odd torsion. The finite-field fibre has a uniform even generator; the integral object records how all prime fibres are glued. A global weight theory must account for both patterns without treating reduction modulo p as termwise reduction of Bökstedt's integral homotopy groups.

3.6 Cyclotomic structure and topological cyclic homology

The circle action remembers rotations, but arithmetic asks for more. For every prime p, the p-fold covering of the circle compares a loop with its C_p-fixed repetition. The Tate construction converts this covering operation into a Frobenius map. A cyclotomic spectrum is exactly the datum in which these prime-indexed comparisons are retained.

Definition 3.12 (Cyclotomic spectrum). [ESTABLISHED]

A **cyclotomic spectrum** in the Nikolaus–Scholze sense is a spectrum X with an S^1-action and, for every prime p, an S^1-equivariant map

$$\varphi_p : X \longrightarrow X^{tC_p}.$$

The target has the residual action of S^1/C_p, identified with S^1 through the p-th-root isomorphism. No equivalences among the maps φ_p are demanded in this definition. A p-cyclotomic spectrum retains only the map for the chosen prime. For a ring spectrum A, $\mathrm{THH}(A)$ has a canonical cyclotomic structure.

Set

$$\mathrm{TC}^-(X) = X^{hS^1}, \qquad \mathrm{TP}(X) = X^{tS^1}.$$

There is always a canonical map $\mathrm{can} : X^{hS^1} \to X^{tS^1}$. The cyclotomic Frobenius induces a second map after the Tate-orbit comparison. The familiar equaliser formula is correct prime by prime; integrally, the product over all primes must not be suppressed.

Theorem 3.13 (The Nikolaus–Scholze formula for topological cyclic homology). [ESTABLISHED]

Let X be a bounded-below cyclotomic spectrum. There is a natural fibre sequence

$$\mathrm{TC}(X) \longrightarrow X^{hS^1} \xrightarrow{(\mathrm{can}_p - \varphi_p)_p} \prod_p \left(X^{tS^1}\right)^\wedge_p .$$

For a fixed prime p, after p-completion this becomes the homotopy equaliser

$$\mathrm{TC}(X;p) \simeq \mathrm{Eq}\left(\mathrm{TC}^-(X)^\wedge_p \underset{\varphi_p}{\overset{\mathrm{can}}{\rightrightarrows}} \mathrm{TP}(X)^\wedge_p\right).$$

Thus the two arrows are the canonical homotopy-fixed-point-to-Tate map and the map induced by the cyclotomic Frobenius. Writing a single integral equaliser $\mathrm{TC}^- \rightrightarrows \mathrm{TP}$ is harmless only when the primewise product and completions are understood.

Proof of Theorem 3.13.

Nikolaus and Scholze define $\mathrm{TC}(X)$ as the mapping spectrum from the cyclotomic sphere to X. Unwinding a map of cyclotomic spectra gives an S^1-homotopy-fixed point together with, for each p, a homotopy identifying its canonical image in the Tate construction with the image induced by φ_p. Hence $\mathrm{TC}(X)$ is the fibre of the difference of these two maps into the product of primewise targets.

Write $\mathbb{T} = S^1$. Before invoking the Tate-orbit lemma, the common target of the two primewise maps is

$$\left(X^{tC_p}\right)^{h(\mathbb{T}/C_p)}.$$

The canonical map is obtained in three named steps. Restriction of a $\mathbb{T}$-homotopy-fixed point gives a C_p-homotopy-fixed point,

$$X^{h\mathbb{T}} \longrightarrow \left(X^{hC_p}\right)^{h(\mathbb{T}/C_p)}.$$

The norm cofiber sequence for the finite group C_p gives

$$X_{hC_p} \xrightarrow{N} X^{hC_p} \xrightarrow{\mathrm{can}_{C_p}} X^{tC_p}.$$

Taking residual $\mathbb{T}/C_p$-homotopy fixed points therefore gives

$$\mathrm{can}_p : X^{h\mathbb{T}} \longrightarrow \left(X^{tC_p}\right)^{h(\mathbb{T}/C_p)}.$$

The second map starts from the cyclotomic structure map

$$\varphi_p : X \longrightarrow X^{tC_p},$$

which is equivariant for the $\mathbb{T}$-action on the source and the residual $\mathbb{T}/C_p$-action on the target, identified through the p-th-root isomorphism $\mathbb{T}/C_p \simeq \mathbb{T}$. Taking homotopy fixed points gives

$$\varphi_p^{h\mathbb{T}} : X^{h\mathbb{T}} \longrightarrow \left(X^{tC_p}\right)^{h(\mathbb{T}/C_p)}.$$

Thus the equaliser condition has parallel arrows with the same source and target:

$$X^{h\mathbb{T}} \underset{\varphi_p^{h\mathbb{T}}}{\overset{\mathrm{can}_p}{\rightrightarrows}} \left(X^{tC_p}\right)^{h(\mathbb{T}/C_p)}.$$

For bounded-below X, the Tate-orbit lemma identifies the relevant homotopy fixed points of X^{tC_p} with the p-completion of X^{tS^1}. This yields the displayed integral fibre sequence. Retaining one prime and completing at p identifies a fibre of a difference with the homotopy equaliser of the two maps. This is the Nikolaus–Scholze theorem (Nikolaus and Scholze 2018).

□

The formula separates two operations which should not be conflated. The map can comes from equivariant homotopy theory and exists for every circle spectrum. The map φ_p is arithmetic information supplied by the cyclotomic structure. Topological cyclic homology consists of classes on which these two ways of entering periodic theory agree.

For $\mathrm{THH}(\mathbf{F}_p)$, the polynomial class σ feeds the Tate and homotopy-fixed-point spectral sequences and ultimately produces the Witt-vector phenomena characteristic of $\mathrm{TC}(\mathbf{F}_p)$. For $\mathrm{THH}(\mathbf{Z})$, the groups $\mathbf{Z}/i$ contribute at every prime dividing i; the integral product in Theorem 3.13 is therefore not a formal embellishment. It is the mechanism by which a single integral trace remembers all prime supports.

Over $X/\mathbf{F}_q$ the control problem is simpler. There is a distinguished geometric Frobenius and the Grothendieck–Lefschetz trace computes its fixed points. One does not need to manufacture a global endomorphism from all primes. Over $\mathfrak{X}_{\mathbf{Z}}$, by contrast, the maps φ_p live on different finite charts and the circle/scaling directions live at infinity. Cyclotomic structure is the proposed receptacle for assembling them, but the assertion that its local maps coincide with chôral localisation terms is Axiom Axiom 3.4, not a theorem of Nikolaus and Scholze.

3.7 Weight structures without a motivic heart

A t-structure organises a stable category by cohomological truncations and produces an abelian heart. A weight structure reverses the orthogonality and organises objects by extensions of pure pieces. Its heart is additive and Karoubian, but need not be abelian. This is weaker than a motivic t-structure, and the weakness is deliberate: traces require exact triangles and dualisability, not kernels and cokernels in a conjectural category of mixed motives.

Definition 3.14 (Bondarko weight structure). [ESTABLISHED]

A **weight structure** w on a stable ∞-category C is a pair of full subcategories $(C_{w\le 0}, C_{w\ge 0})$ satisfying the following conditions.

First, both subcategories are closed under equivalences and retracts. Secondly,

$$C_{w\le 0} \subset C_{w\le 0}[1], \qquad C_{w\ge 0}[1] \subset C_{w\ge 0}.$$

Thirdly, if $A \in C_{w\le 0}$ and $B \in C_{w\ge 1} := C_{w\ge 0}[1]$, then there is no morphism $A \to B$ in the homotopy category, equivalently $\pi_0 \operatorname{Map}_C(A, B) = 0$. Finally, every object M occurs in a cofiber sequence

$$A \longrightarrow M \longrightarrow B$$

with $A \in C_{w\le 0}$ and $B \in C_{w\ge 1}$.

Set $C_{w\le n} = C_{w\le 0}[n]$ and $C_{w\ge n} = C_{w\ge 0}[n]$. The **weight heart** is the idempotent-complete additive category

$$\mathcal{H}w = C_{w=0} = C_{w\le 0} \cap C_{w\ge 0}.$$

The structure is **bounded** if every object lies in $C_{w\ge a} \cap C_{w\le b}$ for some integers $a \le b$.

These are Bondarko's conventions (Bondarko 2010). A weight decomposition is generally not functorial and not unique. The non-uniqueness is controlled: successive choices produce weight Postnikov towers whose associated complexes agree in the appropriate homotopy category. A treatment which writes "the weight filtration" as though it were a canonical filtration by subobjects has silently imported a t-structure and is not using Bondarko's notion.

The basic model is the homotopy category of bounded complexes in an additive idempotent-complete category. The stupid weight structure has $C_{w\le 0}$ represented by complexes concentrated in degrees ≥ 0 and $C_{w\ge 0}$ represented by complexes concentrated in degrees ≤ 0, with heart the original additive category in degree zero. The opposite-looking inequalities are exactly what make weight decompositions peel off terms rather than cohomology objects.

Bondarko's weight-complex theorem. Let C be a stable ∞-category with a bounded weight structure. A finite choice of compatible weight decompositions for M produces a complex

$$t_w(M) = \left(\cdots \longrightarrow P^{n-1} \xrightarrow{d^{n-1}} P^n \xrightarrow{d^n} P^{n+1} \longrightarrow \cdots \right), \qquad P^n \in \mathcal{H}w,$$

well defined up to the weak homotopy relation of weight complexes. With the standard enhancement hypotheses, this construction refines to an exact strong weight-complex functor

$$t_w : C \longrightarrow K^b(\mathcal{H}w).$$

The construction does not require an abelian heart.

Bondarko proves the triangulated form and its functoriality properties (Bondarko 2010). Stable enhancements remove part of the ambiguity by retaining the coherent choices. The theorem does not say that t_w is fully faithful, nor that every extension in C can be reconstructed from the differential in an ordinary complex. It says that every bounded object has a finite pure ledger on which additive invariants can be computed.

3.7.1 Construction of the tower

Choose integers $a \le b$ with $M \in \mathcal{C}_{w\ge a} \cap \mathcal{C}_{w\le b}$. A weight decomposition at $b-1$ yields a triangle

$$w_{\le b-1}M \longrightarrow M \longrightarrow P_b[-b]$$

with $P_b \in \mathcal{H}w$. Repeat on $w_{\le b-1}M$, continuing downwards. After finitely many steps one obtains a tower

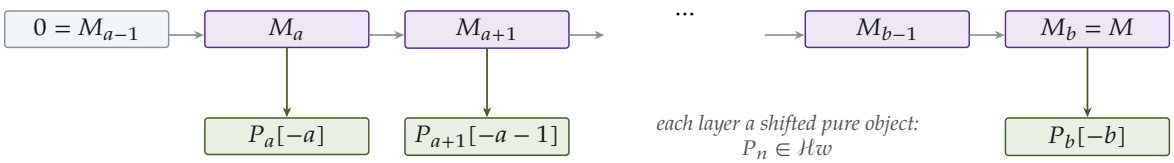
$$0 = M_{a-1} \longrightarrow M_a \longrightarrow \cdots \longrightarrow M_b = M$$

whose cofibers satisfy

$$\operatorname{cofib}(M_{n-1} \to M_n) \simeq P_n[-n], \qquad P_n \in \mathcal{H}w.$$

The connecting morphisms between successive triangles induce the differentials of $t_w(M)$. The relation $d^{n+1}d^n = 0$ follows from the octahedral axiom, or from the corresponding coherent cofiber diagram in the stable enhancement.

The system-level dependencies of the construction are displayed in Plate 3.2.



ADDITIVITY: THE TOWER BECOMES AN ALTERNATING SUM [THEOREM]

$$\tau(M,f) = \sum_{n=a}^{b} \tau(G_n,f_n) = \sum_{n=a}^{b} (-1)^n \tau(P_n,\tilde{f}_n), \qquad G_n = \operatorname{cofib}(M_{n-1} \to M_n) \simeq P_n[-n]$$

KEPT IN THE TOWER
the extension data: how the layers are glued, in the stable category

EVALUATED ON THE PURE LAYERS
every additive invariant: the trace, its local terms and their global sum

Plate 3.2. The weight tower is a finite succession of cofiber sequences. Each layer is a shifted pure object, and additivity of trace converts the tower into an alternating sum. The architecture keeps extension data in the tower while allowing every additive invariant to be evaluated on its pure layers.

What it does not show: the extension classes themselves: the trace and every other additive invariant are blind to them, and a non-additive invariant may still see them.

3.8 Additivity of trace along a weight filtration

Additivity is the point at which weights and traces meet. It is not a statement that a trace ignores extensions. Rather, an invariant trace on an exact triangle is constrained so that the middle trace is the sum of the outer traces. The extension class can affect non-additive invariants, but not this one.

Theorem 3.15 (Trace additivity along a weight tower). [THEOREM]

Let $\mathcal{C}$ be a symmetric monoidal stable ∞-category and let τ be a trace theory satisfying Axiom 3.3. Suppose a dualisable object M has a finite weight tower

$$0 = M_{a-1} \to M_a \to \cdots \to M_b = M$$

which is preserved by an endomorphism f, and suppose every layer $G_n = \operatorname{cofib}(M_{n-1} \to M_n)$ is dualisable. Then

$$\tau(M,f) = \sum_{n=a}^{b} \tau(G_n,f_n).$$

If $G_n \simeq P_n[-n]$ with $P_n \in \mathcal{H}w$, then

$$\tau(M,f) = \sum_{n=a}^{b} (-1)^n \tau(P_n,\tilde{f}_n).$$

When the localisation datum of Axiom 3.4 applies functorially to the tower, the same identities hold for every local term and hence for their global sum.

Proof of Theorem 3.15.

For each n there is an f-equivariant cofiber sequence

$$M_{n-1} \longrightarrow M_n \longrightarrow G_n.$$

The additivity axiom for τ, which in classical stable settings is May's additivity theorem (May 2001), gives

$$\tau(M_n,f|_{M_n}) = \tau(M_{n-1},f|_{M_{n-1}}) + \tau(G_n,f_n).$$

Starting with $\tau(0) = 0$ and applying this identity successively from a to b gives the first formula by induction.

It remains to compute the trace of a shift. In a stable symmetric monoidal category, the suspension object is tensoring with the invertible sphere S^1. Its Euler characteristic is -1: the cofiber sequence $\mathbf{1} \to 0 \to \mathbf{1}[1]$ and additivity give $0 = 1 + \chi(\mathbf{1}[1])$. Multiplicativity of trace therefore yields

$$\tau(P[-n],f[-n]) = (-1)^n \tau(P,f).$$

Substitution gives the alternating formula. Finally, functorial localisation carries each equivariant cofiber sequence to the corresponding cofiber sequence of local trace terms. The identical induction proves the local assertion.

□

The distinguished-triangle calculation can be seen explicitly in perfect complexes. Let $u : A^\bullet \to B^\bullet$ be a chain map and let $C^\bullet = \operatorname{Cone}(u)$, so

$$C^n = B^n \oplus A^{n+1}.$$

Suppose f_A and f_B form an endomorphism of u up to a specified chain homotopy h. The induced endomorphism of the cone has upper-triangular form

$$f_C = \begin{pmatrix} f_B & h \\ 0 & f_A[1] \end{pmatrix}.$$

The off-diagonal entry contributes nothing to the trace. Taking the supertrace term by term gives

$$\begin{aligned}\operatorname{tr}(f_C) &= \sum_n (-1)^n \operatorname{tr}(f_B^n) + \sum_n (-1)^n \operatorname{tr}(f_A^{n+1}) \\ &= \operatorname{tr}(f_B) - \operatorname{tr}(f_A).\end{aligned}$$

Therefore

$$\operatorname{tr}(f_B) = \operatorname{tr}(f_A) + \operatorname{tr}(f_C).$$

This proof also shows why a non-split extension causes no correction term: the extension and the chosen homotopy occur off the diagonal.

May's additivity theorem proves that the same identity holds in every stable symmetric monoidal setting in which the three objects are dualisable (May 2001). Its proof replaces the displayed matrix by the endomorphism of the universal filtered object $0 \subset A \subset B$ and uses the triangle identities to cancel the two off-diagonal string diagrams. Axiom Axiom 3.3 imports precisely this established theorem into the primitive trace theory. The induction in Theorem 3.15 is consequently a complete proof from the chôral axioms, while the cone calculation exhibits the identity in the basic algebraic model.

This is the promised theorem that reciprocity-relevant traces can be computed without a motivic t-structure. The theorem is formal once a bounded weight structure and an additive trace exist. Those two hypotheses are not formal in the arithmetic application. In particular, neither the bounded weight structure on the proposed category of arithmetic gauges nor its compatibility with the absolute Frobenius has been constructed.

3.8.1 Worked computation: a torsion perfect complex

Take $\mathcal{C} = \operatorname{Perf}(\mathbf{Z})$ with the stupid weight structure and consider the two-term cochain complex

$$M = [\mathbf{Z} \xrightarrow{2} \mathbf{Z}],$$

with the first copy in degree zero and the second in degree one. Multiplication by 3 on both terms defines an endomorphism f of the complex. Its weight complex is already displayed: the two pure terms are copies of $\mathbf{Z}$ in adjacent degrees. The theorem gives

$$\operatorname{tr}_M(f) = 3 - 3 = 0.$$

The cohomology of M is $H^0(M) = 0$ and $H^1(M) = \mathbf{Z}/2$. On that torsion module, multiplication by 3 is the identity. There is no ordinary trace of this endomorphism in the category of finitely generated projective $\mathbf{Z}$-modules because $\mathbf{Z}/2$ is not projective. The stable categorical trace nevertheless exists and is computed exactly by the weight terms. This small example isolates the methodological point: passage to an abelian heart can destroy the domain on which trace is naturally defined, whereas the weight complex preserves it.

The same phenomenon is visible in $\mathrm{THH}(\mathbf{Z})$. Its positive homotopy consists of torsion $\mathbf{Z}/i$, but the trace object is defined before those groups are split into cohomological pieces. Bökstedt's computation is therefore closer in spirit to a weight-complex calculation than to a semisimple eigenvalue calculation. Cyclotomic and scaling structures are needed before arithmetic weights can be read from it.

3.8.2 A weight structure and a t-structure on $\mathrm{Perf}(\mathbf{Z})$

The small category $\mathrm{Perf}(\mathbf{Z})$ carries both structures, so their difference can be computed rather than described abstractly. Its stupid weight structure has heart

$$\mathcal{H}w = \mathrm{Proj}^{\mathrm{fg}}(\mathbf{Z}),$$

the additive category of finitely generated free abelian groups. A complex is in weight range $[a, b]$ when it is homotopy equivalent to a complex of finite free groups concentrated in the corresponding cohomological degrees, with the sign convention of Definition 3.14.

The standard t-structure inherited from $D(\mathbf{Z})$ also restricts to $\mathrm{Perf}(\mathbf{Z})$. Its heart is the abelian category of finitely generated abelian groups. Indeed every such group has a finite free resolution of length at most one, so it is a perfect complex concentrated in t-degree zero. The two hearts are therefore

$$\mathrm{Proj}^{\mathrm{fg}}(\mathbf{Z}) \subsetneq \mathrm{Mod}^{\mathrm{fg}}(\mathbf{Z}).$$

Consider again multiplication by two,

$$2 : \mathbf{Z} \longrightarrow \mathbf{Z}.$$

In the weight heart its cokernel does not exist as an object of the heart: $\mathbf{Z}/2$ is not projective. The morphism is retained through the two-term weight complex $[\mathbf{Z} \xrightarrow{2} \mathbf{Z}]$. In the t-heart the cokernel exists and is exactly $\mathbf{Z}/2$, while the kernel is zero. Functorial t-truncation sends the two-term complex to its cohomology object $\mathbf{Z}/2$ in the appropriate degree. Weight decomposition instead peels off the two free terms and keeps multiplication by two as the connecting differential.

This example identifies the information supplied by a conjectural motivic t-structure. It would provide an abelian category of mixed motives with functorial subobjects, quotients, images, kernels and cokernels. Extension classes would be groups

$$\mathrm{Ext}^1_{\mathcal{M}}(M, N)$$

inside that heart rather than merely morphisms in a surrounding triangulated category. Realisation functors could be required to be t-exact, so a mixed motive would have canonical cohomology motives. Under additional rigidity, finite-dimensionality and a fibre functor, one could apply Tannakian reconstruction to obtain a motivic Galois group for the mixed category.

Bondarko's weight structure gives none of these conclusions. It gives a Karoubian additive pure heart, noncanonical weight decompositions, weight complexes and spectral sequences. That is enough for the additive trace formula because the trace of the two-term complex is $3 - 3 = 0$ in the example above. It is not enough to classify the torsion cohomology object or its extensions. The framework's claim is therefore limited: weights replace the missing heart for trace computations, not for the full theory of mixed motives.

Over $X/\mathbf{F}_q$, the realised category of mixed ℓ-adic complexes has a standard t-structure and Deligne's weight filtration on its cohomology sheaves. This is established. The motivic source of that structure is not known in the desired generality. Over $\mathfrak{X}_{\mathbf{Z}}$, even the category on which both structures would be compared remains conjectural.

3.9 Scaling and the eigenvalue meaning of weight

The Bondarko structure gives a finite tower; it does not by itself produce an eigenvalue decomposition. The latter requires an additional split scaling action, represented geometrically by a Rees $\mathbf{G}_m$-direction and at the archimedean chart by the split factor $\mathbf{R}_{>0} \subset \mathbf{C}^\times$. Confusing these two ingredients would turn a design requirement into a theorem.

Definition 3.16 (Scaling-split weight and chôral purity). [DESIGN]

Let (C, w, Φ) be a coefficient-linear stable category with a bounded weight structure, an exact endomorphism Φ, and a scaling action $\rho : \mathbf{G}_m \to \mathrm{Aut}(C)$ commuting with Φ. Assume the trace of every object under consideration becomes a finite direct sum after extension to a characteristic-zero coefficient field K.

A mixed object M is **scaling-split** if its trace has a decomposition

$$\tau(M, \Phi) \otimes K \simeq \bigoplus_m T_m(M)$$

such that $a \in \mathbf{G}_m$ acts on $T_m(M)$ by the character $a \mapsto a^m$, and the summand $T_m(M)$ is the trace of the weight-graded layer $\mathrm{gr}^W_m M$. This is the precise sense in which the weight filtration is an eigenvalue decomposition of the trace under scaling.

A dualisable Φ-equivariant object M is **chôrally pure of weight** w if:

1. $M \in C_{w=w}$;
2. its trace is scaling-split and $T_m(M) = 0$ for $m \neq w$;
3. for every finite-field realisation at a closed point x with $q_x = |\kappa(x)|$, every eigenvalue α of geometric Frobenius on $H^i(R_x M)$ is algebraic and satisfies $|\iota(\alpha)| = q_x^{(w+i)/2}$ for every embedding $\iota : \overline{\mathbf{Q}} \hookrightarrow \mathbf{C}$;
4. at an archimedean realisation, the split scaling character on the weight w piece has metric exponent $w/2$, compatibly with Verdier duality sending weight w to weight $-w$ after the prescribed Tate normalisation.

Conditions (3) and (4) are realisation conditions, not consequences of (1).

The definition separates structural purity from metric purity. Condition (1) is Bondarko's categorical notion. Condition (2) says that the Rees action actually diagonalises the trace. Condition (3) is Deligne's absolute-value condition, including the cohomological shift. Condition (4) is the demanded archimedean analogue. Over a finite field all but the archimedean clause are classical. Over $\mathfrak{X}_{\mathbf{Z}}$, the scaling action and its compatibility with all local Frobenii are part of the proposed base and remain unproved.

Semisimplicity is deliberately present in the phrase *scaling-split*. Without it, one obtains generalised eigenspaces and a nilpotent logarithm. The weight filtration can still exist, but the trace is not the direct sum of characters. That nilpotent remainder is the natural home for monodromy and, in the Langlands application, for an Arthur SL_2. Purity is therefore not synonymous with the existence of a filtration; it is the absence of mixed characters and nontrivial monodromy after the stated normalisation.

3.10 Deligne purity as the control theorem

The finite-field case fixes the meaning of every word in the preceding definition. Let X be a scheme of finite type over $\mathbf{F}_q$, let $\ell \neq p$, and let K be a mixed constructible $\overline{\mathbf{Q}}_\ell$-complex. At a closed point x of degree r, an algebraic number α is a Weil number of weight m if every complex embedding satisfies

$$|\iota(\alpha)| = q^{rm/2}.$$

A lisse sheaf is pointwise pure of weight m if the eigenvalues of geometric Frobenius at every closed point are Weil numbers of weight m. A complex K is pure of weight w in Deligne's cohomological convention if $\mathcal{H}^i(K)$ is pointwise pure of weight $w + i$. Mixedness means that each cohomology sheaf admits a finite filtration with pure graded pieces. These are metric conditions on algebraic eigenvalues; they are not consequences of the formal existence of Frobenius.

Theorem 3.17 (Comparison with Deligne purity and the Weil formalism). [THEOREM]
Let $X/\mathbf{F}_q$ be separated of finite type. Suppose a stable category $\mathcal{C}(X)$ has a bounded weight structure, a Frobenius endomorphism, an additive localising trace, and an ℓ-adic realisation

$$R_\ell : \mathcal{C}(X) \longrightarrow D^b_m(X_{\overline{\mathbf{F}}_q}, \overline{\mathbf{Q}}_\ell)$$

which is weight-exact, commutes with Frobenius and the six operations, and identifies scaling character m with Deligne weight m. Assume moreover that R_ℓ detects the two weight bounds on the thick subcategory generated by pure objects. Then an object in that subcategory is chôrally pure of weight w if and only if its realisation is pure of weight w in Deligne's sense.

For a smooth proper d-dimensional $X/\mathbf{F}_q$, applied to the constant object of weight zero, the trace formalism gives

$$Z(X,t) = \prod_{i=0}^{2d} \det\left(1 - t\,\mathrm{Frob}_q \mid H^i(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell)\right)^{(-1)^{i+1}}.$$

Finite-dimensionality implies rationality. Poincaré duality implies the functional equation up to the standard monomial and sign. Deligne purity, namely $|\iota(\alpha)| = q^{i/2}$ for every eigenvalue on H^i, implies the Riemann-hypothesis part of the Weil conjectures. The first two conclusions are formal consequences of trace and duality; the last is the non-formal purity input.

Proof of Theorem 3.17.

By weight-exactness, an object in $\mathcal{C}(X)_{w=w}$ is carried to a complex whose ith cohomology sheaf has weight $w + i$. Compatibility of the scaling character with Deligne weight turns condition (2) of Definition 3.16 into the statement that no other weight occurs. Compatibility with Frobenius turns condition (3) into Deligne's closed-point absolute-value condition. Thus chôral purity implies Deligne purity. Conversely, if $R_\ell(M)$ is pure and the realisation detects the weight bounds on the subcategory generated by pure objects, then $M \in \mathcal{C}_{w=w}$; the scaling compatibility shows that only character w occurs. This proves the comparison. The detection hypothesis is essential: a non-conservative realisation cannot rule out an invisible mixed summand. Bondarko's construction supplies the motivic weight structure in its established domain (Bondarko 2010), while Deligne supplies the absolute-value theorem (Deligne 1980).

For the zeta formula, the Grothendieck–Lefschetz theorem gives, for every $n \geq 1$,

$$|X(\mathbf{F}_{q^n})| = \sum_i (-1)^i \,\mathrm{Tr}\left(\mathrm{Frob}_q^n \mid H^i(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell)\right).$$

Taking the exponential generating series and using

$$\exp\left(\sum_{n\geq 1}\operatorname{Tr}(F^n)\frac{t^n}{n}\right)=\det(1-tF)^{-1}$$

for a finite-dimensional endomorphism F yields the displayed product. Rationality follows because it is a finite product of polynomials and their inverses.

For smooth proper X, Poincaré duality is a perfect Frobenius-equivariant pairing

$$H^i(X_{\overline{\mathbf{F}}_q},\mathbf{Q}_\ell)\otimes H^{2d-i}(X_{\overline{\mathbf{F}}_q},\mathbf{Q}_\ell)\longrightarrow\mathbf{Q}_\ell(-d).$$

If α is an eigenvalue in degree i, the dual eigenvalue in degree $2d-i$ is q^d/α. Pairing the determinant factors in complementary degrees transforms t into $q^{-d}t^{-1}$ and produces only a monomial in t, a power of q, and the determinant sign. This is the functional equation. Finally, Deligne's theorem says that every α in degree i has all complex absolute values $q^{i/2}$ (Deligne 1974a, 1980). Hence every reciprocal zero contributed by degree i has absolute value $q^{-i/2}$. That is exactly the Riemann-hypothesis assertion for the zeta function.

□

The proof gives an exact accounting. The determinant identity is linear algebra. The trace formula is Grothendieck's theorem. Rationality follows from finite-dimensionality. The functional equation follows from duality. The location of the zeros follows from Deligne's purity theorem and from nothing formal in the trace axioms. A framework which claims to prove purity merely by constructing a trace has confused the easy three quarters of the Weil conjectures with the hard quarter.

3.10.1 Worked computation: the projective line

For $X=\mathbf{P}^1_{\mathbf{F}_q}$, the only nonzero cohomology groups are

$$H^0(X_{\overline{\mathbf{F}}_q},\mathbf{Q}_\ell)=\mathbf{Q}_\ell,\qquad H^2(X_{\overline{\mathbf{F}}_q},\mathbf{Q}_\ell)=\mathbf{Q}_\ell(-1).$$

Geometric Frobenius acts by 1 in degree zero and by q in degree two. The trace formula therefore gives

$$|\mathbf{P}^1(\mathbf{F}_{q^n})|=1+q^n,$$

and

$$Z(\mathbf{P}^1,t)=\exp\left(\sum_{n\geq 1}(1+q^n)\frac{t^n}{n}\right)=\frac{1}{(1-t)(1-qt)}.$$

The two factors have weights zero and two. Under a split scaling action the trace has two characters, not one: the cohomological trace of a pure weight-zero object is mixed because cohomological degree shifts the weights. This is why Definition 3.16 places H^i in weight $w+i$ rather than demanding that all cohomology carry the same character.

The weight tower has two pure layers. Trace additivity reads

$$\operatorname{Tr}(\mathrm{Frob}_q\mid R\Gamma(X))=1+q,$$

with no degree-one term. The local side consists of one contribution for each of the $q+1$ rational points. This is the simplest complete instance of the chôral programme: global trace, local fixed points, weight decomposition and purity all agree, and every assertion is a theorem.

3.10.2 Worked computation: an elliptic curve over $\mathbf{F}_5$

Consider

$$E : y^2 = x^3 + x + 1$$

over $\mathbf{F}_5$. The quadratic residues are $0, 1, 4$. For $x = 0, 1, 2, 3, 4$, the right-hand side is respectively $1, 3, 1, 1, 4$. These values contribute respectively $2, 0, 2, 2, 2$ affine points. Adding the point at infinity gives

$$|E(\mathbf{F}_5)| = 9.$$

Thus

$$a_5 = 5 + 1 - |E(\mathbf{F}_5)| = -3.$$

The Frobenius polynomial on H^1 is

$$P_1(T) = T^2 - a_5 T + 5 = T^2 + 3T + 5,$$

whose roots are

$$\alpha = \frac{-3 + i\sqrt{11}}{2}, \qquad \beta = \frac{-3 - i\sqrt{11}}{2}.$$

They satisfy $\alpha\beta = 5$ and $|\alpha| = |\beta| = \sqrt{5}$. The zeta function is therefore

$$Z(E, t) = \frac{1 + 3t + 5t^2}{(1 - t)(1 - 5t)}.$$

The point counts over extensions are

$$|E(\mathbf{F}_{5^n})| = 5^n + 1 - \alpha^n - \beta^n.$$

For example, $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 10 = -1$, so $|E(\mathbf{F}_{25})| = 25 + 1 - (-1) = 27$.

Here H^0, H^1, and H^2 have weights zero, one, and two. Duality pairs 1 with 5 and pairs α with $5/\alpha = \beta$. Rationality and the functional equation follow before the absolute values are known. The equality $|\alpha| = \sqrt{5}$ is purity. In genus one it also follows from Hasse's bound, but Deligne's theorem supplies the corresponding assertion in all dimensions.

The example also shows what an arithmetic scaling action would have to do. It would place the trace contribution $\alpha + \beta = -3$ in scaling weight one, while the degree-zero and degree-two terms occupy weights zero and two. Merely knowing the integer point count 9 cannot recover this decomposition; the categorical trace must retain the graded action before decategorification.

3.10.3 The same elliptic curve at three primes

The curve $E : y^2 = x^3 + x + 1$ has discriminant $-16 \cdot 31$, so it has good reduction at 5, 7, and 11. Direct enumeration gives the following table.

p	$\lvert E(\mathbf{F}_p)\rvert$	$a_p = p + 1 - \lvert E(\mathbf{F}_p)\rvert$	$2\sqrt{p}$
5	9	-3	4.472 …
7	5	3	5.291 …
11	14	-2	6.633 …

For $p = 7$, the values of $x^3 + x + 1$ for $x = 0, 1, \dots, 6$ are

$$1, 3, 4, 3, 6, 5, 6.$$

Only 1 and 4 are nonzero quadratic residues among these values. They give four affine points, and the point at infinity gives $|E(\mathbf{F}_7)| = 5$. Hence $a_7 = 3$ and

$$P_{1,7}(T) = T^2 - 3T + 7, \qquad \alpha_7, \beta_7 = \frac{3 \pm i\sqrt{19}}{2}.$$

Their product is 7 and both have absolute value $\sqrt{7}$.

For $p = 11$, the values at $x = 0, \dots, 10$ are

$$1, 3, 0, 9, 3, 10, 3, 10, 4, 2, 10.$$

The nonzero quadratic residues are $1, 3, 4, 5, 9$. Counting two square roots for each nonzero residue, one root for zero, and the point at infinity gives 14 points. Thus $a_{11} = -2$ and

$$P_{1,11}(T) = T^2 + 2T + 11, \qquad \alpha_{11}, \beta_{11} = -1 \pm i\sqrt{10},$$

again with absolute value $\sqrt{11}$.

The Hasse–Weil inequality is numerically visible:

$$|a_5| = 3 < 2\sqrt{5}, \qquad |a_7| = 3 < 2\sqrt{7}, \qquad |a_{11}| = 2 < 2\sqrt{11}.$$

At each prime the trace of geometric Frobenius on H^1 is a_p, while the point count is

$$1 - a_p + p.$$

The three computations therefore verify the Lefschetz trace formula and the weight-one bound in actual fibres. What varies with p is the Frobenius conjugacy class; what remains fixed is the purity exponent one. This is the pattern the arithmetic scaling action is required to express globally.

3.11 Purity and Ramanujan bounds

The passage from Weil weights to Ramanujan bounds is a normalisation statement. At an unramified place v, an automorphic representation has a semisimple Satake parameter s_v in the dual group. Given a finite-dimensional representation r of the dual group, the local Euler factor is

$$L_v(s, \pi, r) = \det\left(1 - r(s_v) q_v^{-s}\right)^{-1}.$$

Temperedness says, after the unitary normalisation, that the eigenvalues of $r(s_v)$ have complex absolute value one. Geometric purity says that before normalisation they have a prescribed power of q_v. The two statements are the same after dividing by that power.

Criterion 3.18 (Purity criterion for Ramanujan-type bounds). [THEOREM]

Assume an unramified Langlands parameter is realised by a scaling-split chôral object M which is pure of weight w, and assume the chosen representation r identifies the local Frobenius eigenvalues with the Satake eigenvalues. Then every eigenvalue α of $r(s_v)$ satisfies

$$|\iota(\alpha)| = q_v^{w/2}.$$

After the normalisation $r(s_v) \mapsto q_v^{-w/2} r(s_v)$, all eigenvalues have absolute value one, so the parameter is tempered whenever the standard semisimple temperedness criterion applies. Conversely, temperedness together with algebraicity, semisimplicity, and the stated compatibility of scaling weights implies the absolute-value clause of purity, but not by itself the existence of a pure object or a weight structure.

Proof of Criterion 3.18.

The first assertion is condition (3) of Definition 3.16 in degree zero. Dividing each eigenvalue by $q_v^{w/2}$ gives absolute value one under every complex embedding. For an unramified semisimple parameter, the image is bounded in the complex dual group exactly when all characters occurring in finite -dimensional representations have unit absolute value; this is the usual Satake criterion for temperedness. The converse recovers the numerical absolute-value statement after restoring the factor $q_v^{w/2}$. It does not construct a Bondarko decomposition, prove algebraicity, or eliminate nilpotent monodromy, so those hypotheses must remain explicit.

□

For GL_n over a function field, Laurent Lafforgue's global correspondence and Deligne's purity technology give the required bounds for cuspidal representations with finite-order central character (Lafforgue 2002; Deligne 1980). In that control case the criterion is realised by the cohomology of shtuka stacks. For general reductive groups, Vincent Lafforgue constructs semisimple parameters by excursion operators (Lafforgue 2018), but one must distinguish that parametrisation from a blanket theorem of temperedness in all cases.

Over number fields the criterion is conditional twice over. The arithmetic chôra and its scaling action have not been constructed, and general Ramanujan is not known. The criterion does not prove the conjecture; it identifies the specific geometric statement which would imply it. Failure of purity can occur through a mixture of weights, through non-semisimple scaling, or through a nonzero monodromy operator. In the proposed spectral picture these failures should be supported on singular loci and should encode the Arthur SL_2, but that identification is conjectural.

3.12 The integral trace as a running diagnostic

The two Bökstedt calculations can now be passed through every structure introduced above. Doing so prevents several plausible but false identifications. The first is the diagonal error already corrected. The second is the belief that a circle action can be read from graded homotopy groups alone. The third is that a weight decomposition is the same thing as a decomposition by topological degree. None survives the comparison between $\mathbf{Z}$ and $\mathbf{F}_p$.

3.12.1 What the circle action adds

The graded group $\pi_*\,\mathrm{THH}(A)$ forgets the S^1-action. Even when two S^1-spectra have isomorphic homotopy groups, their homotopy fixed points and Tate constructions may differ. The homotopy-fixed-point spectral sequence has the form

$$E_2^{s,t} = H^s(BS^1; \pi_t X) \Longrightarrow \pi_{t-s}(X^{hS^1}),$$

while the Tate spectral sequence extends the cohomological index in both directions. Differentials and multiplicative extensions depend on the action, not merely on $\pi_* X$. Thus Bökstedt's formulas are the input to a cyclotomic computation, not the computation of TC itself.

For $A = \mathbf{F}_p$, the class σ in degree two generates the entire nonequivariant homotopy ring. The Tate spectral sequence introduces the periodic class coming from the cohomology of the circle, and the cyclotomic Frobenius relates powers of σ to this periodic direction. This is why Witt vectors arise from a polynomial algebra which, before equivariance, looks almost too simple. The arithmetic information lies in the comparison of the canonical and Frobenius maps, not in the existence of the polynomial generator alone.

For $A = \mathbf{Z}$, the class in degree $2i-1$ has order i. At a fixed prime p, its p-primary component has order $p^{v_p(i)}$. Consequently the p-completed theory sees infinitely many integral degrees: precisely those i divisible by p, with increasing exponents when i is divisible by high powers of p. This is a concrete reason for retaining the full formula in Theorem 3.13. Passing directly from integral THH to a single residue field loses the intersections among different prime supports; passing directly to rational coefficients kills every positive-degree group. The cyclotomic object sits between these destructive simplifications.

The local-to-global pattern is already visible in elementary arithmetic. The finite group $\mathbf{Z}/i$ decomposes as the direct sum of its p-primary parts,

$$\mathbf{Z}/i \simeq \bigoplus_{p \mid i} \mathbf{Z}/p^{v_p(i)},$$

but this decomposition depends on the integral group from which the local parts arose. A product of independently specified p-complete spectra need not carry an integral lattice. Likewise, the proposed arithmetic chôra cannot be defined as an unstructured product of its finite-place traces. The halo and the rational diagonal are supposed to supply the integral gluing which the Bökstedt groups exhibit in their simplest form.

3.12.2 Reduction modulo p is not the comparison

There is a natural map

$$\mathrm{THH}(\mathbf{Z}) \otimes^{\mathbf{L}}_{\mathbf{Z}} \mathbf{F}_p \longrightarrow \mathrm{THH}(\mathbf{F}_p),$$

but it should not be mistaken for an isomorphism obtained by reducing the listed homotopy groups termwise. Derived base change contributes Tor terms, and topological Hochschild homology itself participates in a base-change spectral sequence. On the target, one polynomial generator creates a copy of $\mathbf{F}_p$ in every nonnegative even degree. On the source, the odd torsion classes interact with the derived tensor product and with extensions. The contrast is therefore not a contradiction; it records the fact that the arithmetic diagonal changes under specialisation.

Geometrically, the diagonal of $\operatorname{Spec} \mathbf{F}_p$ over the sphere is more singular, in a periodic way, than the integral diagonal. The class σ is a stable-homotopical direction invisible to ordinary Kähler differentials. This is exactly the kind of direction sought by the arithmetic Ran proposal. Yet the existence of this direction at each residue field does not construct a global Ran stack, any more than knowing every fibre of a family constructs its descent datum.

3.12.3 Trace additivity on the Bökstedt Postnikov tower

A bounded weight tower was used in Theorem 3.15. The Postnikov tower of $\mathrm{THH}(\mathbf{Z})$ is not bounded above, so the theorem cannot simply be applied to the whole spectrum. Finite truncations do, however, provide a controlled approximation. Let $\tau_{\leq 2N-1}\,\mathrm{THH}(\mathbf{Z})$ be the connective truncation. Its positive homotopy groups are

$$\mathbf{Z}/2, \mathbf{Z}/3, \dots, \mathbf{Z}/N$$

in degrees $3, 5, \dots, 2N-1$. If a compatible finite filtration by perfect layers is available after a chosen coefficient or completion, additivity expresses the trace of an endomorphism of the truncation as the alternating sum of the traces on those layers. Passing to the limit requires continuity of the trace. This is automatic neither for arbitrary inverse limits nor for the large automorphic categories of the main conjecture.

This elementary observation is the prototype of the nuclearity problem. A finite weight tower and a bounded-below cyclotomic spectrum are manageable. An analytic category with continuous spectrum may involve completed products, unbounded filtrations and noncompact operators. The formula $\mathrm{TC} = \mathrm{Eq}(\mathrm{can}, \varphi)$ remains a formal expression, but its convergence and dualisability can fail. The standing framework therefore restricts established trace statements to the compactly generated or otherwise continuously dualisable range.

3.12.4 Weight is extra structure

Suppose one assigns Rees character i to σ^i in $\mathrm{THH}(\mathbf{F}_p)$. Then the polynomial multiplication is compatible with addition of weights, and the trace decomposes into one-dimensional scaling eigenspaces. This is a useful model of a scaling-split object. It is still a choice of normalisation: topological degree $|\sigma^i| = 2i$ and Rees weight i differ by a factor of two, reflecting the convention that a Weil number of weight m has absolute value $q^{m/2}$.

No analogous polynomial basis exists integrally. The group $\mathbf{Z}/i$ in degree $2i-1$ cannot be made into a one-dimensional character space over a characteristic-zero field because it vanishes after such extension. Integral weights must therefore be formulated through filtered or graded objects before rationalisation, not recovered from eigenvalues after all torsion is killed. This is one reason the chôral definition contains both a Bondarko weight structure and a scaling realisation. The former can retain integral extension data; the latter supplies eigenvalue magnitudes where a field-valued realisation exists.

The running example thus enforces four separations. The tensor square is not the self-intersection. The trace spectrum is not its graded homotopy group. The circle action is not the scaling action. The weight structure is not the Postnikov t-structure. The proposed arithmetic geometry needs all four pairs, and any construction which identifies a pair prematurely will fail either the integral calculation or the finite-field control.

3.13 The function-field control case in full

The control case should be stated at the same categorical level as the arithmetic proposal. Let $X/\mathbf{F}_q$ be a smooth projective geometrically connected curve, and let F denote geometric Frobenius on $X_{\overline{\mathbf{F}}_q}$. The category $D_c^b(X_{\overline{\mathbf{F}}_q}, \mathbf{Q}_\ell)$ has Verdier duality, the six operations and a Frobenius pullback. On the constructible subcategory with the usual finiteness conditions, the cohomological trace of Frobenius is defined and the Lefschetz–Verdier theorem localises it on the fixed-point scheme. Deligne's theory supplies weights. Every ingredient requested of the arithmetic chôra therefore has a theorem-level model here, except that the language of a single primitive THH-valued trace is a modern repackaging rather than the historical route to the results.

3.13.1 From rational points to closed points

For the constant sheaf, the trace formula counts rational points over every finite extension. Reorganising those counts by closed points gives the Euler product. If $|x| = q^{\deg x}$, then a closed point of degree r contributes to $X(\mathbf{F}_{q^n})$ exactly when r divides n, in which case it contributes r geometric points. Therefore

$$|X(\mathbf{F}_{q^n})| = \sum_{r \mid n} r N_r,$$

where N_r is the number of closed points of degree r. It follows that

$$\begin{aligned} \log Z(X,t) &= \sum_{n \geq 1} |X(\mathbf{F}_{q^n})| \frac{t^n}{n} \\ &= \sum_{r \geq 1} N_r \sum_{m \geq 1} \frac{t^{rm}}{m} \\ &= -\sum_{r \geq 1} N_r \log(1 - t^r), \end{aligned}$$

whence

$$Z(X,t) = \prod_{x \in |X|} (1 - t^{\deg x})^{-1}.$$

The equality between this local Euler product and the cohomological determinant is already a global trace formula. The local side is indexed by fixed closed orbits of Frobenius; the spectral side is indexed by eigenvalues on cohomology.

For a lisse sheaf $\mathcal{F}$, the same reorganisation gives

$$L(X, \mathcal{F}, t) = \prod_{x \in |X|} \det\left(1 - t^{\deg x} \operatorname{Frob}_x \mid \mathcal{F}_{\bar{x}}\right)^{-1}$$

and

$$L(X, \mathcal{F}, t) = \prod_i \det\left(1 - t \operatorname{Frob}_q \mid H_c^i(X_{\overline{\mathbf{F}}_q}, \mathcal{F})\right)^{(-1)^{i+1}}.$$

This is precisely the passage from local terms to a global trace required by Axiom 3.4. In the arithmetic setting, Euler products are known analytically in many cases, but there is no constructed global fixed-point figure whose categorical trace theorem yields them all at once.

3.13.2 Proper and nonproper weights

Purity behaves best under proper smooth pushforward. For nonproper varieties, Deligne's theorem gives inequalities rather than a single weight. If $\mathcal{F}$ is pure of weight w, the eigenvalues on $H^i_c(X, \mathcal{F})$ have weights at most $w + i$ under the standard hypotheses, while ordinary cohomology has the complementary lower bounds. The boundary contributes mixed weights. The weight filtration records these contributions, and trace additivity separates them.

A simple example is $\mathbf{G}_m$. Its point count is $|\mathbf{G}_m(\mathbf{F}_{q^n})| = q^n - 1$, so

$$Z(\mathbf{G}_m, t) = \frac{1-t}{1-qt}.$$

Compactly supported cohomology has a degree-one contribution producing the factor $1 - t$ and a degree-two contribution producing $(1 - qt)^{-1}$. The negative sign is a shift sign in the categorical trace; it is not a negative multiplicity of points. Compactification $\mathbf{G}_m \subset \mathbf{P}^1$ and the two-point boundary give an exact triangle. Additivity of trace turns $Z(\mathbf{P}^1, t)$ into the quotient by the two boundary factors, recovering the displayed formula. This is localisation along a geometric filtration and is formally identical to localisation along a weight tower.

The example matters for automorphic geometry because Bun_G is nonproper and stratified by Harder–Narasimhan type. Boundary and Eisenstein contributions should not be discarded. They create mixed weights exactly as the missing points 0 and ∞ alter the trace of $\mathbf{G}_m$. Over a function field, compactifications and the theory of weights control these terms. Over the global arithmetic base, the compatible compactification and weight bounds are not known.

3.13.3 Frobenius trace and categorical Langlands

The function-field Langlands mechanism adds categories of sheaves on Bun_G. The categorical trace of Frobenius on the automorphic category is identified with automorphic functions in the programme on automorphic functions as the trace of Frobenius. Shtuka cohomology and excursion operators then act on this trace, and Vincent Lafforgue extracts semisimple global parameters (Lafforgue 2018). For GL_n, Laurent Lafforgue's theorem gives the full cuspidal correspondence and purity gives Ramanujan (Lafforgue 2002).

The role of the circle is transparent in this setting. A categorical trace of an endofunctor is the value on its mapping torus. Rotation of the mapping torus provides cyclic symmetry, while Frobenius labels the longitudinal direction. There is one distinguished Frobenius, so ordinary twisted Hochschild homology is sufficient for the basic trace. Topological cyclic homology becomes necessary in the number-field proposal because there is no single global Frobenius. The family of cyclotomic maps is intended to replace it.

This marks the exact departure. In the function-field case, the base curve, its diagonal, its Ran space, its Frobenius, its sheaf theory, its weight theory and its fixed-point formula exist independently and are linked by theorems of Grothendieck, Deligne, Drinfeld and the Lafforgues. In the arithmetic case:

1. the global base is specified but not constructed;
2. the diagonal calculation exists spectrally, but its geometric gluing does not;
3. the local cyclotomic Frobenii exist on THH, but their identification with placewise localisation terms is conjectural;
4. the scaling action at infinity is a design requirement, not a proved part of a six-functor formalism;
5. a bounded weight structure on arithmetic gauges has not been constructed;
6. no arithmetic analogue of Deligne's purity theorem is known.

These six gaps are not variants of one missing estimate. They are distinct construction and comparison problems. The chapter's formal theorems show what would follow if they were solved; they do not solve them.

3.13.4 Control ledger for the principal assertions

The comparison can be summarised without weakening the distinctions among the results.

Assertion	Over $X/\mathbf{F}_q$	Over $\mathfrak{X}_{\mathbf{Z}}$
trace of an endomorphism	formal for dualisable categories; developed through higher trace theory	formal only after dualisability of the arithmetic category is proved
cyclicity and the circle action	formal; the trace is a labelled mapping circle	formal on any trace object which exists
localisation on fixed points	Grothendieck–Lefschetz–Verdier, with categorical refinements	Axiom Axiom 3.4; not constructed
weights and purity	Deligne's Weil I and Weil II	Definition Definition 3.16; conjectural
weight additivity	May additivity and Bondarko towers	Theorem Theorem 3.15, conditional on the axioms
shtuka legs and partial Frobenii	Drinfeld and the Lafforgues, using X^I and the Ran space	dependent on the absolute product and diagonal correction
automorphic parameters from traces	Laurent and Vincent Lafforgue, with excursion operators	part of arithmetic trace reciprocity; conjectural

The first two rows are categorical and survive change of base. The next three require geometry, finiteness and weights on the chosen base. The last two use the actual curve structure: independent moving legs, partial Frobenii and factorisation. This is why the function-field theorem is more than an analogy. It is a complete model which separates formal trace identities from geometric theorems.

There is also a precise reason that ordinary Frobenius trace suffices over $\mathbf{F}_q$ while cyclotomic structure is proposed over the integers. A curve over $\mathbf{F}_q$ has one global Frobenius whose powers account for all extensions $\mathbf{F}_{q^n}$. The prime-indexed maps φ_p in a cyclotomic spectrum do not replace different powers of that one endomorphism; they replace the absence of a single endomorphism common to all residue characteristics. At infinity, the split scaling action supplies metric weight data which finite-field Frobenius already carries through its absolute values. The arithmetic construction must therefore glue more kinds of structure than the function-field construction, not merely repeat it at infinitely many primes.

3.14 Zeta objects and the localisation of determinants

A trace is linear; an Euler product is multiplicative. The bridge is the exponential identity used in Theorem 3.17. It is useful to state the construction at the level of a general coefficient field. Suppose that a realisation of $\tau(M,f^n)$ is a scalar for every $n \geq 1$. Define the formal trace zeta series

$$Z(M,f;t) = \exp\left(\sum_{n\geq 1} \tau(M,f^n)\frac{t^n}{n}\right).$$

If M is represented by a finite-dimensional graded vector space and f by a graded endomorphism, then

$$Z(M,f;t) = \prod_i \det(1 - tf \mid H^i(M))^{(-1)^{i+1}}.$$

This formula is not a new axiom. It follows by applying $-\log\det(1-tf) = \sum_{n\geq 1} \mathrm{Tr}(f^n)t^n/n$ in every degree.

Now let the hypotheses of Theorem 3.15 hold for every power f^n. Additivity gives

$$\tau(M,f^n) = \sum_j \tau(G_j,f_j^n).$$

Substitution in the exponential shows

$$Z(M,f;t) = \prod_j Z(G_j,f_j;t).$$

If $G_j \simeq P_j[-j]$, this becomes

$$Z(M,f;t) = \prod_j Z(P_j,\tilde{f_j};t)^{(-1)^j}.$$

Thus localisation of traces along a weight tower becomes factorisation of zeta objects. The extension classes in the tower do not alter this product, although they can alter the object M and every non-additive invariant of it.

This observation clarifies the slogan that rationality is formal. If there are finitely many weight pieces and each pure piece has a finite-dimensional realisation, every factor on the right is a rational function. The product is therefore rational. Nothing in this argument bounds the absolute values of its zeros and poles. Those are determined by purity. Nor does the argument prove that a geometric Euler product converges in an analytic half-plane. It proves an identity of formal power series, after which analytic continuation is an additional theorem.

3.14.1 Duality and reciprocal weights

Assume Verdier duality carries a pure object of weight w to a pure object of weight $-w$ before a Tate twist, and let an oriented proper figure of dimension d provide the dualising twist $(-d)[-2d]$. A weight-$w+i$ eigenvalue α in degree i then pairs with

$$\frac{q^d}{\alpha}$$

in degree $2d-i$. The product of their absolute values is q^d. This is the weight-theoretic origin of the line of symmetry in a functional equation. Again the decomposition of labour is exact: duality supplies the reciprocal pairing; purity tells us that both members lie on their prescribed circles.

In a scaling-split chôra, the same statement should be visible without choosing a finite-field realisation. The dual of the character a^m is a^{-m}, while the orientation contributes the character a^{2d} in the convention where cohomological degree two has weight two. Hence the trace characters occur in pairs m and $2d-m$. A regularised determinant of the scaling generator should then satisfy a reflection formula. At the archimedean place this is the desired formal shape of the functional equation for gamma factors. The construction of the determinant and the proof that it equals the classical gamma factor are not consequences of weight theory alone.

The example $\mathrm{THH}(\mathbf{Z})$ makes the limitation visible. Integral torsion has no nonzero characteristic-zero eigenvalue, so a determinant formed only after tensoring with $\mathbf{Q}$ sees none of the positive homotopy. An integral functional equation would need a determinant or Euler characteristic defined in a category which retains finite groups. This is one reason topological cyclic homology, rather than periodic cyclic homology over $\mathbf{Q}$, is the natural arithmetic receptacle.

3.14.2 Local terms and weight factors

Suppose the fixed-point figure X^f is a finite disjoint union of components Z_c, and localisation gives

$$\tau(M,f) = \sum_c \mathrm{loc}_c(M,f).$$

If the weight tower restricts to every component, then

$$\mathrm{loc}_c(M,f) = \sum_j \mathrm{loc}_c(G_j,f_j).$$

There are therefore two finite summations, over fixed components and over weights, and functoriality permits them to be interchanged. In a finite-field trace formula this interchange corresponds to sorting the local Frobenius contributions by weight before summing over points. In an automorphic trace formula it would sort orbital terms by the weight of their categorical source.

This interchange is formal only for finite decompositions or convergent sums. The Arthur–Selberg trace formula involves continuous spectrum, truncation and regularisation. A chôral localisation theorem for it must explain why the two orders of summation agree or identify the defect when they do not. Weight additivity does not solve convergence; it supplies the algebraic ledger in which a convergence theorem would be stated.

The distinction also appears in topological cyclic homology. The product over primes in Theorem 3.13 is a local decomposition of the cyclotomic condition, while the Bökstedt grading gives a homotopical filtration. Interchanging prime completion with an unbounded homotopy limit requires hypotheses. For bounded below cyclotomic spectra Nikolaus and Scholze provide them. For the proposed large automorphic categories, the corresponding continuity theorem is among the unresolved analytic foundations.

3.15 Computing a weight complex

A weight complex is not computed by taking cohomology. It is computed by repeatedly solving weight-decomposition problems. In geometric categories the solver may be a Chow–Künneth decomposition, a stratification argument, a purity theorem, or an explicit complex of projectives. The algorithm below separates the formal recursion from that geometric oracle.

Finite weight-tower extraction (formalisation target). Input. An object M in a stable category with a bounded weight structure; integers $a \le b$ certifying $M \in \mathcal{C}_{w\ge a} \cap \mathcal{C}_{w\le b}$; and an oracle which, for $N \in \mathcal{C}_{w\le n}$, returns a weight decomposition

$$N' \longrightarrow N \longrightarrow P_n[-n]$$

with $N' \in \mathcal{C}_{w\le n-1}$ and $P_n \in \mathcal{H}w$.

Output. A finite tower with layers $P_n[-n]$ and a weight complex $t_w(M) = (P^\bullet, d)$, together with certificates that every P_n lies in the weight heart and every consecutive composite of differentials is null.

```
N := M
for n from b down to a:
    (Nprime, P[n], triangle[n]) := WeightDecompose(N, n)
    store P[n] and triangle[n]
    N := Nprime
assert N is zero
```

```
for n from a to b - 1:
    d[n] := ConnectingMap(triangle[n], triangle[n + 1])
    certify d[n + 1] * d[n] is null
return (P[a..b], d[a..b-1], triangle[a..b])
```

If an endomorphism f preserves the returned tower, apply it to every P_n and compute its trace by the alternating formula of Theorem 3.15.

The oracle is the mathematical content. In $K^b(\operatorname{Proj} R)$ it truncates a complex and is completely explicit. For geometric motives it uses the existence of the Chow weight structure. For mixed sheaves it uses weight truncation results derived from Deligne's theory. For arithmetic gauges no such oracle has been constructed. Marking the algorithm as a formalisation target records that, once the decompositions and certificates are supplied, the remaining recursion is finite and machine-checkable.

The corresponding algorithmic and verification flow is summarised in Plate 3.3.

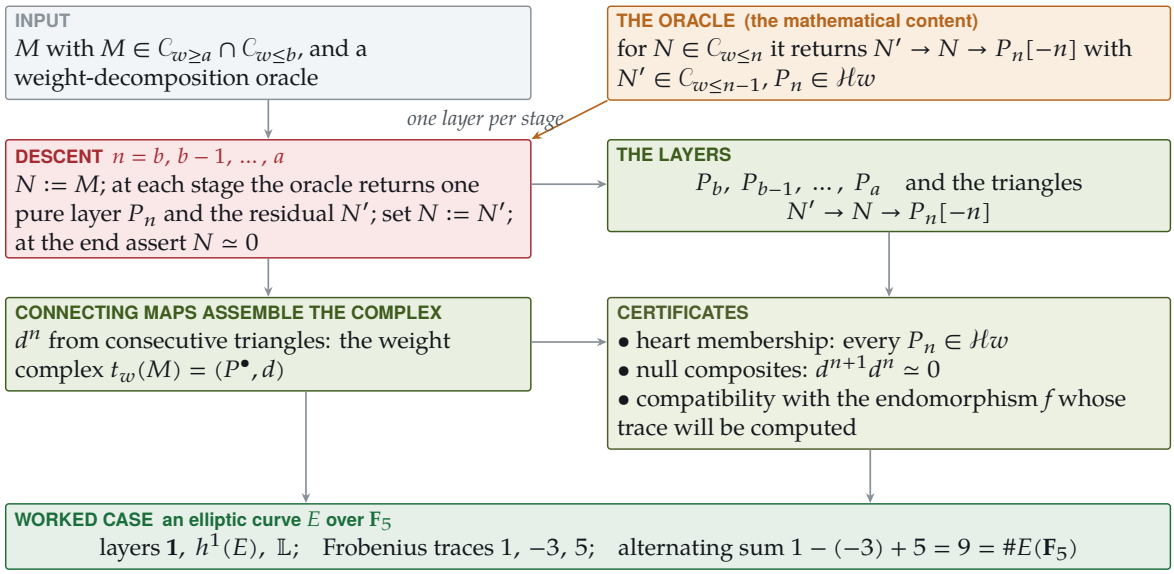


Plate 3.3. The algorithm descends from the highest permitted weight to the lowest. At each stage an oracle returns one pure layer and a residual object; connecting maps assemble the layers into a complex, while certificates record heart membership, null composites and compatibility with the endomorphism whose trace will be computed.

What it does not show: an oracle for arithmetic gauges: the weight decompositions are supplied by the geometry (Chow–Künneth, purity, explicit complexes), and for arithmetic gauges no such oracle has been constructed.

For the complex $[\mathbf{Z} \xrightarrow{2} \mathbf{Z}]$, the oracle returns the second copy of $\mathbf{Z}$ as the weight-one layer and the first copy as the weight-zero layer. The connecting map is multiplication by two. For $\mathbf{P}^1$, a motivic oracle returns the unit motive and the Lefschetz motive. For the elliptic curve over $\mathbf{F}_5$, it returns the unit, the weight-one motive $h^1(E)$, and the Lefschetz motive; Frobenius has traces $1, -3, 5$ on these pieces, and their alternating cohomological placement gives the point count 9.

The last example also shows that the weight complex need not split the motive as a direct sum inside every integral category. The trace calculation only requires the tower and equivariance. A decomposition after semisimplification is useful for eigenvalues but is stronger than additivity.

3.16 What is lost without a t-structure

Status and Caveat. Replacing a motivic t-structure by a weight structure preserves exactly what an additive trace needs, but it does not preserve everything arithmetic geometry normally asks of a category of mixed motives. The heart $\mathcal{H}w$ is additive and idempotent-complete, not abelian. There are no intrinsic kernels, cokernels, images or Jordan–Hölder filtrations in it. A weight decomposition is generally noncanonical, so a morphism need not admit a strictly functorial filtration. The weight complex records pure layers and connecting maps, but it need not classify the original object or its extension classes.

This loss has concrete consequences. One cannot define motivic Galois groups by applying neutral Tannakian duality to the weight heart. One cannot formulate subobjects of a mixed motive as subobjects in an abelian category. Perverse truncations, intermediate extension and nearby-cycle exactness belong naturally to t-structured categories and must be imported through realisations rather than recovered from weights alone. Questions about integral lattices and torsion extensions can be invisible after passage to a semisimple pure heart.

The trace formalism survives because it is stable and additive. Reciprocity as a trace identity may therefore require less than a motivic heart. Classification of mixed objects, deformation theory of extensions, and a Tannakian description of the full motivic Galois group require more. The framework claims only the former economy. If later arguments use kernels, exact sequences in a heart, or canonical filtrations without specifying a realisation carrying a t-structure, they exceed Axiom Axiom 3.7.

There is a second loss at the interface with analysis. A weight tower is finite for bounded constructible objects, whereas automorphic categories contain continuous and Eisenstein spectrum. A bounded weight structure on their compact objects would not automatically control completed direct integrals or regularised traces. The absent t-structure is therefore not the sole gap; continuity of weights and traces on large analytic categories is an independent problem.

The caveat does not reverse the methodological choice. No unconditional mixed motivic t-structure is available, whereas Bondarko's weight structures exist in substantial motivic settings. The sound procedure is to use the weaker structure for the conclusions it proves, and to name every place where the stronger structure would add information.

3.17 Arithmetic purity: the unresolved assertion

The chôral definition Definition 3.16 is designed so that a future arithmetic purity theorem can be stated without ambiguity. Such a theorem would require a category of constructible arithmetic gauges on $\mathfrak{X}_{\mathbf{Z}}$, a bounded weight structure, an absolute endo-correspondence Φ, a split scaling action, and compatible realisations at every place. It would assert that objects arising from proper smooth arithmetic figures, after the correct shifts and Tate twists, are pure.

No theorem presently supplies this package. At a finite prime, prismatic F-crystals, syntomic gauges and p-adic Hodge theory provide weight-like filtrations and Frobenius slopes, but slopes are not automatically Deligne weights. At an ℓ-adic place different from p, one has Galois representations and weight-monodromy results in many geometric cases, but not a single global category joining them. At infinity, pure Hodge structures and polarised twistor structures provide the correct metric notion, but the required archimedean six-functor formalism has not been constructed. The product formula relates norms across places, yet it does not manufacture the common object whose eigenvalues they measure.

The arithmetic theorem would also need an integral statement strong enough to retain the torsion in $\mathrm{THH}(\mathbf{Z})$. Deligne purity is formulated over characteristic-zero coefficient fields and controls absolute values of algebraic eigenvalues. The Bökstedt groups are finite and disappear rationally. A complete arithmetic weight theory may therefore have two layers: an integral weight filtration controlling torsion and a field-valued metric realisation controlling eigenvalue sizes. Definition Definition 3.16 anticipates this separation but does not construct it.

Finally, purity must be compatible with localisation. Over a nonproper figure, boundary strata contribute lower or higher weights. In the automorphic setting Eisenstein series and residual spectrum are precisely such boundary contributions. A useful arithmetic purity theorem should not declare the whole automorphic category pure. It should identify a pure cuspidal or proper part, measure the mixed boundary, and make trace additivity recover the full formula. That is a sharper and more plausible goal than a blanket purity assertion.

3.17.1 Slopes are not weights

At a finite prime one must also distinguish Newton slopes from Weil weights. If a crystalline Frobenius eigenvalue has p-adic valuation λ, that valuation measures a Newton slope. Deligne weight measures the complex absolute values of all conjugates. The two are related by global algebraicity and comparison theorems, but neither determines the other locally. An ordinary elliptic curve over $\mathbf{F}_p$ has slopes zero and one on crystalline cohomology while both complex Frobenius eigenvalues on H^1 have Weil weight one. A supersingular elliptic curve has two slopes one half and still has Weil weight one. Thus a proposed arithmetic weight structure cannot be replaced by a slope filtration on each syntomic chart.

The Nygaard filtration is likewise not a Weil-weight filtration. It controls divisibility of Frobenius and mediates among crystalline, de Rham and Hodge–Tate specialisations. Its graded pieces are indispensable local data, but purity requires a further statement relating them to algebraic Frobenius eigenvalues and to the archimedean metric. The global scaling action is meant to encode that further statement. A successful comparison should send a single pure chôral object to several local filtrations: Newton and Nygaard at p, monodromy and inertia away from p, and Hodge or twistor weight at infinity.

This distinction prevents a common shortcut. Agreement of Newton and Hodge polygons is an ordinarity statement, not a proof of Weil purity. Conversely, Weil purity does not force a unique Newton polygon. The chôral trace must retain enough structure for both invariants and

for their comparison. In the function-field case, crystalline–étale comparison and Deligne's theorem give this compatibility for geometric objects. Over the proposed global arithmetic curve it is another missing comparison theorem.

3.18 Twisted traces, centralisers, and secondary circles

The S^1-action must be stated with a qualification. The trace of the identity endomorphism has a canonical circle action because every rotation preserves the labelling of the circle. For a general endomorphism F of a category $\mathcal{C}$, the twisted trace

$$\mathrm{Tr}(F \mid \mathcal{C})$$

is Hochschild homology with coefficients in the bimodule defined by F. Its bar construction is generally paracyclic rather than cyclic: after one full rotation the coefficient has been transported through the twist. A genuine circle action therefore requires either $F = \mathrm{id}$, a chosen trivialisation of the accumulated twist, or an equivariant mapping-torus construction in which the monodromy is part of the geometry.

This distinction is essential for Frobenius. Over a finite field, the trace of Frobenius is a twisted categorical trace, and its numerical fixed-point formula does not arise merely by taking S^1-homotopy fixed points of ordinary THH. The Frobenius labels the mapping torus. Rotation of that mapping torus commutes with the monodromy, and the relevant symmetry is the centraliser of the labelled endomorphism. When Frobenius is promoted to a cyclotomic structure, the prime-power covering maps add comparisons among these twisted circles.

For an automorphism F, the mapping torus may be described as

$$\mathcal{C} \times_F S^1.$$

A point on the base circle chooses a cut and identifies the fibre with $\mathcal{C}$; moving once around applies F. The trace is independent of the cut, but the return map remembers the monodromy. This is the correct geometric picture behind the notation $\mathrm{Tr}(F \mid \mathcal{C})$. It also explains why the finite-field trace is not identical to $\mathrm{THH}(\mathcal{C})$: the latter is the special case with trivial monodromy.

3.18.1 Commuting endomorphisms

Let F and G be commuting endomorphisms of a fully dualisable object in a symmetric monoidal 2-category. The commutation supplies an endomorphism of $\mathrm{Tr}(F)$ induced by G. One may take its trace again. The result is represented by a torus whose two fundamental cycles are labelled by F and G. Exchanging the cycles by an orientation-preserving diffeomorphism of the torus gives the secondary-trace symmetry

$$\mathrm{tr}\,(G \mid \mathrm{Tr}(F)) \simeq \mathrm{tr}\,(F \mid \mathrm{Tr}(G))$$

under the usual dualisability hypotheses. This is the two-dimensional analogue of Theorem 3.2.

The arithmetic relevance is direct. Frobenius and scaling are required to commute. A finite-place Frobenius φ_p supplies one cycle; circle rotation or an additional Hecke endomorphism supplies another. Compatibility of local factors with global scaling should be a secondary-trace statement. The topology predicts the symmetry, but the arithmetic categories must still be shown fully dualisable and the endomorphisms commuting.

The torus also clarifies the difference between cyclicity and commutativity. The equality $\mathrm{tr}(fg) = \mathrm{tr}(gf)$ does not assert $fg = gf$. It closes a word into a circle, after which only

cyclic order remains. A secondary trace does require a specified commutation between two monodromies, because the fundamental group of the torus is abelian. Topology therefore tells us exactly which compatibility is formal and which must be supplied as data.

3.18.2 Multiplicativity

If X, Y are dualisable and f, g are endomorphisms, the tensor product duality gives

$$\operatorname{tr}_{X\otimes Y}(f\otimes g) = \operatorname{tr}_X(f)\operatorname{tr}_Y(g).$$

Graphically, the trace of the tensor product is a disjoint union of two labelled circles, and the symmetric monoidal product sends disjoint union to multiplication. This elementary fact is the categorical origin of the multiplicativity of Euler factors for direct sums and tensor constructions of Galois representations.

For zeta objects, direct sums give products and tensor products give more subtle convolutions of eigenvalues. If f has eigenvalues α_i and g has eigenvalues β_j, then $f\otimes g$ has eigenvalues $\alpha_i\beta_j$. Weights add:

$$|\alpha_i| = q^{a/2}, \quad |\beta_j| = q^{b/2} \quad\Longrightarrow\quad |\alpha_i\beta_j| = q^{(a+b)/2}.$$

Thus compatibility of a weight structure with tensor product is not cosmetic. It is what makes purity stable under the representation-theoretic operations used to define L-functions.

3.19 Weight spectral sequences and invariance of supertrace

Every homological functor applied to a weight tower produces a weight spectral sequence. If H is a homological functor from C to an abelian category and $t_w(M) = P^\bullet$, then the first page is assembled from the pure terms,

$$E_1^{p,q} = H^q(P^{-p}) \Longrightarrow H^{p+q}(M),$$

with indexing adjusted to the convention of Definition 3.14. The spectral sequence is functorial from a sufficiently early page even when the individual weight decompositions are not canonical. It is the main device by which a weight structure exports information to a t-structured realisation.

Suppose an endomorphism f preserves the tower and all terms are finite-dimensional after realisation. On every page define the total supertrace

$$S_r(f) = \sum_{p,q} (-1)^{p+q} \operatorname{Tr}\left(f \mid E_r^{p,q}\right).$$

The passage from E_r to E_{r+1} takes cohomology with respect to a differential commuting with f. For any finite complex with a commuting endomorphism, the supertrace on the complex equals the supertrace on its cohomology: boundaries and their shifted copies cancel. Hence

$$S_r(f) = S_{r+1}(f).$$

At the first page this is the alternating trace on the pure weight terms; at the limiting page it is the alternating trace on the associated graded of the realised cohomology; finite filtrations do not alter trace, so it equals the trace on $H^*(M)$. This is the spectral-sequence form of Theorem 3.15.

The argument explains why a noncanonical weight tower can still yield a canonical trace. Different towers may give different first pages and differentials, but each spectral sequence

converges to the same realised object, and supertrace is invariant from page to page. The trace forgets the choices because every discrepancy occurs in cancelling exact pairs.

Purity can force degeneration. If different columns of E_1 have disjoint Frobenius weights, a differential between them would have to preserve Frobenius eigenvalues while changing weight. No nonzero map can do so after semisimplification. The spectral sequence then degenerates at the first or a controlled later page. This is one of Deligne's most effective uses of weights: weight separation turns a potentially complicated extension problem into a vanishing statement. The same argument would be available in a chôra only if the scaling characters were semisimple and faithfully detected by the chosen realisation.

For $\mathbf{P}^1$, the weight spectral sequence has two nonzero entries and no possible differential. For the open curve $\mathbf{G}_m$, the boundary triangle produces entries of different weights whose signs yield the numerator and denominator of the zeta function. For the perfect complex $[\mathbf{Z} \xrightarrow{2} \mathbf{Z}]$, an integral homological functor sees a torsion differential, so degeneration fails even though the supertrace remains zero. These three examples display, respectively, pure splitting, mixed boundary, and torsion extension.

3.19.1 Compatibility with the six operations

A useful weight formalism must prescribe how the six operations change weight. Over finite fields, Deligne's theory supplies the model. Smooth pullback shifts weights according to the chosen cohomological and Tate normalisation; proper pushforward preserves purity for pure complexes arising from smooth proper maps; arbitrary pushforward and extension by zero satisfy upper or lower weight bounds; Verdier duality reverses inequalities and applies the dualising twist; tensor products add weights; internal Hom subtracts them.

In Bondarko language these become weight-exactness or one-sided weight-exactness statements. A functor F is left weight-exact if it preserves $w \leq 0$, right weight-exact if it preserves $w \geq 0$, and weight-exact if it does both. A proper smooth pushforward in the pure range should carry a heart object, after the appropriate shift, to a heart object. An open pushforward need not: boundary weights appear. The asymmetry between $f_!$ and f_* is therefore measured by weight bounds as well as by support.

For the arithmetic chôra these compatibilities cannot be placed silently in the definition of a weight structure. They are additional axioms or theorems for the six-functor formalism. At finite primes, comparison with étale and prismatic realisations gives candidates. At infinity, compatibility with mixed Hodge or twistor modules is required. Across the halo, the local weight bounds must descend to a global bound. Until this is proved, a global weight tower cannot be inferred from separate local towers.

The relation with ambidexterity is especially delicate. If $f_! \simeq f_*$, then the lower bounds naturally associated with one and the upper bounds associated with the other meet. In favourable proper smooth situations this can force purity. If ambidexterity fails, the gap between the bounds is a range of mixed weights supported at the boundary. This gives a precise weight interpretation of the principle that failure of properness creates Eisenstein and residual terms.

3.20 From THH to de Rham–Witt phenomena

The calculation $\pi_* \mathrm{THH}(\mathbf{F}_p) = \mathbf{F}_p[\sigma]$ is only the first page of a richer arithmetic story. Homotopy fixed points, Tate constructions and cyclotomic Frobenius convert the polynomial class into Witt-vector and de Rham–Witt structures. Hesselholt and Madsen's work identifies degree-zero parts of topological restriction homology with Witt vectors in the appropriate settings (Hesselholt and Madsen 1997). Bhatt, Morrow and Scholze show how Nygaard filtrations and integral p-adic Hodge-theoretic invariants emerge from THH and its cyclotomic relatives (Bhatt et al. 2019).

This provides a finite-place model for the chôral claim that trace and weight interact. The Nygaard filtration is not the same as a Bondarko weight structure, but it is a filtered structure on a cyclotomic invariant whose graded pieces have geometric meaning. Any global arithmetic construction must explain the comparison rather than identify the two by terminology. The Bondarko structure organises motives by purity; the Nygaard filtration organises p-adic periods by divisibility and Frobenius. Their compatibility is a theorem in specific realisations, not a formal property of THH.

For $\mathbf{Z}$ the situation is more global. Each torsion class $\mathbf{Z}/i$ has contributions at every prime dividing i. Its cyclotomic images enter the corresponding p-typical theories. The integral TC equaliser asks for simultaneous agreement of canonical and Frobenius maps. In this sense $\mathrm{THH}(\mathbf{Z})$ is a universal meeting place for finite-prime information. It does not, by itself, contain the archimedean metric needed for purity. That requires the separate scaling action of the global base.

The contrast with $\mathbf{F}_p$ is now complete. A single residue field has a single characteristic and a polynomial Bökstedt class; its cyclotomic theory builds the p-typical Witt tower. The integers have torsion distributed across all primes and no positive rational homotopy; their cyclotomic theory must glue all p-typical conditions integrally. The proposed arithmetic chôra adds the archimedean scaling character to this integral object. Exactly this last step has no established construction.

3.21 Algebraic shadows of the spectral trace

The abstract construction specialises to familiar traces, but the target changes before the familiar scalar is obtained. Let A be an ordinary associative ring and let P be a finitely generated projective right A-module. Its dual is $P^* = \mathrm{Hom}_A(P, A)$. Choosing a dual basis (p_i, λ_i) with $p = \sum_i p_i \lambda_i(p)$, the categorical trace of an endomorphism $f : P \to P$ is represented by

$$\sum_i \lambda_i(f(p_i)) \in A/[A, A] = HH_0(A).$$

This is the Hattori–Stallings trace. It is independent of the chosen dual basis because it is defined by evaluation and coevaluation. If A is commutative, the quotient by commutators is A and the expression becomes the ordinary trace of a projective-module endomorphism.

For maps $f : P \to Q$ and $g : Q \to P$, the two expressions for the traces of gf and fg need not be equal as elements of a noncommutative ring A, but their difference is a sum of commutators. They are equal in $HH_0(A)$. This is the algebraic shadow of Theorem 3.2: closing the diagram forces the target to forget the choice of a starting point, and algebraically this means quotienting by commutators. Higher Hochschild homology retains the higher coherences of the same operation.

If $P^\bullet$ is a bounded complex of finite projectives and f is a chain endomorphism, then

$$\operatorname{tr}(f \mid P^\bullet) = \sum_n (-1)^n \operatorname{tr}(f^n \mid P^n)$$

in $HH_0(A)$. The formula is independent of the chosen perfect representative. A contractible summand contributes zero because its two adjacent terms have equal traces with opposite signs. This is the elementary model for both May additivity and the weight-tower formula.

The target can be refined from $HH_0(A)$ to the spectrum $\mathrm{THH}(A)$. A scalar trace is then obtained by applying π_0, and even that may land in a Hochschild quotient rather than the centre of A. Thus the slogan "trace is a number" is true only after commutativity, decategorification and a coefficient choice. The arithmetic framework delays all three operations so that circle actions, torsion and Frobenius remain visible.

3.21.1 Relative base matters

For the ordinary ring $\mathbf{Z}$ relative to itself,

$$HH^{\mathbf{Z}}_*(\mathbf{Z}) = \mathbf{Z}$$

concentrated in degree zero. Relative to the sphere spectrum, $\mathrm{THH}(\mathbf{Z})$ has the Bökstedt torsion in every relevant odd degree. The difference is not caused by replacing homology with cohomology; it is caused by changing the base from $H\mathbf{Z}$ to $\mathbf{S}$. The absolute diagonal over the sphere has derived directions which the relative diagonal over $H\mathbf{Z}$ does not see.

This makes the base in Proposition 3.10 indispensable. Writing $\mathbf{Z} \otimes \mathbf{Z}$ without a subscript suppresses the geometry. Over $\mathbf{Z}$ the tensor square is $\mathbf{Z}$ and the diagonal is trivial. Over $\mathbf{S}$ the tensor square is spectral, and its derived self-intersection is $\mathrm{THH}(\mathbf{Z})$. Shtuka legs require the second setting because independent legs must move relative to a base below $\operatorname{Spec}\mathbf{Z}$.

3.21.2 Traces as functions on loop and fixed-point figures

For a suitably perfect derived stack Y, the trace of the identity on $\mathrm{QCoh}(Y)$ is identified with functions on the derived loop stack

$$LY = Y \times_{Y \times Y} Y.$$

The two maps to $Y \times Y$ are the diagonal, so this is again a derived self-intersection. The circle acts by rotating loops. This geometric realisation of Hochschild homology is developed in derived character theory and integral-transform formalisms (Ben-Zvi et al. 2010).

For an endomorphism $f : Y \to Y$, the twisted loop or fixed-point figure is

$$Y^f = Y \times_{Y \times Y, (\mathrm{id}, f)} Y.$$

Under properness and perfectness hypotheses, the categorical trace of the induced functor is represented by sheaf-theoretic data on Y^f. A Lefschetz formula then pushes that data to the point and decomposes it into local contributions. The formula in Axiom 3.4 is therefore not arbitrary: it is the abstract shape shared by derived loops, classical fixed points and categorical traces.

Over $\mathbf{F}_q$, taking f to be Frobenius makes Y^f the arithmetic fixed-point stack, and its rational points recover the objects defined over the finite field. This is the geometric reason the trace of Frobenius on a sheaf category produces functions on $Y(\mathbf{F}_q)$. Over $\mathfrak{X}_{\mathbf{Z}}$, there is no single constructed endomorphism whose fixed points simultaneously encode all places. Cyclotomic structure proposes a primewise replacement, while the archimedean scaling flow supplies

a continuous component. Proving that their combined equaliser is represented by a global fixed-point figure is beyond the Nikolaus–Scholze theorem and remains part of the arithmetic conjecture.

The derived-loop viewpoint also explains why local terms can carry more than multiplicities. If a fixed point is non-transverse, its derived intersection has higher homotopy and the local trace records a complex or spectrum. Numerical intersection multiplicities, signs and Behrend-type weights are shadows of this derived local object. In the function-field Lefschetz formula the required intersection theory is established. In the chôral arithmetic setting its existence depends on the unconstructed derived geometry of the global base.

3.22 Normalisations of Frobenius, twists, and weight

Weight statements are meaningless until conventions are fixed. This chapter uses geometric Frobenius in trace formulae. On $\mathbf{Q}_\ell(1)$ geometric Frobenius acts by q^{-1}, and on $\mathbf{Q}_\ell(-1)$ it acts by q. Hence the latter has weight two. A pure lisse sheaf of weight w has closed-point eigenvalues of absolute value $q_x^{w/2}$, and proper smooth cohomology in degree i has weight $w + i$. These conventions make $H^2(\mathbf{P}^1) = \mathbf{Q}_\ell(-1)$ pure of weight two and make the roots on H^1 of an elliptic curve have absolute value $\sqrt{q}$.

Arithmetic Frobenius is the inverse automorphism on residue fields. If it is used instead, every eigenvalue is inverted and every displayed local factor must be adjusted. The categorical trace theorem itself is indifferent to this choice, but purity and the variable in the zeta function are not. The book therefore uses geometric Frobenius whenever comparing with Deligne and writes an inverse explicitly when an arithmetic convention from representation theory is imported.

The Rees scaling character is normalised integrally: weight m means that $a \in \mathbf{G}_m$ acts by a^m. The metric absolute value associated with that character is $|a|^{m/2}$ in the Weil convention. This factor of two is why the Bökstedt generator σ, of topological degree two, can naturally be assigned Rees weight one in a finite-field model without identifying degree and weight. A Tate twist (1) lowers Weil weight by two, while a cohomological shift $[1]$ raises the weight index of cohomology by one under Deligne's complex convention. The Bondarko shift conventions in Definition 3.14 are chosen independently and are converted through the realisation functor.

For unramified Langlands parameters, one often inserts the half-sum of positive roots or a power of q_v so that tempered parameters become unitary. The criterion Criterion 3.18 is stated before that unitary normalisation: a weight-w eigenvalue has magnitude $q_v^{w/2}$. Dividing by this factor gives magnitude one. For representations r of the dual group, the relevant w depends on the motivic normalisation of r and on any Tate twist built into the local Langlands correspondence. A claim of Ramanujan which does not state this normalisation is incomplete.

The four conventions may be summarised as follows.

Operation	Effect on the convention used here
geometric Frobenius on $\mathbf{Q}_\ell(1)$	eigenvalue q^{-1}
Tate twist (1)	weight decreases by two
cohomological degree i	realised weight increases by i
Rees character a^m	Weil metric exponent $m/2$

These choices reconcile the examples, Deligne's theorem and the Satake criterion. They do not assert that the global scaling action exists. They state what its characters must mean if the arithmetic chôra is constructed.

A final test concerns independence of the auxiliary prime ℓ. Deligne purity is meaningful because characteristic polynomials of Frobenius in geometric situations have algebraic coefficients and compatible complex absolute values, independent of the chosen ℓ-adic realisation. A chôral purity theorem over $\mathfrak{X}_{\mathbf{Z}}$ must similarly show that its finite-place eigenvalues, its p-adic slopes and its archimedean scaling characters arise from one algebraic datum. Separate purity statements in every realisation are insufficient if no comparison identifies their eigenvalues. The primitive trace is intended to be that common datum, but $\mathrm{THH}(\mathbf{Z})$ alone does not prove compatibility. The comparison maps from the integral cyclotomic trace to étale, crystalline, de Rham and archimedean realisations would have to be constructed and shown conservative on pure objects. This independence requirement is the arithmetic analogue of the compatibility built into the Weil cohomology formalism, and it is another exact point at which the function-field theorem exceeds the present number-field proposal.

3.23 What a trace can and cannot detect

Purity is stronger than a statement about one numerical trace. Let V be an n-dimensional vector space and F an endomorphism with eigenvalues $\alpha_1, \dots, \alpha_n$ in an algebraic closure. The single number $\mathrm{Tr}(F) = \sum_i \alpha_i$ permits extensive cancellation. Even a zero trace says nothing about the magnitudes of the individual eigenvalues. A pure two-dimensional representation with eigenvalues α and $-\alpha$ has trace zero, while a mixed representation can also have trace zero. Consequently the fixed-point count for one field extension cannot establish purity.

The full sequence of power traces contains more information. Newton's identities express the elementary symmetric functions of the eigenvalues in terms of

$$\mathrm{Tr}(F), \mathrm{Tr}(F^2), \dots, \mathrm{Tr}(F^n).$$

For known finite dimension, the first n power traces recover the characteristic polynomial. Equivalently, the formal zeta series

$$\exp\left(\sum_{m\geq 1} \mathrm{Tr}(F^m)\frac{t^m}{m}\right) = \det(1 - tF)^{-1}$$

recovers the multiset of eigenvalues with multiplicity. This is why point counts over all extensions, rather than only over $\mathbf{F}_q$, determine the zeta function.

Even the characteristic polynomial does not recover the endomorphism. It does not detect Jordan blocks, extension classes or the category in which F acts. A semisimple pure representation and a nontrivial self-extension can have the same characteristic polynomial and all the same power traces. The nilpotent part becomes visible only through additional structure, such as monodromy, a derived fixed-point complex, or higher operations on the trace. This is the reason Definition 3.16 requires structural purity and split scaling, not merely the correct absolute values in a determinant.

Categorical traces retain more than numerical traces but remain additive invariants. Morita-equivalent categories have the same THH, and distinct objects can have equal classes in every additive invariant. A weight tower can therefore be used to compute a trace without being classified by that trace. The extension information remains in the tower and its connecting maps. This is a virtue for computation and a limitation for reconstruction.

The integral example is sharper. Every positive homotopy group of $\mathrm{THH}(\mathbf{Z})$ vanishes after tensoring with $\mathbf{Q}$. Any rational numerical trace of those groups is therefore zero, although the integral spectrum is far from trivial. Passing to all p-completions detects the primary parts, but an integral gluing condition is still required to know that they arise from $\mathbf{Z}/i$. Thus detection depends on the coefficient system as well as on the family of traces.

Over a finite field, Deligne purity avoids these ambiguities by imposing the absolute-value condition at every closed point and on every cohomology sheaf, with algebraicity and compatibility across embeddings. The theorem is not inferred from a few trace values. The traces are consequences of the pure sheaf, and the zeta function packages them. In the arithmetic proposal the logical order must be the same: construct the weighted object, prove purity, and then deduce bounds for its trace. Defining purity solely by the desired Ramanujan inequality would reverse implication and conceal the missing geometry.

This also explains the role of non-degeneracy in Axiom 3.6. A degenerate trace can annihilate a nonzero summand, making an impure or torsion object invisible. Non-degeneracy prevents that failure on the chosen coefficient realisation, but no single realisation is expected to detect every integral phenomenon. A credible arithmetic theorem must therefore state both the class of objects on which the trace is non-degenerate and the family of realisations used to test purity.

3.24 Circle rotation is not weight scaling

The compact and split actions must finally be kept apart. The canonical S^1-action on THH rotates a loop. Its parameter has unit modulus, and its existence follows from cyclic symmetry. A weight action is an action of $\mathbf{G}_m$, or analytically of the split group $\mathbf{R}_{>0}$, whose characters have nontrivial magnitudes. It grades a Rees construction and supplies the exponents from which absolute-value bounds are read. Rotation creates cyclicity; scaling creates weight.

At the archimedean chart the real points of the Deligne torus decompose as

$$\mathbf{C}^\times \cong \mathbf{R}_{>0} \times S^1.$$

This makes it plausible that the two actions are factors of one archimedean symmetry, but a polar decomposition of groups is not an identification of their roles. Taking S^1-homotopy fixed points produces negative cyclic information; it does not diagonalise a weight filtration. Conversely, a $\mathbf{G}_m$-grading can exist on an algebra with no coherent cyclic trace. The arithmetic chôra requires the actions to commute and to satisfy the specified comparison with finite Frobenii.

The Bökstedt examples demonstrate the distinction. The circle action exists canonically on both $\mathrm{THH}(\mathbf{Z})$ and $\mathrm{THH}(\mathbf{F}_p)$. A scaling character assignment for σ^i is an additional grading choice, and no global split grading of the integral torsion groups has been constructed. Thus canonical cyclicity survives where a weight decomposition is absent.

This separation also protects the purity claim from a false formal proof. One cannot argue that, because THH has an S^1-action, its trace is pure. Compact rotations have eigenvalues of complex absolute value one for representation-theoretic reasons, but those eigenvalues are Fourier modes of the circle action, not Frobenius eigenvalues normalised by $q^{w/2}$. Purity requires the split action, algebraicity and comparison at closed points.

In the function-field case the distinction is supplied externally. Frobenius provides arithmetic eigenvalues, Deligne weights control their magnitudes, and the cyclic symmetry of the categorical trace explains invariance under moving the cut. Over $\mathfrak{X}_{\mathbf{Z}}$, the framework proposes that the compact and split factors meet at infinity and that cyclotomic Frobenii meet them at finite places. Establishing this common action is part of constructing the base, not a consequence of the abstract trace theory developed here.

Summary. The categorical trace is the value of a labelled circle, and its cyclicity is rotation of that circle. For an identity endomorphism the trace is THH, whose cyclic bar construction carries the

canonical S^1-action; cyclotomic Frobenii refine it, and Nikolaus–Scholze TC equalises the canonical and Frobenius maps prime by prime. Bökstedt's calculations show that the arithmetic diagonal is a derived self-intersection: polynomial and even over $\mathbf{F}_p$, torsion and odd over $\mathbf{Z}$. Bondarko weights replace the unavailable motivic heart for the purposes of trace, because additivity expresses the trace of a weight tower as the alternating sum of its pure layers. Deligne's theorem proves the resulting purity statement over finite fields and yields the Weil bounds; over $\mathfrak{X}_{\mathbf{Z}}$, the base, scaling action, weight structure, localisation theorem and purity comparison remain open.

3.25 Notes and references

The categorical trace used here belongs to the general theory of duality in symmetric monoidal higher categories. Ponto and Shulman develop traces through bicategorical shadows (Ponto and Shulman 2012), while Hoyois, Scherotzke and Sibilla relate higher traces, noncommutative motives and the categorified Chern character (Hoyois et al. 2017). Lurie's work on rotation invariance makes explicit the relation between trace coherence and circle actions (Lurie 2015). May's theorem is the classical stable reference for additivity (May 2001).

The identification of Hochschild homology with the trace of the identity and its geometric interpretation as functions on a derived loop space run through modern derived algebraic geometry. Ben-Zvi, Francis and Nadler identify centres, loops and integral transforms in this setting (Ben-Zvi et al. 2010). Toën and Vezzosi develop equivariant traces and Chern characters (Toën and Vezzosi 2015). The formulation of THH as the universal recipient of coherent cyclic trace data should be read with the qualification made in Theorem 3.9: universal localising invariants are noncommutative motives, not THH itself (Blumberg et al. 2013).

Bökstedt's computations originated in foundational work circulated before the modern publication conventions of the subject. The available bibliography records the Bökstedt–Hsiang–Madsen cyclotomic trace paper (Bökstedt et al. 1993). Bhatt, Morrow and Scholze provide a modern setting in which THH, TC and integral p-adic Hodge theory interact (Bhatt et al. 2019). The Nikolaus–Scholze definition of cyclotomic spectra and their formula for TC are the normative reference (Nikolaus and Scholze 2018). Hesselholt's account of periodic topological cyclic homology and zeta functions is the closest established precedent for the regularised-determinant ambitions of the arithmetic framework (Hesselholt 2018).

Bökstedt's calculation should be understood historically as the discovery that Hochschild homology over the sphere is arithmetically larger than algebraic Hochschild homology over $\mathbf{Z}$. The polynomial class for $\mathbf{F}_p$ became the point of departure for the restriction, Frobenius and Verschiebung operators studied by Bökstedt, Hsiang, Madsen, Hesselholt and Madsen. Their work connected topological cyclic constructions to Witt vectors and made the cyclotomic trace from algebraic K-theory computationally effective (Hesselholt and Madsen 1997). The modern account does not replace those calculations; it reorganises their equivariant input.

Nikolaus and Scholze's contribution is especially important for the logical architecture of this chapter. Earlier definitions used genuine equivariant fixed-point systems and restriction maps indexed by all finite cyclic subgroups. Their theorem packages the same bounded-below information as an S^1-spectrum with maps $\varphi_p : X \to X^{tC_p}$. This makes the sources and targets of the canonical and Frobenius maps visible and turns TC into a homotopy equaliser. The simplification is conceptual, not an assertion that the primewise completions may be omitted.

Ponto and Shulman's bicategorical shadow formalism is the appropriate reference for the claim that cyclicity belongs to the geometry of a closed string rather than to a list of scalar identities. A shadow is exactly the structure which allows a composite to be closed and then rotated. Their formalism also clarifies why twisted traces and secondary traces require extra

centraliser or commutation data. The higher-categorical refinements cited above supply the coherent S^1-action which an equality in an ordinary homotopy category would forget.

Weight structures are due to Bondarko (Bondarko 2010). Their principal advantage here is not that they duplicate a motivic t-structure, but that they do not. They provide bounded pure towers and weight complexes in settings where no abelian category of mixed motives is known. Deligne's two Weil papers supply the control theorem for metric purity (Deligne 1974a, 1980). The Grothendieck–Lefschetz–Verdier trace formula is recorded in SGA 5 (Grothendieck and Illusie 1977). Laurent Lafforgue's theorem for GL_n and Vincent Lafforgue's excursion formalism show how these structures enter function-field Langlands (Lafforgue 2002; Lafforgue 2018).

Bondarko's construction belongs to a line of ideas in which complexes of pure objects replace hypothetical abelian categories of mixed objects. Its distinctive point is the reversal of the orthogonality convention familiar from t-structures. This reversal makes the heart negative and allows successive pure terms to assemble into a weight complex. The example of $\mathrm{Perf}(\mathbf{Z})$ in this chapter is elementary, but it shows exactly what is retained and lost: projective terms and their differential are retained, while the torsion cokernel belongs only to the t-heart.

Deligne's work supplies a different ingredient. A weight structure labels pieces; Weil II proves absolute-value bounds for Frobenius eigenvalues and controls how those bounds behave under the six operations. The rationality and functional equation of zeta functions do not contain that metric theorem. This distinction, already present in the Weil conjectures, is the reason the chapter refuses to derive purity from trace formalism. The function-field Langlands theorems inherit their Ramanujan bounds from this geometric purity input, not from excursion operators alone.

The chôral synthesis adds a conjectural identification: primitive trace is to be represented by THH, localisation by cyclotomic structure, and weight by the split scaling action on the global arithmetic curve. The first identification is established under dualisability and finiteness; the second and third are not established for the proposed arithmetic categories. The correction from $\mathbf{Z} \otimes_{\mathbf{S}} \mathbf{Z}$ to $\Delta^*\Delta_* O$ follows the framework's explicit warning and must be maintained throughout the remainder of the book.

3.26 Exercises

Exercise 3.19. [DESIGN]

Let V and W be finite-dimensional vector spaces and let $f : V \to W$ and $g : W \to V$. Derive $\mathrm{Tr}(gf) = \mathrm{Tr}(fg)$ first from matrices and then from Theorem 3.2. Identify exactly which choice of basis disappears in the categorical proof.

Exercise 3.20. [DESIGN]

Using only the cofiber sequence $\mathbf{1} \to 0 \to \mathbf{1}[1]$, additivity and multiplicativity of trace, prove that $\mathrm{tr}(f[n]) = (-1)^n \mathrm{tr}(f)$ for every dualisable object and integer n.

Exercise 3.21. [DESIGN]

List $\pi_n \mathrm{THH}(\mathbf{Z})$ for $0 \le n \le 15$, decompose each nonzero finite group into its primary parts, and record which degrees survive after completion at 2, at 3, and after rationalisation.

Exercise 3.22. [DESIGN]

For a commutative R-algebra A, prove directly from affine base change that $\Delta^*\Delta_* O \simeq A \otimes_{A \otimes_R A} A$. Explain why $A \otimes_R A$ and the derived self-intersection have opposite variances in the two occurrences of the diagonal.

Exercise 3.23. [DESIGN]

Apply the finite weight-tower extraction procedure above to the three-term complex $[\mathbf{Z} \xrightarrow{2} \mathbf{Z} \xrightarrow{0} \mathbf{Z}]$. Compute the trace of scalar multiplication by m and compare it with any trace visible on its cohomology modules.

Exercise 3.24. [DESIGN]

Derive $Z(\mathbf{G}_m, t) = (1-t)/(1-qt)$ both by point counting and by the localisation triangle for $\mathbf{G}_m \subset \mathbf{P}^1$. Identify the weight and the cohomological sign of every factor.

Exercise 3.25. [DESIGN]

For the elliptic curve computed above, use the recurrence $s_n = -3s_{n-1} - 5s_{n-2}$ with $s_0 = 2$ and $s_1 = -3$ to compute $|E(\mathbf{F}_{5^n})|$ for $n = 2, 3, 4$. Check the answers against the zeta function.

Exercise 3.26. [DESIGN]

For an unramified representation of GL_2 with Satake parameters α, β and $\alpha\beta = q^w$, prove the criterion in Criterion 3.18 explicitly. Give an example showing that unit absolute values alone do not supply algebraicity or a weight filtration.

Exercise 3.27. [DESIGN]

Using $\pi_*\,\mathrm{THH}(\mathbf{F}_p[x]) \cong \mathbf{F}_p[x] \otimes \Lambda(dx) \otimes \mathbf{F}_p[\sigma]$, compute the groups through degree eight, their products by dx and σ, and the Hilbert series after assigning rank one to $\mathbf{F}_p[x]$. Repeat for $\mathbf{F}_p[x, y]$, including the class $dx \wedge dy$.

Exercise 3.28. [DESIGN]

Let $R = \mathbf{F}_{p^r}$. Compute $\pi_*\,\mathrm{THH}(R)$ through degree ten and compare it with the polynomial-ring calculation. Prove directly that $\Omega^1_{R/\mathbf{F}_p} = 0$, and explain why vanishing of the full cotangent complex for a perfect algebra is stronger than this elementary calculation.

Exercise 3.29. [DESIGN]

Carry out the mod-two base-change test in §3.4. Use $H_*(H\mathbf{Z}; \mathbf{F}_2) \cong \mathbf{F}_2[\zeta_1^2, \zeta_2, \dots]$ and Bökstedt's integral groups to compute degree two on both sides. Conclude without using ring structure that $H\mathbf{Z} \otimes_{\mathbf{S}} H\mathbf{Z}$ and $\mathrm{THH}(\mathbf{Z})$ are not equivalent.

Exercise 3.30. [DESIGN]

For $E : y^2 = x^3 + x + 1$, reproduce the point counts at 5, 7, and 11. For each prime compute the Frobenius polynomial, its two complex roots, and their absolute values. Then use the recurrence determined by that polynomial to compute $|E(\mathbf{F}_{p^2})|$ and verify the Weil bound for the trace of Frob_p^2.

Chapter 4

Dualisability, Ambidexterity, and Compactness

Synopsis. Every trace in this book is a trace of an endomorphism of a dualisable object. If the object is not dualisable there is no trace, and the identity that the book is organised around — spectral side equals automorphic side, both read off from a single categorical trace — does not merely become hard to prove: it fails to have a meaning. This chapter therefore treats dualisability as a hypothesis to be earned rather than assumed. We develop duality data and the triangle identities in a symmetric monoidal ∞-category; we pass to dualisability of categories themselves, where compact generation and Efimov's retract criterion take over; we characterise smoothness and properness by left and right adjointability, following the circle of ideas around Lurie's cobordism hypothesis and the Ben-Zvi–Francis–Nadler theory of perfect stacks; and we review ambidexterity in the sense of Hopkins and Lurie, together with its m-semiadditivity hierarchy. The chapter's centre of gravity, however, is a failure. Over $\mathbf{F}_q$, the stack $\mathrm{Bun}_G(X)$ is not proper, and the repairs that make its category of sheaves dualisable — Harder–Narasimhan truncation, miraculous duality, and in the Fargues–Scholze setting ℓ-cohomological smoothness — are substantial theorems, not formalities. Over an arithmetic base none of these repairs is available in the form in which they are known. We state precisely, as a conjecture, what must hold; we exhibit an explicit chôra whose trace does not exist, computed rather than asserted; and we work GL_1 over $\mathbf{Q}$ completely by hand, where the ambidexterity defect turns out to be a rank-two vector space that one can point at, and which coincides with the two poles of Tate's zeta integral.

4.1 What dualisability is for

A trace is not an operation one can perform on an arbitrary endomorphism. In linear algebra over a field this is easy to forget, because every finite-dimensional vector space is dualisable and one learns the trace as a formula — the sum of the diagonal entries — rather than as a construction. The formula conceals the hypothesis. As soon as one leaves finite dimensions the hypothesis becomes visible: the identity endomorphism of an infinite-dimensional vector space has no trace, not because the sum $1 + 1 + 1 + \cdots$ diverges in some remediable analytic sense, but because the categorical operation that would produce the number does not exist. There is no coevaluation map.

This book asserts an equality of two quantities, one spectral and one automorphic, and asserts that both are values of a single categorical trace. Every theorem downstream of that assertion, and every conjecture, is conditional on the trace being defined. It is therefore not enough to say that the relevant objects are "expected to be" dualisable. We must say what dualisability means in the ∞-categorical setting where the objects live, we must isolate the criteria that can actually be checked, and — this is the part that is usually skipped — we must be candid about the cases where the criteria fail and no repair is known.

The organising observation of this chapter is that a single structural failure, stated in one line, accounts for three phenomena that the theory of automorphic forms has treated as unrelated difficulties for half a century. Let $\mathfrak{X}$ be an automorphic chôra in the sense of Chapter 2, with structure morphism $f\colon \mathfrak{X} \to *$. Ambidexterity for f is the assertion that a canonical natural transformation

$$\mathrm{Nm}_f\colon f_! \longrightarrow f_*$$

is an equivalence. When it is, sheaf theory on $\mathfrak{X}$ behaves as though $\mathfrak{X}$ were both smooth and proper: compactly supported cohomology agrees with cohomology, Poincaré duality holds in the naive form, and the trace of an endomorphism can be computed by a Lefschetz count over fixed points. When it is not, all three of these statements fail simultaneously, and they fail by the *same* amount, namely the cofibre of Nm_f.

The three classical symptoms are these.

First, the continuous spectrum. The Hilbert space $L^2(G(F)\backslash G(\mathbf{A}))$ does not decompose as a direct sum of irreducibles. Langlands's theory of Eisenstein series produces a continuous part indexed by the proper parabolic subgroups of G together with the split centre, and the Arthur–Selberg trace formula acquires, alongside its geometric side, a family of terms which are not orbital integrals at all (Arthur 2013). One does not count fixed points; one counts fixed points and then corrects.

Second, the cohomological defect. For an arithmetic group the cohomology of the associated locally symmetric space is not concentrated in a single degree but spread over an interval of length $\ell_0 = \operatorname{rank} G(F \otimes \mathbf{R}) - \operatorname{rank} K_\infty Z_\infty$. When $\ell_0 = 0$, as for GL_2 over $\mathbf{Q}$, the classical Taylor–Wiles method applies verbatim. When $\ell_0 > 0$, as for GL_2 over an imaginary quadratic field or GL_3 over $\mathbf{Q}$, it does not, and the repairs of Calegari–Geraghty (Calegari and Geraghty 2018) and Galatius–Venkatesh (Galatius and Venkatesh 2018) are needed — repairs which, read correctly, replace a ring by a differential graded ring precisely of amplitude ℓ_0.

Third, the failure of Poincaré duality. A locally symmetric space of positive $\mathbf{Q}$-rank is not compact. Its Borel–Serre compactification has boundary, the natural map $H^*_c \to H^*$ is not an isomorphism, and interior cohomology is a proper subquotient of both. The Hecke algebra acts on all three, and the discrepancy between them is again indexed by proper parabolics and the split centre.

The three symptoms are one symptom. The three phenomena above are the same phenomenon read in three languages: spectral decomposition, homological amplitude, and duality. Each is measured by $\mathrm{cofib}(\mathrm{Nm}_f)$ for a single morphism f. This is a design principle rather than a theorem: it asserts that a certain reorganisation of the subject is the right one. Its content is that the reorganisation is possible at all, and this chapter's later sections make it precise in the two cases — GL_1 over $\mathbf{Q}$, and Bun_G over a function field — where every term can be identified.

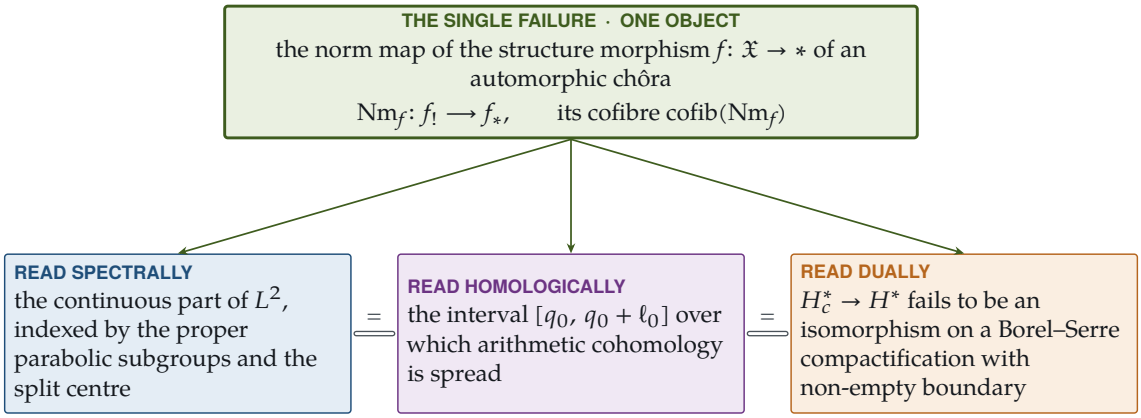


Plate 4.1. The single failure and its three classical symptoms. The centre of the plate is the norm map $\mathrm{Nm}_f\colon f_! \to f_*$ for the structure morphism of an automorphic chôra; the three radiating panels show what its cofibre becomes when the diagram is read spectrally (the continuous part of L^2, indexed by proper parabolics and the split centre), homologically (the interval $[q_0, q_0 + \ell_0]$ over which arithmetic cohomology is spread), and dually (the failure of $H^*_c \to H^*$ to be an isomorphism on a Borel–Serre compactification with non-empty boundary). The reader is asked to notice that the three panels are not three theorems but three coordinate systems on one object, and that the arrows between them are equalities rather than analogies.

What it does not show: a proof that the three readings are one object over an arithmetic base: that is the last sentence of the Ambidexterity–Compactness Principle, marked [C].

Plate 4.1 fixes the picture that the rest of the chapter makes precise. What must be supplied is, in order: a definition of dualisability stable enough to survive the passage from objects to categories; a criterion for it that can be verified in practice; a construction of Nm_f that does not presuppose what it is meant to establish; and an explicit account of the cases in which Nm_f is not an equivalence and is not going to become one.

We proceed in that order. Sections §4.2 and §4.3 treat duality for objects and for categories; section §4.4 characterises smoothness and properness by adjointability; section §4.5 constructs the norm map and reviews the semiadditivity hierarchy; section §4.6 states the principle towards which the whole chapter is directed; section §4.7 is the critical section, on the failure of dualisability for Bun_G over an arithmetic base; and sections §4.8 and §4.9 are computational.

4.2 Duality data in a symmetric monoidal ∞-category

Throughout, $(C, \otimes, \mathbf{1})$ denotes a symmetric monoidal ∞-category in the sense of (Lurie 2017, sect. 2). We write Map for mapping spaces and reserve Hom for internal mapping objects when they exist. No completeness or presentability hypothesis is imposed until section §4.3, where it becomes essential.

Definition 4.1 (Dualisable object). [ESTABLISHED]
An object $X \in \mathcal{C}$ is *dualisable* if there exist an object $X^\vee \in \mathcal{C}$ and morphisms

$$\mathrm{coev} \colon \mathbf{1} \longrightarrow X \otimes X^\vee, \qquad \mathrm{ev} \colon X^\vee \otimes X \longrightarrow \mathbf{1},$$

called *coevaluation* and *evaluation*, such that the two composites

$$X \xrightarrow{\mathrm{coev} \otimes \mathrm{id}} X \otimes X^\vee \otimes X \xrightarrow{\mathrm{id} \otimes \mathrm{ev}} X, \qquad X^\vee \xrightarrow{\mathrm{id} \otimes \mathrm{coev}} X^\vee \otimes X \otimes X^\vee \xrightarrow{\mathrm{ev} \otimes \mathrm{id}} X^\vee$$

are homotopic to the respective identities. These homotopies, together with the coherence data they generate, are the *triangle identities*.

Two points deserve emphasis, because they are exactly the points at which the ∞-categorical situation differs from the classical one, and both matter later.

The first is that in an ∞-category the triangle identities are not conditions but *data*: one must specify the homotopies, and then coherences between them. It is a theorem, not an observation, that this apparently infinite tower collapses.

Proposition 4.2 (Duality is a property). [ESTABLISHED]
Let $X \in \mathcal{C}$. The space of duality data $(X^\vee, \mathrm{ev}, \mathrm{coev}, \text{triangle identities})$ for X is either empty or contractible. In particular $X^\vee$, when it exists, is determined up to a contractible space of choices, and dualisability is a property of X rather than a structure on it.

Proof of Proposition 4.2.
This is (Lurie 2017, sect. 4.6.1); we recall the mechanism because the argument recurs in section §4.5. Duality data for X exhibit $X^\vee$ as a right adjoint to X in the one-object 2-category, or rather $(\infty,2)$-category, $B\mathcal{C}$ obtained by delooping $\mathcal{C}$ along its monoidal structure: objects of $\mathcal{C}$ become 1-morphisms, $\otimes$ becomes composition, coev becomes the unit of an adjunction and ev the counit, and the triangle identities become the triangle identities of an adjunction. The space of adjoints to a fixed 1-morphism in an $(\infty,2)$-category is empty or contractible, by the usual argument that a right adjoint corepresents a functor and representing objects form a contractible space when they exist. The tower of higher coherences is therefore not an obstruction: it is generated, uniquely up to contractible choice, by the first two levels. □

The second point is that dualisability is preserved by symmetric monoidal functors, and this triviality is the engine of the entire subject. If $F \colon \mathcal{C} \to \mathcal{D}$ is symmetric monoidal then F carries duality data to duality data; hence $F(X)$ is dualisable whenever X is, and $F(X)^\vee \simeq F(X^\vee)$. Every comparison theorem in this book is, at bottom, an application of this sentence to a functor whose construction is the real work.

Definition 4.3 (Trace of an endomorphism). [ESTABLISHED]
Let $X \in \mathcal{C}$ be dualisable and let $\varphi \colon X \to X$. The *trace* $\mathrm{tr}(\varphi) \in \mathrm{End}_{\mathcal{C}}(\mathbf{1})$ is the composite

$$\mathbf{1} \xrightarrow{\mathrm{coev}} X \otimes X^\vee \xrightarrow{\varphi \otimes \mathrm{id}} X \otimes X^\vee \xrightarrow{\sigma} X^\vee \otimes X \xrightarrow{\mathrm{ev}} \mathbf{1},$$

where σ is the symmetry. We write $\dim(X) := \mathrm{tr}(\mathrm{id}_X)$ for the *dimension* or *Euler characteristic* of X.

The symmetry σ is not decorative. It is the reason traces exist in symmetric (or at least braided) monoidal categories and not in merely monoidal ones, and it is the reason that the trace of an endomorphism is a *circle*: the composite above is the value on φ of a functor defined on the free symmetric monoidal ∞-category on a dualisable object, whose endomorphisms of the unit are computed by the circle. This is the point of contact with topological field theory to which we return in section §4.4.

Proposition 4.4 (Formal properties of the trace). [ESTABLISHED]

Let $X, Y \in \mathcal{C}$ be dualisable, $\varphi\colon X \to X$, $\psi\colon Y \to Y$.

(*i*) Symmetry. *If $u\colon X \to Y$ and $v\colon Y \to X$ then* $\operatorname{tr}(v \circ u) = \operatorname{tr}(u \circ v)$.

(*ii*) Monoidality. *$X \otimes Y$ is dualisable and* $\operatorname{tr}(\varphi \otimes \psi) = \operatorname{tr}(\varphi) \cdot \operatorname{tr}(\psi)$.

(*iii*) Functoriality. *If $F\colon \mathcal{C} \to \mathcal{D}$ is symmetric monoidal then* $\operatorname{tr}(F\varphi) = F(\operatorname{tr}(\varphi))$ *in* $\operatorname{End}_{\mathcal{D}}(\mathbf{1})$.

(*iv*) Additivity. *If $\mathcal{C}$ is stable and*

$$X' \to X \to X''$$

is a fibre sequence of dualisable objects compatible with an endomorphism φ of the sequence, then $\operatorname{tr}(\varphi) = \operatorname{tr}(\varphi') + \operatorname{tr}(\varphi'')$.

Proof of Proposition 4.4.

We prove (i)–(iii) by manipulating the triangle identities, and cite (iv).

For (i), write out both composites. By definition $\operatorname{tr}(v \circ u)$ is

$$\mathbf{1} \xrightarrow{\operatorname{coev}_X} X \otimes X^\vee \xrightarrow{(vu)\otimes \mathrm{id}} X \otimes X^\vee \xrightarrow{\sigma} X^\vee \otimes X \xrightarrow{\operatorname{ev}_X} \mathbf{1}.$$

Insert the identity of Y in the middle by factoring vu as $X \xrightarrow{u} Y \xrightarrow{v} X$, and then use the naturality of the symmetry σ together with the *duality transpose*: for $u\colon X \to Y$ there is $u^\vee\colon Y^\vee \to X^\vee$ characterised by $\operatorname{ev}_X \circ (u^\vee \otimes \mathrm{id}_X) = \operatorname{ev}_Y \circ (\mathrm{id}_{Y^\vee} \otimes u)$. Applying this identity once converts $\operatorname{ev}_X \circ \sigma \circ ((vu) \otimes \mathrm{id}_{X^\vee}) \circ \operatorname{coev}_X$ into $\operatorname{ev}_Y \circ \sigma \circ ((uv) \otimes \mathrm{id}_{Y^\vee}) \circ \operatorname{coev}_Y$, which is $\operatorname{tr}(u \circ v)$. The one place where symmetry rather than mere braiding is used is in identifying σ^2 with the identity when the two insertions are undone.

For (ii), duality data for $X \otimes Y$ are given by $(X \otimes Y)^\vee = Y^\vee \otimes X^\vee$ with $\operatorname{coev}_{X\otimes Y}$ the composite $\mathbf{1} \xrightarrow{\operatorname{coev}_X} X \otimes X^\vee \xrightarrow{\mathrm{id} \otimes \operatorname{coev}_Y \otimes \mathrm{id}} X \otimes Y \otimes Y^\vee \otimes X^\vee$ and $\operatorname{ev}_{X\otimes Y}$ the analogous composite of evaluations. The triangle identities for the composite follow from those for X and Y by inserting one inside the other. With these data, $\operatorname{tr}(\varphi \otimes \psi)$ is a composite of two disjoint circles, and since $\operatorname{End}_{\mathcal{C}}(\mathbf{1})$ is a commutative monoid under composition and tensor by Eckmann–Hilton, the two circles multiply.

For (iii), a symmetric monoidal functor F carries coev to a map $\mathbf{1}_{\mathcal{D}} \simeq F(\mathbf{1}) \to F(X \otimes X^\vee) \simeq F(X) \otimes F(X^\vee)$ and similarly for ev, and the triangle identities are carried along because F preserves the associativity and unit constraints. Hence $F(X)$ is dualisable with dual $F(X^\vee)$, and the four-term composite defining $\operatorname{tr}(F\varphi)$ is the image of the four-term composite defining $\operatorname{tr}(\varphi)$; the symmetry is preserved because F is symmetric monoidal and not merely monoidal, which is the reason the hypothesis cannot be weakened.

Part (iv) is the additivity of traces, which is a theorem and not a formality: in the 1-categorical setting it is due to May (May 2001), and the coherent statement, including the compatibility data required to make "compatible with an endomorphism" precise, is established in (Hoyois et al. 2017). We shall use (iv) in the form: a filtration of a dualisable object by dualisable subquotients induces an additive decomposition of the trace. That is the shape of the Eisenstein defect filtration in section §4.6.

□

Additivity is worth pausing over. It is the reason a trace can be computed by stratifying a space, and therefore the reason that the Harder–Narasimhan stratification of Bun_G has anything to do with the trace formula. It is also the property whose failure is most easily overlooked: additivity requires the subquotients to be dualisable, and in the situations of section §4.7 they are dualisable while the total object is not, which is precisely why truncation repairs the theory.

We now record the non-example that everything else in this chapter is arranged around.

Example 4.5 (The identity with no trace). [ESTABLISHED]

Let k be a field and $\mathcal{C} = \mathrm{Mod}_k$ with its usual symmetric monoidal structure. Let $V = \bigoplus_{n\in\mathbf{N}} k$, of countably infinite rank. Then V is not dualisable, and id_V has no trace.

Suppose duality data existed. The evaluation map must be the usual pairing $V^\vee \otimes V \to k$, since ev is adjoint to a map $V^\vee \to V^\vee$ and the triangle identities force it to be the identity. Consider the canonical map

$$\iota\colon V \otimes V^\vee \longrightarrow \mathrm{End}_k(V), \qquad \iota(v\otimes\lambda)(w) = \lambda(w)\,v.$$

Its image consists exactly of the endomorphisms of finite rank: any element of $V \otimes V^\vee$ is a *finite* sum $\sum_{i=1}^{m} v_i \otimes \lambda_i$, and the associated endomorphism has rank at most m. The first triangle identity says that $\iota(\mathrm{coev}(1)) = \mathrm{id}_V$. But id_V has infinite rank. Hence no coev exists and V is not dualisable.

The obstruction is not that a numerical series diverges. Nothing diverges, because nothing is ever written down. The naive formula $\sum_{n\in\mathbf{N}} 1$ is the shadow of an absent morphism, and no regularisation of the series produces the morphism. This distinction — between a divergent expression and a missing construction — is the distinction that section §4.7 insists upon in the case of Bun_G.

Finite rank as compactness. The proof above used only one thing about V: that the image of ι consists of *finite-rank*, that is *compact*, endomorphisms. Read in that form, the argument generalises verbatim: in a compactly generated stable ∞-category, the objects one can hope to dualise are the compact ones, and the failure of dualisability is always a failure of some canonical object to be compact. Section §4.3 turns this reading into a criterion.

Before leaving the module case it is worth naming the criterion that governs it, since it is the one instance in which dualisability admits a completely explicit answer. Let A be an E_1-ring and $\mathcal{C} = \mathrm{Mod}_A$. An object M is dualisable if and only if it is *perfect*, that is, if it lies in the smallest stable subcategory containing A and closed under retracts. One direction is immediate: A itself is dualisable, and dualisable objects are closed under finite colimits and retracts, so every perfect module is dualisable. The converse uses the argument of Example 4.5 in its general form. If M is dualisable then $M \otimes_A -$ preserves products, and a module whose tensor functor preserves products is finitely presented in the derived sense; an induction on the Postnikov tower then writes M as a retract of a finite cell module. For A an ordinary Noetherian ring this recovers the classical statement that the dualisable objects of the derived category are the perfect complexes — bounded complexes of finitely generated projectives up to quasi-isomorphism — and not the coherent ones. The distinction matters: over a singular ring the structure sheaf of a closed point is coherent but not perfect, hence has no trace, and this is the elementary origin of the singular-support conditions that reappear in section §4.7. One should therefore expect, and this expectation is borne out throughout the book, that every dualisability question in geometry is a question about which sheaves are perfect rather than merely coherent, and that the answer is controlled by the singularities of the object being sheafified.

A second consequence of the same argument deserves recording because it is used silently later. Since dualisability is detected by the existence of a coevaluation, and coevaluation is a map out of the unit, dualisable objects are closed under all the operations that preserve the unit: finite limits, finite colimits, retracts, and base change along symmetric monoidal functors. They are *not* closed under infinite coproducts, and this single fact is the seed of every failure discussed in this chapter. The infinite coproduct $\bigoplus_n A$ is not dualisable while each summand is; the infinite disjoint union $\coprod_{d\in\mathbf{Z}} \mathrm{Pic}^d$ is not cohomologically proper while

each term is; the colimit over level structures of automorphic categories is not compactly generated while each level is. The same sentence explains all three.

4.3 Dualisable categories, compact generation, and THH

The objects whose traces this book computes are not modules but *categories*: the category of sheaves on a chôra, with its Hecke endomorphisms. To trace an endomorphism of a category one needs the category to be a dualisable object of a symmetric monoidal ∞-category of categories. We therefore fix

$$\mathcal{C}\mathrm{at} \;:=\; \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}},$$

the ∞-category of presentable stable ∞-categories and colimit-preserving functors, with the Lurie tensor product (Lurie 2017, sect. 4.8) and unit the category Sp of spectra. Everything below is stated relative to a base E_∞-ring R when convenient, with $\mathcal{C}\mathrm{at}_R := \mathrm{Mod}_{\mathrm{Mod}_R}$ and unit Mod_R.

Recall that $X \in \mathcal{C}$ is *compact* if $\mathrm{Map}(X, -)$ commutes with filtered colimits, that $\mathcal{C}$ is *compactly generated* if it is presentable and generated under colimits by a set of compact objects, and that a functor is *proper* in this context if it preserves compact objects.

Theorem 4.6 (Compact-Generation Criterion). [ESTABLISHED]
Let $\mathcal{C}$ be a compactly generated stable ∞-category. Then $\mathcal{C}$ is dualisable in $\mathcal{C}$at, with dual

$$\mathcal{C}^\vee \;\simeq\; \mathrm{Ind}\big((\mathcal{C}^\omega)^{\mathrm{op}}\big),$$

where $\mathcal{C}^\omega \subseteq \mathcal{C}$ is the full subcategory of compact objects. The evaluation is induced by the mapping-spectrum pairing $(\mathcal{C}^\omega)^{\mathrm{op}} \times \mathcal{C}^\omega \to \mathrm{Sp}$, and the coevaluation sends the sphere to the object $\bigoplus_i c_i^\vee \otimes c_i$ determined by a set of compact generators $\{c_i\}$ and the diagonal.

Proof of Theorem 4.6.
See (Lurie 2017, sect. 4.8.4) and, for the identification of the dual, (Ben-Zvi et al. 2010, sect. 3). We indicate the mechanism only. If $\mathcal{C} = \mathrm{Ind}(\mathcal{C}_0)$ for a small stable idempotent-complete $\mathcal{C}_0$, then for any $\mathcal{D}$ one has $\mathrm{Fun}^{\mathrm{L}}(\mathcal{C}, \mathcal{D}) \simeq \mathrm{Fun}^{\mathrm{ex}}(\mathcal{C}_0, \mathcal{D}) \simeq \mathrm{Ind}(\mathcal{C}_0^{\mathrm{op}}) \otimes \mathcal{D}$, the second equivalence because a right-exact functor out of a small stable category is determined by a bimodule. Taking $\mathcal{D} = \mathrm{Sp}$ identifies $\mathrm{Ind}(\mathcal{C}_0^{\mathrm{op}})$ as a candidate dual, and the triangle identities are checked on generators, where they reduce to the statement that a compact object is recovered from its mapping spectra out of the generators. □

The criterion is powerful but it is not sharp, and pretending otherwise would misrepresent the state of the art. There are dualisable categories that are not compactly generated — indeed the categories that arise from analytic geometry are typically of this kind, having no non-zero compact objects at all — and the correct general statement was found only recently.

Theorem 4.7 (Retract Criterion). [ESTABLISHED]
For a presentable stable ∞-category $\mathcal{C}$ the following are equivalent.
(*i*) *$\mathcal{C}$ is dualisable in $\mathcal{C}$at.*
(*ii*) *The colimit functor* $\mathrm{colim}\colon \mathrm{Ind}(\mathcal{C}) \to \mathcal{C}$ *admits a left adjoint.*
(*iii*) *$\mathcal{C}$ is a retract, in $\mathcal{C}$at, of a compactly generated stable ∞-category.*

Proof of Theorem 4.7.

This is Efimov's theorem (Efimov 2024). The equivalence (i) $\Leftrightarrow$ (ii) is the conceptual heart: the putative coevaluation, evaluated on the unit, must be an object of $C \otimes C^{\vee}$ playing the role of a "diagonal", and one shows that constructing it is the same as constructing a left adjoint to colim, which one may think of as a canonical way of writing an object as a filtered colimit. The implication (iii) $\Rightarrow$ (i) is formal, since dualisable objects are closed under retracts; (i) $\Rightarrow$ (iii) is the substantial direction and proceeds by exhibiting C as a retract of $\mathrm{Ind}(C^{\kappa})$ for suitable κ using the left adjoint produced in (ii). □

The Retract Criterion is what makes the analytic categories of Clausen–Scholze usable (Clausen and Scholze 2019, 2024). It is also what licenses the strategy of this book: we do not need our automorphic categories to be compactly generated, which they are not, but only to be retracts of compactly generated ones, which is a strictly weaker and more flexible demand. When we speak below of *certifying* dualisability we shall mean, in practice, exhibiting such a retraction.

It is worth being explicit about how much the Retract Criterion concedes. A compactly generated category is determined by a small amount of data — the category of its compact objects — and one can often write that data down. A retract of a compactly generated category need not be so determined, and the retraction itself is extra information that has to be produced. What one gains is that the criterion is stable under the operations that destroy compact generation: completion, passage to a closed subcategory, and the formation of categories of quasi-coherent sheaves on non-Noetherian or analytic objects. Efimov's own motivation was continuous K-theory, which extends the invariant of (Blumberg et al. 2013) from small stable categories to dualisable presentable ones and is localising for the sequences that arise in analytic geometry; the same extension applies to THH and hence, by Theorem 4.8, to the trace itself. In other words, the class of categories for which the traces of this book are defined is exactly the class for which continuous K-theory is defined, and one may take that coincidence as a check on the choice of framework rather than as an accident.

Theorem 4.8 (The trace of a category is topological Hochschild homology). [ESTABLISHED]
Let C be a dualisable object of Cat_R and $F\colon C \to C$ a colimit-preserving endofunctor. Then

$$\mathrm{tr}(F) \;\simeq\; \mathrm{HH}(C/R; F),$$

the Hochschild homology of C with coefficients in the bimodule determined by F. In particular $\mathrm{tr}(\mathrm{id}_C) \simeq \mathrm{THH}(C)$ *when $R = \mathbb{S}$, and if $C = \mathrm{Mod}_A^{\mathrm{perf}}$ for an E_1-ring A then* $\mathrm{tr}(\mathrm{id}) \simeq \mathrm{THH}(A)$.

Proof of Theorem 4.8.
Unwinding Definition 4.3 with the duality data of Theorem 4.6, the composite $\mathrm{Mod}_R \to C \otimes C^{\vee} \to C^{\vee} \otimes C \to \mathrm{Mod}_R$ is the two-sided bar construction on the bimodule F, which is by definition Hochschild homology; see (Toën and Vezzosi 2015) and (Hoyois et al. 2017, sect. 4). The case $C = \mathrm{Mod}_A^{\mathrm{perf}}$ is the classical identification $\mathrm{HH}(\mathrm{Mod}_A^{\mathrm{perf}}) \simeq \mathrm{THH}(A)$. □

This is the theorem that converts the abstract programme into computable mathematics: a categorical trace is a Hochschild-homology spectrum, and Hochschild-homology spectra can be computed. Two computations will be used repeatedly, and we record them now for reference rather than proving them.

Bökstedt's computation. For the ring of integers, $\pi_0\,\mathrm{THH}(\mathbf{Z}) = \mathbf{Z}$, $\pi_{2i-1}\,\mathrm{THH}(\mathbf{Z}) = \mathbf{Z}/i$ for $i \geq 1$, and $\pi_{2i}\,\mathrm{THH}(\mathbf{Z}) = 0$ for $i > 0$. Thus the first few homotopy groups are

$\mathbf{Z}, 0, 0, \mathbf{Z}/2, 0, \mathbf{Z}/3, 0, \mathbf{Z}/4, \ldots$, and $\mathrm{THH}(\mathbf{Z})$ is rationally trivial above degree zero, all of its higher homotopy being torsion. The computation is Bökstedt's; the cyclotomic structure that makes it the starting point of arithmetic trace methods is due to Bökstedt, Hsiang and Madsen (Bökstedt et al. 1993), and the modern formulation is (Nikolaus and Scholze 2018). The presence of the torsion groups $\mathbf{Z}/i$ — rather than a single copy of $\mathbf{Z}$, as an overly optimistic analogy with $\mathbf{F}_q$ might suggest — is the first indication that the arithmetic trace has more structure than the geometric one, and that the extra structure is exactly a Frobenius-like operator rather than a Frobenius.

For comparison, if k is a perfect field of characteristic p then $\mathrm{THH}(k) \simeq k[u]$ with $|u| = 2$, a polynomial ring; the trace is therefore "as large as possible" in a sense that will recur. By contrast $\mathrm{THH}(\mathbf{F}_q)$ carries a Frobenius whose homotopy fixed points recover $\mathbf{Z}_p$, and it is this recovery — not any naive point-counting — that the arithmetic theory must imitate.

We close the section with the observation that makes the compactness hypothesis unavoidable rather than convenient. Let $\mathcal{C}$ be a presentable stable ∞-category which is *not* dualisable, and let $F\colon \mathcal{C} \to \mathcal{C}$ be a colimit-preserving endofunctor. Then there is no Sp-valued invariant $\mathrm{tr}(F)$ satisfying the conclusions of Proposition 4.4; and in particular no choice of regularisation, zeta-function normalisation, or analytic continuation supplies one. To see this, suppose such an assignment existed for all pairs $(\mathcal{C}, F)$ in a class closed under tensor products and containing a non-dualisable $\mathcal{C}$. By functoriality Proposition 4.4, applied to the symmetric monoidal inclusion of the full subcategory on dualisable objects, the assignment restricts to the honest trace there. By symmetry Proposition 4.4, tr descends to a map from π_0 of the *cyclic* mapping object, that is from $\pi_0 \mathrm{Map}(\mathbf{1}, \mathcal{C} \otimes_{\mathcal{C}\otimes\mathcal{C}^{\mathrm{op}}} \mathcal{C})$; but this target is defined only once $\mathcal{C}$ has been identified with a $\mathcal{C} \otimes \mathcal{C}^{\mathrm{op}}$-module admitting a dual, which is Theorem 4.7. The obstruction is thus not a value but a domain: there is no object on which the putative trace is a function. The situation is exactly that of Example 4.5 one categorical level up.

> **Status and Caveat.** The reader should resist a natural misreading of the previous paragraph. It does not say that non-dualisable categories carry no interesting invariants; they carry many, including K-theory, which is defined without any dualisability hypothesis whatsoever (Blumberg et al. 2013). It says that the *trace*, with the four properties of Proposition 4.4, is not among them. When later chapters speak of "the trace of Frobenius on the automorphic category", the phrase is meaningful only after dualisability has been certified, and the certification is not automatic.

4.4 Smooth and proper by adjointability

Dualisability of a category is an absolute notion. What one actually verifies in geometry is a relative notion, attached to a morphism, and the translation between the two is the content of this section.

Fix a six-functor formalism in the sense of Chapter 3, following (Heyer and Mann 2024) and (Scholze 2025): to each object $\mathcal{X}$ of a geometric category is assigned a symmetric monoidal stable ∞-category $\mathcal{D}(\mathcal{X})$, and to each morphism $f\colon \mathcal{X} \to \mathcal{Y}$ a string of adjoints

$$f_! \dashv f^!, \qquad f^* \dashv f_*,$$

together with a natural transformation $f_! \to f_*$ when f is nice enough, base change, and the projection formula.

Definition 4.9 (Cohomologically smooth and cohomologically proper). [ESTABLISHED]
A morphism $f\colon \mathfrak{X} \to \mathfrak{Y}$ is

(i) *cohomologically proper* if the natural transformation $f_! \to f_*$ is an equivalence;
(ii) *cohomologically smooth* if $f^!$ is equivalent to $f^*(-) \otimes \omega_f$ for an invertible object $\omega_f \in \mathcal{D}(\mathfrak{X})$, the *dualising object* of f, and this equivalence is compatible with base change.

The names are chosen to warn the reader that these are not the classical notions. Cohomological properness is implied by, but does not imply, properness of the underlying morphism of stacks; a quotient stack $[*/G]$ with G finite of order invertible on the base is cohomologically proper without being proper in any scheme-theoretic sense, and this is the elementary case of ambidexterity treated in section §4.5. Cohomological smoothness is the notion under which Fargues and Scholze prove that Bun_G over the Fargues–Fontaine curve behaves like a smooth stack of dimension zero (Fargues and Scholze 2026), although Bun_G is not smooth in any classical sense and its underlying diamond has no tangent bundle.

Four small examples fix the meanings, and the reader should keep them in view whenever the words are used later.

The affine line $\mathbf{A}^1_k \to \operatorname{Spec} k$ is smooth and cohomologically smooth, with $\omega = k[2](1)$ in the ℓ-adic setting, but it is not proper and not cohomologically proper: $H^*_c(\mathbf{A}^1, k)$ is $k(-1)$ in degree 2 while $H^*(\mathbf{A}^1, k)$ is k in degree 0, and there is no map between them in either direction that could be an equivalence. Here the coevaluation exists and the evaluation does not, exactly as the slogan predicts.

A projective nodal cubic over k is proper, hence cohomologically proper, but not smooth; its structure morphism has $f_! = f_*$ yet $f^!$ is not f^* twisted by an invertible object, because the dualising complex is not invertible at the node. Here the evaluation exists and the coevaluation does not. The category of perfect complexes is not self-dual, and the trace of the identity — which would be the Euler characteristic — exists but is computed by a formula involving the singularity.

The classifying stack $[*/G]$ for G finite of order invertible on the base is both cohomologically smooth, with $\omega = \mathbf{1}$, and cohomologically proper, by Example 4.14; its category of sheaves is dualisable and its trace of the identity is the class function algebra, of dimension the number of conjugacy classes. If the order of G is not invertible the properness fails and, by the same example, fails infinitely.

The classifying stack $[*/\mathbf{G}_m]$ over a field is cohomologically smooth of dimension -1 but is not cohomologically proper: $H^*([*/\mathbf{G}_m], k) = k[c]$ with $|c| = 2$ is infinite dimensional, while compactly supported cohomology is concentrated in negative degrees. This is the example to hold in mind when reading the gerbe factor of Example 4.25, where the automorphism group $\mathbf{G}_m$ contributed the harmless denominator $q - 1$ only because the base field was finite and the gerbe was neutral.

The reformulation we want is in terms of adjointability of squares. Recall that a commutative square of ∞-categories is *left adjointable* if the horizontal functors admit left adjoints and the resulting Beck–Chevalley transformation is an equivalence, and *right adjointable* if the same holds for right adjoints.

Theorem 4.10 (Smooth-and-Proper Criterion). [ESTABLISHED]
Let $f\colon \mathfrak{X} \to \mathfrak{Y}$ be a morphism in a six-functor formalism and suppose $\mathcal{D}(\mathfrak{Y})$ is dualisable. Then $\mathcal{D}(\mathfrak{X})$ is dualisable as a $\mathcal{D}(\mathfrak{Y})$-module, with dual $\mathcal{D}(\mathfrak{X})$ itself, if and only if f is both cohomologically smooth and cohomologically proper. In that case the evaluation and coevaluation are

$$\mathrm{ev} = f_! \circ \Delta^*, \qquad \mathrm{coev} = \Delta_* \circ f^!,$$

with $\Delta\colon \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ *the diagonal, and for* φ *an endomorphism given by a kernel* K_φ *on* $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ *one has the Lefschetz formula*

$$\operatorname{tr}(\varphi) \;\simeq\; f_! \, \Delta^* \, K_\varphi .$$

Proof of Theorem 4.10.

For the case of perfect stacks over a field of characteristic zero this is (Ben-Zvi et al. 2010, Thm. 1.2 and §4): there, $\mathcal{X}$ perfect implies $\mathrm{QCoh}(\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}) \simeq \mathrm{Fun}^{\mathrm{L}}_{\mathcal{D}(\mathcal{Y})}(\mathrm{QCoh}\,\mathcal{X}, \mathrm{QCoh}\,\mathcal{X})$, so that all colimit-preserving endofunctors are integral transforms, and the duality data displayed above satisfy the triangle identities by base change along Δ. In the generality of an abstract six-functor formalism the statement and proof are (Scholze 2025, sect. 4) and (Heyer and Mann 2024), where cohomological smoothness is precisely the condition making coev well defined — one needs $f^!$ to be f^* twisted by an invertible object, so that $\Delta_* f^!$ lands in the right place — and cohomological properness is precisely the condition making ev satisfy the second triangle identity, since it is what allows $f_!$ to be replaced by f_* when one composes. The converse direction extracts the two conditions from the two triangle identities in turn. □

Theorem Theorem 4.10 is the technical statement that the rest of the book leans on, and its shape should be memorised: *smooth gives coevaluation, proper gives evaluation.* When one of the two fails, exactly one half of the duality data is missing, and one can often say which half. In the automorphic setting it is properness that fails and evaluation that is missing; in the Fargues–Scholze setting smoothness is available and properness is not, which is why the theory there is organised around compact generation rather than around duality.

Why this is the cobordism hypothesis in disguise. A fully dualisable object of a symmetric monoidal (∞, n)-category determines a framed n-dimensional topological field theory, by Lurie's proof of the cobordism hypothesis (Lurie 2009b). The evaluation and coevaluation of Theorem 4.10 are the values of that theory on the elementary cobordisms, and the trace is its value on the circle. The reason this book insists on dualisability rather than on some weaker finiteness condition is that dualisability is exactly the input the cobordism hypothesis requires: it is not a convenient hypothesis but the correct one, in the precise sense that weakening it destroys the field theory. When section §4.7 reports that Bun_G over an arithmetic base may fail to be dualisable, the report should be heard as saying that the corresponding field theory does not exist, and that constructing a substitute is a research programme rather than a lemma.

Proposition 4.11 (Verdier self-duality from ambidexterity). [THEOREM]

Let $f\colon \mathcal{X} \to *$ *be cohomologically smooth with dualising object* ω_f *and cohomologically proper. Then for* $\mathcal{F} \in \mathcal{D}(\mathcal{X})$ *dualisable there is a natural equivalence*

$$f_* \, \mathcal{H}\mathrm{om}(\mathcal{F}, \omega_f) \;\simeq\; \mathrm{Hom}(f_! \mathcal{F}, \mathbf{1}),$$

that is, Poincaré–Verdier duality in its naive form, with no correction term.

Proof of Proposition 4.11.

The adjunction $f_! \dashv f^!$ gives $\mathrm{Hom}(f_!\mathcal{F}, \mathbf{1}) \simeq f_* \, \mathcal{H}\mathrm{om}(\mathcal{F}, f^!\mathbf{1})$ formally, and $f^!\mathbf{1} \simeq \omega_f$ by cohomological smoothness. This much requires no properness. Properness enters when one wishes to replace f_* by $f_!$ on the left-hand side, which is what makes the duality an involution on a single category rather than a pairing between two. The corrected statement in the non-proper case replaces the equivalence by a fibre sequence whose third term is $\mathrm{cofib}(\mathrm{Nm}_f)$ applied to $\mathcal{H}\mathrm{om}(\mathcal{F}, \omega_f)$; this is the precise sense in which the third classical symptom of section §4.1 is measured by the norm map. □

4.5 Ambidexterity and the semiadditivity hierarchy

Ambidexterity is the assertion that a left adjoint and a right adjoint coincide. Stated so baldly it sounds either trivial or absurd; the content is that there is a *canonical* comparison map between them, constructed with no choices, and that asking whether this specific map is an equivalence is a meaningful and often decidable question. The construction is due to Hopkins and Lurie, in their work on ambidexterity in $K(n)$-local stable homotopy theory, and it is the single most important technical import into this book from homotopy theory. Because that paper has no entry in our bibliography we attribute it here in prose and, throughout, by name.

4.5.1 The norm map

Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a morphism for which the adjunctions $f_! \dashv f^* \dashv f_*$ all exist. There is no reason in general for $f_!$ and f_* to be comparable at all. The Hopkins–Lurie insight is that a comparison can be built by induction along the *diagonal tower* of f.

Write $\delta_f\colon \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ for the diagonal. If f is, in a suitable sense, m-truncated, then δ_f is $(m-1)$-truncated, and one may hope to have already solved the problem for δ_f by induction.

Definition 4.12 (The norm map). [ESTABLISHED]

Suppose inductively that δ_f is *ambidextrous*, so that there is a chosen equivalence $\mathrm{Nm}_{\delta_f}\colon (\delta_f)_! \xrightarrow{\sim} (\delta_f)_*$. Define the *integration transformation*

$$\mu_f\colon f_! f^* \longrightarrow \mathrm{id}$$

as the composite obtained from the counit of $(\delta_f)_! \dashv \delta_f^*$ after transporting along $\mathrm{Nm}_{\delta_f}^{-1}$ and applying base change to the square defining $\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. The *norm map* of f is then

$$\mathrm{Nm}_f\colon\ f_! \xrightarrow{f_!\eta} f_! f^* f_* \xrightarrow{\mu_f f_*} f_*,$$

where $\eta\colon \mathrm{id} \to f^* f_*$ is the unit. We say f is *ambidextrous* if Nm_f is an equivalence, and *weakly ambidextrous* if merely δ_f is ambidextrous, so that Nm_f is defined.

The induction is anchored: for f a (-2)-truncated map, that is an equivalence, both $f_!$ and f_* are inverse to f^* and Nm_f is the identity. What must be checked at each stage — and this is where the real labour of the Hopkins–Lurie paper lies — is that the chosen equivalences are coherently compatible with composition and base change, so that "ambidextrous" is a property of f and not extra structure. It is worth saying what that verification involves, since a reader who takes it on faith will misjudge the difficulty of transporting the theory to a new setting. One must show, for a composable pair $g \circ f$, that the norm of the composite is the composite of the norms; for a base change square, that the norm is compatible with the Beck–Chevalley transformations in both directions; and, at each stage of the induction, that the equivalence Nm_{δ_f} chosen for the diagonal is the one induced by the equivalences already chosen for the diagonals of the diagonal. The last condition is the delicate one: the diagonal tower of a map is not a tower of maps of the same kind, and the coherence data required to compare consecutive levels form, a priori, an infinite system. Hopkins and Lurie show that this system is generated by its first two levels, by an argument formally parallel to the collapse in Proposition 4.2 — ambidexterity, like dualisability, turns out to be a property — but the parallel is a theorem, not an observation, and it does not follow from Proposition 4.2. We do not reproduce the verification; the reader who intends to build on it should read the original, and should be warned that the coherence is not a formality.

Definition 4.13 (The semiadditivity hierarchy). [ESTABLISHED]
A space is (-2)-*finite* if it is contractible, (-1)-*finite* if it is empty or contractible, 0-*finite* if it is a finite set, and m-*finite* for $m \geq 1$ if it has finitely many components, each of which is m-truncated with finite homotopy groups. A space is π-*finite* if it is m-finite for some m.

An ∞-category $\mathcal{C}$ with all limits and colimits is m-*semiadditive* if for every m-finite space A the norm map $\mathrm{Nm}_A \colon \mathrm{colim}_A \to \lim_A$ is an equivalence, and ∞-*semiadditive* if it is m-semiadditive for all m.

The hierarchy begins with familiar cases. A (-1)-semiadditive ∞-category is one in which the initial and terminal objects agree, that is, a pointed one. A 0-semiadditive ∞-category is one in which finite coproducts and finite products agree, that is, a semiadditive one in the classical sense; every stable ∞-category is 0-semiadditive. The first new condition is 1-semiadditivity, which concerns finite groupoids, that is, classifying spaces BG of finite groups, and which is therefore the condition that the norm map for a finite group action be an equivalence — a Tate-vanishing statement.

Example 4.14 (Ambidexterity for a finite group, and its failure). [THEOREM]
Let G be a finite group, k a commutative ring, and consider $\mathcal{C} = \mathrm{Mod}_k$ with $f \colon BG \to *$, so that $f_! = (-)_{hG}$ and $f_* = (-)^{hG}$. For $X \in \mathrm{Mod}_k$ with G-action, the norm map on π_0 is induced by

$$\mathrm{Nm} \colon X_G \longrightarrow X^G, \qquad [x] \longmapsto \sum_{g \in G} g \cdot x.$$

If $|G|$ is invertible in k, then Nm is an equivalence, with inverse $y \mapsto [\,|G|^{-1} y\,]$: indeed $\mathrm{Nm}([\,|G|^{-1} y\,]) = \sum_g g \cdot |G|^{-1} y = |G|^{-1} \sum_g y = y$ for y invariant, and conversely $[\,|G|^{-1} \sum_g gx\,] = [x]$ in the coinvariants since gx and x have the same class. The higher homotopy of X_{hG} and X^{hG} vanishes for the same reason, so Nm_f is an equivalence of spectra and f is ambidextrous.

If $|G|$ is not invertible the map is not merely non-invertible; it can be zero. Take $G = \mathbf{Z}/p$, $k = \mathbf{F}_p$, $X = k$ with trivial action. Then $\mathrm{Nm} = \sum_g g = p \cdot \mathrm{id} = 0$. The cofibre of the norm is the Tate construction, and

$$\widehat{H}^n(\mathbf{Z}/p; \mathbf{F}_p) = \mathbf{F}_p \quad \text{for every } n \in \mathbf{Z},$$

so the failure is not a single group in a single degree but an infinite 2-periodic (for $p = 2$) or 2-periodic-up-to-Bockstein pattern spreading in both directions. Ambidexterity fails maximally.

The last assertion is worth carrying out with the group ring written down, since the smallest cases are where the mechanism is visible.

Example 4.15 (The group ring at $p = 2$ and at $p = 3$). [THEOREM]
Let $G = \mathbf{Z}/2 = \langle \sigma \rangle$ and $k = \mathbf{F}_2$. In characteristic 2 one has $\sigma^2 - 1 = (\sigma + 1)^2$, so with $\epsilon := \sigma + 1$ the group ring is

$$R := \mathbf{F}_2[G] = \mathbf{F}_2[\sigma]/(\sigma^2 - 1) = \mathbf{F}_2[\epsilon]/(\epsilon^2),$$

a local ring of $\mathbf{F}_2$-dimension 2 with maximal ideal (ϵ) and residue field k. The minimal free resolution of the trivial module $k = R/(\epsilon)$ is

$$\cdots \longrightarrow R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \longrightarrow k \longrightarrow 0,$$

infinite and 1-periodic. Applying $\mathrm{Hom}_R(-,k)$ kills every differential, because ϵ acts as zero on k; hence

$$H^n(\mathbf{Z}/2;\mathbf{F}_2) = \mathbf{F}_2 \quad (n \geq 0), \qquad H_n(\mathbf{Z}/2;\mathbf{F}_2) = \mathbf{F}_2 \quad (n \geq 0),$$

and splicing the resolution with its dual gives the complete resolution $\cdots \to R \xrightarrow{\epsilon} R \xrightarrow{\epsilon} R \to \cdots$, whence

$$\widehat{H}^n(\mathbf{Z}/2;\mathbf{F}_2) = \mathbf{F}_2 \quad \text{for all } n \in \mathbf{Z}, \qquad \widehat{H}^*(\mathbf{Z}/2;\mathbf{F}_2) = \mathbf{F}_2[u^{\pm 1}],\ |u| = 1.$$

The norm element is $N = 1 + \sigma = \epsilon$, which acts as zero on $k = R/(\epsilon)$, confirming $\mathrm{Nm} = 0$ directly.

At $p = 3$ the pattern changes shape but not conclusion. With $G = \mathbf{Z}/3$, $k = \mathbf{F}_3$ and $\epsilon := \sigma - 1$ one has $\sigma^3 - 1 = (\sigma - 1)^3$ in characteristic 3, so $R = \mathbf{F}_3[\epsilon]/(\epsilon^3)$, of dimension 3. The minimal resolution is now 2-periodic with alternating differentials,

$$\cdots \longrightarrow R \xrightarrow{\epsilon^2} R \xrightarrow{\epsilon} R \xrightarrow{\epsilon^2} R \xrightarrow{\epsilon} R \longrightarrow k \longrightarrow 0,$$

giving $\widehat{H}^n(\mathbf{Z}/3;\mathbf{F}_3) = \mathbf{F}_3$ for all $n \in \mathbf{Z}$, with ring structure $\Lambda(a) \otimes \mathbf{F}_3[b^{\pm 1}]$, $|a| = 1$, $|b| = 2$. The norm element is $N = 1 + \sigma + \sigma^2 = \epsilon^2$, again zero on k.

Finally the invertible case, to see the contrast concretely. Take $G = \mathbf{Z}/2$ and $k = \mathbf{F}_5$. Since 2 is invertible, with $2^{-1} = 3$ in $\mathbf{F}_5$, the elements $e_\pm = 3(1 \pm \sigma)$ are orthogonal idempotents summing to 1, so

$$\mathbf{F}_5[\mathbf{Z}/2] \;\cong\; \mathbf{F}_5 \times \mathbf{F}_5,$$

a semisimple ring. Every module is projective, all Tate cohomology vanishes, and the norm map is an isomorphism with inverse multiplication by 3. The three computations exhibit the whole dichotomy: the group ring is either semisimple, in which case ambidexterity holds and the defect is zero, or local of positive nilpotency degree, in which case the defect is non-zero in every degree of $\mathbf{Z}$. There is no intermediate behaviour, and no size of $|G|$ interpolates between them.

The example is worth holding onto because it exhibits, in the smallest possible case, the phenomenon that section §4.7 will meet in the largest: the norm map is not approximately an equivalence, and its failure is not concentrated. Either one has an invertibility statement or one has an infinite defect.

4.5.2 Hopkins–Lurie and height

Theorem 4.16 (Hopkins–Lurie ambidexterity). [ESTABLISHED]
For each $n \geq 0$ and each prime p, the ∞-category $\mathrm{Sp}_{K(n)}$ of $K(n)$-local spectra is ∞-semiadditive: for every π-finite space A, the norm map $\mathrm{Nm}_A \colon \mathrm{colim}_A \to \lim_A$ is an equivalence in $\mathrm{Sp}_{K(n)}$.

This is the main theorem of Hopkins and Lurie's *Ambidexterity in $K(n)$-local stable homotopy theory*; we mark it E and attribute it by name, having no bibliography key for it. Subsequent work of Carmeli, Schlank and Yanovski established the converse-flavoured statements that make the hierarchy a genuine invariant: an ∞-semiadditive ∞-category has a well-defined *height*, computed by the invertibility of certain "cardinality" elements $|A| \in \pi_0 \mathrm{End}(\mathbf{1})$, and the height of $\mathrm{Sp}_{K(n)}$ is n. The reader should note that height is *not* the same as the truncation level m of m-semiadditivity: m measures which spaces one may integrate over, height measures how degenerate the integration is.

What ambidexterity replaces. The classical route to Poincaré duality runs through manifolds: one needs a fundamental class, hence an orientation, hence a tangent bundle, hence a smooth structure. Ambidexterity produces the same conclusion — $f_! \simeq f_*$, hence duality — with no manifold anywhere. The only input is a tower of diagonals. This is why the notion is the right one for arithmetic, where $\operatorname{Spec} \mathbf{Z}$ has no tangent bundle in any sense that supports a fundamental class, and where every attempt to manufacture one has foundered. Ambidexterity is duality without manifolds.

That observation invites an analogy which we state as a conjecture precisely so that it can be attacked, and which we flag as the place where the present framework is most likely to be wrong.

Conjecture 4.17 (Height is arithmetic dimension). [CONJECTURE]

Let $\mathfrak{X}_{\mathbf{Z}}$ be the arithmetic chôra of Chapter 2 and let $\mathcal{D}(\mathfrak{X}_{\mathbf{Z}})$ be its category of sheaves. Then $\mathcal{D}(\mathfrak{X}_{\mathbf{Z}})$ is ∞-semiadditive after suitable completion, and its height in the sense of Carmeli–Schlank–Yanovski equals the arithmetic dimension of $\mathfrak{X}_{\mathbf{Z}}$; in particular for $\mathfrak{X}_{\mathbf{Z}} = \operatorname{Spec} \mathbf{Z}$ compactified at the archimedean place, the height is 1 and the semiadditive cardinality of $B\mathbf{Z}/p$ is a unit times p.

> **Status and Caveat.** Conjecture Conjecture 4.17 is the most speculative statement in this chapter and possibly in the book. The analogy "chromatic height = arithmetic dimension" is seductive and has produced correct predictions in the function-field case, where the Fargues–Scholze picture assigns dimension zero to Bun_G and the corresponding categories are indeed 1-semiadditive after ℓ-adic completion. But the two hierarchies are indexed by different data — one by a prime and a chromatic integer, the other by a ramification behaviour at the archimedean place — and no functor is known relating them. A reader who finds this conjecture false should not conclude that the rest of the chapter falls; nothing later depends on it. It is included because a framework that cannot say what would refute it is not a framework.

Proposition 4.18 (Norm Criterion). [THEOREM]

Let $f\colon \mathfrak{X} \to \mathfrak{Y}$ be weakly ambidextrous. The following are equivalent.

(i) f is ambidextrous.
(ii) f is cohomologically proper in the sense of Definition 4.9.
(iii) The functor $f_!$ preserves all limits.
(iv) For every $\mathcal{F} \in \mathcal{D}(\mathfrak{X})$ the object $\operatorname{cofib}(\mathrm{Nm}_f)(\mathcal{F})$ vanishes.

Moreover, if f is ambidextrous and cohomologically smooth then $\mathcal{D}(\mathfrak{X})$ is dualisable over $\mathcal{D}(\mathfrak{Y})$ and the Lefschetz formula of Theorem 4.10 holds.

Proof of Proposition 4.18.

(i) $\Leftrightarrow$ (iv) is a restatement, and (i) $\Leftrightarrow$

(ii) holds because Nm_f *is* the natural transformation $f_! \to f_*$ of the six-functor formalism whenever both are defined; this compatibility is part of the construction in (Heyer and Mann 2024, sect. 4) and is the reason Definition 4.9 was phrased as it was. For (i) $\Rightarrow$ (iii): a right adjoint preserves limits, and under ambidexterity $f_!$ is equivalent to the right adjoint f_*. For (iii) $\Rightarrow$ (i): if $f_!$ preserves limits then, being a colimit-preserving functor between presentable categories that also preserves limits, it admits a left adjoint and is itself a right adjoint; the uniqueness of adjoints identifies it with f_*. To see that the identification is Nm_f and not some other equivalence, evaluate both on the unit and use the triangle identity for $f^* \dashv f_*$: the composite $f_! \xrightarrow{f_!\eta} f_!f^*f_* \xrightarrow{\mu_f f_*} f_*$ becomes, after applying f^* and using base change along the diagonal, the counit of $(\delta_f)_! \dashv \delta_f^*$ transported through $\mathrm{Nm}_{\delta_f}^{-1}$, which is an equivalence by weak ambidexterity; since a natural transformation between right adjoints is determined by its effect after applying

f^* when f^* is conservative, and f^* is conservative on the essential image of $f_!$, the two identifications agree. The final clause is Theorem 4.10 combined with (ii).

□

The Norm Criterion is the form in which ambidexterity will be checked in practice, because (iii) is often accessible: one asks whether compactly supported cohomology commutes with infinite products. For a quasi-compact space it does; for a space with infinitely many components of unbounded complexity it does not; and Bun_G has infinitely many components.

4.6 The Ambidexterity–Compactness Principle

We can now state the principle towards which the chapter has been moving. It is the organising claim of Part I, and later chapters will refer to it by name.

Principle 4.19 (Ambidexterity–Compactness Principle). [CONJECTURE]
Let $\mathfrak{X}$ be an automorphic chôra with structure morphism $f\colon \mathfrak{X} \to *$ over an arithmetic base, and let $\mathcal{D}(\mathfrak{X})$ be its category of sheaves. Then the following are equivalent, and the classical failures of the automorphic theory are the failure of these conditions.

(i) *f is ambidextrous: $\mathrm{Nm}_f\colon f_! \to f_*$ is an equivalence.*
(ii) *$\mathcal{D}(\mathfrak{X})$ is dualisable in $\mathcal{C}\mathrm{at}$ and the unit object is compact.*
(iii) *The categorical trace of every Hecke endomorphism of $\mathcal{D}(\mathfrak{X})$ exists and is computed by the Lefschetz formula $\mathrm{tr}(\varphi) \simeq f_!\Delta^* K_\varphi$.*

Moreover, when these fail, the defect $\mathrm{cofib}(\mathrm{Nm}_f)$ is a single object which simultaneously measures the continuous spectrum, the cohomological amplitude ℓ_0, and the failure of Poincaré duality.

The status mark deserves comment. The implications (ii) ⇔ (iii) and (i) ⇒ (ii) are theorems in the generality of section §4.4 — they are Theorem 4.10 and Proposition 4.18 — and if the Principle asserted only those, it would be marked *T*. What makes it a conjecture is the final sentence, together with the implicit claim that the notion of "automorphic chôra over an arithmetic base" has been defined precisely enough for the equivalences to be checkable. Neither the base-change compatibilities needed for Proposition 4.18 nor the identification of the defect with the continuous spectrum has been established in the arithmetic case. We therefore mark the Principle *C* and, in the remainder of the section, extract from it the one part that can be proved.

The filtration drawn in the middle of Plate 4.2 is not decorative; it is a theorem in the function-field case, and its shape is what makes the arithmetic conjecture plausible.

Theorem 4.20 (Eisenstein defect filtration). [THEOREM]
Let X be a smooth projective geometrically connected curve over $\mathbf{F}_q$, G a split reductive group over $\mathbf{F}_q$, and $f\colon \mathrm{Bun}_G(X) \to *$. Let A_G denote the maximal split central torus of G. Then the defect $\mathrm{cofib}(\mathrm{Nm}_f)$ admits a finite filtration whose associated graded pieces are indexed by the conjugacy classes of proper parabolic subgroups $P \subsetneq G$ together with the contribution of A_G, and whose P-th graded piece is

$$\mathrm{gr}_P \;\simeq\; \mathrm{Ind}_P^G\big(\mathrm{cofib}(\mathrm{Nm}_{f_M})\big),$$

where M is the Levi quotient of P, f_M the structure morphism of $\mathrm{Bun}_M(X)$, and Ind_P^G the parabolic-induction functor obtained as $p_!\,q^*$ from the correspondence $\mathrm{Bun}_G \xleftarrow{p} \mathrm{Bun}_P \xrightarrow{q} \mathrm{Bun}_M$. The A_G-contribution is the defect of the morphism $\mathrm{Bun}_{A_G}(X) \to *$, which is the constant-degree direction.

Proof of Theorem 4.20.

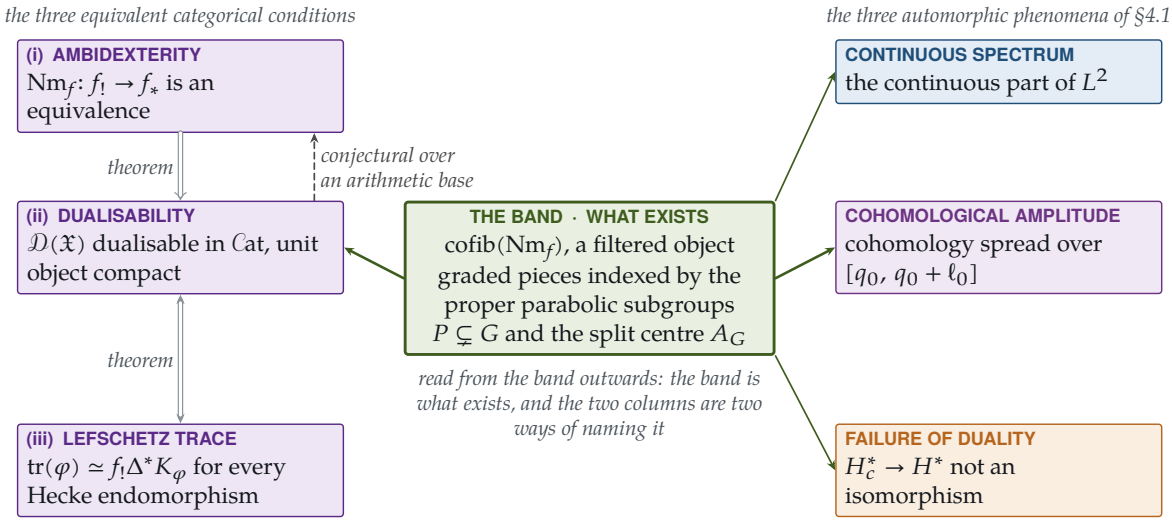


Plate 4.2. The architecture of the Ambidexterity–Compactness Principle. The left column lists the three equivalent categorical conditions — ambidexterity of the structure morphism, dualisability of the sheaf category with compact unit, and existence of the Lefschetz trace — with double arrows marking the implications that are theorems and single dashed arrows those that remain conjectural over an arithmetic base. The right column lists the three automorphic phenomena of section §4.1, and the horizontal band joining the columns is the single object $\mathrm{cofib}(\mathrm{Nm}_f)$, drawn as a filtered object whose graded pieces are indexed by the proper parabolic subgroups of G together with the split centre A_G. The plate should be read from the band outwards: the band is what exists, and the two columns are two ways of naming it.

What it does not show: that the band exists over an arithmetic base: the filtration of $\mathrm{cofib}(\mathrm{Nm}_f)$ *is a theorem in the function-field case only, and the Principle that joins the columns through it is marked* [C].

The input is the Harder–Narasimhan stratification. Every G-bundle on X has a canonical reduction to a parabolic P with Levi M such that the induced M-bundle is semistable and the associated slope vector is dominant regular; the resulting decomposition of Bun_G into locally closed substacks $\mathrm{Bun}_G^{(P,\mu)}$ is due, in the form we use, to Harder–Narasimhan for vector bundles, Behrend for general reductive G, and Schieder for the compatibility with degeneration. The open substacks $\mathrm{Bun}_G^{\leq \mu}$ obtained by bounding the slope are quasi-compact, by Drinfeld–Simpson, and their structure morphisms are ambidextrous after ℓ-adic completion.

Now filter Bun_G by the $\mathrm{Bun}_G^{\leq \mu}$ and apply the localisation sequences $j_! j^* \to \mathrm{id} \to i_* i^*$ to both $f_!$ and f_*. The norm map is compatible with these sequences — this is base change, which holds for Nm by construction in Definition 4.12 — so $\mathrm{cofib}(\mathrm{Nm}_f)$ inherits the filtration. On each open stratum the norm is an equivalence, by quasi-compactness and the previous paragraph, so those graded pieces vanish. What survives is indexed by the strata at infinity, that is by the pairs (P, μ) with P proper, together with the direction along which μ is unbounded even for fixed P, which is the A_G-direction. On the (P, μ)-stratum, the identification of the graded piece with Ind_P^G applied to the M-defect is hyperbolic localisation in the sense of Braden, in the form used by Drinfeld–Gaitsgory and Gaitsgory–Kazhdan–Rozenblyum–Varshavsky; the key point is that the correspondence $\mathrm{Bun}_G \leftarrow \mathrm{Bun}_P \to \mathrm{Bun}_M$ has contracting fibres in the direction of increasing slope, so that $p_! q^*$ and $p_* q^!$ agree up to a shift on the stratum. Finiteness of the filtration is finiteness of the set of conjugacy classes of parabolics.

□

Where the defect lives for a torus. A torus T has no proper parabolic subgroups. Theorem Theorem 4.20 therefore predicts that the entire defect of Bun_T comes from the $A_T = T$ direction

— the degree, or in the arithmetic case the Arakelov degree. This is correct and is verified by direct computation in Example 4.25 and §4.8. Readers who have seen the Eisenstein filtration stated for semisimple groups, where the central term is absent, should note that the central term cannot be dropped in general: for GL_n it is the determinant direction and it carries the poles of the standard L-function at $s = 0$ and $s = 1$.

We record the consequence for the trace, which is the form in which the filtration will be used.

Corollary 4.21 (Additive decomposition of the corrected trace). [THEOREM]
In the situation of Theorem 4.20, let φ be a Hecke endomorphism preserving the Harder–Narasimhan filtration. Then the truncated traces $\mathrm{tr}(\varphi|_{\leq\mu})$ exist for all μ, and

$$\mathrm{tr}(\varphi|_{\leq\mu'}) - \mathrm{tr}(\varphi|_{\leq\mu}) \;=\; \sum_{\mu<\nu\leq\mu'} \mathrm{tr}(\varphi|_{\mathrm{gr}_\nu}),$$

each term being the trace of φ on an Ind_P^G-induced piece. The limit $\lim_{\mu\to\infty}\mathrm{tr}(\varphi|_{\leq\mu})$ exists if and only if all but finitely many terms vanish.

Proof of Corollary 4.21.
Each $\mathrm{Bun}_G^{\leq\mu}$ is quasi-compact with ambidextrous structure morphism, so Proposition 4.18 and Theorem 4.10 apply and the truncated traces exist. The displayed identity is additivity of traces, Proposition 4.4, applied to the fibre sequences of the stratification, whose terms are dualisable even though the total object is not. The final sentence is the definition of convergence of a filtered colimit of traces in a discrete abelian group.
□

Corollary Corollary 4.21 is the categorical shadow of Arthur's truncation. Arthur truncates automorphic forms by a parameter T, obtains a convergent distribution J^T, and shows that J^T is a polynomial in T whose constant term is the invariant trace formula (Arthur 2013). Here we truncate the stack by a slope parameter μ, obtain a well-defined trace, and observe that the μ-dependence is governed by the graded pieces. That the two truncations agree is not proved in this book; we state the expected comparison so that it is on record.

The comparison, if it holds, would explain a feature of the trace formula that has always looked like an artefact of the method. Arthur's J^T is a polynomial in T, and the invariant distribution is extracted as its constant term. A polynomial dependence is exactly what Corollary 4.21 predicts when the graded pieces of the filtration are indexed by a lattice of rank equal to the number of independent unbounded directions: each direction contributes one degree to the polynomial, and the constant term is the value one would obtain if the object were proper. On this reading the polynomial is not a computational device but a record of the geometry at infinity, and the number of its coefficients counts the proper parabolic classes together with the split centre. The prediction is testable in the smallest non-trivial case, $G = \mathrm{GL}_2$ over $\mathbf{Q}$, where the Borel is the only proper parabolic and the split centre is one-dimensional, so that the degree should be one; and it is indeed one.

A second consequence is worth stating because it changes what one should look for. If the defect carries the continuous spectrum, then a trace formula which discards the defect discards precisely the Eisenstein information, and the resulting identity relates only the discrete parts of the two sides. That is the classical situation, and it is why the classical trace formula requires a separate treatment of the continuous contributions. The categorical formulation of this chapter suggests the opposite discipline: keep the defect, since it is a single object with a finite filtration, and read the classical continuous terms off its graded pieces. Whether that

discipline can be carried out is Conjecture 4.22; that it is the right discipline is the design decision recorded in Principle 4.19.

Conjecture 4.22 (Defect poles are intertwining poles). [CONJECTURE]
Let $\mathfrak{X}$ be the automorphic chôra of G over a number field F, and let $\mathcal{M}$ denote the Mellin transform of the μ-dependence of $\mathrm{tr}(\varphi|_{\leq\mu})$ along the A_G-direction. Then $\mathcal{M}$ extends meromorphically, and its poles coincide, with multiplicity, with the poles of the standard intertwining operators $M(w,\lambda)$ appearing in Langlands's constant-term formula for Eisenstein series. In particular the P-graded piece of Theorem 4.20 contributes exactly the poles associated with the Weyl-group elements w with $w^{-1}Pw \neq P$.

> **Status and Caveat.** The framework as first stated gives the Eisenstein-defect theorem and the pole statement together, marked *T*. We have split them, and downgraded the second half to *C*. The reason is that the filtration argument of Theorem 4.20 is a formal consequence of stratification plus base change plus hyperbolic localisation, all of which are available in the function-field case, whereas the identification of the Mellin poles with intertwining poles requires an analytic input — the meromorphic continuation of Eisenstein series — which has no categorical proof at present. The divergence from that first statement is deliberate and is recorded here rather than left silent.

There is one further consequence worth stating, because it is the bridge to the deformation-theoretic material of Part III.

Conjecture 4.23 (Arithmetic Koszul-duality patching). [CONJECTURE]
Let R_∞ be the patched deformation ring and M_∞ the patched module of Calegari–Geraghty type, in the setting where the defect ℓ_0 is positive. Then M_∞ is a compact generator of the category of modules over the Koszul dual of R_∞, and the resulting equivalence identifies the amplitude ℓ_0 with the cofibre of the norm map of Principle 4.19 restricted to the archimedean place.

The point of stating Conjecture 4.23 here, rather than in Part III where it belongs, is that it is the concrete payoff of dualisability. If M_∞ is a compact generator, then the patched category is compactly generated, hence dualisable by Theorem 4.6, hence carries a trace; and the trace is what the $R = T$ theorems compute. The chain of implications runs from compactness to duality to trace to modularity, and it breaks at the first link if compactness fails.

4.7 Bun_G over an arithmetic base is not obviously dualisable

This is the section the chapter exists for. Everything above has been preparation; what follows is an account of a gap, and the account is meant to be uncomfortable.

4.7.1 What goes wrong

Three distinct things go wrong, and it is important to separate them, because they have different repairs and different degrees of hopelessness.

Non-quasi-compactness. The stack Bun_G is not quasi-compact. Already for $G = \mathrm{GL}_1$ over a curve X it has $\pi_0 = \mathrm{Pic}(X)/\mathrm{Pic}^0(X) \cong \mathbf{Z}$, so infinitely many connected components; for general G it has, in addition, unbounded Harder–Narasimhan type within each component of π_1. Non-quasi-compactness alone would be survivable — a disjoint union of dualisable categories indexed by a set is dualisable if one takes the product, and the issue is only whether one takes the sum or the product. But $f_!$ takes the sum and f_* takes the product, and these disagree precisely when the index set is infinite. The elementary form of the failure is thus already visible: for $f\colon \coprod_{i\in I} * \to *$ with I infinite, $f_!\mathcal{F} = \bigoplus_I \mathcal{F}_i$ while $f_*\mathcal{F} = \prod_I \mathcal{F}_i$, and Nm_f is the inclusion of the sum in the product, which is never an equivalence.

Infinite type of the automorphic side. The automorphic side is not merely infinite; it is infinite in a way that resists reindexing. The space of automorphic forms on $G(F)\backslash G(\mathbf{A})$

is a smooth representation of infinite length, and its decomposition involves a continuous parameter. Categorically, the object one wants to be compact — the "automorphic sheaf" whose endomorphisms form the Hecke algebra — is a colimit over level structures, and compactness would require the colimit to stabilise. It does not: raising the level produces new vectors at every stage. Over $\mathbf{F}_q$ this is controlled because the relevant cohomology, after truncation, is finite-dimensional; over $\mathbf{Q}$ there is an additional archimedean direction along which nothing is finite, and this is exactly the direction that produces ℓ_0.

No properness statement to invoke. Here the assignment's usual phrasing needs correcting, and we correct it explicitly because the loose version has caused confusion. It is sometimes said that $\mathrm{Bun}_G(X)$ over $\mathbf{F}_q$ is proper, and that the arithmetic case fails because the analogous properness is unavailable. This is not right: $\mathrm{Bun}_G(X)$ over $\mathbf{F}_q$ is *not* proper. It is a smooth algebraic stack, locally of finite type, with affine diagonal, and it is neither quasi-compact nor separated in the sense that would give properness. What is true — and what does the work in the function-field theory — is a package of three substitutes, none of which is properness and all of which are theorems:

1. *Truncation.* The Harder–Narasimhan strata assemble into an exhaustion of Bun_G by quasi-compact open substacks $\mathrm{Bun}_G^{\leq \mu}$. This is due to Harder–Narasimhan in the vector-bundle case, Behrend in general, with the requisite finiteness and the reduction to Levi subgroups established by Drinfeld–Simpson and, for the degeneration behaviour, by Schieder.
2. *Compact generation of the truncated categories.* Each $\mathcal{D}(\mathrm{Bun}_G^{\leq \mu})$, in the appropriate ℓ-adic or ind-coherent sense, is compactly generated (Gaitsgory 2013; Arinkin and Gaitsgory 2015), hence dualisable by Theorem 4.6. Compactness is delicate: for ind-coherent sheaves one must impose nilpotent singular support, and the resulting category $\mathcal{D}_{\mathcal{N}}(\mathrm{Bun}_G)$ is the one that is compactly generated, not the naive one.
3. *Miraculous duality.* Drinfeld and Gaitsgory constructed the pseudo-identity functor $\mathrm{Ps\text{-}Id}\colon \mathcal{D}(\mathrm{Bun}_G) \to \mathcal{D}(\mathrm{Bun}_G)$, given by $\Delta_! \Delta^!$ on the diagonal, and conjectured — with the conjecture now established as part of the proof of geometric Langlands (Arinkin, Beraldo, Chen, et al. 2024) — that it is an equivalence. The statement that Ps-Id is an equivalence is precisely the statement that Bun_G is *cohomologically* proper even though it is not proper: it identifies $f_!$ with f_* up to the twist that miraculous duality supplies. This is the real replacement for properness, and it is a hard theorem.

That is the function-field picture. The reader should notice that in none of the three items does properness appear.

> **Status and Caveat.** It is tempting to regard the arithmetic case as the function-field case with harder estimates. It is not. Each of the three substitutes above fails in a *structural* way over an arithmetic base, not a quantitative one. There is no Harder–Narasimhan stratification of the automorphic chôra at the archimedean place, because there is no notion of slope that takes values in a discrete set — the Arakelov degree is real, not integral, and the strata form a continuum rather than a locally finite family. There is no known compact generator of the arithmetic automorphic category, and Conjecture 4.23 is precisely the assertion that one exists. And there is no miraculous duality: the diagonal of the arithmetic chôra is not known to be cohomologically smooth, so $\Delta_! \Delta^!$ is not even defined in the required generality. Reporting the arithmetic case as "expected to work similarly" would be a misrepresentation.

4.7.2 What Fargues–Scholze does and does not supply

The one p-adic instance in which a Bun_G has been mastered is that of Fargues and Scholze (Fargues and Scholze 2026), where X is the Fargues–Fontaine curve associated with a local field E and Bun_G is the stack of G-bundles on it. Their results are exactly of the type this chapter requires, and it is worth being precise about their scope.

They prove that Bun_G is a small v-stack which is ℓ-cohomologically smooth of dimension zero; that its Harder–Narasimhan, here Newton, stratification has strata $\mathrm{Bun}_G^b \cong [*/\widetilde{G}_b(E)]$ for $b \in B(G)$, each a classifying stack of a locally profinite group; and that the category $D_{\mathrm{lis}}(\mathrm{Bun}_G, \Lambda)$ of lisse-étale sheaves is compactly generated. Compact generation plus Theorem 4.6 gives dualisability, and cohomological smoothness gives the coevaluation half of Theorem 4.10. Hecke operators are then defined by correspondences, and the spectral action is constructed.

What is *not* supplied is a properness statement. The stratification is infinite (indexed by $B(G)$, which is infinite for non-trivial G), and $f_!$ still differs from f_*. The theory is organised so that this does not matter: one works with compact objects and their duals rather than with a trace of the identity, and the "trace" that appears is a trace of a Hecke operator on a *finite* piece. Read through the lens of this chapter, Fargues–Scholze establishes clause (ii) of Principle 4.19 without clause (i), and lives with the consequence.

This is instructive. It suggests that the arithmetic programme should aim first at dualisability, which is a compactness statement, and only afterwards at ambidexterity, which is a properness statement. It also suggests that ambidexterity may simply be false in the arithmetic case, and that the correct object is not a trace but a trace-with-defect. That is the reading under which Theorem 4.20 is not a regrettable correction but the theorem.

There is a third strategy in the literature which avoids the problem rather than solving it, and which deserves mention because it is the only one that has produced unconditional theorems over a global function field. Vincent Lafforgue's construction of the automorphic-to-Galois direction of the Langlands correspondence for a reductive group over a function field (Lafforgue 2018) proceeds through *excursion operators*: cohomology classes of moduli of shtukas with several legs, which are assembled into an algebra acting on the space of cuspidal automorphic forms. The construction never forms the trace of an endomorphism of $\mathcal{D}(\mathrm{Bun}_G)$. It works with the cuspidal part, which is finite dimensional after fixing a level and a central character, and it obtains the Galois representations from the algebra of excursion operators rather than from a Lefschetz count. Read in the language of this chapter, Lafforgue replaces a trace on a non-dualisable object by a family of traces on dualisable subquotients, indexed by the number of legs, and extracts the arithmetic content from the compatibilities among them.

The lesson for the arithmetic case is double-edged. On the one hand it shows that theorems can be proved without resolving the dualisability question, which is encouraging. On the other hand it shows that cuspidality is doing essential work: the passage to the cuspidal part is exactly the passage to the locus where the Harder–Narasimhan truncation is unnecessary, because cusp forms are compactly supported modulo the centre. In the arithmetic case there is no reason to expect the cuspidal part alone to carry the trace one wants, since the phenomena of (Calegari and Geraghty 2018) and (Galatius and Venkatesh 2018) — torsion classes, degree shifts, derived structure — are visible already in the cuspidal range. Avoiding the defect is not the same as understanding it, and this book has chosen to try to understand it.

4.7.3 What must hold

We state the requirement as a conjecture, in the sharpest form we can justify.

Conjecture 4.24 (Arithmetic dualisability). [CONJECTURE]
Let F be a number field, G a reductive group over F, and $\mathrm{Bun}_G^{\mathrm{ar}}$ the arithmetic automorphic chôra of Chapter 2, with structure morphism f. Then:

(*i*) (Truncation.) *There is an exhaustion of $\mathrm{Bun}_G^{\mathrm{ar}}$ by open substacks $\mathfrak{U}_\mu$, indexed by μ in a directed poset, such that each $\mathcal{D}(\mathfrak{U}_\mu)$ is dualisable and each transition functor preserves compact objects.*

(*ii*) (Dualisability.) *$\mathcal{D}(\mathrm{Bun}_G^{\mathrm{ar}})$ is a retract in $\mathcal{C}\mathrm{at}$ of a compactly generated stable ∞-category, hence dualisable by Theorem 4.7.*

(*iii*) (Controlled defect.) *The morphism f is weakly ambidextrous, and $\mathrm{cofib}(\mathrm{Nm}_f)$ admits a finite filtration as in Theorem 4.20, with graded pieces indexed by the proper parabolic subgroups of G over F together with the split centre and the archimedean direction, the last contributing an object of amplitude exactly ℓ_0.*

(*iv*) (Trace with defect.) *For every Hecke endomorphism φ compatible with the exhaustion, the truncated traces $\mathrm{tr}(\varphi|_{\mathfrak{U}_\mu})$ exist and their μ-dependence is governed by (iii).*

Clause (ii) alone suffices for the existence of a categorical trace of φ; clauses (iii) and (iv) are what identify that trace with the automorphic side.

Three remarks on the shape of the conjecture. First, we have deliberately *not* conjectured ambidexterity. Conjecturing Nm_f to be an equivalence would contradict Example 4.26 below, where the defect is computed and is non-zero. Second, clause (ii) demands only a retraction, using Theorem 4.7 rather than Theorem 4.6; this is a weaker and, we believe, more realistic demand, and it is the one that the analytic-geometry machinery of Clausen and Scholze (Clausen and Scholze 2024) is designed to supply. Third, clause (iii) contains the only new prediction: that the archimedean contribution has amplitude exactly ℓ_0. That number is computable — $\ell_0 = \operatorname{rank} G(F \otimes \mathbf{R}) - \operatorname{rank} K_\infty Z_\infty$, so $\ell_0 = 0$ for $\mathrm{GL}_2/\mathbf{Q}$, $\ell_0 = 1$ for GL_2 over an imaginary quadratic field and for $\mathrm{GL}_3/\mathbf{Q}$, and $\ell_0 = \lfloor (n-1)/2 \rfloor$ for $\mathrm{GL}_n/\mathbf{Q}$ — so the prediction is testable against the known structure of the cohomology of arithmetic groups, where it matches the interval $[q_0, q_0 + \ell_0]$ of (Calegari and Geraghty 2018) and (Galatius and Venkatesh 2018).

What would refute the conjecture. If someone exhibits a reductive G over a number field for which $\mathcal{D}(\mathrm{Bun}_G^{\mathrm{ar}})$ has no non-zero compact objects and is not a retract of a compactly generated category, clause (ii) is false and the programme of this book has no object to trace. The most likely source of such an example is the archimedean place, where the relevant categories are categories of $(\mathfrak{g}, K)$-modules or their derived analogues, and where compactness of the trivial module already fails for $\mathrm{SL}_2(\mathbf{R})$ in the naive formulation. We regard the identification of the correct archimedean category as the central open problem of Part I.

4.8 Explicit computations

Nothing above will convince a sceptical reader who suspects that the failure of dualisability is an artefact of insufficient care with definitions. This section therefore computes. The first computation exhibits a chôra whose trace does not exist, and the divergence is exhibited numerically rather than asserted. The second works GL_1 over $\mathbf{Q}$ completely by hand and identifies the defect with an object one can point at. The third does rank two over the projective line, where the Harder–Narasimhan strata can be listed and their masses summed exactly. The fourth descends to a finite Hecke algebra, presented by generators and relations, where dualisability holds but semisimplicity fails and one can watch the two part company.

4.8.1 A chôra with no trace

Example 4.25 (The divergent mass of $\mathrm{Bun}_{\mathbf{G}_m}$). [THEOREM]

Let E be the elliptic curve $y^2 + y = x^3 - x^2 - 10x - 20$ over $\mathbf{Q}$, of conductor 11, and let $X = E_{\mathbf{F}_5}$ be its reduction modulo 5, which is good since $5 \nmid 11$. Since $-10x - 20 \equiv 0$ modulo 5, the reduction is $y^2 + y = x^3 - x^2$ over $\mathbf{F}_5$. Complete the square: $y^2 + y = c$ has solutions if and only if $4c + 1$ is a square in $\mathbf{F}_5$, and then it has two, or one when $4c + 1 = 0$. The squares in $\mathbf{F}_5$ are $\{0, 1, 4\}$. Tabulating $c = x^3 - x^2$ and $4c + 1$ in $\mathbf{F}_5$:

x	0	1	2	3	4
$c = x^3 - x^2$	0	0	4	3	3
$4c + 1$	1	1	2	3	3
$\#\{y\}$	2	2	0	0	0

There are four affine points and one point at infinity, so

$$\#X(\mathbf{F}_5) = 5, \qquad a_5 = 5 + 1 - 5 = 1,$$

in agreement with the weight-two newform of level 11, whose q-expansion begins $q - 2q^2 - q^3 + 2q^4 + q^5 + \cdots$.

The zeta function of X is

$$Z(X, t) = \frac{1 - a_5 t + 5t^2}{(1 - t)(1 - 5t)} = \frac{1 - t + 5t^2}{(1 - t)(1 - 5t)},$$

and $\#\operatorname{Pic}^0(X)(\mathbf{F}_5) = P(1) = 1 - 1 + 5 = 5$.

Now consider $\mathrm{Bun}_{\mathbf{G}_m}(X) = \operatorname{Pic}(X)$ as an algebraic stack over $\mathbf{F}_5$. It is a $\mathbf{G}_m$-gerbe over the Picard scheme, so every point has automorphism group $\mathbf{G}_m(\mathbf{F}_5) = \mathbf{F}_5^\times$, of order 4. The groupoid mass of the degree-d component is therefore

$$|\operatorname{Pic}^d(X)(\mathbf{F}_5)| = \frac{\#\operatorname{Pic}^0(X)(\mathbf{F}_5)}{\#\mathbf{F}_5^\times} = \frac{5}{4}$$

for every $d \in \mathbf{Z}$, since Pic^d is a torsor under Pic^0 and is non-empty for all d because X has a rational point. Hence

$$|\mathrm{Bun}_{\mathbf{G}_m}(X)(\mathbf{F}_5)| = \sum_{d \in \mathbf{Z}} \frac{5}{4} = +\infty.$$

This is the promised non-example. The "trace of Frobenius on $\mathcal{D}(\mathrm{Bun}_{\mathbf{G}_m}(X))$" would, if it existed, be given by the Lefschetz formula of Theorem 4.10 as the groupoid mass just computed, and that mass is a divergent sum of equal positive terms. Note carefully what kind of divergence this is. It is not a sum of decaying terms that fails to converge absolutely, so no summation method applies; it is a sum of the constant $5/4$ over $\mathbf{Z}$. And — exactly as in Example 4.5 — the divergence is a symptom rather than the disease. The disease is that Nm_f for $f\colon \operatorname{Pic}(X) \to *$ is the inclusion $\bigoplus_{d \in \mathbf{Z}} \to \prod_{d \in \mathbf{Z}}$ of the sum of the degree components into their product, which is not an equivalence, so that f is not ambidextrous, $\mathcal{D}(\operatorname{Pic} X)$ is not dualisable in the required relative sense, and no trace exists to be regularised.

By contrast, each truncation $\operatorname{Pic}^{[-N,N]}(X) = \coprod_{|d| \le N} \operatorname{Pic}^d$ is quasi-compact, its category is dualisable, and its trace is the finite number $(2N + 1) \cdot 5/4$. Corollary Corollary 4.21 is visible

here in its simplest form: the graded pieces are the individual degrees, each contributing $5/4$, and the limit fails to exist because infinitely many of them are non-zero.

The projective line, and why the defect has rank two. The same computation for $X = \mathbf{P}^1_{\mathbf{F}_q}$ gives $\mathrm{Pic}(X) \cong \mathbf{Z}$ with mass $1/(q-1)$ per degree, again divergent. But here one can see the *shape* of the defect. The zeta function

$$Z(\mathbf{P}^1, t) = \frac{1}{(1-t)(1-qt)}$$

has exactly two factors in the denominator and none in the numerator. Those two factors are the two cohomology groups H^0 and H^2, and they are precisely the two "trivial" contributions that the degree direction of Pic fails to kill. We shall meet the same rank-two object in the arithmetic case, where it appears as the two poles of Tate's zeta integral, and the coincidence is not a coincidence: both are $\mathrm{cofib}(\mathrm{Nm})$ for a one-dimensional non-compact direction.

4.8.2 GL_1 over $\mathbf{Q}$, completely

We now do by hand the smallest arithmetic case, where every term can be named.

Example 4.26 (The ambidexterity defect of $\mathrm{GL}_1/\mathbf{Q}$). [THEOREM]
The automorphic chôra of GL_1 over $\mathbf{Q}$ is the idele class group

$$\mathfrak{X} = \mathbf{Q}^\times \backslash \mathbf{A}^\times,$$

regarded as a chôra with its natural topology and its $\mathbf{Q}$-analytic structure. Because $\mathbf{Q}$ has class number one and unit group $\{\pm 1\}$, strong approximation gives a canonical isomorphism

$$\mathbf{Q}^\times \backslash \mathbf{A}^\times \;\cong\; \mathbf{R}_{>0} \times \widehat{\mathbf{Z}}^\times,$$

the first factor being the image of the absolute value $|\cdot| \colon \mathbf{A}^\times \to \mathbf{R}_{>0}$ and the second the finite ideles modulo the rationals, which is $\widehat{\mathbf{Z}}^\times$ by unique factorisation.

The second factor is compact, and this is the whole point. Let ℓ be a prime and take coefficients in $k = \overline{\mathbf{Q}}_\ell$. Smooth representations of a profinite group over a field of characteristic zero are semisimple; hence for $g \colon [*/\widehat{\mathbf{Z}}^\times] \to *$ the norm map is an equivalence and g is ambidextrous. Everything that could go wrong is therefore concentrated in the first factor $\mathbf{R}_{>0} \cong \mathbf{R}$.

So compute the norm map for $q \colon \mathbf{R}^r \to *$ with constant coefficients k, for $r \geq 1$. One has

$$q_! k \simeq k[-r], \qquad q_* k \simeq k,$$

the first because compactly supported cohomology of $\mathbf{R}^r$ is k in degree r, the second because $\mathbf{R}^r$ is contractible. The norm map is a morphism $k[-r] \to k$ in $D(k)$, and

$$\mathrm{Hom}_{D(k)}(k[-r], k) = H^{-r}(k) = 0 \qquad \text{for } r \geq 1.$$

Therefore $\mathrm{Nm}_q = 0$. Not merely non-invertible: zero. Its cofibre is

$$\mathrm{cofib}(\mathrm{Nm}_q) \;\simeq\; k \,\oplus\, k[1-r],$$

a rank-two object. For $r = 1$, the case at hand, this is $k \oplus k$ in degree 0: a two-dimensional vector space.

Assembling, the ambidexterity defect of GL_1 over $\mathbf{Q}$ is

$$\mathrm{cofib}(\mathrm{Nm}_f) \;\simeq\; k^{\oplus 2} \otimes_k O(\widehat{\mathbf{Z}}^\times) \quad \text{concentrated in degree } 0,$$

where the second tensor factor is ambidextrous and contributes no defect. The defect is a rank-two object over the Hecke algebra of the compact factor.

Two identifications complete the picture, and both are exact rather than analogical.

First: the defect is the pair of poles of Tate's zeta integral. For a Schwartz–Bruhat function f on $\mathbf{A}$, Tate's integral

$$Z(f,s) = \int_{\mathbf{A}^\times} f(x)\,|x|^s\,d^\times x$$

converges for $\mathrm{Re}(s) > 1$ and continues meromorphically to $\mathbf{C}$ with exactly two simple poles, at $s = 0$ with residue $-\kappa f(0)$ and at $s = 1$ with residue $\kappa \hat{f}(0)$, where $\kappa = \mathrm{vol}(\mathbf{Q}^\times\backslash\mathbf{A}^1)$; and it satisfies the functional equation $Z(f,s) = Z(\hat{f}, 1-s)$ (Tate 1950). The two poles are the two summands of $\mathrm{cofib}(\mathrm{Nm}_q)$ for $r = 1$. The regularisation that Tate performs — splitting the integral at $|x| = 1$ and adding the two boundary terms — is, in the language of this chapter, the replacement of the missing evaluation map by an explicit correction, and the correction has rank two because the non-compact direction has dimension one.

Second: the defect is the kernel of the Artin map. Class field theory for $\mathbf{Q}$, in the form of the Kronecker–Weber theorem, gives

$$\Gamma_{\mathbf{Q}}^{\mathrm{ab}} \;\cong\; \widehat{\mathbf{Z}}^\times,$$

and the Artin reciprocity map $\mathbf{Q}^\times\backslash\mathbf{A}^\times \to \Gamma_{\mathbf{Q}}^{\mathrm{ab}}$ is the projection onto the second factor. Its kernel is exactly $\mathbf{R}_{>0}$: exactly the ambidexterity defect. Thus the failure of ambidexterity for $\mathrm{GL}_1/\mathbf{Q}$ is not an accident of the analysis but the precise measure of the discrepancy between the idele class group and the Galois group — the discrepancy which class field theory handles by fiat, taking connected components, and which the categorical theory must handle by carrying the defect.

Finally, the archimedean direction $\mathbf{R}_{>0}$ is the split factor of the Deligne torus $\mathrm{Res}_{\mathbf{C}/\mathbf{R}}\,\mathbf{G}_m$ in its archimedean chart, and it is the continuous replacement for the discrete degree map $\deg\colon \mathrm{Pic}(X) \to \mathbf{Z}$ of Example 4.25. Over $\mathbf{F}_q$ the defect direction is $\mathbf{Z}$ and the divergence is a sum; over $\mathbf{Q}$ the defect direction is $\mathbf{R}$ and the divergence is an integral. This is the whole of the difference in the case GL_1, and it is the reason GL_1 can be done by hand.

Proposition 4.27 (The Tate defect). [THEOREM]

Let F be a number field with r_1 real and r_2 complex places, and let $f\colon \mathbf{G}_m^{\mathrm{ar}}(F) \to *$ be the structure morphism of the idele-class chôra. Then f is weakly ambidextrous, $\mathrm{Nm}_f = 0$, and

$$\mathrm{cofib}(\mathrm{Nm}_f) \;\simeq\; (k \oplus k[1-r]) \otimes_k C^\infty(C_F^1), \qquad r = 1,$$

where C_F^1 is the norm-one idele class group, compact by the unit theorem, and the factor $k \oplus k$ corresponds to the two poles of the Dedekind zeta function of F completed at all places — at $s = 0$ and $s = 1$. In particular the rank of the defect is 2 independently of r_1 and r_2.

Proof of Proposition 4.27.

The idele class group $C_F = F^\times\backslash\mathbf{A}_F^\times$ sits in an exact sequence $1 \to C_F^1 \to C_F \xrightarrow{|\cdot|} \mathbf{R}_{>0} \to 1$, split by choosing an idele of absolute value t at one archimedean place. Compactness of C_F^1 is Dirichlet's unit theorem together with finiteness of the class number. By Example 4.14 and the semisimplicity of smooth representations of a compact group in characteristic zero, the C_F^1-direction is ambidextrous and contributes no defect; by the computation in Example 4.26 with $r = 1$, the $\mathbf{R}_{>0}$ direction has zero norm map and rank-two cofibre. The identification of the two summands with the poles at $s = 0$ and $s = 1$ is Tate's thesis (Tate 1950). Weak

ambidexterity, that is the existence of Nm_f, holds because the diagonal of f is the diagonal of a group, hence a pullback of the identity, hence ambidextrous by the anchoring of the induction in Definition 4.12. Independence of the rank from r_1 and r_2 is the statement that $C_F/C_F^1 \cong \mathbf{R}_{>0}$ for every number field, the higher-rank unit directions having been absorbed into the compact factor.

□

The proposition is the most satisfying evidence for Principle 4.19 that we can offer, because in it every clause of the Principle is verified and the defect is not merely bounded but identified. It also sets the pattern for what a proof in higher rank would have to look like: exhibit the non-compact directions, compute their norm maps, and identify the cofibres with the poles of the relevant L-functions. For GL_n the non-compact directions are the split centre and the proper parabolics, which is Theorem 4.20; what is missing is the archimedean analysis, which is Conjecture 4.24.

4.8.3 Rank two over the projective line

The $\mathbf{G}_m$ computation of Example 4.25 used only the degree direction. The next case exhibits the parabolic direction as well, and it is small enough that every stratum can be named and every automorphism group counted.

Example 4.28 (The mass of $\mathrm{Bun}_{\mathrm{GL}_2}(\mathbf{P}^1)$). [THEOREM]

Let $X = \mathbf{P}^1_{\mathbf{F}_q}$ and $G = \mathrm{GL}_2$. By Grothendieck's splitting theorem every rank-two vector bundle on $\mathbf{P}^1$ over a field is isomorphic to $\mathcal{O}(a) \oplus \mathcal{O}(b)$ with $a \geq b$, and the pair (a, b) is determined by the bundle. The degree is $d = a + b$ and the Harder–Narasimhan type is (a, b): the bundle is semistable exactly when $a = b$, and otherwise its canonical parabolic reduction is to the Borel, with $\mathcal{O}(a)$ the destabilising line subbundle.

We first count automorphisms. For $E = \mathcal{O}(a) \oplus \mathcal{O}(b)$ an endomorphism is a matrix

$$\begin{pmatrix} \alpha & \phi \\ \psi & \beta \end{pmatrix}, \qquad \alpha, \beta \in H^0(\mathcal{O}) = \mathbf{F}_q, \quad \phi \in H^0(\mathcal{O}(a-b)), \quad \psi \in H^0(\mathcal{O}(b-a)).$$

If $a > b$ then $b - a < 0$ and $H^0(\mathcal{O}(b-a)) = 0$, while $\dim_{\mathbf{F}_q} H^0(\mathcal{O}(a-b)) = a - b + 1$. Such a matrix is invertible exactly when $\alpha, \beta \in \mathbf{F}_q^\times$, so

$$\#\,\mathrm{Aut}(\mathcal{O}(a) \oplus \mathcal{O}(b)) = (q-1)^2\, q^{a-b+1} \qquad (a > b),$$

whereas for $a = b$ the endomorphism ring is $M_2(\mathbf{F}_q)$ and

$$\#\,\mathrm{Aut}(\mathcal{O}(a)^{\oplus 2}) = \#\,\mathrm{GL}_2(\mathbf{F}_q) = (q^2-1)(q^2-q).$$

Now sum. Fix an even degree $d = 2m$. The isomorphism classes are $(a, b) = (m + k,\ m - k)$ for $k \geq 0$, so $a - b = 2k$, and the groupoid mass is

$$|\mathrm{Bun}^d_{\mathrm{GL}_2}(\mathbf{P}^1)(\mathbf{F}_q)| = \frac{1}{(q^2-1)(q^2-q)} + \sum_{k \geq 1} \frac{1}{(q-1)^2\, q^{2k+1}}.$$

The sum is geometric with ratio q^{-2}:

$$\sum_{k \geq 1} q^{-(2k+1)} = q^{-1} \cdot \frac{q^{-2}}{1 - q^{-2}} = \frac{1}{q^3 - q} = \frac{1}{q(q^2-1)}.$$

Using $(q^2-1)(q^2-q) = q(q+1)(q-1)^2$ and $(q-1)^2 q(q^2-1) = q(q+1)(q-1)^3$, the two terms combine:

$$\frac{1}{q(q+1)(q-1)^2} + \frac{1}{q(q+1)(q-1)^3} = \frac{1}{q(q+1)(q-1)^2} \cdot \frac{q}{q-1} = \frac{1}{(q+1)(q-1)^3}.$$

For odd degree $d = 2m+1$ there are no semistable bundles at all: $a = b$ is impossible. The classes are $(a,b) = (m+k,\, m+1-k)$ with $a-b = 2k-1$, $k \geq 1$, and

$$|\mathrm{Bun}^d_{\mathrm{GL}_2}(\mathbf{P}^1)(\mathbf{F}_q)| = \sum_{k\geq 1} \frac{1}{(q-1)^2 q^{2k}} = \frac{1}{(q-1)^2} \cdot \frac{q^{-2}}{1-q^{-2}} = \frac{1}{(q-1)^2(q^2-1)} = \frac{1}{(q+1)(q-1)^3}.$$

The two answers agree. Hence

$$|\mathrm{Bun}^d_{\mathrm{GL}_2}(\mathbf{P}^1)(\mathbf{F}_q)| = \frac{1}{(q+1)(q-1)^3} \qquad \text{for every } d \in \mathbf{Z},$$

and the total mass $\sum_{d\in\mathbf{Z}}$ diverges exactly as in Example 4.25.

The reader should check the case $q = 2$ by hand, where every number is small. Then $\#\,\mathrm{GL}_2(\mathbf{F}_2) = 3 \cdot 2 = 6$, and for $d = 0$ the strata are $(0,0)$ with mass $1/6$, then $(1,-1)$ with $\#\,\mathrm{Aut} = 1 \cdot 2^3 = 8$, then $(2,-2)$ with $\#\,\mathrm{Aut} = 2^5 = 32$, then $2^7 = 128$, and so on. The unstable part contributes $1/8+1/32+1/128+\cdots = (1/8)/(1-1/4) = 1/6$, and the total is $1/6+1/6 = 1/3$, in agreement with $1/((q+1)(q-1)^3) = 1/3$. For $q = 3$: $\#\,\mathrm{GL}_2(\mathbf{F}_3) = 8 \cdot 6 = 48$, the unstable strata contribute $\sum_{k\geq 1} 1/(4\cdot 3^{2k+1}) = \frac{1}{108}\cdot\frac{9}{8} = \frac{1}{96}$, and $1/48+1/96 = 1/32 = 1/((3+1)(3-1)^3)$.

Three features of the computation are the content of the chapter in miniature. First, the mass is independent of d, so the divergence is of the same crude kind as in Example 4.25: infinitely many equal terms, admitting no summation. Second, within a fixed degree the sum over Harder–Narasimhan strata *converges*, and converges because the automorphism groups grow like q^{a-b} — the unipotent radical of the Borel grows with the instability. This is why truncation by slope repairs the theory and truncation by degree does not. Third, the split answer $1/6 + 1/6$ at $q = 2$ displays the two sources of defect of Theorem 4.20 side by side: the semistable term is the would-be trace, and the sum over unstable strata is the parabolic contribution, here of the same size as the trace itself.

4.8.4 A finite Hecke algebra, by generators and relations

The last computation leaves geometry entirely. It is the smallest setting in which dualisability holds, ambidexterity holds or fails depending on a congruence, and every object can be written as a matrix.

Example 4.29 (Trivial and Steinberg over a finite field). [THEOREM]

Let $G = \mathrm{GL}_2(\mathbf{F}_q)$ and $B \subset G$ the Borel of upper triangular matrices, so that $\#(G/B) = q+1$. Let k be a field and consider

$$\mathcal{H} := \mathrm{End}_{k[G]}(\mathrm{Ind}_B^G \mathbf{1}).$$

By the Bruhat decomposition $G = B \sqcup BwB$ with w the non-trivial Weyl element, $\mathcal{H}$ has k-basis $\{1, T\}$ with T the double coset operator of BwB, and the single relation

$$T^2 = (q-1)T + q, \qquad \text{that is} \qquad (T-q)(T+1) = 0.$$

Thus $\mathcal{H} \cong k[T]/(T^2 - (q-1)T - q)$, a commutative k-algebra of dimension 2.

Everything can now be computed as a 2×2 matrix. In the basis $\{1, T\}$, multiplication by T has matrix

$$M_T = \begin{pmatrix} 0 & q \\ 1 & q-1 \end{pmatrix}, \qquad \operatorname{tr} M_T = q - 1, \qquad \det M_T = -q,$$

consistent with the roots q and -1 of the relation, whose sum is $q-1$ and whose product is $-q$. The two characters $\chi_{\mathrm{triv}} \colon T \mapsto q$ and $\chi_{\mathrm{St}} \colon T \mapsto -1$ correspond to the two constituents of the induced representation,

$$\operatorname{Ind}_B^G \mathbf{1} \;\cong\; \mathbf{1} \oplus \mathrm{St}, \qquad 1 + q = q + 1 = \dim \operatorname{Ind}_B^G \mathbf{1},$$

so the Steinberg representation has dimension exactly q. For $q = 2$ this is $\mathrm{GL}_2(\mathbf{F}_2) \cong S_3$ acting on the three cosets, decomposing as the trivial character plus the two-dimensional irreducible, and $\mathcal{H} = k[T]/(T-2)(T+1)$.

Dualisability is immediate: $\mathcal{H}$ is a finite-dimensional k-algebra, so $\mathrm{Mod}_{\mathcal{H}}$ is compactly generated by $\mathcal{H}$ itself and hence dualisable by Theorem 4.6. The trace of the identity is $\mathrm{HH}_0(\mathcal{H}) = \mathcal{H}/[\mathcal{H}, \mathcal{H}] = \mathcal{H}$, of dimension 2, since $\mathcal{H}$ is commutative.

Ambidexterity, by contrast, depends on a congruence. The discriminant of $T^2 - (q-1)T - q$ is $(q-1)^2 + 4q = (q+1)^2$, so the two roots coincide precisely when $q + 1 = 0$ in k, that is when $\operatorname{char} k$ divides $q + 1$. If $\operatorname{char} k \nmid q(q+1)$ the algebra is étale, $\mathcal{H} \cong k \times k$, every module is projective, and the norm map for the G-action is invertible. If $\operatorname{char} k = \ell$ divides $q + 1$ — the *non-banal* case — then $(T - q)(T + 1) = (T + 1)^2$ and

$$\mathcal{H} \;\cong\; k[\epsilon]/(\epsilon^2), \qquad \epsilon = T + 1,$$

which is the same local ring that appeared in Example 4.15. Its residue field $k = \mathcal{H}/(\epsilon)$ has infinite minimal free resolution, hence is not perfect, hence is not dualisable in $\mathrm{Mod}_{\mathcal{H}}$ and has no trace; and the Tate cohomology of the corresponding two-element group scheme is non-zero in every degree. The smallest instance is $q = 2$, $\ell = 3$: then $T^2 - T - 2 = (T+1)^2$ in $\mathbf{F}_3[T]$, and the Hecke algebra of S_3 at the Borel becomes the dual numbers over $\mathbf{F}_3$.

The example is the arithmetic phenomenon in miniature. The category is dualisable in every case, so a trace of every Hecke operator exists; what fails in the non-banal case is that the trivial module is no longer traceable, so the *Lefschetz* interpretation of the trace as a count of fixed points on the trivial representation breaks down. That is precisely the distinction, drawn abstractly in Theorem 4.10, between having a dualisable category and having a cohomologically proper structure morphism.

4.8.5 The control case, assertion by assertion

It is a discipline of this book that every arithmetic assertion be checked against the function-field case, where the corresponding statement is either a theorem with a name attached or an open problem one can point to. We carry out that check for the principal assertions of the present chapter. Throughout, X is a smooth projective geometrically connected curve over $\mathbf{F}_q$ and G is a split reductive group.

For Theorem 4.6 and Theorem 4.7 there is nothing to check: both are statements about abstract ∞-categories, due to Lurie and to Efimov respectively, and they hold over any base whatever. For Theorem 4.8 the same applies, the identification of the categorical trace with Hochschild homology being due to Toën–Vezzosi and to Hoyois–Scherotzke–Sibilla.

For Theorem 4.10 the function-field instance is the theorem of Ben-Zvi, Francis and Nadler that a perfect stack has QCoh dualisable with all endofunctors given by kernels; $\mathrm{Bun}_G^{\leq \mu}(X)$ is perfect, so the theorem applies stratum by stratum. Over an arithmetic base there is no perfect-stack statement, and this is one of the two places where the transfer is not routine.

For Proposition 4.18 the function-field instance is the statement that $\mathrm{Bun}_G^{\leq \mu}$ is cohomologically proper after ℓ-adic completion. This is a theorem: it follows from the quasi-compactness of the truncation, due to Behrend and Drinfeld–Simpson, together with the properness of the relevant Artin–Schreier-type maps used in the ℓ-adic formalism.

For Theorem 4.16 the function-field analogue is the ambidexterity statement proved as the fourth instalment of the recent work on the geometric Langlands conjecture (Arinkin, Beraldo, Chen, et al. 2024), whose title is precisely *Ambidexterity*; the compatibility with parabolic induction that underlies Theorem 4.20 is the third instalment (Campbell et al. 2024). Neither has an arithmetic analogue.

For Theorem 4.20 the function-field case is the theorem proved in section §4.6. The Harder–Narasimhan input is Harder–Narasimhan for vector bundles and Behrend in general; the hyperbolic localisation is Braden's, in the form used by Drinfeld–Gaitsgory; the parabolic induction compatibility is (Campbell et al. 2024). Over a number field the stratification itself is missing, so the assertion is Conjecture 4.24 and nothing more.

For Principle 4.19 the function-field version — that ambidexterity, dualisability and the existence of a Lefschetz trace are equivalent for $\mathrm{Bun}_G(X)$, and that the defect is the Eisenstein part — is, taken clause by clause, known: (i) after truncation is Behrend and Drinfeld–Simpson, (ii) is Arinkin–Gaitsgory together with Theorem 4.6, and (iii) is the categorical trace formalism of Gaitsgory–Kazhdan–Rozenblyum–Varshavsky. It is the *conjunction over the untruncated stack*, and the identification of the defect with the continuous spectrum, that remains open even over $\mathbf{F}_q$; V. Lafforgue's theorem (Lafforgue 2018) proves the arithmetic consequence one wants without proving the categorical statement.

For Proposition 4.27 the function-field analogue is exactly Example 4.25: the defect of $\mathrm{Bun}_{\mathbf{G}_m}(X)$ is the degree direction, of rank two after Fourier transform, matching the two trivial zeroes of $Z(X,t)$ at $t = 1$ and $t = q^{-1}$. The arithmetic statement is a theorem here because Tate proved it (Tate 1950); it is the one principal assertion of the chapter whose arithmetic case is *better* understood than its geometric one.

Finally, for Conjecture 4.17 there is no function-field control case at all, because chromatic height has no established meaning for ℓ-adic sheaves on a curve. The absence of a control case is the reason that conjecture is the most doubtful statement here.

4.9 Certifying compact generation

Because dualisability is the hypothesis on which everything depends, and because Theorem 4.7 reduces it to a retraction onto a compactly generated category, it is worth setting down explicitly the procedure by which a certificate of compact generation is produced. The procedure is elementary; its value is that it is checkable, and that its failure points are exactly the three failures of section §4.7.

Algorithm 4.30 (Certifying compact generation). [FORMAL TARGET]

Input: a presentable stable ∞-category C presented as $\mathcal{D}(\mathfrak{X})$ for a chôra $\mathfrak{X}$, together with a candidate exhaustion $\{\mathfrak{U}_\mu\}$ by open substacks and a candidate generating set $\{c_i\}$.

Output: either a certificate that C is dualisable, or a located obstruction.

```
1  for each mu in the directed poset:
2      check U_mu quasi-compact  -> else FAIL(qc, mu)
```

```
3      check D(U_mu) generated by {c_i restricted}
4                                -> else FAIL(gen, mu)
5      check each c_i|U_mu compact:
6          Map(c_i, -) commutes with filtered colimits
7                                -> else FAIL(cpt, i, mu)
8  check transition functors preserve compact objects
9                                -> else FAIL(trans)
10 form colim_mu D(U_mu) and the comparison
11     Phi : colim_mu D(U_mu) -> D(X)
12 if Phi is an equivalence:  CERTIFY compact generation
13 else if Phi has a section: CERTIFY dualisable (retract)
14 else:                      FAIL(retract)
```

Soundness: lines 1–7 verify the hypothesis of Theorem 4.6 for each $\mathfrak{U}_\mu$; line 8 ensures the colimit in line 10 is taken in the subcategory of $\mathcal{C}$at on compact-object-preserving functors, where dualisable objects are closed under filtered colimits; lines 12–13 appeal to Theorem 4.6 and Theorem 4.7 respectively.

The four failure modes are not hypothetical. `FAIL(qc)` is what happens if one forgets to truncate Bun_G; `FAIL(cpt)` is what happens for ind-coherent sheaves without a nilpotent-singular-support condition (Arinkin and Gaitsgory 2015); `FAIL(trans)` is the observation that the inclusions $\mathrm{Bun}_G^{\leq \mu} \hookrightarrow \mathrm{Bun}_G^{\leq \mu'}$ have !-pushforwards that do not preserve compactness without a properness input; and `FAIL(retract)` is the arithmetic situation of Conjecture 4.24, where no section is known.

The decision tree of Plate 4.3 should be read as a specification for a formalisation target rather than as an implemented algorithm; the predicates on lines 2, 3, 5 and 8 are not decidable in general, and the status mark *F* records this. What the specification buys is that a claim of dualisability in later chapters can be audited against a fixed list of obligations.

Summary. Dualisability of an object in a symmetric monoidal ∞-category is the existence of evaluation and coevaluation maps satisfying the triangle identities; it is a property, not a structure (Proposition 4.2), and it is exactly what is needed for a trace to exist (Definition 4.3, Proposition 4.4). For categories, compact generation implies dualisability (Theorem 4.6) and Efimov's retract criterion characterises it exactly (Theorem 4.7); the resulting trace is Hochschild homology (Theorem 4.8). Relatively, a morphism gives a dualisable module category precisely when it is cohomologically smooth and cohomologically proper (Theorem 4.10): smooth supplies coevaluation, proper supplies evaluation. Properness in this cohomological sense is ambidexterity, the invertibility of the Hopkins–Lurie norm map (Definition 4.12, Proposition 4.18), which sits in a hierarchy of m-semiadditivity whose flagship example is the ∞-semiadditivity of $K(n)$-local spectra (Theorem 4.16). The Ambidexterity–Compactness Principle (Principle 4.19) asserts that the classical failures of the automorphic theory — continuous spectrum, cohomological amplitude ℓ_0, failure of Poincaré duality — are one failure, measured by $\mathrm{cofib}(\mathrm{Nm}_f)$. Over a function field the defect is filtered by proper parabolics and the split centre (Theorem 4.20); over an arithmetic base, dualisability is open (Conjecture 4.24), the three function-field repairs — Harder–Narasimhan truncation, compact generation with nilpotent singular support, miraculous duality — all failing structurally rather than quantitatively. The failure is real: $\mathrm{Bun}_{\mathbb{G}_m}$ over an elliptic curve has divergent groupoid mass (Example 4.25) and no trace, while GL_1 over $\mathbf{Q}$ has a defect of rank exactly two, equal to the two poles of Tate's zeta integral and to the kernel of the Artin map (Example 4.26, Proposition 4.27).

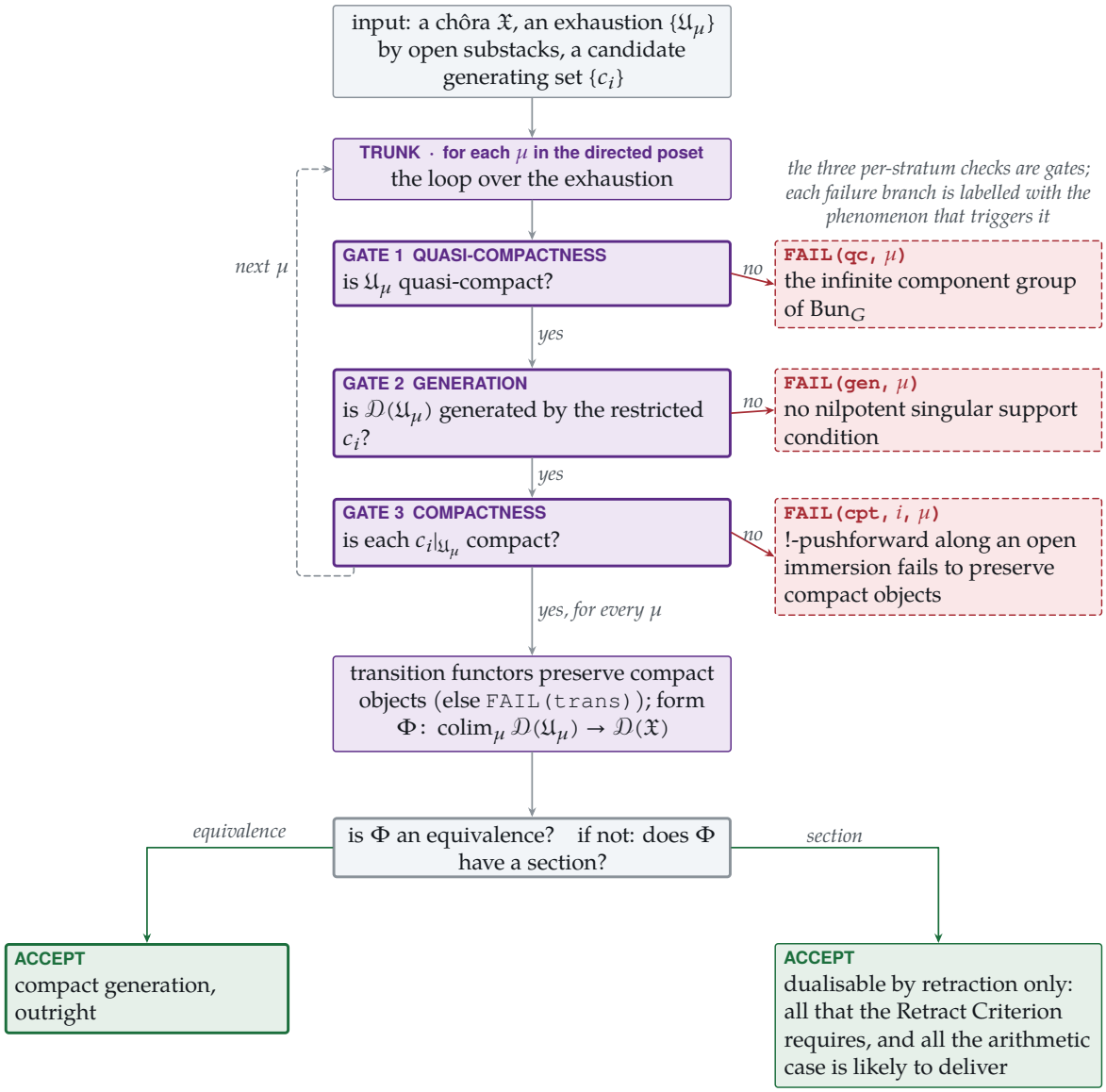


Plate 4.3. The certification procedure as a decision tree. The trunk is the loop over the exhaustion, with the three per-stratum checks — quasi-compactness, generation, compactness of the generators — drawn as gates; the three corresponding failure branches are labelled with the mathematical phenomena that trigger them, namely the infinite component group of Bun_G, the absence of a nilpotent singular support condition, and the failure of !-pushforward to preserve compact objects along an open immersion. The two accepting states at the foot of the tree are distinguished: the left certifies compact generation outright, the right certifies only dualisability by retraction, which is all that Theorem 4.7 requires and all that the arithmetic case is likely to deliver.

What it does not show: an implemented algorithm: the tree is a specification for a formalisation target, and the predicates at its gates are not decidable in general.

4.10 Notes and references

The theory of dualisable objects in a symmetric monoidal category is older than the ∞-categorical language and was developed in the 1970s under the name of *strong dualisability*, chiefly by Dold and Puppe and by Lewis, May and Steinberger, in the setting of equivariant stable homotopy theory. Their motivation was the Lefschetz fixed-point theorem, and it is

worth remembering that the entire apparatus of this chapter was invented to prove a fixed-point theorem; the arithmetic applications are a late and somewhat surprising inheritance. The ∞-categorical treatment is in (Lurie 2017, sect. 4.6), where Proposition 4.2 appears, and the reader who wants the 2-categorical argument in full should consult that section directly, since the delooping trick used in our proof is stated there with the care it needs.

Additivity of traces, Proposition 4.4, has a longer and more interesting history than its statement suggests. The naive argument — filter, take traces, add — is wrong, because the trace of a filtered object depends on a choice of splitting and the choices need not be compatible. May's paper (May 2001) isolates exactly what must be assumed, and the higher-categorical version (Hoyois et al. 2017) shows that the required compatibility is automatic once one works with coherent rather than strict data, which is one of the clearest cases where the ∞-categorical framework simplifies rather than complicates. The bicategorical refinement due to Ponto and Shulman (Ponto and Shulman 2012), with its notion of *shadow*, is the right framework for traces of 1-morphisms rather than objects, and it is what one needs if the endomorphism being traced is itself a correspondence — which is the case for every Hecke operator in this book. We have avoided the shadow formalism in the main text to keep the notation manageable, but the reader who wants to check the coherence of the Lefschetz formula in Theorem 4.10 should work in it.

Efimov's theorem (Efimov 2024) is recent and has already reorganised the subject. Before it, one either assumed compact generation, which excludes analytic geometry, or worked with ad-hoc finiteness conditions. The recognition that dualisability of a presentable category is equivalent to the existence of a left adjoint to $\operatorname{colim}\colon \operatorname{Ind}(C) \to C$ — a condition with no reference to compact objects at all — is what makes the continuous K-theory of non-compactly generated categories possible, and it is what allows this book to state Conjecture 4.24 in a form that has a chance of being true. The analytic categories to which it applies are those of Clausen and Scholze (Clausen and Scholze 2019, 2024); a reader coming from arithmetic geometry should note that these categories have no non-zero compact objects whatsoever, so that any programme resting on compact generation excludes them by fiat.

Ben-Zvi, Francis and Nadler (Ben-Zvi et al. 2010) introduced perfect stacks and proved that for them all colimit-preserving endofunctors of the category of quasi-coherent sheaves are integral transforms. That theorem is the reason Theorem 4.10 can be stated with kernels rather than with abstract functors, and it is the reason the Hecke operators of this book can be presented by correspondences. The connection to topological field theory sketched in section §4.4 is Lurie's (Lurie 2009b); the reader should be aware that the cobordism hypothesis requires *full* dualisability, which is dualisability at every categorical level, and that the present chapter has treated only the first level. What full dualisability would mean for an automorphic chôra is not known and is not conjectured here.

Ambidexterity in the sense of section §4.5 is due to Hopkins and Lurie, in *Ambidexterity in $K(n)$-local stable homotopy theory*; the paper has no entry in our bibliography and we have attributed it by name throughout, which the reader should regard as a defect of the bibliography rather than of the attribution. The inductive construction of the norm map along the diagonal tower is the paper's central technical device, and the coherence verification that makes "ambidextrous" a property rather than a structure occupies a substantial portion of it; we have not reproduced it and would caution against assuming it is routine. The height theory — semiadditive cardinality, the invariance of height, the classification of ∞-semiadditive localisations — is due to Carmeli, Schlank and Yanovski in a series of papers, likewise absent from our bibliography. Conjecture Conjecture 4.17, which proposes that chromatic height

matches arithmetic dimension, is our own and is offered in the expectation that it will be either proved or decisively refuted; the analogy is attractive and no functor supports it.

On the function-field side, the Harder–Narasimhan stratification of Bun_G is due to Harder and Narasimhan for vector bundles and to Behrend in general; the quasi-compactness of the truncations and the reduction to Levi subgroups rest on Drinfeld–Simpson, and the degeneration behaviour on Schieder's work. Miraculous duality and the pseudo-identity functor are due to Drinfeld and Gaitsgory; the proof that the pseudo-identity is an equivalence is part of the recent series establishing the geometric Langlands conjecture (Gaitsgory and Raskin 2024a, 2024b; Arinkin, Beraldo, Campbell, et al. 2024; Campbell et al. 2024; Arinkin, Beraldo, Chen, et al. 2024), of which the fourth instalment is explicitly devoted to ambidexterity and is the closest existing analogue of what this chapter asks for in the arithmetic case. The singular-support condition without which nothing is compactly generated is (Arinkin and Gaitsgory 2015), resting on the ind-coherent formalism of (Gaitsgory 2013) and (Gaitsgory and Rozenblyum 2017a, 2017b). Preygel's work on singular support and matrix factorisations, and the Gaitsgory–Kazhdan–Rozenblyum–Varshavsky treatment of trace formalisms for categories with monodromy, are the sources for the hyperbolic-localisation step in the proof of Theorem 4.20; neither is in our bibliography.

The Fargues–Scholze geometrisation (Fargues and Scholze 2026) is the one place where a Bun_G over a p-adic base has been brought fully under control, and section §4.7.2 has tried to describe its scope accurately. It is worth emphasising once more what is proved there: ℓ-cohomological smoothness of dimension zero, compact generation of D_{lis}, and a Newton stratification with strata $[*/\widetilde{G}_b(E)]$. It is not proved, and is not true, that the structure morphism is cohomologically proper. Readers who take Fargues–Scholze as evidence that the arithmetic case will yield to the same methods should note that the Fargues–Fontaine curve is a curve over a *local* field, and that the global arithmetic case has no analogue of it; the "curve" whose existence would settle everything is precisely what Chapter 2 is trying to construct.

Tate's thesis (Tate 1950) remains, seventy years on, the cleanest place to see the rank-two defect of Proposition 4.27 computed, and the reader who wants to feel the force of the identification should read Tate's splitting of the integral at $|x| = 1$ with the categorical statement in hand: the two boundary terms he adds are the two summands of the cofibre. The relation of the archimedean defect to the invariants ℓ_0 and q_0 is Calegari–Geraghty (Calegari and Geraghty 2018) and Galatius–Venkatesh (Galatius and Venkatesh 2018), with the derived Hecke perspective of (Venkatesh 2019); these are the sources for Conjecture 4.23, which is stated in Part III in its proper setting. For the classical automorphic background against which the Ambidexterity–Compactness Principle should be measured, Arthur's book (Arthur 2013) is the standard reference, and Langlands's *Beyond endoscopy* (Langlands 2004) is the essay that first suggested that the continuous spectrum is not an annoyance but a carrier of information — a suggestion of which Theorem 4.20 is one formalisation.

Finally, on the computations of section §4.8. The mass computation of Example 4.28 descends from Siegel's work of the 1930s on the mass formula for quadratic forms, transported to bundles on curves by Harder and Narasimhan in 1975 in the paper that introduced the filtration bearing their names; the reformulation as a point count on a stack is due to the Grothendieck school and became standard only after Behrend's thesis of 1991 supplied the Lefschetz formalism for algebraic stacks. The particular fact that the mass of $\mathrm{Bun}^{d}_{\mathrm{GL}_2}(\mathbf{P}^1)$ is independent of d is a shadow of the Atiyah–Bott picture, in which the equivariant cohomology of the space of connections is a free module over the cohomology of the classifying space, and the degree merely translates the answer. The Hecke algebra of Example 4.29 is the rank-one case of Iwahori's presentation of 1964, and the distinction between the banal and non-banal

cases, which we have used to separate dualisability from traceability, is Vignéras's, from her work of the 1990s on modular representations of p-adic groups; the block-theoretic framework in which it sits is Bernstein's. The divergence in Example 4.25 is well known to anyone who has tried to write the Lefschetz trace formula for Bun_G without truncating, and versions of it appear in the literature on the Behrend–Dhillon formula for the motive of Bun_G. What we have tried to add is the insistence that the divergence is a symptom of a missing morphism rather than a badly behaved series, and the explicit comparison with Example 4.5. The reader who accepts that comparison will find that most of the "convergence problems" of the trace formula literature are visible, before any analysis, as failures of a coevaluation map to exist. The arithmetic side of the same story is older still: the two poles of Proposition 4.27 are the two poles Tate isolated in his 1950 thesis (Tate 1950), and the reader who wishes to see the whole of section §4.8 in one page may read that thesis as the statement that $\mathrm{Bun}_{\mathbf{G}_m}$ over $\mathbf{Q}$ fails to be dualisable in exactly two directions, one archimedean and one discrete.

4.11 Exercises

4.1. Let $\mathcal{C}$ be a symmetric monoidal ∞-category and $X \in \mathcal{C}$ dualisable. Show that the functor $- \otimes X$ is both left and right adjoint to $- \otimes X^{\vee}$, and deduce that $- \otimes X$ preserves all limits and colimits that exist in $\mathcal{C}$. Conversely, give an example of an object X in a presentable symmetric monoidal ∞-category for which $- \otimes X$ preserves limits but X is not dualisable.

4.2. Verify directly that in Mod_k the trace of an endomorphism φ of a finite-dimensional vector space, as defined in Definition 4.3, agrees with the sum of the diagonal entries of a matrix representing φ. Then explain, in one sentence, why the symmetry σ is what makes the two agree.

4.3. Let G be a finite group and k a field of characteristic p dividing $|G|$. Compute $\mathrm{cofib}(\mathrm{Nm})$ for the trivial G-module k when $G = \mathbf{Z}/p^2$, and compare with the case $G = \mathbf{Z}/p$ of Example 4.14. Is the defect larger, smaller, or incomparable?

4.4. Show that a filtered colimit of dualisable categories along functors preserving compact objects is dualisable, and give an example showing that the hypothesis on the transition functors cannot be dropped. (This is the content of line 8 of Algorithm 4.30.)

4.5. Let X be a smooth projective curve of genus g over $\mathbf{F}_q$ with $\#X(\mathbf{F}_q) = N$. Compute the groupoid mass of $\mathrm{Pic}^d(X)$ in terms of the zeta function of X, and show that it is independent of d. Deduce that the total mass diverges for every X, so that Example 4.25 is not special to the curve chosen there.

4.6. Redo the computation of Example 4.26 for $q\colon \mathbf{R}^2 \to *$ and confirm that $\mathrm{cofib}(\mathrm{Nm}_q) \simeq k \oplus k[-1]$. Which number field, if any, produces this as its idele-class defect? Explain why the answer is "none", using Proposition 4.27.

4.7. Let $F = \mathbf{Q}(\sqrt{-1})$. Compute ℓ_0 for GL_2/F and for GL_1/F, and state what Conjecture 4.24 predicts in each case about the amplitude of the archimedean graded piece.

4.8. Prove that if $f\colon \mathfrak{X} \to \mathfrak{Y}$ and $g\colon \mathfrak{Y} \to \mathfrak{Z}$ are ambidextrous then so is $g \circ f$, assuming the coherence of the norm construction with composition. Then show by example that the converse fails.

4.9. Take $\mathcal{C} = \mathrm{Sp}_{K(1)}$ at the prime 2. Using Theorem 4.16, compute the semiadditive cardinality of $B\mathbf{Z}/2$ and compare it with the naive cardinality 2. What does the discrepancy say about Conjecture 4.17?

4.10. Formulate precisely the assertion that Arthur's truncation parameter T and the Harder–Narasimhan parameter μ of Corollary 4.21 are the same parameter, for $G = \mathrm{GL}_2$ over

Q, and identify the one step in the identification that requires the meromorphic continuation of Eisenstein series. (This is the accessible fragment of Conjecture 4.22.)

4.11. Show that a stable ∞-category is always 0-semiadditive, and give an example of a stable ∞-category which is not 1-semiadditive. Then show that Mod_k for k of characteristic zero *is* 1-semiadditive but not 2-semiadditive, and locate the obstruction.

4.12. Suppose that clause (ii) of Conjecture 4.24 holds but clause (iii) fails, in the sense that the defect admits no finite filtration. What survives of Principle 4.19? Write down the strongest statement about traces that would remain available, and decide whether it suffices to formulate the Main Conjecture of Chapter 1.

Chapter 5

Correspondences, Kernels, and the Calculus of Traces

Synopsis. This chapter constructs the computational engine used throughout the book. Starting from an infinity-category with pullbacks, it builds the $(\infty, 2)$-category of correspondences, explains the distinct contributions of Gaitsgory–Rozenblyum, Liu–Zheng, and Mann, and derives the six operations, base change, and projection formula from functoriality. It then identifies continuous functors with kernels under dualisability, develops convolution and the Ponto–Shulman shadow formalism, and proves the fibre-product formula for the trace of a composite. Two computations anchor the formalism: the spherical GL_2 Hecke correspondence at one place and Frobenius on a variety over $\mathbf{F}_q$. The corresponding construction over $\mathfrak{X}_{\mathbf{Z}}$ is isolated exactly where it remains conjectural.

5.1 Why a calculus of correspondences is necessary

A trace identity is useful only when it can be manipulated without returning at every step to the geometric construction from which it arose. The later parts of this book compare automorphic and spectral traces, insert Hecke operators, move a Frobenius past a transform, separate local factors, and compose interfaces. Each operation has a familiar classical formula, but the classical formula suppresses its hypotheses. A pull–push expression such as $q_! p^*$ composes only because a certain square is Cartesian; an integral kernel composes only because base change and the projection formula are compatible; a cyclic trace is defined only after dualisability has supplied coevaluation and evaluation. The purpose of the present chapter is to put all these conditions into one machine.

The machine has three levels. At the geometric level a morphism from X to Y is replaced by a span $X \leftarrow Z \rightarrow Y$. At the categorical level a span acts by pullback along one leg and pushforward along the other. At the trace level an endospan is closed into a loop, so that its value is an integral over a derived fixed-point fibre product. Composition at the first level becomes convolution at the second and a Lefschetz-type integral at the third.

The governing mathematical relations are collected in Plate 5.1.

There is a terminological point that will matter later. A *categorical trace* is produced from duality in a symmetric monoidal category. A *shadow*, in the sense of Ponto and Shulman, is the cyclic receptacle in which bicategorical traces take values. A *primitive trace* in a chôra includes such cyclic data as part of the geometry, together with additivity, localisation, and the chosen comparison with fixed points. These notions are related, but they are not synonyms. Confusing them would make the strongest axiom of the book look formal when it is not.

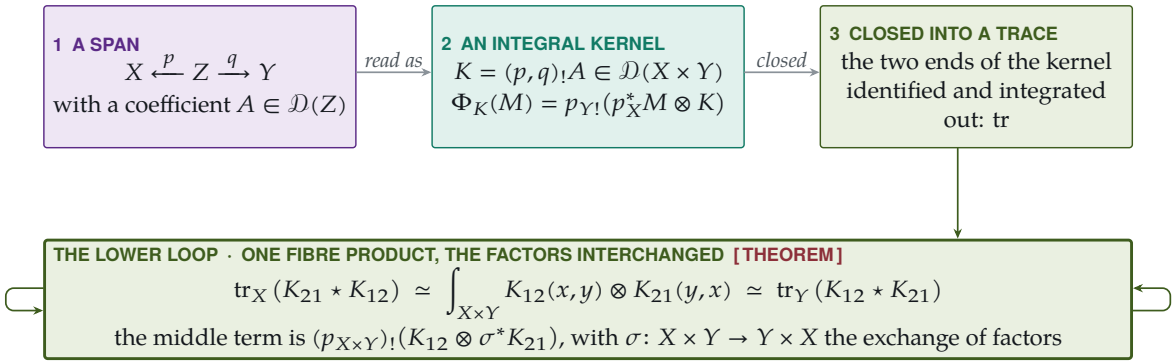


Plate 5.1. A correspondence is first read as a span, then as an integral kernel, and finally closed into a cyclic trace. The lower loop records that the trace of $K_2 \circ K_1$ is computed on the same fibre product as the trace of $K_1 \circ K_2$, with the two factors interchanged.

What it does not show: that the trace is primitive: cyclicity of the trace of a composite follows here from duality and base change, whereas the primitive trace of a chôra includes cyclic data, additivity and localisation as part of the geometry.

5.2 The correspondence context

We begin in a form broad enough to cover ordinary spans, proper correspondences, sheaf-theoretic pull–push functors, and the two-sided correspondence categories used in six-functor formalisms.

Definition 5.1 (Correspondence context). [DESIGN]

A **correspondence context** is a triple $(\mathcal{C}, \mathcal{L}, \mathcal{R})$ with the following data and properties.

1. $\mathcal{C}$ is an infinity-category admitting pullbacks.
2. $\mathcal{L}$ and $\mathcal{R}$ are subcategories containing every object and every equivalence, and each is closed under composition.
3. Pullback of a morphism in $\mathcal{L}$ along a morphism in $\mathcal{R}$ exists and again belongs to $\mathcal{L}$; pullback of a morphism in $\mathcal{R}$ along a morphism in $\mathcal{L}$ exists and again belongs to $\mathcal{R}$.
4. If a symmetric monoidal structure is required, $\mathcal{C}$ is Cartesian monoidal and both classes are closed under products.

The one-sided notation $\mathrm{Corr}(\mathcal{C}, E)$ means $\mathcal{L} = \mathrm{Mor}(\mathcal{C})$ and $\mathcal{R} = E$. Our orientation convention is that a correspondence from X to Y is $X \xleftarrow{p} Z \xrightarrow{q} Y$, with $p \in \mathcal{L}$ and $q \in \mathcal{R}$.

The convention is chosen so that the associated functor will be $q_! p^*$. Other authors reverse the legs or pass to one of the two opposites. Nothing mathematical depends on that choice, but formulas do, so the convention will not change within this book.

Definition 5.2 (The correspondence (infinity,2)-category). [ESTABLISHED]

For a correspondence context $(\mathcal{C}, \mathcal{L}, \mathcal{R})$, the $(\infty, 2)$-category $\mathrm{Corr}_{\mathcal{L},\mathcal{R}}(\mathcal{C})$ has:

- objects the objects of $\mathcal{C}$;
- a mapping infinity-category from X to Y given by the full subcategory of $\mathcal{C}_{/X\times Y}$ on maps $Z \to X \times Y$ whose projections $p : Z \to X$ and $q : Z \to Y$ lie in $\mathcal{L}$ and $\mathcal{R}$;
- one-morphisms represented by spans $X \xleftarrow{p} Z \xrightarrow{q} Y$;
- two-morphisms from $(p, q) : Z$ to $(p', q') : Z'$ represented by maps $a : Z \to Z'$ equipped with the coherent equivalences $p \simeq p'a$ and $q \simeq q'a$.

Composition of $X \xleftarrow{p} Z \xrightarrow{q} Y$ and $Y \xleftarrow{r} W \xrightarrow{s} T$ is represented by

$$X \xleftarrow{p\pi_Z} Z \times_Y W \xrightarrow{s\pi_W} T.$$

This construction, including its higher coherences, is the correspondence machinery developed systematically by Gaitsgory and Rozenblyum (Gaitsgory and Rozenblyum 2017a).

The displayed definition describes the mapping infinity-categories and the binary composite, but an $(\infty, 2)$-category also contains all associators, unitors, and their coherences. One may construct them by a complete Segal object whose n-simplices are strings of n composable correspondences and whose structure maps are induced by iterated pullback. This is the point at which a picture of spans ceases to be a proof and the Gaitsgory–Rozenblyum construction becomes indispensable.

Proposition 5.3 (Composition by pullback). [ESTABLISHED]
The composite in Definition 5.2 is again an admissible correspondence. It is associative up to a canonical contractible space of coherent equivalences, and the span $X \xleftarrow{\mathrm{id}} X \xrightarrow{\mathrm{id}} X$ is a unit.

Proof of Proposition 5.3.
Let $p \in \mathcal{L}$, $q \in \mathcal{R}$, $r \in \mathcal{L}$, and $s \in \mathcal{R}$. In the Cartesian square

$$\begin{array}{ccc} Z \times_Y W & \xrightarrow{\pi_W} & W \\ \downarrow \pi_Z & & \downarrow r \\ Z & \xrightarrow{q} & Y, \end{array}$$

π_Z is the pullback of r along q, hence belongs to $\mathcal{L}$, while π_W is the pullback of q along r, hence belongs to $\mathcal{R}$. Closure under composition gives $p\pi_Z \in \mathcal{L}$ and $s\pi_W \in \mathcal{R}$.

For three spans, the two parenthesisations have middle objects $(Z \times_Y W) \times_T V$ and $Z \times_Y (W \times_T V)$. Both are limits of the same zigzag $Z \to Y \leftarrow W \to T \leftarrow V$. The infinity-category of limit cones on a fixed diagram is either empty or contractible; it is nonempty here. Hence there is a canonical equivalence between the two middle objects, and every higher comparison among such equivalences is drawn from a contractible space. The same argument with the terminal zigzags $Z \to X \leftarrow X$ and $X \to X \leftarrow Z$ proves the unit assertions. The complete Segal construction packages these contractible choices into the required coherent associators and unitors. This proves the claim.

□

When $\mathcal{C}$ has finite products, the Cartesian product of spans makes correspondences symmetric monoidal. More surprisingly, spans contain their own adjoints.

Proposition 5.4 (Adjoints and self-duality in correspondences). [ESTABLISHED]
Assume $\mathcal{L} = \mathcal{R} = \mathrm{Mor}(\mathcal{C})$. Reversing a span defines an equivalence

$$\mathrm{Corr}(\mathcal{C})^{2\text{-op}} \simeq \mathrm{Corr}(\mathcal{C}).$$

Every map $f : X \to Y$, regarded as the span $X \xleftarrow{\mathrm{id}} X \xrightarrow{f} Y$, is left adjoint to the reversed span $Y \xleftarrow{f} X \xrightarrow{\mathrm{id}} X$. Under the Cartesian symmetric monoidal structure every object X is dual to itself, with coevaluation and evaluation represented by the diagonal and the terminal map.

Proof of Proposition 5.4.
Reversal exchanges source and target and sends a map of spans to the same map of middle objects, so it is involutive. For the adjunction, the unit is the map of spans from the identity of X to the composite

$$X \leftarrow X \times_Y X \to X$$

induced by the diagonal $X \to X \times_Y X$. The counit is the map from the composite $Y \xleftarrow{f} X \xrightarrow{f} Y$ to the identity span of Y, induced by f. The two triangle composites reduce to the two

projections from a pullback followed by its diagonal; the universal property of the pullback identifies each with the identity. For object duality, take

$$\mathrm{coev}_X : * \leftarrow X \xrightarrow{\Delta} X \times X, \qquad \mathrm{ev}_X : X \times X \xleftarrow{\Delta} X \to *.$$

Their zigzag composites are the spans obtained from the pullbacks of Δ along itself and collapse to the identity by the same diagonal calculation. The coherent triangle identities follow because the relevant spaces of pullback identifications are contractible.

□

The system-level dependencies of the construction are displayed in Plate 5.2.

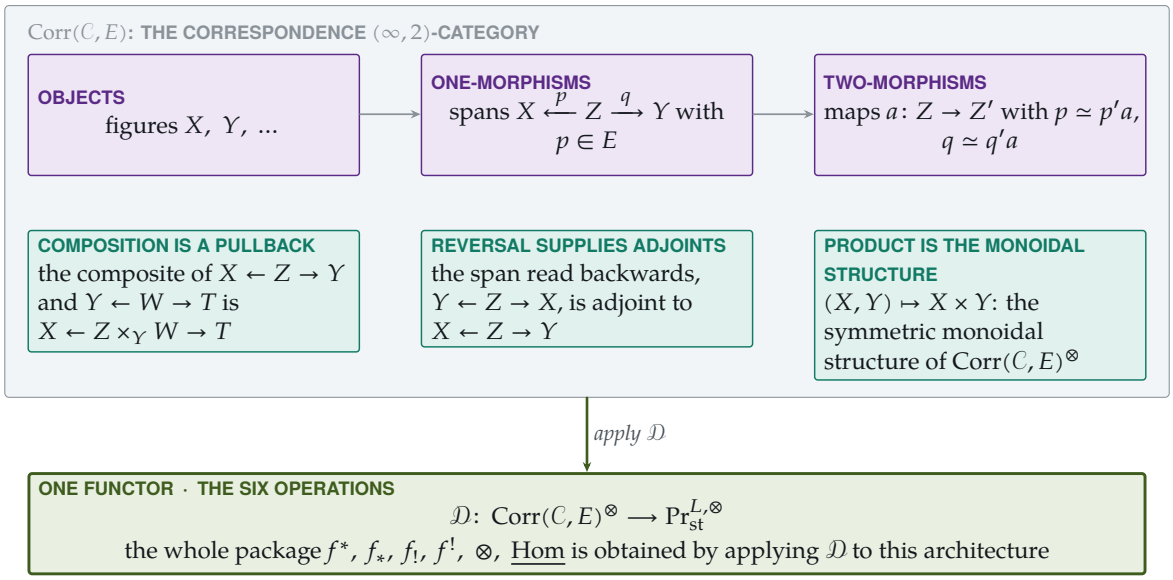


Plate 5.2. Objects, one-morphisms, and two-morphisms in the correspondence $(\infty, 2)$-category. Horizontal composition is a pullback, reversal supplies adjoints, and Cartesian product supplies the symmetric monoidal structure; the entire six-operation package is obtained by applying one functor to this architecture.

What it does not show: which functor: the architecture fixes what a six-functor formalism must be, and the arithmetic sheaf theory to which it is applied is constructed only in later chapters, conditionally.

5.3 The universal property and the six operations

The slogan that a six-functor formalism is a functor out of correspondences needs qualification. Gaitsgory and Rozenblyum construct the higher correspondence categories and use them to organise sheaf theories and IndCoh. Liu and Zheng construct enhanced six operations for higher Artin stacks by multisimplicial descent, proving the coherences needed for pullback, proper pushforward, and the Lefschetz–Verdier formalism. Mann makes the functor out of a correspondence category the axiomatic interface of a six-functor formalism and applies it in rigid and p-adic geometry (Mann 2022); Heyer and Mann carry the same organisation into smooth representation theory (Heyer and Mann 2024). These are related formulations, not interchangeable citations.

The universal property can be stated without choosing a model for $(\infty, 2)$-categories. Write $i_!$ for a map regarded covariantly and j^* for a map regarded contravariantly.

Theorem 5.5 (Universal property of correspondences). [ESTABLISHED]
Let $(\mathcal{C}, \mathcal{L}, \mathcal{R})$ be a correspondence context and let $\mathcal{B}$ be an $(\infty, 2)$-category. To give a functor

$$F : \mathrm{Corr}_{\mathcal{L},\mathcal{R}}(\mathcal{C}) \longrightarrow \mathcal{B}$$

is equivalent to giving:

1. *an object $F(X)$ for every $X \in \mathcal{C}$;*
2. *a covariant one-morphism $r_! : F(X) \to F(Y)$ for every $r : X \to Y$ in $\mathcal{R}$;*
3. *a contravariant one-morphism $l^* : F(Y) \to F(X)$ for every $l : X \to Y$ in $\mathcal{L}$;*
4. *coherent composition and unit equivalences for the two assignments;*
5. *for every Cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{r'} & Y' \\ \downarrow{\scriptstyle l'} & & \downarrow{\scriptstyle l} \\ X & \xrightarrow{r} & Y \end{array}$$

with $l, l' \in \mathcal{L}$ and $r, r' \in \mathcal{R}$, an invertible Beck–Chevalley two-morphism

$$l^* r_! \xrightarrow{\sim} r'_! l'^*,$$

coherent under pasting of Cartesian squares.

Under this equivalence the span $X \xleftarrow{p} Z \xrightarrow{q} Y$ is sent to $q_! p^*$. If the data are symmetric monoidal, the equivalence holds in the symmetric monoidal setting.

Proof of Theorem 5.5.

A functor F restricts along the embeddings $r \mapsto (X \xleftarrow{\mathrm{id}} X \xrightarrow{r} Y)$ and $l \mapsto (Y \xleftarrow{l} X \xrightarrow{\mathrm{id}} X)$, producing the covariant and contravariant assignments. A Cartesian square identifies two composites of elementary spans:

$$(Y' \xleftarrow{l} Y \xrightarrow{r} X) \quad \text{and} \quad (Y' \xleftarrow{r'} X' \xrightarrow{l'} X),$$

with orientations adjusted to source and target. Applying F gives the Beck–Chevalley equivalence. Associativity in the correspondence category implies its compatibility with pasted squares.

Conversely, factor every span uniquely in the form

$$X \xleftarrow{p} Z \xrightarrow{\mathrm{id}} Z, \qquad Z \xleftarrow{\mathrm{id}} Z \xrightarrow{q} Y,$$

and assign to it $q_! p^*$. For two composable spans the desired comparison between the assigned composite and the functor assigned to the pullback span is exactly the Beck–Chevalley equivalence for the defining pullback square, followed by the composition constraints for $(-)^*$ and $(-)_!$. For three spans, the two comparisons are equal because both are obtained by pasting the same three Cartesian squares; this is precisely the assumed pasting coherence. Units are treated by the identity constraints. Maps of spans act by the natural two-morphisms induced from the covariant and contravariant data. Thus the construction extends to a functor, and restriction followed by extension, or extension followed by restriction, is equivalent to the identity. The symmetric monoidal assertion is obtained by applying the same argument to products of all diagrams. The higher-coherent version of this argument is the universal property proved by the correspondence constructions of Gaitsgory–Rozenblyum; Mann's formulation takes this property as the interface for six operations.

□

Definition 5.6 (Six-functor formalism). [DESIGN]
Let (C, E) be a correspondence context. A **six-functor formalism** on it is a lax symmetric monoidal functor

$$\mathcal{D} : \mathrm{Corr}(C, E)^{\otimes} \longrightarrow \mathrm{Pr}^{L,\otimes}_{\mathrm{st}}$$

such that for every map $f : X \to Y$ the pullback f^* has a right adjoint f_*, and for every $f \in E$ the functor $f_!$ has a right adjoint $f^!$. The closed symmetric monoidal structure on $\mathcal{D}(X)$ supplies $\otimes$ and $\underline{\mathrm{Hom}}$.

This definition is KH.1–KH.2 in operational form. It does not assert that such a functor exists for the conjectural arithmetic base. It says what must be constructed, and it ensures that once it has been constructed, the usual compatibilities are not a second independent task.

Proposition 5.7 (Base change from correspondence functoriality). [ESTABLISHED]
For a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

with $f \in E$, a six-functor formalism supplies canonical equivalences

$$g^* f_! \simeq f'_! g'^*, \qquad g'_* f'^! \simeq f^! g_*.$$

They are compatible with horizontal and vertical pasting of Cartesian squares.

Proof of Proposition 5.7.
The first equivalence is the image under $\mathcal{D}$ of the canonical equivalence between the two composite correspondences represented by the square. The second is its right mate under the adjunctions $f_! \dashv f^!$ and $f'_! \dashv f'^!$, together with $g^* \dashv g_*$ and $g'^* \dashv g'_*$. Pasting compatibility holds before applying $\mathcal{D}$ because iterated pullbacks compute the same limit; a functor preserves the resulting coherent equivalences. Taking mates preserves pasting, so the second formula has the same compatibility.

□

5.3.1 Beck–Chevalley coherence written out

The assertion that base-change maps are compatible with pasting contains the coherence needed in every later multi-square calculation. We spell it out for a horizontal composite; the vertical case is the same limit calculation with the roles of rows and columns exchanged.

Consider two adjacent Cartesian squares

$$\begin{array}{ccccc} X'' & \xrightarrow{a'} & X' & \xrightarrow{b'} & X \\ \downarrow u'' & & \downarrow u' & & \downarrow u \\ Y'' & \xrightarrow{a} & Y' & \xrightarrow{b} & Y. \end{array}$$

Assume that the horizontal and vertical arrows lie in the classes required by the correspondence context. There are two ways to transform the composite pull–push expression associated with the outside rectangle. The direct route uses the Beck–Chevalley map for the outside rectangle:

$$(ba)^* u_! \longrightarrow u''_! (b'a')^*.$$

The stepwise route first applies base change to the right square and then to the left square:

$$a^* b^* u_! \longrightarrow a^* u'_! b'^* \longrightarrow u''_! a'^* b'^*.$$

The composition constraints identify the source and target of the stepwise route with those of the direct route. We must show that the resulting natural transformations agree.

Represent $u_!$ by the graph span

$$Y \xleftarrow{u} X \xrightarrow{\mathrm{id}} X$$

with the orientation adjusted to the convention of Definition 5.2, and represent pullback by the reversed graph. Composing with the reversed graph of b replaces the middle object by $X \times_Y Y'$, which is canonically X'. Composing once more with the reversed graph of a replaces it by

$$(X \times_Y Y') \times_{Y'} Y''.$$

The stepwise Beck–Chevalley map is induced by the canonical comparison from this iterated pullback to X''. The direct outside rectangle instead produces

$$X \times_Y Y''.$$

Both objects are limits of the same diagram

$$X \longrightarrow Y \longleftarrow Y' \longleftarrow Y''.$$

More explicitly, a test object T maps to either pullback by the same data: a map $T \to X$, a map $T \to Y''$, and a homotopy between their composites in Y. The intermediate map to Y' is forced to be the composite $T \to Y'' \to Y'$. Hence the two pullbacks represent equivalent functors of T. Yoneda gives a canonical equivalence

$$(X \times_Y Y') \times_{Y'} Y'' \simeq X \times_Y Y''.$$

Under this equivalence, the maps to X and Y'' agree. Therefore the two composite spans are not merely equivalent after applying a sheaf theory; they are the same one-morphism in the correspondence $(\infty, 2)$-category up to the contractible space of limit identifications. Applying $\mathcal{D}$ makes the direct and stepwise base-change transformations equal.

For three or more adjacent squares, every parenthesisation produces a limit of the same rectangular diagram. The infinity-groupoid of limit cones is contractible, so all comparison equivalences among parenthesisations and all higher homotopies among those comparisons are uniquely determined. This contractibility is the higher associativity datum that a diagram chase in an ordinary derived category suppresses.

The compatibility with mates is equally concrete. Suppose the first base-change map is

$$g^* f_! \xrightarrow{\sim} f'_! g'^*.$$

Its right mate is obtained by composing with the units of $f_! \dashv f^!$ and $g'^* \dashv g'_*$, and the counits of the other two adjunctions. If two Cartesian squares are pasted, the mate of the pasted base-change map contains the units and counits for the composite adjunctions. Those units and counits are themselves the composites of the units and counits for the constituent adjunctions. The two triangle identities cancel the adjacent pairs, leaving exactly the composite of the two individual right-mate base-change maps. Thus taking mates commutes with pasting.

Finally, symmetric monoidal compatibility follows by taking the product of the entire diagram with a fixed figure Z. Pullback commutes with that product, and the product of a limit cone with Z is again a limit cone because the monoidal structure is Cartesian. Hence the Beck–Chevalley equivalence for the product square is the external product of the original equivalence with the identity on Z. This is the coherence used when base change is applied inside a tensor product of kernels.

Proposition 5.8 (Projection formula). [ESTABLISHED]
Let $f : X \to Y$ belong to E. For $M \in \mathcal{D}(X)$ and $N \in \mathcal{D}(Y)$ there is a natural equivalence

$$f_!(M \otimes f^*N) \simeq f_!M \otimes N.$$

Its right-adjoint form is

$$f_*\underline{\mathrm{Hom}}(M, f^!N) \simeq \underline{\mathrm{Hom}}(f_!M, N).$$

Proof of Proposition 5.8.
The lax symmetric monoidal structure gives a map $f_!M \otimes N \to f_!(M \otimes f^*N)$. In a correspondence formalism the product of the span for $f_!$ with the identity span of Y, followed by pullback along the diagonal $Y \to Y \times Y$, is canonically the span for $f_!$ applied to $M \otimes f^*N$. The Beck–Chevalley equivalence for the resulting Cartesian square identifies the displayed lax-monoidal map with an equivalence. Equivalently, test against $P \in \mathcal{D}(Y)$:

$$\begin{aligned}
\mathrm{Map}(f_!(M \otimes f^*N), P) &\simeq \mathrm{Map}(M \otimes f^*N, f^!P) \\
&\simeq \mathrm{Map}(M, \underline{\mathrm{Hom}}(f^*N, f^!P)) \\
&\simeq \mathrm{Map}(M, f^!\underline{\mathrm{Hom}}(N, P)) \\
&\simeq \mathrm{Map}(f_!M, \underline{\mathrm{Hom}}(N, P)) \\
&\simeq \mathrm{Map}(f_!M \otimes N, P).
\end{aligned}$$

The middle equivalence is the mate of the same monoidal base-change square. Yoneda gives the first formula. Taking its right adjoint gives the second.

□

5.3.2 Three constructions, three claims

The literature cited above answers three logically separate questions. It is worth recording the separation because later chapters use all three.

First, one must construct an $(\infty, 2)$-category whose one-morphisms are correspondences. The difficulty is not binary composition: the pullback formula for two spans is elementary. The difficulty is to produce a complete family of mapping infinity-categories and to verify the Segal conditions that make all iterated pullbacks coherent. Gaitsgory and Rozenblyum solve this problem in a form adapted to derived algebraic geometry. Their formalism permits designated vertical and horizontal classes and the several opposites needed to encode $*$- and !-variance. It also makes adjunctions visible at the level of correspondences before a sheaf theory is applied.

Second, for a particular sheaf theory, one must actually construct the functors and prove descent. Liu and Zheng's multisimplicial method addresses this problem for enhanced operations on higher stacks. Starting with a basis of geometric situations on which pullback and pushforward are known, they organise iterated Čech nerves so that descent produces a functor with all higher comparison maps. Their result is therefore a construction theorem for

a specific formalism, including the coherences required by base change and Lefschetz–Verdier theory. It is not merely the observation that spans compose. In particular, when this book attributes the enhanced étale operations on higher Artin stacks to Liu–Zheng, it is invoking their descent theorem, not a new definition of the word correspondence.

Third, one needs an interface by which an arbitrary sheaf theory can be called a six-functor formalism and compared with another. Mann's formulation treats a lax symmetric monoidal functor from a suitable correspondence category as that interface. The geometric work then lies in verifying that a proposed theory extends to the functor and that the chosen classes of morphisms satisfy the required base-change and adjointability properties. This viewpoint is especially effective in analytic and p-adic geometry because it permits the formal consequences to be proved once, while the geometry-specific work is concentrated in existence, descent, and finiteness.

There are consequently two universal properties in play. The first is the universal property of the span category in Theorem 5.5: a functor out of spans is covariant data, contravariant data, and coherent Beck–Chevalley equivalences. The second, stronger assertion sometimes called a universal six-functor theorem says that a partially defined sheaf theory has a canonical extension to such a functor. The first is formal once the correspondence category exists. The second depends on descent, compactification, or other geometric input and can fail. The arithmetic programme requires both: KH.1 specifies the first interface, while the construction of a sheaf theory on $\mathfrak{X}_{\mathbf{Z}}$ would have to prove the second kind of theorem.

The role of adjoints has a parallel division. In the unrestricted span category, reversing a graph span gives a formal adjoint as in Proposition 5.4. After applying a sheaf theory, this formal adjunction becomes $f^* \dashv f_*$ or $f_! \dashv f^!$ only when the target functors and units exist. Properness is not needed to write the reversed span, but it may be needed to identify $f_!$ with f_*. Smoothness is not needed to define $f^!$, but it is needed to identify $f^!$ with a shifted and twisted f^*. Thus the correspondence category organises the operations; it does not erase the geometric theorems that compare them.

A useful coherence audit follows. Every time a later calculation moves a pullback across a pushforward, it invokes Proposition 5.7. Every time it moves a coefficient across a pushforward, it invokes Proposition 5.8. Every time it erases parentheses from an iterated integral, it invokes functoriality of $(-)_!$ and the contractibility of iterated pullback choices. Every time it rotates a closed diagram, it invokes a shadow. These four moves exhaust the formal manipulations in the trace calculus. Any step not reducible to one of them is a geometric or analytic input and must be stated separately.

5.4 Integral transforms and kernels

Correspondences become computational once their action can be represented by objects rather than by unnamed functors. The passage from a functor to a kernel is not automatic for arbitrary infinity-categories. Its exact hypothesis is dualisability in the tensor category of presentable categories.

Let $\mathrm{Pr}^L_{\mathrm{st}}$ denote the symmetric monoidal infinity-category of presentable stable infinity-categories and colimit-preserving functors. Its tensor product is characterised by

$$\mathrm{Fun}^L(\mathcal{A} \otimes \mathcal{B}, \mathcal{E}) \simeq \mathrm{Fun}^{L,L}(\mathcal{A} \times \mathcal{B}, \mathcal{E}).$$

An object C is dualisable if there are a category $C^\vee$ and continuous functors

$$\mathrm{coev} : \mathrm{Sp} \to C \otimes C^\vee, \qquad \mathrm{ev} : C^\vee \otimes C \to \mathrm{Sp}$$

satisfying the two triangle identities. Compactly generated presentable stable categories are dualisable; if $\mathcal{C}^c$ is their small subcategory of compact objects, then $\mathcal{C}^\vee \simeq \mathrm{Ind}((\mathcal{C}^c)^{\mathrm{op}})$. Large automorphic categories need not be compactly generated, which is why the book later imposes nuclearity or another continuous finiteness condition.

Definition 5.9 (Integral transform). [DESIGN]
Let X and Y be figures for which the sheaf theory admits external products and a push-forward along $p_Y : X \times Y \to Y$. A **kernel** from X to Y is an object $K \in \mathcal{D}(X \times Y)$. Its integral transform is

$$\Phi_K : \mathcal{D}(X) \longrightarrow \mathcal{D}(Y), \qquad \Phi_K(M) = p_{Y!}(p_X^* M \otimes K).$$

When a correspondence $X \xleftarrow{p} Z \xrightarrow{q} Y$ carries a coefficient $A \in \mathcal{D}(Z)$, its kernel is

$$K_{Z,A} = (p,q)_! A \in \mathcal{D}(X \times Y),$$

and base change gives $\Phi_{K_{Z,A}}(M) \simeq q_!(p^* M \otimes A)$.

The last equality is the bridge between geometric correspondences and Fourier–Mukai language. Indeed,

$$\begin{aligned} \Phi_{K_{Z,A}}(M) &= p_{Y!}(p_X^* M \otimes (p,q)_! A) \\ &\simeq p_{Y!}(p,q)_!((p,q)^* p_X^* M \otimes A) \\ &\simeq q_!(p^* M \otimes A), \end{aligned}$$

where the second line is the projection formula and the third uses $p_Y(p,q) = q$ and $p_X(p,q) = p$.

Theorem 5.10 (Continuous functors as kernels). [ESTABLISHED]
Let $\mathcal{C}$ and $\mathcal{D}$ be presentable stable infinity-categories, and suppose that $\mathcal{C}$ is dualisable in $\mathrm{Pr}^L_{\mathrm{st}}$. Then there is a canonical equivalence

$$\mathrm{Fun}^L(\mathcal{C}, \mathcal{D}) \simeq \mathcal{C}^\vee \otimes \mathcal{D}.$$

Under this equivalence a tensor $c^\vee \otimes d$ represents the functor $c \mapsto \mathrm{ev}(c^\vee \otimes c) \otimes d$. For perfect stacks X, Y for which $\mathrm{QCoh}(X)^\vee \simeq \mathrm{QCoh}(X)$ and $\mathrm{QCoh}(X) \otimes \mathrm{QCoh}(Y) \simeq \mathrm{QCoh}(X \times Y)$, this becomes

$$\mathrm{Fun}^L(\mathrm{QCoh}(X), \mathrm{QCoh}(Y)) \simeq \mathrm{QCoh}(X \times Y),$$

with the equivalence realised by integral transforms (Ben-Zvi et al. 2010).

Proof of Theorem 5.10.
For any $K \in \mathcal{C}^\vee \otimes \mathcal{D}$ define

$$\Theta(K) : \mathcal{C} \xrightarrow{K \otimes \mathrm{id}} \mathcal{C}^\vee \otimes \mathcal{D} \otimes \mathcal{C} \xrightarrow{\mathrm{ev} \otimes \mathrm{id}} \mathcal{D},$$

with the symmetry used to place the factors in the displayed order. This is colimit-preserving. Conversely, for $F \in \mathrm{Fun}^L(\mathcal{C}, \mathcal{D})$, set

$$\Xi(F) = (\mathrm{id}_{\mathcal{C}^\vee} \otimes F)\,\mathrm{coev}(\mathbb{S}) \in \mathcal{C}^\vee \otimes \mathcal{D}.$$

Apply Θ to $\Xi(F)$. The resulting composite contains coev followed by ev on the $\mathcal{C}$-factor; the triangle identity contracts these two maps and leaves F. Conversely, $\Xi\Theta(K)$ inserts

a coevaluation and then contracts the adjacent $C^\vee \otimes C$ pair. The other triangle identity leaves K. Thus Θ and Ξ are inverse equivalences. The statement for perfect stacks follows by substituting the two displayed geometric identifications. Those identifications are the substantive geometric input; the equivalence of functors with the tensor product is purely formal.

□

The theorem explains both the power and the limitation of kernels. A kernel is not merely a convenient formula for a functor. It is the functor itself under a duality equivalence. When dualisability fails, a putative kernel may still produce a functor, but there is no reason that every continuous functor arises in this way, nor that traces exist.

5.4.1 Dualisability diagnostics for kernel categories

The abstract theorem is most transparent for module categories. Let A be an associative algebra object in spectra and let Mod_A be the presentable category of right A-modules. Its dual in $\mathrm{Pr}^L_{\mathrm{st}}$ is $\mathrm{Mod}_{A^{\mathrm{op}}}$, with evaluation

$$\mathrm{Mod}_{A^{\mathrm{op}}} \otimes \mathrm{Mod}_A \longrightarrow \mathrm{Sp}, \qquad (N, M) \longmapsto N \otimes_A M.$$

The coevaluation is the A–A bimodule A, regarded as an object of

$$\mathrm{Mod}_A \otimes \mathrm{Mod}_{A^{\mathrm{op}}} \simeq \mathrm{Mod}_{A \otimes A^{\mathrm{op}}}.$$

Consequently

$$\mathrm{Fun}^L(\mathrm{Mod}_A, \mathrm{Mod}_B) \simeq \mathrm{Mod}_{A^{\mathrm{op}} \otimes B}.$$

A continuous functor is therefore an A–B bimodule, and composition is relative tensor product. The trace of an A–A bimodule M is

$$M \otimes_{A \otimes A^{\mathrm{op}}} A = HH(A; M).$$

This algebraic model contains every formula of the chapter: relative tensor product is convolution, the A bimodule is the diagonal kernel, and the cyclic equivalence

$$HH(A; M \otimes_B N) \simeq HH(B; N \otimes_A M)$$

is the Ponto–Shulman shadow rotation.

For a perfect stack X, the geometric substitute for the regular bimodule is the diagonal object $\Delta_{X!}\mathbb{1}_X$. The evaluation pairing is

$$\mathcal{D}(X)^\vee \otimes \mathcal{D}(X) \longrightarrow \mathcal{D}(*)$$

obtained by external product on $X \times X$, restriction to the diagonal, and integration over X. If Verdier duality identifies $\mathcal{D}(X)^\vee$ with $\mathcal{D}(X)$, the pairing takes the familiar form

$$(M, N) \longmapsto \int_X \mathbb{D}_X M \otimes N.$$

The triangle identity is then the assertion that convolution with the diagonal kernel is the identity. This is why diagonal, duality, and identity kernel always appear together.

Four checks are required before using a geometric kernel theorem.

First, **dual generation**: the category must possess enough compact, nuclear, or otherwise dualisable objects for the evaluation pairing to be defined. Compact generation is sufficient but not necessary. Second, **product compatibility**: the external-product functor

$$\mathcal{D}(X) \otimes \mathcal{D}(Y) \longrightarrow \mathcal{D}(X \times Y)$$

must be an equivalence on the chosen class of figures. Third, **diagonal admissibility**: Δ_X must belong to the class along which the required pushforward exists. Fourth, **integrability**: the structure map of the fixed-point or convolution fibre product must admit the supported pushforward used in the trace.

These checks are independent. For example, a sheaf category may be dualisable while the external product fails to identify its tensor product with sheaves on the geometric product. A diagonal kernel may exist while the structure map is non-proper, leaving its trace divergent. A functor may have a kernel while the kernel is not trace-class. The later analytic theory addresses these failures by replacing compactness with nuclearity and ordinary pushforward with renormalised pushforward; none of those replacements is formal at the level of Theorem 5.10.

A particularly useful test is the identity functor. Under Theorem 5.10, the identity corresponds to the coevaluation object. If a proposed geometric equivalence sends that object to something other than $\Delta_{X!}\mathbb{1}_X$, then either the duality identification or the product Künneth equivalence has been normalised incorrectly. Likewise, its trace must be

$$\operatorname{tr}(\mathrm{id}_{\mathcal{D}(X)}) \simeq \int_X \Delta_X^* \Delta_{X!} \mathbb{1}_X,$$

which is Hochschild homology in the algebraic model and the derived self-intersection of the diagonal in geometry. This diagnostic will recur when the arithmetic diagonal is discussed: the relevant invariant is the self-intersection measured by THH, not the underived absolute tensor square by itself.

Proposition 5.11 (Convolution of kernels). [ESTABLISHED]

Let X, Y, Z be figures for which the indicated operations exist. For kernels $K_{12} \in \mathcal{D}(X \times Y)$ and $K_{23} \in \mathcal{D}(Y \times Z)$, define

$$K_{23} \star K_{12} := p_{13!}(p_{12}^* K_{12} \otimes p_{23}^* K_{23}) \in \mathcal{D}(X \times Z).$$

Then

$$\Phi_{K_{23} \star K_{12}} \simeq \Phi_{K_{23}} \circ \Phi_{K_{12}}.$$

Convolution is associative up to its canonical coherent equivalence, and the kernel $\Delta_{X!}\mathbb{1}_X$ is a unit.

Proof of Proposition 5.11.

For $M \in \mathcal{D}(X)$, expand the composite:

$$\Phi_{K_{23}} \Phi_{K_{12}}(M) = p_{Z!}\left(p_Y^* p_{Y!}(p_X^* M \otimes K_{12}) \otimes K_{23}\right).$$

Place the calculation on $X \times Y \times Z$. Proper or supported base change for the square formed by the two projections to Y rewrites the inner pullback–pushforward as a pushforward from the triple product. The projection formula moves K_{23} inside that pushforward. Fubini, namely functoriality of $(-)_!$ for the composite projection, gives

$$p_{Z!}p_{23!}(p_{12}^*(p_X^* M \otimes K_{12}) \otimes p_{23}^* K_{23}) \simeq p_{Z!}p_{13!}(p_X^* M \otimes p_{12}^* K_{12} \otimes p_{23}^* K_{23}).$$

Applying the projection formula to p_{13} yields

$$p_{Z!}(p_X^* M \otimes p_{13!}(p_{12}^* K_{12} \otimes p_{23}^* K_{23})),$$

which is $\Phi_{K_{23} \star K_{12}}(M)$. For three kernels, both parenthesisations push forward the tensor $p_{12}^* K_{12} \otimes p_{23}^* K_{23} \otimes p_{34}^* K_{34}$ from $X_1 \times X_2 \times X_3 \times X_4$ to $X_1 \times X_4$; base change and Fubini identify them coherently. The diagonal calculation

$$p_{2!}(p_1^* M \otimes \Delta_! \mathbb{1}) \simeq M$$

proves the unit statement.

□

5.4.2 Associativity of convolution without suppressed steps

Let

$$K_{12} \in \mathcal{D}(X_1 \times X_2), \quad K_{23} \in \mathcal{D}(X_2 \times X_3), \quad K_{34} \in \mathcal{D}(X_3 \times X_4).$$

We compare the two parenthesised convolutions. Expanding the left-associated one gives

$$K_{34} \star (K_{23} \star K_{12}) = q_{14!} \left(q_{13}^* (K_{23} \star K_{12}) \otimes q_{34}^* K_{34} \right)$$

on $X_1 \times X_3 \times X_4$, where the symbols q_{ij} denote the relevant projections. Substitute the definition of the inner convolution:

$$K_{23} \star K_{12} = p_{13!}(p_{12}^* K_{12} \otimes p_{23}^* K_{23})$$

on $X_1 \times X_2 \times X_3$. Form the Cartesian square

$$\begin{array}{ccc} X_1 \times X_2 \times X_3 \times X_4 & \xrightarrow{r_{123}} & X_1 \times X_2 \times X_3 \\ \downarrow r_{134} & & \downarrow p_{13} \\ X_1 \times X_3 \times X_4 & \xrightarrow{q_{13}} & X_1 \times X_3. \end{array}$$

Beck–Chevalley replaces $q_{13}^* p_{13!}$ by $r_{134!} r_{123}^*$. The left-associated convolution is therefore

$$q_{14!} \left(r_{134!} r_{123}^* (p_{12}^* K_{12} \otimes p_{23}^* K_{23}) \otimes q_{34}^* K_{34} \right).$$

Apply the projection formula to r_{134}. Pulling back the last factor gives $r_{34}^* K_{34}$, while the first two factors become $r_{12}^* K_{12}$ and $r_{23}^* K_{23}$. Functoriality of supported pushforward then composes $q_{14!} r_{134!}$ into the projection $r_{14!}$ from the fourfold product. Hence

$$K_{34} \star (K_{23} \star K_{12}) \simeq r_{14!} \left(r_{12}^* K_{12} \otimes r_{23}^* K_{23} \otimes r_{34}^* K_{34} \right).$$

Now expand the right-associated convolution. One starts with

$$(K_{34} \star K_{23}) \star K_{12} = q'_{14!} \left({q'_{12}}^* K_{12} \otimes {q'_{24}}^* (K_{34} \star K_{23}) \right).$$

Substitution of the inner convolution, base change for the square obtained by forgetting X_1, projection formula, and composition of pushforwards give

$$(K_{34} \star K_{23}) \star K_{12} \simeq r_{14!} \left(r_{12}^* K_{12} \otimes r_{23}^* K_{23} \otimes r_{34}^* K_{34} \right).$$

The two expressions are therefore canonically equivalent. This proves binary associativity, but a monoidal operation requires the pentagon. For four kernels, each of the five parenthesisations reduces by the same procedure to one pushforward from the fivefold product of the tensor

$$r_{12}^{*}K_{12} \otimes r_{23}^{*}K_{23} \otimes r_{34}^{*}K_{34} \otimes r_{45}^{*}K_{45}.$$

Each edge of the pentagon is the comparison obtained from a pasting of Cartesian projection squares. By the coherence proved in §5.3.1, any two paths around the pentagon are the images of the same equivalence between two limit cones. The space of such equivalences is contractible. Hence the pentagon commutes, including all higher homotopies.

The unit coherence is similar but simpler. Let $I_X = \Delta_{X!}\mathbb{1}_X$. In $K \star I_X$, base change along the diagonal identifies the middle fibre product with the original product supporting K. Projection formula removes the factor $\mathbb{1}_X$, and the remaining pushforward is the identity projection. Thus $K \star I_X \simeq K$. The proof of $I_Y \star K \simeq K$ uses the other diagonal. For a pair of unit insertions, the two cancellations are induced by the same pullback of two diagonals, so the triangle coherence follows from the uniqueness of its limit cone.

This expanded proof shows exactly where the hypotheses enter. If one of the projection squares lacks Beck–Chevalley, the two parenthesisations need not agree. If projection formula fails, a coefficient cannot be moved into the fourfold integral. If pushforwards do not compose, the iterated integral has no Fubini theorem. Associativity of convolution is therefore not an independent miracle; it is the joint content of these three structural properties.

The coefficient carried by a correspondence is not incidental. It is the categorified analogue of a density or matrix entry, and its composition law is forced by the same pullback square as the underlying span.

Proposition 5.12 (Composition of weighted correspondences). [THEOREM]
Let

$$X \xleftarrow{p} Z \xrightarrow{q} Y, \qquad Y \xleftarrow{r} W \xrightarrow{s} T$$

be admissible correspondences with coefficients $A \in \mathcal{D}(Z)$ and $B \in \mathcal{D}(W)$. Their composite is the correspondence with middle object $U = Z \times_Y W$ and coefficient

$$C = \pi_Z^{*}A \otimes \pi_W^{*}B \in \mathcal{D}(U).$$

The associated pull–tensor–push functors satisfy

$$s_!(r^{*}(-) \otimes B) \circ q_!(p^{*}(-) \otimes A) \simeq (s\pi_W)_!((p\pi_Z)^{*}(-) \otimes C).$$

The kernel of the composite is $K_{W,B} \star K_{Z,A}$.

Proof of Proposition 5.12.
For $M \in \mathcal{D}(X)$, begin with

$$s_!(r^{*}q_!(p^{*}M \otimes A) \otimes B).$$

Beck–Chevalley for the defining square of U gives $r^{*}q_! \simeq \pi_{W!}\pi_Z^{*}$. Substitute this and apply the projection formula for π_W:

$$\begin{aligned} s_!(\pi_{W!}\pi_Z^{*}(p^{*}M \otimes A) \otimes B) &\simeq s_!\pi_{W!}(\pi_Z^{*}(p^{*}M \otimes A) \otimes \pi_W^{*}B) \\ &\simeq (s\pi_W)_!((p\pi_Z)^{*}M \otimes \pi_Z^{*}A \otimes \pi_W^{*}B). \end{aligned}$$

This is the stated functor. Applying the correspondence-to-kernel calculation of Definition 5.9 identifies its kernel with the convolution. Associativity follows because on a triple fibre product the coefficient is simply the tensor product of the three pulled-back coefficients; the two parenthesisations are identified by associativity of tensor product and the canonical equivalence of iterated pullbacks.

□

In the finite-set model, A and B are functions on Z and W, and the coefficient on $(z, w) \in Z \times_Y W$ is $A(z)B(w)$. Pushing forward sums over all pairs with matching middle point. The proposition is therefore the exact sheaf-theoretic lift of weighted matrix multiplication, including the measure factor that later becomes an orbital integral or a transfer weight.

The corresponding algorithmic and verification flow is summarised in Plate 5.3.

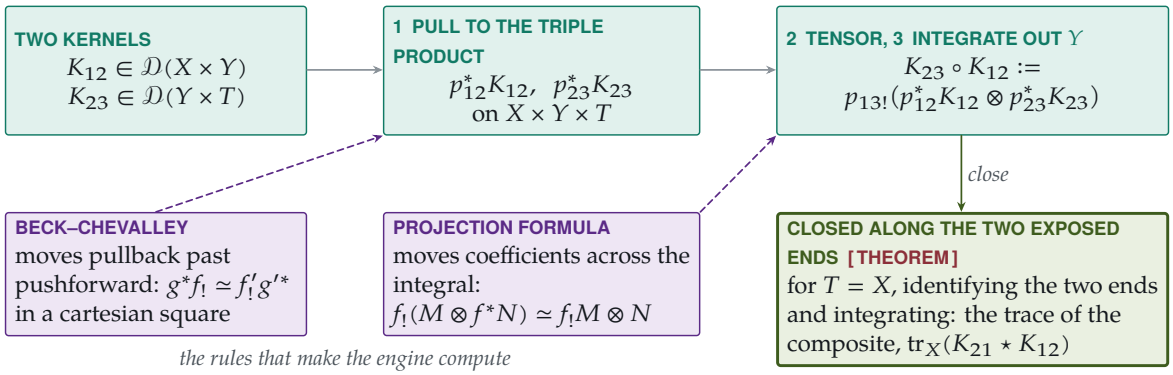


Plate 5.3. The convolution engine pulls two kernels to a triple product, tensors them, and integrates out the middle variable. Beck–Chevalley moves pullback past pushforward, the projection formula moves coefficients across the integral, and the same diagram, closed along its two exposed ends, computes the trace of the composite.

What it does not show: the coherence of the engine: associativity of convolution up to coherent equivalence rests on the correspondence machinery of Gaitsgory and Rozenblyum, which a diagram of spans does not prove.

5.5 Shadows and primitive traces

A bicategory has horizontal composition but need not have a symmetry that would permit one to rotate a composite. Ponto and Shulman isolate exactly the extra structure required for rotation (Ponto and Shulman 2012). Their notion is the proper home for Hochschild homology of algebras, Reidemeister traces, and traces of bimodules.

Definition 5.13 (Shadow). [ESTABLISHED]

Let $\mathcal{B}$ be a bicategory and $\mathcal{T}$ a category. A **shadow** on $\mathcal{B}$ consists of functors

$$\langle\!\langle - \rangle\!\rangle_A : \mathcal{B}(A, A) \longrightarrow \mathcal{T}$$

for every object A, together with natural cyclicity isomorphisms

$$\theta_{M,N} : \langle\!\langle M \odot N \rangle\!\rangle_A \xrightarrow{\sim} \langle\!\langle N \odot M \rangle\!\rangle_B$$

for $M : A \to B$ and $N : B \to A$. These are required to be unital and to satisfy the cyclic hexagon: rotating three composable one-morphisms by one place and then by one place again agrees with the direct two-place rotation.

If $M : A \to B$ is right dualisable with right dual $M^\vee : B \to A$, and $f : Q \odot M \to M \odot P$ is a two-morphism with $Q : A \to A$ and $P : B \to B$, its bicategorical trace is the composite

$$\begin{aligned} \langle\!\langle Q \rangle\!\rangle_A &\xrightarrow{\text{coev}} \langle\!\langle Q \odot M \odot M^\vee \rangle\!\rangle_A \\ &\xrightarrow{f} \langle\!\langle M \odot P \odot M^\vee \rangle\!\rangle_A \\ &\xrightarrow{\theta} \langle\!\langle P \odot M^\vee \odot M \rangle\!\rangle_B \\ &\xrightarrow{\text{ev}} \langle\!\langle P \rangle\!\rangle_B. \end{aligned}$$

For rings and bimodules, the shadow of an A–A bimodule M is $HH_0(A; M)$, or its derived Hochschild complex. The cyclic isomorphism is the familiar passage from $M \otimes_B N$ modulo commutators to $N \otimes_A M$ modulo commutators. For presentable categories and kernels, the shadow of an endokernel is its restriction to the diagonal followed by integration. This is the geometric formula proved below.

Proposition 5.14 (A shadow supplies the cyclic core of a primitive trace). [THEOREM]

Let $\mathfrak{C}$ be a chôra whose one- and two-morphisms form a bicategory $\mathcal{B}$ of dualisable kernels or correspondences. A shadow on $\mathcal{B}$, together with:

1. *additivity for exact triangles of endomorphisms;*
2. *symmetric monoidality for external products; and*
3. *a localisation equivalence from the shadow of a geometric endomorphism to an integral over its derived fixed-point figure,*

defines the primitive trace theory required by KH.6–KH.7. Conversely, the cyclicity datum of any such primitive trace restricts to a Ponto–Shulman shadow on endomorphism one-categories.

Proof of Proposition 5.14.

Given a shadow, the trace of a two-morphism is defined by the coevaluation, cyclic rotation, and evaluation composite in Definition 5.13. Naturality of that construction gives invariance under equivalence and conjugation. The cyclic hexagon gives compatibility with iterated composition. The three added conditions supply precisely the parts not contained in the definition of a shadow: exact additivity, multiplicativity across the monoidal product, and a geometric identification of the abstract cyclic value with local fixed-point terms. These are the data demanded of the primitive trace in the axiom system, so they define it.

Conversely, let τ be a primitive trace. For $M : A \to B$ and $N : B \to A$, cyclic invariance gives an equivalence $\tau_A(M \odot N) \simeq \tau_B(N \odot M)$. The associativity coherence of τ gives the cyclic hexagon, and its unit coherence gives the shadow unit axiom. Restriction to each $\mathcal{B}(A, A)$ is functorial. Hence these restrictions form a shadow. The argument also shows why a shadow is exactly the *cyclic core* of a primitive trace, but not the whole primitive trace: localisation is additional geometric data and cannot be recovered from cyclic rotation alone.

□

The established part of this proposition is Ponto–Shulman's shadow and trace formalism. The theorem-of-framework part is the identification of its cyclic core with the primitive trace axiom. This is a definition-level theorem, not a claim that the arithmetic localisation datum has been constructed.

5.6 The trace of a composite

We now close the convolution diagram. Write $\int_X A$ for $p_{X!}A$, valued in the coefficient category at a point. Assume throughout this section that the figures and kernels are dualisable and that all displayed pushforwards, base-change maps, and projection formulas are defined.

Theorem 5.15 (Composition-of-traces theorem). [THEOREM]
Let $K_{12} \in \mathcal{D}(X \times Y)$ and $K_{21} \in \mathcal{D}(Y \times X)$ be kernels. Then

$$\mathrm{tr}_X(K_{21} \star K_{12}) \simeq \int_{X\times Y} K_{12}(x,y) \otimes K_{21}(y,x) \simeq \mathrm{tr}_Y(K_{12} \star K_{21}).$$

In invariant notation, if $\sigma : X \times Y \to Y \times X$ exchanges the factors, the middle term is

$$(p_{X\times Y})_!(K_{12} \otimes \sigma^* K_{21}).$$

If $K_{12} = (p_1, p_2)_! A$ comes from $X \xleftarrow{p_1} Z_{12} \xrightarrow{p_2} Y$ and $K_{21} = (q_2, q_1)_! B$ comes from $Y \xleftarrow{q_2} Z_{21} \xrightarrow{q_1} X$, then the trace is

$$\int_{Z_{12}\times_{X\times Y} Z_{21}^{\mathrm{op}}} \pi_{12}^* A \otimes \pi_{21}^* B,$$

where $Z_{21}^{\mathrm{op}} \to X \times Y$ is (q_1, q_2). In particular the trace is computed by a fibre-product integral.

Proof of Theorem 5.15.
The trace of an endokernel $L \in \mathcal{D}(X \times X)$ is the shadow

$$\mathrm{tr}_X(L) = \int_X \Delta_X^* L.$$

Apply this to

$$L = K_{21} \star K_{12} = p_{13!}(p_{12}^* K_{12} \otimes p_{23}^* K_{21})$$

on $X \times Y \times X$. Let $i : X \times Y \to X \times Y \times X$ be $i(x,y) = (x,y,x)$. The square

$$\begin{array}{ccc} X \times Y & \xrightarrow{i} & X \times Y \times X \\ \downarrow p_X & & \downarrow p_{13} \\ X & \xrightarrow{\Delta_X} & X \times X \end{array}$$

is Cartesian. Beck–Chevalley gives $\Delta_X^* p_{13!} \simeq p_{X!} i^*$. Therefore

$$\begin{aligned} \mathrm{tr}_X(K_{21} \star K_{12}) &= \int_X \Delta_X^* p_{13!}(p_{12}^* K_{12} \otimes p_{23}^* K_{21}) \\ &\simeq \int_X p_{X!} i^* (p_{12}^* K_{12} \otimes p_{23}^* K_{21}) \\ &\simeq \int_{X\times Y} K_{12} \otimes \sigma^* K_{21}. \end{aligned}$$

The last equality uses $p_{12} i = \mathrm{id}_{X\times Y}$ and $p_{23} i = \sigma$. Repeating the calculation with X and Y interchanged gives the second trace and the same integral after applying the symmetry of the tensor product. This proves cyclicity.

For correspondence kernels, substitute $K_{12} = (p_1, p_2)_! A$ and $\sigma^* K_{21} = (q_1, q_2)_! B$. The tensor product of these two pushforwards is, by base change and the projection formula, the pushforward from their derived fibre product over $X \times Y$ of $\pi_{12}^* A \otimes \pi_{21}^* B$. Integrating the resulting pushforward gives the stated fibre-product integral. Every equivalence used is one of Proposition 5.7 or Proposition 5.8, so the proof also establishes its coherence under further composition.

□

Corollary 5.16 (Kernel form of the Lefschetz trace). [THEOREM]
Let $f : X \to X$ and let $K_f = (\mathrm{id}, f)_! A$ be the kernel of the graph correspondence. Then

$$\mathrm{tr}(K_f) \simeq \int_{X^f} i^* A, \qquad X^f := X \times_{X \times X, (\mathrm{id}, f)} X.$$

Thus the categorical trace is supported on the derived fixed-point figure. If the fixed points are isolated and the sheaf theory carries a Lefschetz local term, decategorification gives the sum of those local terms.

Proof of Corollary 5.16.
The trace is $\int_X \Delta^* (\mathrm{id}, f)_! A$. Apply base change to the Cartesian square defining X^f. It identifies $\Delta^* (\mathrm{id}, f)_! A$ with the pushforward from X^f of $i^* A$. Composing pushforwards gives the displayed integral. The local-term statement is exactly the localisation comparison of the chosen trace theory; for constructible sheaves it is the Lefschetz–Verdier theorem of SGA 5 (Grothendieck and Illusie 1977), with refinements such as (Hoyois 2014).

□

Proposition 5.17 (Trace as intersection of two correspondences). [THEOREM]
Let

$$X \xleftarrow{p_1} Z_1 \xrightarrow{p_2} Y, \qquad Y \xleftarrow{q_2} Z_2 \xrightarrow{q_1} X$$

carry coefficients A_1 and A_2. Suppose that the derived fibre product

$$I = Z_1 \times_{X \times Y} Z_2,$$

where the maps are (p_1, p_2) and (q_1, q_2), is proper over the point. Then

$$\mathrm{tr}_X \left((q_1)_! (q_2^* (-) \otimes A_2) \circ (p_2)_! (p_1^* (-) \otimes A_1) \right) \simeq R\Gamma_c \left(I, \pi_1^* A_1 \otimes \pi_2^* A_2 \right).$$

If I is a finite transverse set of points and both coefficients are constant, the numerical trace is $\#I$; with Weil complexes it is the sum of the local Frobenius traces of the tensor coefficient.

Proof of Proposition 5.17.
The first assertion is the correspondence form of Theorem 5.15. Properness permits the integral over I to be identified with compactly supported global sections. If I is finite and transverse, its derived structure is discrete and $R\Gamma_c(I, -)$ is the direct sum of the stalks. The Euler trace of a direct sum is the sum of the Euler traces, giving $\#I$ for the constant rank-one coefficient. For Weil complexes, the same direct-sum decomposition commutes with Frobenius, so its trace is the sum of the stalk traces. No additional fixed-point theorem is used in this transverse case; the local terms are literally the stalks on the derived intersection.

□

This proposition separates the formal trace from the difficult local-term problem. When the intersection is transverse, the local term is one. When it is not transverse, the derived fibre product retains an excess normal complex. A Lefschetz local term is a rule for extracting from that complex the correct coefficient, often involving a determinant of $1 - df$, an orientation, or a vanishing-cycle complex. The shadow tells us *where* the local term lives; it does not compute that term without the localisation theory of the sheaf formalism.

For ordinary finite sets the proposition can be checked without category theory. Let $K_{12}(x, y)$ and $K_{21}(y, x)$ be scalar kernels. Their composite has diagonal value

$$(K_{21} K_{12})(x, x) = \sum_y K_{12}(x, y) K_{21}(y, x).$$

Summing over x gives the middle integral in Theorem 5.15. Interchanging the order of the two finite sums gives the trace of the reverse composite. In geometry, Beck–Chevalley is the replacement for this interchange of sums, and properness or nuclearity is the condition under which the interchange is legitimate.

5.6.1 From derived intersections to local terms

The derived fibre product is essential even when the final answer is a number. Suppose X is smooth and $f : X \to X$ has an isolated fixed point x. The classical intersection of Γ_f with Δ_X remembers only the point. The derived intersection also remembers the two-term tangent complex

$$T_x X \xrightarrow{1-df_x} T_x X.$$

If $1 - df_x$ is invertible, the complex is acyclic and the intersection is transverse. The local term for the constant sheaf is then one in the ordinary étale Lefschetz formula. In coherent or oriented theories, the determinant of $1 - df_x$ can survive as a refined local index. If the map is not invertible, the cohomology of the tangent complex records the excess fixed directions, and a naive count of classical points loses precisely the information needed for the trace.

The kernel proof therefore has two stages. The first, entirely formal, is

$$\operatorname{tr}(K_f) \simeq \int_{X^f} \mathcal{L}_f,$$

where $\mathcal{L}_f$ is the pullback of the kernel coefficient to the derived fixed-point figure. The second is a localisation theorem that rewrites the integral as a sum of contributions supported on connected components or points. The first stage requires duality, base change, and projection formula. The second can require purity, orientations, vanishing cycles, deformation to the normal cone, or a theory of characteristic classes. Calling both stages "the trace formula'' obscures where the geometry enters.

For two correspondences the excess complex has the same form. At a point (z_1, z_2) of $Z_1 \times_{X \times Y} Z_2$, the tangent complex is the fibre of

$$T_{z_1} Z_1 \oplus T_{z_2} Z_2 \longrightarrow T_x X \oplus T_y Y,$$

where the arrow is the difference of the derivatives of the two maps to $X \times Y$. Transversality means that this arrow is surjective with the expected kernel. When it fails, the derived structure is the correct intersection multiplicity before decategorification. This observation is the geometric reason that the composition-of-traces theorem is stated with a derived fibre product rather than an ordinary set-theoretic intersection.

There is also a measure-theoretic analogue. For finite sets, Fubini permits the order of summation to be exchanged. For locally compact spaces, the same operation requires absolute integrability or a trace-class hypothesis. In a stable analytic category, nuclearity is intended to play that role. Thus the three phrases `proper correspondence''`, trace-class kernel'', and "nuclear functor'' express versions of one demand at different levels: the middle variable may be integrated out, and the resulting integral may be closed into a trace without dependence on the order of operations.

This is the point at which the present calculus interfaces with the later Arthur–Selberg theory. Orbital integrals are local terms on the geometric fixed-point side; characters and multiplicities are local terms on the spectral side. The categorical theorem identifies the ambient trace in which both decompositions should live. It does not prove convergence

of either decomposition, nor the equality of unstable and stabilised groupings. Those are additional analytic and endoscopic theorems. The gain is exact bookkeeping: once both decompositions are defined as localisations of the same shadow, their equality is no longer a separate formal manipulation.

The middle integral in Theorem 5.15 is the precise categorical shadow of the elementary matrix identity

$$\operatorname{tr}(BA) = \sum_{x,y} A_{xy} B_{yx} = \operatorname{tr}(AB).$$

The theorem replaces the finite index set by a derived fibre product, the matrix entries by sheaf-theoretic kernels, and the finite sum by supported pushforward. Nothing else changes. This is why the Lefschetz formula is not an analogy for the trace calculus: it is its geometric instance.

5.7 A normal form for trace calculations

A later argument may begin with a long composite of pullbacks, tensor products, and pushforwards whose geometry is difficult to see. The following reduction puts such an expression into a form to which the trace rules apply.

First, replace every pull–tensor–push functor by its weighted correspondence. This step records the source, the target, the middle figure, and the coefficient separately. Second, compose adjacent correspondences by forming their derived fibre product and pulling all coefficients to it, as in Proposition 5.12. At this stage no trace has been taken; one has a single span with a single coefficient. Third, pass to the kernel on the product of the exposed source and target. This makes external products and tensor factors visible. Fourth, if the source and target agree, close the kernel by pulling it to the diagonal. Base change then replaces the closure by the derived intersection of the original correspondence with the diagonal. Fifth, integrate the coefficient on that intersection. Only after this object has been formed should one decategorify, decompose into local terms, or take an ordinary numerical trace.

In symbols, a composite

$$X_0 \leftarrow Z_1 \rightarrow X_1 \leftarrow Z_2 \rightarrow \cdots \leftarrow Z_n \rightarrow X_n$$

with $X_n = X_0$ reduces to

$$Z := Z_1 \times_{X_1} Z_2 \times_{X_2} \cdots \times_{X_{n-1}} Z_n,$$

carrying the coefficient

$$A := \pi_1^* A_1 \otimes \cdots \otimes \pi_n^* A_n.$$

If $a, b : Z \to X_0$ are the exposed legs, the trace is

$$\int_{Z \times_{X_0 \times X_0} X_0} A,$$

where $X_0 \to X_0 \times X_0$ is the diagonal and the fibre product uses (a, b). This is the normal form of every trace expression in the chapter. Cyclicity says that choosing another place at which to cut the circular chain produces a canonically equivalent integral.

The order of operations is a safeguard. Taking stalk traces before composing kernels can discard derived intersection multiplicities. Replacing ! by $*$ before proving ambidexterity

can delete boundary contributions. Passing to isomorphism classes before imposing the groupoid measure can lose automorphism weights. Taking a numerical trace before proving nuclearity can turn a defined trace object into a divergent series. The normal form delays each irreversible operation until its hypotheses have been checked.

For practical use, a trace calculation should therefore carry a short audit: which maps supply the pullbacks; which maps admit !-pushforward; which Cartesian squares supply Beck–Chevalley; which objects or categories are dualisable; which structure map makes the final integral; and which theorem identifies that integral with local terms. In the function-field examples all six entries are established. In the arithmetic applications the first missing entry must be stated rather than hidden by formal notation.

5.8 Trace calculus rules

The following rules are the identities used later without rederivation. Each is stated in the stable symmetric monoidal setting. When a rule is applied to kernels, the hypotheses include the existence of the pushforwards and the base-change equivalences appearing in §5.4.

1. **Cyclicity.**

Proposition 5.18 (Trace rule: cyclicity). [THEOREM]

Let $\mathcal{C}$ and $\mathcal{D}$ be dualisable objects of a symmetric monoidal $(\infty,2)$-category with shadow. If $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are composable dualisable one-morphisms, then

$$\mathrm{tr}_{\mathcal{C}}(G \circ F) \simeq \mathrm{tr}_{\mathcal{D}}(F \circ G).$$

More generally, the trace of a cyclic word is invariant under cyclic permutation of its factors.

Proof of Proposition 5.18.

The first equivalence is the shadow rotation $\theta_{F,G}$ of Definition 5.13. For three factors, the cyclic hexagon says that two successive one-place rotations agree with a direct two-place rotation. Induction gives the assertion for any finite cyclic word. In the kernel model, this is exactly Theorem 5.15.

□

2. **Additivity in exact triangles.**

Proposition 5.19 (Trace rule: additivity). [ESTABLISHED]

Let $X \to Y \to Z$ be a cofibre sequence of dualisable objects in a stable symmetric monoidal infinity-category, and let f_X, f_Y, f_Z be a compatible endomorphism of this triangle. Then

$$\mathrm{tr}(f_Y) = \mathrm{tr}(f_X) + \mathrm{tr}(f_Z)$$

in $\pi_0 \mathrm{End}(\mathbb{1})$. At spectrum level there is a canonical additivity path whose image on π_0 is the displayed equality. The same statement holds for a cofibre sequence of endokernels.

Proof of Proposition 5.19.

May's additivity theorem proves the statement for triangulated traces under the standard compatibility axioms (May 2001). In stable infinity-categories the cofibre square is also a pushout square, and the trace is an additive invariant of endomorphisms: applying the universal additive invariant sends the square to a cofibre sequence, while the trace map is natural on that invariant. Hoyois, Scherotzke, and Sibilla give the higher trace and categorified Chern-character form of this argument (Hoyois et al. 2017). Therefore the path exists before passage to π_0, and its decategorification is May's equality. For kernels, apply the exact functor Δ^* followed by the exact pushforward to a cofibre sequence of kernels.

□

3. **Multiplicativity under tensor product.**

Proposition 5.20 (Trace rule: multiplicativity). [ESTABLISHED]
Let X, Y be dualisable and let $f : X \to X$, $g : Y \to Y$. Then

$$\operatorname{tr}(f \otimes g) \simeq \operatorname{tr}(f) \otimes \operatorname{tr}(g).$$

If the target endomorphisms of the unit form a commutative ring spectrum, this is multiplication in that ring spectrum.

Proof of Proposition 5.20.
The dual of $X \otimes Y$ is $X^\vee \otimes Y^\vee$. Its coevaluation is the tensor product of the two coevaluations, followed by the symmetry that places dual factors next to their partners. Its evaluation is the tensor product of the two evaluations, preceded by the inverse symmetry that places each dual factor beside its primal factor. Substituting these maps into the definition of trace separates the composite into the tensor product of the trace composite for f and that for g. Symmetric-monoidal coherence identifies the two expressions.

□

4. **Behaviour under adjunction.**

Proposition 5.21 (Trace rule: adjunction and mates). [THEOREM]
Let $M : A \to B$ be right dualisable with right adjoint one-morphism $M^\vee : B \to A$. Let $Q : A \to A$ and $P : B \to B$, and let

$$f : Q \odot M \Longrightarrow M \odot P$$

be a two-morphism. Its right mate is

$$f^\# : M^\vee \odot Q \Longrightarrow P \odot M^\vee.$$

The bicategorical trace computed from f agrees with the trace computed from $f^\#$ after the canonical shadow rotations. In particular, if $F : \mathcal{C} \to \mathcal{D}$ is an equivalence, then for every endomorphism h of $\mathcal{C}$,

$$\operatorname{tr}_{\mathcal{D}}(FhF^{-1}) \simeq \operatorname{tr}_{\mathcal{C}}(h).$$

Proof of Proposition 5.21.
By definition, $f^\#$ is obtained by adjoining the coevaluation $\mathrm{id}_B \to M \odot M^\vee$, applying $M^\vee \odot f \odot M^\vee$ in the appropriate factors, and then applying the evaluation $M^\vee \odot M \to \mathrm{id}_A$. Insert this expression into the trace of $f^\#$. The resulting closed string contains an adjacent coevaluation–evaluation pair on each side. The two triangle identities cancel those pairs. What remains is the closed string defining the trace of f, with only a cyclic rotation of factors; the shadow hexagon identifies that rotation canonically. For an equivalence take $M = F$ and $M^\vee = F^{-1}$, with $Q = h$ and $P = FhF^{-1}$. The identity $Fh \simeq (FhF^{-1})F$ supplies f, and the general mate statement gives conjugation invariance.

□

5. **The Künneth rule.**

Proposition 5.22 (Trace rule: Künneth). [ESTABLISHED]
Let $K \in \mathcal{D}(X \times X)$ and $L \in \mathcal{D}(Y \times Y)$ be trace-class endokernels. Their external product is an endokernel on $X \times Y$, and

$$\operatorname{tr}_{X\times Y}(K \boxtimes L) \simeq \operatorname{tr}_X(K) \otimes \operatorname{tr}_Y(L).$$

For endomorphisms $f : X \to X$ and $g : Y \to Y$ one has

$$(X \times Y)^{f\times g} \simeq X^f \times Y^g,$$

and the Lefschetz integral factors accordingly.

Proof of Proposition 5.22.
The diagonal of a product is $\Delta_{X\times Y} = \Delta_X \times \Delta_Y$. Hence

$$\Delta^*_{X\times Y}(K \boxtimes L) \simeq \Delta^*_X K \boxtimes \Delta^*_Y L.$$

The Künneth equivalence for supported pushforward gives

$$\int_{X\times Y} (\Delta^*_X K \boxtimes \Delta^*_Y L) \simeq \left(\int_X \Delta^*_X K\right) \otimes \left(\int_Y \Delta^*_Y L\right).$$

This is the trace formula. The fixed-point equivalence follows by distributing finite limits over products: both sides are the pullback of $\Gamma_f \times \Gamma_g$ along $\Delta_X \times \Delta_Y$. Substitution in Corollary 5.16 gives the final assertion.

□

6. **Base change of traces.**

Proposition 5.23 (Trace rule: base change). [THEOREM]
Let $S' \xrightarrow{g} S$, let $p : X \to S$, and let $f : X \to X$ be an S-endomorphism. Put $X' = X \times_S S'$ and let f' be the base change of f. Suppose the graph kernels are trace-class and the relevant Beck–Chevalley maps are equivalences. Then

$$g^* \operatorname{tr}_S(K_f) \simeq \operatorname{tr}_{S'}(K_{f'}).$$

Moreover,

$$(X')^{f'} \simeq X^f \times_S S',$$

and the trace equivalence is the pullback of the fixed-point integral.

Proof of Proposition 5.23.
The fixed-point figure X^f is the pullback of the graph $(\mathrm{id}, f) : X \to X \times_S X$ along the relative diagonal $\Delta_{X/S}$. Base-changing this entire diagram along g gives the graph of f' and the diagonal of X'/S'. Since finite limits commute with base change, the resulting fixed-point object is $X^f \times_S S'$. By Corollary 5.16,

$$\operatorname{tr}_S(K_f) = p^f_! \mathbb{1}_{X^f},$$

with the pulled-back coefficient inserted when present. Proper or supported base change for the square

$$\begin{array}{ccc} X^f \times_S S' & \longrightarrow & X^f \\ \downarrow & & \downarrow p^f \\ S' & \xrightarrow{g} & S \end{array}$$

identifies $g^{*}p^{f}_{!}$ with $p^{f'}_{!}\,g^{*}_{f}$. This is the asserted trace identity. Compatibility under iterated base change follows from the pasting coherence of Proposition 5.7.

□

7. **Transfer along a non-invertible morphism.**

Proposition 5.24 (Trace rule: non-invertible transfer). [ESTABLISHED]
Let $M : A \to B$ be right dualisable in a bicategory with shadow, not necessarily invertible. For endomorphisms $Q : A \to A$ and $P : B \to B$, every two-morphism

$$f : Q \odot M \Longrightarrow M \odot P$$

induces a canonical trace transfer

$$\operatorname{tr}(f) : \langle\!\langle Q \rangle\!\rangle_{A} \longrightarrow \langle\!\langle P \rangle\!\rangle_{B}.$$

Transfers compose under horizontal composition of right-dualisable one-morphisms. If M is not invertible, the transfer need not be an equivalence; there is in general no equality $\operatorname{tr}(Q) = \operatorname{tr}(P)$.

Proof of Proposition 5.24.
The transfer is the coevaluation–f–rotation–evaluation composite in Definition 5.13. If $N : B \to C$ is right dualisable and $h : P \odot N \Rightarrow N \odot R$, substitute the definition of $\operatorname{tr}(f)$ into that of $\operatorname{tr}(h)$. The evaluation of M and the coevaluation of N can be regrouped, by associativity and the shadow hexagon, into the coevaluation and evaluation for the composite $M \odot N$, whose right dual is $N^{\vee} \odot M^{\vee}$. The resulting closed string is the trace of the horizontally pasted two-morphism

$$Q \odot M \odot N \Longrightarrow M \odot N \odot R.$$

Thus transfers compose. Ponto and Shulman's functoriality theorem for bicategorical traces proves this in full coherence (Ponto and Shulman 2012). Invertibility was used in Proposition 5.21 only to turn the transfer into conjugation invariance. Without an inverse there is no map in the opposite direction unless additional adjoint or ambidexterity data are supplied, so no trace equivalence follows.

□

The distinction between multiplicativity and Künneth is practical. Proposition 5.20 is an abstract statement about dualisable objects. Proposition 5.22 is its geometric realisation, including the identification of the fixed-point figure and the factorisation of the integral.

5.9 Special and degenerate cases

The formalism is easiest to test at its extremes. These cases also prevent several common errors: confusing an empty trace with a divergent trace, confusing a classical fixed-point set with its derived intersection, and assuming cyclicity only after diagonalising an operator.

5.9.1 A complete two-point matrix calculation

Let $X = Y = \{1, 2\}$ and take kernels with values in vector spaces whose Euler characteristics form the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

The entry A_{ij} is the coefficient carried by the correspondence from i to j. Convolution sums over the middle point. Direct multiplication gives

$$BA = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}, \qquad AB = \begin{pmatrix} -2 & 5 \\ -4 & 11 \end{pmatrix}.$$

The diagonal traces are both nine. The fibre-product integral of Theorem 5.15 gives the same value without first multiplying:

$$\begin{aligned} \sum_{i,j} A_{ij} B_{ji} &= A_{11}B_{11} + A_{12}B_{21} + A_{21}B_{12} + A_{22}B_{22} \\ &= (1)(0) + (2)(-1) + (3)(1) + (4)(2) \\ &= 9. \end{aligned}$$

Reversing the factors gives

$$\sum_{j,i} B_{ji} A_{ij} = 9,$$

which is literally the same four summands in another order. If the entries are complexes rather than integers, multiplication is tensor product, addition is direct sum, and each integer above is replaced by an Euler characteristic. The calculation then proves cyclicity after decategorification while the shadow equivalence proves it before decategorification.

Take instead

$$A' = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B' = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

There is no matching middle index, so $B'A' = A'B' = 0$. Geometrically the fibre product of the two correspondences is empty. Its compactly supported integral is the zero object. This is an empty trace, not a failure of trace-classness. By contrast, an infinite identity matrix has a nonempty diagonal but a divergent numerical trace; categorically it signals failure of the required compactness or nuclearity. The distinction between emptiness and divergence is load-bearing in the automorphic setting.

5.9.2 Identity kernels and Hochschild self-intersection

For any admissible figure X, the identity kernel is $I_X = \Delta_! \mathbb{1}_X$. Closing it gives

$$\operatorname{tr}(I_X) = \int_X \Delta^* \Delta_! \mathbb{1}_X.$$

If X is a finite discrete set of cardinality n, the diagonal has no derived self-intersection and the answer is the direct sum of n copies of the coefficient unit. Its numerical trace is n. If $X = \operatorname{Spec} A$ is derived affine and the sheaf theory is modules, the same expression is

$$A \otimes_{A \otimes A^{\mathrm{op}}} A = HH(A).$$

Thus the trace of the identity changes from ordinary counting to Hochschild homology precisely when the diagonal acquires derived thickness. This is the local algebra behind the book's insistence that the arithmetic diagonal be measured by THH.

For the dual numbers $A = k[\varepsilon]/(\varepsilon^2)$, the degree-zero Hochschild homology is

$$HH_0(A) = A/[A, A] = A,$$

which has basis $1, \varepsilon$. Even though $\operatorname{Spec} A$ has one classical point, the trace of its identity has two degree-zero directions. A point count would see one; the kernel trace sees the nilpotent self-intersection. Higher Hochschild groups retain further information about the singularity.

5.9.3 A non-transverse fixed point

Let k have characteristic different from two and consider

$$f : \mathbb{A}^1_k \longrightarrow \mathbb{A}^1_k, \qquad f(t) = t + t^2.$$

The fixed-point equation is $t^2 = 0$. Classically there is only the point $t = 0$, but the scheme-theoretic intersection of the graph with the diagonal is

$$\operatorname{Spec} k[t]/(t^2),$$

of length two. The derivative satisfies $df_0 = 1$, so $1 - df_0 = 0$ and transversality fails maximally in the one-dimensional tangent direction. In a coherent intersection theory, the local trace object retains the two-dimensional algebra $k[t]/(t^2)$; in a constructible Lefschetz theory, a localisation theorem applies the appropriate sheaf-theoretic local-term map rather than simply taking its length. The example shows why §5.6.1 separates the derived fixed-point object from its chosen numerical realisation.

Perturb the map to $f_a(t) = at + t^2$ with $a \neq 1$. The fixed equation is

$$t((a - 1) + t) = 0,$$

so there are two distinct fixed points when $a - 1$ is nonzero. At $t = 0$, $1 - df_a = 1 - a$ is invertible; at $t = 1 - a$, $1 - df_a = a - 1$ is invertible. The length-two non-transverse point has split into two transverse points. The total intersection number is preserved. This is the elementary deformation-to-transversality picture behind the stability of Lefschetz traces.

5.9.4 Small residue fields in the Hecke calculations

At $Q = 2$, the spherical GL_2 fibre is $\mathbf{P}^1(\mathbf{F}_2)$ with the three lines

$$[1 : 0], \qquad [0 : 1], \qquad [1 : 1].$$

The normalised Hecke operator sends the constant function to

$$2^{-1/2}(3)\mathbf{1} = (2^{1/2} + 2^{-1/2})\mathbf{1}.$$

At Iwahori level the selected flag removes one of these three lines, leaving two relative-position choices. The unnormalised matrix is

$$\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix},$$

with characteristic polynomial $\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1)$. Its eigenvectors may be taken as $(1, 1)$ for eigenvalue 2 and $(-2, 1)$ for eigenvalue -1. After normalisation the eigenvalues are

$\sqrt{2}$ and $-1/\sqrt{2}$. This smallest residue field already distinguishes the spherical, trivial-Iwahori, and Steinberg traces without any asymptotic argument.

For GL_1 over the same field there remains only one local coset per valuation, so no $Q+1$ or Q multiplicity occurs. The comparison makes the source of the multiplicities visible: they count flags in a two-dimensional residue-space fibre, not powers inserted by normalisation.

5.10 The elementary correspondence for GL_1

For $G = \mathrm{GL}_1$, the bundle stack is the Picard stack $\mathrm{Bun}_1 = \mathrm{Pic}(X)$. At a closed point x of degree d, a lower modification is uniquely

$$L' = L(-x) \subset L, \qquad L/L(-x) \simeq k_x.$$

There is no projective line of choices. The Hecke stack is therefore the graph of the translation

$$t_{-x} : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X), \qquad L \longmapsto L(-x).$$

Its kernel is

$$\mathcal{K}_x^{(1)} = (\mathrm{id}, t_{-x})_! \overline{\mathbf{Q}}_\ell,$$

and the associated transform is simply

$$\mathsf{T}_x^{(1)}(\mathcal{F}) = (t_{-x})_! \mathcal{F}.$$

On functions this reads

$$(T_x^{(1)} f)(L) = f(L(x)),$$

with the sign of x reversed if upper modifications are chosen. Locally, $F_x^\times$ has maximal compact subgroup $O_x^\times$, and the spherical double coset is the single coset

$$O_x^\times \varpi_x O_x^\times = \varpi_x O_x^\times.$$

Thus the spherical Hecke algebra is the Laurent polynomial algebra

$$\mathcal{H}(F_x^\times, O_x^\times) \simeq \overline{\mathbf{Q}}_\ell[z, z^{-1}],$$

where z is the kernel of translation by one valuation. If χ_x is an unramified character and

$$\alpha_x = \chi_x(\varpi_x),$$

then its spherical line is one-dimensional and

$$T_x^{(1)} \mid \chi_x = \alpha_x, \qquad P_x(u) = 1 - \alpha_x u, \qquad L_x(s, \chi) = \frac{1}{1 - \alpha_x Q^{-s}}.$$

Every formula can be checked from the single coset: convolution gives

$$\mathcal{K}_x^{(1)} \star \mathcal{K}_x^{(1)} \simeq (\mathrm{id}, t_{-2x})_! \overline{\mathbf{Q}}_\ell,$$

and the eigenvalue is α_x^2. The inverse kernel is the graph of t_x, and

$$\mathcal{K}_{x,+}^{(1)} \star \mathcal{K}_{x,-}^{(1)} \simeq \Delta_! \overline{\mathbf{Q}}_\ell.$$

Consequently its trace is the trace of the identity on the chosen Picard component.

The unbalanced categorical trace again records components. Translation by $-x$ sends Pic^n to Pic^{n-d}, so its graph does not meet the diagonal and

$$\mathrm{tr}(\mathcal{K}_x^{(1)}) = 0.$$

A degree-zero translation is obtained from two places x, y of equal degree:

$$t_{x-y}(L) = L(x-y).$$

On Pic^0, closing this graph against Frobenius asks for line bundles satisfying

$$\mathrm{Frob}_q^* L \otimes L^{-1} \simeq O_X(x-y).$$

Lang's theorem for the connected algebraic group Pic^0 shows that the map $L \mapsto \mathrm{Frob}_q^* L \otimes L^{-1}$ is surjective and that every fibre is a torsor under $\mathrm{Pic}^0(\mathbf{F}_q)$. Hence the fixed fibre product has exactly $\#\,\mathrm{Pic}^0(\mathbf{F}_q)$ geometric points. For $X = \mathbf{P}^1$, Pic^0 is a point, so the balanced Frobenius–Hecke trace is exactly one. This elementary computation is the rank-one model for component balancing, fixed-point integration, and the appearance of class groups in global trace formulae.

5.11 A complete local calculation: the spherical Hecke kernel for GL_2

Let X be a smooth projective geometrically connected curve over $k = \mathbf{F}_q$, let $x \in |X|$ be a closed point of degree d, and put

$$k_x = \mathbf{F}_{q^d}, \qquad Q = |k_x| = q^d.$$

Write Bun_2 for the stack of rank-two vector bundles on X. We use lower modifications; the opposite convention gives the contragredient operator.

Example 5.25 (The one-place GL2 Hecke correspondence). [ESTABLISHED]

The Hecke stack $\mathcal{H}ecke_x$ classifies exact sequences

$$0 \longrightarrow E' \longrightarrow E \longrightarrow k_x \longrightarrow 0$$

with E, E' rank-two vector bundles and k_x the length-one skyscraper sheaf at x. The two legs are

$$\mathrm{Bun}_2 \xleftarrow{h_0} \mathcal{H}ecke_x \xrightarrow{h_1} \mathrm{Bun}_2, \qquad h_0(E' \subset E) = E, \quad h_1(E' \subset E) = E'.$$

For fixed E, a modification is determined by a nonzero quotient $E_x \twoheadrightarrow k_x$ up to scalar. Consequently

$$h_0^{-1}(E)(k_x) \simeq \mathbf{P}(E_x^\vee)(k_x) \simeq \mathbf{P}^1(k_x), \qquad |h_0^{-1}(E)(k_x)| = Q + 1.$$

The stack is the global form of the minuscule Schubert correspondence for the coweight $(1,0)$. Let $j_x : \mathcal{H}ecke_x \to \mathrm{Bun}_2 \times \mathrm{Bun}_2$ be (h_0, h_1). Choose a square root of q in the coefficient field. Denote by $\mathrm{IC}_x^{\mathrm{norm}}$ the Weil normalisation of the intersection complex whose trace function is $Q^{-1/2}$ on every rational point of the smooth minuscule correspondence. Equivalently, its underlying complex is the constant intersection complex, with the shift, half Tate twist,

and cohomological sign fixed by this trace normalisation. This convention eliminates the otherwise harmless sign variation among geometric Satake references.

The Hecke kernel and Hecke functor are

$$\mathcal{K}_x = j_{x!}\,\mathrm{IC}_x^{\mathrm{norm}}, \qquad \mathsf{T}_x(\mathcal{F}) = h_{1!}(h_0^*\mathcal{F} \otimes \mathrm{IC}_x^{\mathrm{norm}}).$$

By Definition 5.9, $\mathsf{T}_x = \Phi_{\mathcal{K}_x}$. The fibrewise Frobenius trace of the kernel is the actual function

$$k_x(E, E') = Q^{-1/2}\#\{E' \hookrightarrow E : E/E' \simeq k_x\},$$

where the count is interpreted groupoid-theoretically on the stack. On the standard spherical model, with $K_x = \mathrm{GL}_2(O_x)$ and Haar measure $\mathrm{vol}(K_x) = 1$, this is the kernel of

$$T_x = Q^{-1/2}\mathbf{1}_{K_x\left(\begin{smallmatrix}\varpi_x & 0\\ 0 & 1\end{smallmatrix}\right)K_x}.$$

The double coset has the explicit decomposition

$$K_x\left(\begin{smallmatrix}\varpi_x & 0\\ 0 & 1\end{smallmatrix}\right)K_x = \coprod_{a\in k_x}\left(\begin{smallmatrix}\varpi_x & a\\ 0 & 1\end{smallmatrix}\right)K_x \coprod \left(\begin{smallmatrix}1 & 0\\ 0 & \varpi_x\end{smallmatrix}\right)K_x.$$

There are therefore exactly $Q+1$ summands, agreeing with the $\mathbf{P}^1(k_x)$ calculation.

The coset decomposition can be recovered from lattices. Let $\Lambda = O_x^2$. A lower modification of index Q is a lattice $\Lambda' \subset \Lambda$ with $\Lambda/\Lambda' \simeq k_x$. Reduction modulo ϖ_x identifies it with the kernel of a nonzero functional

$$\Lambda/\varpi_x\Lambda \simeq k_x^2 \longrightarrow k_x,$$

and two functionals have the same kernel exactly when they differ by a scalar. Thus the lattices are indexed by $\mathbf{P}((k_x^2)^\vee)$. On the affine chart the functional is $(1, -a)$ and its kernel is generated by $(a, 1)$ modulo ϖ_x; the remaining point is the kernel of the second coordinate. Lifting these generators gives the Q matrices and the final matrix in the displayed double-coset decomposition. This proves geometrically, without a volume computation, that the correspondence has degree $Q+1$ over each unramified source.

The normalising factor is determined by Satake rather than by preference. For the upper triangular Borel with diagonal torus coordinates z_1, z_2, the unnormalised Satake transform of the characteristic function of the double coset is

$$Q^{1/2}(z_1 + z_2).$$

Multiplication by $Q^{-1/2}$ therefore sends the Hecke kernel to the actual character of the standard representation. Without this factor, the eigenvalue would be $Q^{1/2}(\alpha_x + \beta_x)$ and the local polynomial would carry a spurious power of Q. The half Tate twist in the Weil-normalised IC kernel is the sheaf-theoretic implementation of precisely this $Q^{-1/2}$.

There is a second normalisation hidden in the notation for Satake parameters. If χ_1, χ_2 are unramified characters and π_x is normalised parabolic induction, then

$$\alpha_x = \chi_1(\varpi_x), \qquad \beta_x = \chi_2(\varpi_x).$$

For unnormalised induction these entries acquire the factors $Q^{1/2}$ and $Q^{-1/2}$. The convention in Proposition 5.26 is the one for which the standard local factor is

$$\big((1 - \alpha_x Q^{-s})(1 - \beta_x Q^{-s})\big)^{-1}.$$

Stating the convention is necessary: many apparent discrepancies between a geometric Hecke formula and an automorphic one are only shifts between normalised induction, arithmetic versus geometric Frobenius, and IC Tate normalisation.

Proposition 5.26 (Kernel trace and the GL2 eigenvalue relation). [ESTABLISHED]
Let π_x be an unramified representation of $\mathrm{GL}_2(F_x)$ with normalised Satake parameter

$$s(\pi_x) = \mathrm{diag}(\alpha_x, \beta_x) \in \mathrm{GL}_2(\mathbf{C}) /\!/ \mathrm{GL}_2(\mathbf{C}).$$

On its spherical line the kernel $\mathcal{K}_x$ has trace

$$\mathrm{tr}(T_x \mid \pi_x^{K_x}) = \alpha_x + \beta_x.$$

If S_x is the normalised central Hecke operator corresponding to $\mathrm{diag}(\varpi_x, \varpi_x)$, then

$$S_x \mid \pi_x^{K_x} = \alpha_x \beta_x,$$

and the local Hecke polynomial and local standard factor are

$$P_x(u) = 1 - T_x u + S_x u^2,$$

$$P_x(u) \mid \pi_x^{K_x} = (1 - \alpha_x u)(1 - \beta_x u), \qquad L_x(s, \pi, \mathrm{Std}) = P_x(Q^{-s})^{-1}.$$

For the trivial representation, $(\alpha_x, \beta_x) = (Q^{1/2}, Q^{-1/2})$ and

$$T_x \mathbf{1} = Q^{-1/2}(Q+1)\mathbf{1} = (Q^{1/2} + Q^{-1/2})\mathbf{1}.$$

Proof of Proposition 5.26.
The normalised Satake isomorphism sends the double-coset operator T_x to the character $z_1 + z_2$ of the standard representation of the dual group $\mathrm{GL}_2(\mathbf{C})$, and sends S_x to $z_1 z_2$ (Satake 1963). Evaluation at the Satake parameter gives the first two formulas. The factorisation of P_x is then direct multiplication:

$$1 - (\alpha_x + \beta_x)u + \alpha_x \beta_x u^2 = (1 - \alpha_x u)(1 - \beta_x u).$$

Substituting $u = Q^{-s}$ gives the local L-factor. For the trivial representation, the normalised induction parameter is $(Q^{1/2}, Q^{-1/2})$. Independently, the double-coset decomposition has $Q + 1$ terms and each has kernel weight $Q^{-1/2}$, giving $Q^{-1/2}(Q+1)$. The two calculations coincide.

□

5.11.1 Stalks, convolution, and closure of the Hecke kernel

The kernel can be read directly at a pair (E, E'). The fibre of j_x over that pair is the groupoid of injections $E' \hookrightarrow E$ whose cokernel is k_x. Hence proper base change gives

$$(\mathcal{K}_x)_{(E,E')} \simeq R\Gamma_c\left(\{E' \hookrightarrow E : E/E' \simeq k_x\}, \mathrm{IC}_x^{\mathrm{norm}}\right).$$

If there is no such modification, the stalk is zero. If the pair is a geometric point of the open minuscule correspondence and has no extra stabiliser beyond the automorphisms already present in the stack, the stalk is one-dimensional with Frobenius trace $Q^{-1/2}$. This is the sheaf-theoretic origin of the two-valued classical kernel: zero off the Hecke relation and a normalised unit on it.

The action formula on functions follows with no appeal to Satake. For an K_x-spherical automorphic function f and a bundle E,

$$(T_x f)(E) = Q^{-1/2} \sum_{[E' \hookrightarrow E],\ E/E' \simeq k_x} f(E').$$

After choosing a trivialisation of the fibre E_x, the quotient maps are indexed by

$$[1:a] \quad (a \in k_x), \qquad [0:1],$$

so this becomes the explicit formula

$$(T_x f)(E) = Q^{-1/2}\left(\sum_{a\in k_x} f(E'_a) + f(E'_\infty)\right).$$

For the constant function every summand equals one, producing the numerical check in Proposition 5.26. For a Hecke eigenfunction the same sum is forced to be $(\alpha_x + \beta_x)f(E)$.

Convolution gives the quadratic relation geometrically. The middle fibre in $\mathcal{K}_x \star \mathcal{K}_x$ classifies chains

$$E'' \subset E' \subset E, \qquad E/E' \simeq k_x, \qquad E'/E'' \simeq k_x.$$

There are two relative positions for E'' in E. If the two quotient lines in the two-step modification are distinct, the total quotient is the direct sum $k_x \oplus k_x$ and the relative coweight is $(1,1)$, the central modification. If they coincide in the affine-Grassmannian closure, the total relative position is $(2,0)$. Geometric Satake records this stratification as

$$\mathrm{Std} \otimes \mathrm{Std} \simeq \mathrm{Sym}^2(\mathrm{Std}) \oplus \det.$$

Accordingly, after the standard IC normalisations,

$$\mathcal{K}_x \star \mathcal{K}_x \simeq \mathcal{K}_{x,\mathrm{Sym}^2} \oplus \mathcal{K}_{x,\det}.$$

On an eigenline this becomes the elementary identity

$$(\alpha_x + \beta_x)^2 = (\alpha_x^2 + \alpha_x\beta_x + \beta_x^2) + \alpha_x\beta_x.$$

This is a useful test of the claim that kernels compose by tensor-product representation theory: the two-step geometric fibre and the Clebsch–Gordan decomposition give the same two summands.

To close the standard kernel categorically, one must first inspect connected components. Since

$$\deg E' = \deg E - d$$

while Frobenius preserves the degree of a bundle on $X_{\overline{\mathbf{F}}_q}$, the derived intersection

$$\mathcal{H}ecke_x \times_{\mathrm{Bun}_2 \times \mathrm{Bun}_2} \Gamma_{\mathrm{Frob}}$$

is empty for the coweight $(1,0)$. Hence the categorical trace of the unbalanced Frobenius–Hecke composite is zero whenever that trace is defined. This is a computation, not a defect: a trace closes a correspondence only when its total component shift vanishes.

A nonzero closure is obtained by balancing the determinant. Let $\mathcal{K}_{x,\det^{-1}}$ be the central kernel that raises degree by $2d$ and whose Satake eigenvalue is $(\alpha_x\beta_x)^{-1}$. Define

$$\mathcal{B}_x := \mathcal{K}_{x,\det^{-1}} \star \mathcal{K}_x \star \mathcal{K}_x.$$

Its total degree shift is zero. Geometric Satake gives

$$(\mathrm{Std} \otimes \mathrm{Std}) \otimes \overset{-1}{\det} \simeq (\mathrm{Sym}^2 \mathrm{Std} \otimes \overset{-1}{\det}) \oplus \mathbb{1},$$

so on the spherical eigenline

$$\operatorname{tr}(\mathcal{B}_x) = \frac{(\alpha_x + \beta_x)^2}{\alpha_x \beta_x} = \frac{\alpha_x}{\beta_x} + 2 + \frac{\beta_x}{\alpha_x}.$$

The Frobenius closure of $\mathcal{B}_x$ is the degree-balanced two-step shtuka stack at x. By Proposition 5.17, its compactly supported IC cohomology is the categorical trace of $K_{\mathrm{Frob}} \star \mathcal{B}_x$. On a finite-level cuspidal summand on which the operator is trace-class, the spectral trace is

$$\sum_{\pi} m(\pi) \frac{(\alpha_{\pi,x} + \beta_{\pi,x})^2}{\alpha_{\pi,x}\beta_{\pi,x}}.$$

The equality with the Frobenius trace on the balanced shtuka cohomology is the Hecke-inserted categorical trace formula in this component-preserving case. The component check is indispensable: without it, the displayed shtuka stack would be empty and the spectral operator would not be the endomorphism whose trace was being computed.

This computes three distinct traces that are often compressed into one phrase. The sheaf–function trace of $\mathcal{K}_x$ is the function $k_x(E, E')$. Its action on a one-dimensional spherical eigenspace has ordinary linear trace $\alpha_x + \beta_x$. Finally, the categorical trace of the whole Hecke endofunctor requires closing the correspondence, and therefore a trace-class category. On the untruncated stack Bun_2 the standard Hecke operator changes degree and the automorphic category is not finite; an unqualified numerical trace would be meaningless. The correct function-field categorical object is the trace of a component-balanced Frobenius–Hecke composite, which is a shtuka cohomology object rather than a divergent scalar.

For a Hecke eigensheaf $\mathcal{F}_\rho$ attached to a rank-two local system ρ, geometric Satake gives the actual eigenrelation

$$\mathsf{T}_x(\mathcal{F}_\rho) \simeq \mathcal{F}_\rho \otimes \rho_x^{\mathrm{Std}},$$

and taking Frobenius traces gives

$$T_x f_\rho = \operatorname{Tr}(\mathrm{Frob}_x \mid \rho_x^{\mathrm{Std}}) f_\rho = (\alpha_x + \beta_x) f_\rho.$$

Thus the eigenvalue relation is not appended to the correspondence. It is the trace of the standard-representation factor in its Satake kernel.

5.12 A ramified calculation at Iwahori level

Let F_x be the completion at x, let O_x have residue field k_x of order Q, and let

$$I_x = \left\{ \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2(O_x) : c \equiv 0 \pmod{\varpi_x} \right\}$$

be the standard Iwahori subgroup. Geometrically, replacing Bun_2 by the stack Bun_{2,I_x} means remembering a line in the fibre at x. The simple-reflection Hecke correspondence changes that line by relative position

$$s = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right).$$

Its local double coset has the explicit decomposition

$$I_x s I_x = \coprod_{a \in k_x} \left(\begin{smallmatrix} 1 & \tilde{a} \\ 0 & 1 \end{smallmatrix}\right) s I_x,$$

where $\tilde{a}$ is any lift of a to O_x. Thus the fibre has Q points, not $Q+1$: the missing point is the line already selected by the Iwahori flag. Let

$$T_s = \mathbf{1}_{I_x s I_x}, \qquad \mathrm{vol}(I_x) = 1.$$

Counting pairs of relative-position-s flags gives the quadratic relation

$$T_s^2 = (Q-1)T_s + Q\mathbf{1},$$

or equivalently

$$(T_s + 1)(T_s - Q) = 0.$$

This relation can be checked as an actual matrix calculation. On the rank-two free module with basis $(\mathbf{1}, T_s)$, left convolution by T_s has matrix

$$[T_s] = \begin{pmatrix} 0 & Q \\ 1 & Q-1 \end{pmatrix},$$

because $T_s\mathbf{1} = T_s$ and $T_s T_s = Q\mathbf{1} + (Q-1)T_s$. Its characteristic polynomial is

$$\lambda^2 - (Q-1)\lambda - Q = (\lambda - Q)(\lambda + 1).$$

The two one-dimensional eigenspaces therefore have eigenvalues Q and -1. They are the Iwahori-fixed lines of the trivial and Steinberg constituents of the reducible principal series. After the unitary normalisation

$$\widetilde{T}_s = Q^{-1/2} T_s,$$

the relation becomes

$$(\widetilde{T}_s - Q^{1/2})(\widetilde{T}_s + Q^{-1/2}) = 0,$$

and the two traces are $Q^{1/2}$ and $-Q^{-1/2}$.

The sheaf kernel is the Weil-normalised IC complex of the simple-reflection Schubert correspondence on $\mathrm{Bun}_{2,I_x} \times \mathrm{Bun}_{2,I_x}$. Convolution of this kernel with itself has two strata: the identity stratum with multiplicity Q, and the simple-reflection stratum with multiplicity $Q-1$. Taking Frobenius traces gives exactly the displayed quadratic relation. This is the ramified analogue of the spherical $Q+1$ computation: at Iwahori level the kernel is no longer governed by the representation ring of the dual group alone, but by the affine Hecke algebra and its quadratic relation.

5.13 Two distinct places and factorisation

Let $x \neq y$ be closed points with residue cardinalities $Q_x = q^{\deg x}$ and $Q_y = q^{\deg y}$. A modification at x and a modification at y are independent because the completed local rings are disjoint. Given E, choosing both lower modifications amounts to choosing

$$\ell_x \in \mathbf{P}(E_x^\vee)(k_x), \qquad \ell_y \in \mathbf{P}(E_y^\vee)(k_y).$$

Hence the fibre of the composite correspondence contains

$$(Q_x + 1)(Q_y + 1)$$

points. Performing the x-modification and then the y-modification gives the same final subsheaf as performing them in the opposite order:

$$E_{x,y} = \ker\left(E \longrightarrow k_x \oplus k_y\right).$$

The two fibre products of Hecke stacks are therefore canonically equivalent, and the kernels commute:

$$\mathcal{K}_y \star \mathcal{K}_x \simeq \mathcal{K}_x \star \mathcal{K}_y.$$

Their fibrewise trace function is the product

$$k_{x,y} = Q_x^{-1/2} Q_y^{-1/2} k_x^{\mathrm{un}} k_y^{\mathrm{un}},$$

where each unnormalised factor is the indicator of the corresponding local Hecke relation. On the constant function one obtains, step by step,

$$\begin{aligned} T_x T_y \mathbf{1} &= Q_x^{-1/2} Q_y^{-1/2} (Q_x + 1)(Q_y + 1)\mathbf{1} \\ &= (Q_x^{1/2} + Q_x^{-1/2})(Q_y^{1/2} + Q_y^{-1/2})\mathbf{1}. \end{aligned}$$

Let an automorphic representation be unramified at both places, with Satake parameters

$$\operatorname{diag}(\alpha_x, \beta_x), \qquad \operatorname{diag}(\alpha_y, \beta_y).$$

The simultaneous spherical line is one-dimensional, and therefore

$$\operatorname{tr}(T_x T_y) = (\alpha_x + \beta_x)(\alpha_y + \beta_y) = \operatorname{tr}(T_x)\operatorname{tr}(T_y).$$

The local Hecke polynomial also factorises:

$$P_{x,y}(u, v) = (1 - \alpha_x u)(1 - \beta_x u)(1 - \alpha_y v)(1 - \beta_y v).$$

This is the explicit off-diagonal factorisation law. Sheaf-theoretically it is the factorisation structure of the Hecke correspondence away from the diagonal in $X \times X$; categorically it is the Künneth rule Proposition 5.22. For component-balanced kernels $\mathcal{B}_x$ and $\mathcal{B}_y$, the same computation gives the product

$$\frac{(\alpha_x + \beta_x)^2}{\alpha_x \beta_x} \cdot \frac{(\alpha_y + \beta_y)^2}{\alpha_y \beta_y}$$

on the simultaneous eigenline. A global trace still includes a sum over representations and need not be the product of two global traces; what factorises exactly is the local kernel, its modification fibre, and its trace on each simultaneous eigensummand.

5.14 Frobenius as an endo-correspondence

The function-field case is the control because every formal arrow in the trace calculus has a geometric representative. Let X be a separated scheme of finite type over $\mathbf{F}_q$, and write $\overline{X}$ for its base change to $\overline{\mathbf{F}}_q$. We use Frob_q for the Frobenius endomorphism in the convention entering the Grothendieck trace formula. Its graph is the correspondence

$$\overline{X} \xleftarrow{\mathrm{id}} \overline{X} \xrightarrow{\mathrm{Frob}_q} \overline{X}.$$

The associated kernel is

$$K_{\mathrm{Frob}} = (\mathrm{id}, \mathrm{Frob}_q)_! \overline{\mathbf{Q}}_\ell.$$

Theorem 5.27 (Trace of the Frobenius correspondence). [ESTABLISHED]

For $X/\mathbf{F}_q$ separated and of finite type, the derived fixed-point figure of Frob_q is the finite discrete scheme $X(\mathbf{F}_q)$. The trace of its graph kernel is

$$\mathrm{tr}(K_{\mathrm{Frob}}) \simeq R\Gamma_c(X(\mathbf{F}_q), \overline{\mathbf{Q}}_\ell) \simeq \bigoplus_{x \in X(\mathbf{F}_q)} \overline{\mathbf{Q}}_\ell.$$

Its numerical shadow is the Grothendieck–Lefschetz formula

$$\#X(\mathbf{F}_q) = \sum_i (-1)^i \,\mathrm{Tr}\left(\mathrm{Frob}_q \,\middle|\, H^i_c(\overline{X}, \overline{\mathbf{Q}}_\ell)\right).$$

Proof of Theorem 5.27.

By Corollary 5.16, the trace is the compactly supported integral over

$$\overline{X}^{\mathrm{Frob}_q} = \overline{X} \times_{\overline{X}\times\overline{X}, (\mathrm{id}, \mathrm{Frob}_q)} \overline{X}.$$

A geometric point belongs to this fibre product exactly when it is fixed by Frobenius, hence exactly when it descends to $\mathbf{F}_q$. The graph and the diagonal meet transversely: on tangent spaces the derivative of Frobenius is zero, so $1 - d\,\mathrm{Frob}_q$ is the identity. Thus the derived intersection is underived and reduced, a disjoint union of copies of $\mathrm{Spec}\,\mathbf{F}_q$ indexed by $X(\mathbf{F}_q)$. Compactly supported cohomology of that finite scheme is the displayed direct sum in degree zero.

The Lefschetz–Verdier comparison identifies the Euler characteristic of this fixed-point object with the supertrace of Frobenius on $R\Gamma_c(\overline{X}, \overline{\mathbf{Q}}_\ell)$. Taking dimensions on the first description and alternating traces on the second gives the numerical formula. This is the classical theorem of Grothendieck and Verdier in the six-operation formalism (Grothendieck and Illusie 1977).

□

5.14.1 Frobenius on the projective line

For $X = \mathbf{P}^1_{\mathbf{F}_q}$, the fixed points are

$$[1 : a] \quad (a \in \mathbf{F}_q), \qquad [0 : 1],$$

so the fibre-product integral contains $q + 1$ copies of $\overline{\mathbf{Q}}_\ell$. On cohomology,

$$H^0(\mathbf{P}^1_{\overline{\mathbf{F}}_q}) \simeq \overline{\mathbf{Q}}_\ell, \qquad H^1 = 0, \qquad H^2 \simeq \overline{\mathbf{Q}}_\ell(-1),$$

and Frobenius has traces $1, 0, q$. Hence

$$\operatorname{Tr}(\mathrm{Frob}_q \mid H^0) - \operatorname{Tr}(\mathrm{Frob}_q \mid H^1) + \operatorname{Tr}(\mathrm{Frob}_q \mid H^2) = 1 + q = \#\mathbf{P}^1(\mathbf{F}_q).$$

A nontrivial genus-one calculation shows that the fibre-product integral is not merely detecting the two Tate pieces.

Example 5.28 (An explicit elliptic Frobenius trace). [ESTABLISHED]
Let

$$E/\mathbf{F}_5 : \quad y^2 = x^3 - x.$$

For $x = 0, 1, 2, 3, 4$, the right-hand side is respectively $0, 0, 1, 4, 0$. These values contribute respectively $1, 1, 2, 2, 1$ affine points; adding the point at infinity gives

$$\#E(\mathbf{F}_5) = 1 + 1 + 2 + 2 + 1 + 1 = 8.$$

Thus

$$a_5 = 5 + 1 - 8 = -2,$$

and the characteristic polynomial of Frobenius on H^1 is

$$t^2 - a_5 t + 5 = t^2 + 2t + 5.$$

The trace formula reads

$$1 - (-2) + 5 = 8.$$

Geometrically, the graph of Frobenius meets the diagonal in exactly those eight reduced points; categorically, the trace of the graph kernel is an eight-dimensional vector space concentrated in degree zero; cohomologically, the same number is the alternating trace on $H^0 \oplus H^1 \oplus H^2$.

The traces of all powers assemble into the zeta function. This is the simplest instance in which composition of endocorrespondences produces an analytic invariant.

Corollary 5.29 (Zeta function from Frobenius traces). [ESTABLISHED]
For $X/\mathbf{F}_q$ separated of finite type,

$$Z(X, t) := \exp\left(\sum_{n \geq 1} \#X(\mathbf{F}_{q^n}) \frac{t^n}{n}\right)$$

is the exponential of the numerical traces of the powers of the Frobenius kernel. If $R\Gamma_c(\overline{X}, \overline{\mathbf{Q}}_\ell)$ is perfect, then

$$Z(X, t) = \prod_i \det\left(1 - t\,\mathrm{Frob}_q \,\middle|\, H^i_c(\overline{X}, \overline{\mathbf{Q}}_\ell)\right)^{(-1)^{i+1}}.$$

Proof of Corollary 5.29.
Convolution of graph kernels satisfies $K_{\mathrm{Frob}}^{\star n} \simeq K_{\mathrm{Frob}^n}$ because composition of graphs is composition of maps. By Theorem 5.27, its numerical trace is $\#X(\mathbf{F}_{q^n})$. For a finite-dimensional endomorphism A,

$$-\log\det(1 - tA) = \sum_{n \geq 1} \operatorname{Tr}(A^n) \frac{t^n}{n}.$$

Apply this identity to Frobenius on each cohomology group, multiply with sign $(-1)^{i+1}$, and use the Grothendieck–Lefschetz formula for every n:

$$\sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_q^n \mid H_c^i) = \#X(\mathbf{F}_{q^n}).$$

Exponentiation gives the determinant formula.

□

For the projective line this yields

$$Z(\mathbf{P}^1, t) = \frac{1}{(1-t)(1-qt)},$$

because only H^0 and H^2 occur. For the elliptic curve of Example 5.28, the H^1 characteristic polynomial is $1 + 2t + 5t^2$, so

$$Z(E, t) = \frac{1 + 2t + 5t^2}{(1-t)(1-5t)}.$$

The coefficient check at first order is

$$[t] \log Z(E, t) = 8,$$

as required by the direct point count. Thus the zeta function is not an additional construction placed after the trace: it is the generating function of traces of convolution powers of one graph kernel.

This calculation indicates what a global arithmetic trace would have to do. One would need convolution powers, or an appropriate flow of powers, of the absolute endo-correspondence Φ, together with a determinant theory on its trace-class categorical value. The formal logarithm-to-determinant step is available. The arithmetic endomorphism and its convergent trace are not.

The composition theorem also diagnoses component shifts in Frobenius–Hecke traces. For the standard kernel $\mathcal{K}_x$ of §5.11, the formal fibre product is

$$\mathcal{H}ecke_x \times_{\mathrm{Bun}_2 \times \mathrm{Bun}_2} \Gamma_{\mathrm{Frob}}.$$

It classifies a modification $E' \subset E$ and an isomorphism $E' \simeq \mathrm{Frob}_q^* E$. The first condition gives $\deg E' = \deg E - d$, while the second gives $\deg E' = \deg E$; hence the fibre product is empty. Thus the trace integral is zero. After replacing $\mathcal{K}_x$ by the balanced kernel $\mathcal{B}_x$ constructed in §5.11.1, the same fibre-product formula produces the component-preserving two-step shtuka stack and its compactly supported IC cohomology. Closing a Hecke kernel against Frobenius therefore creates a shtuka only after the total coweight has trivial image in the component group, or after an explicit compensating central translation has been inserted.

Theorem 5.30 (Function-field automorphic trace theorem). [ESTABLISHED]

For a reductive group G over a smooth projective curve $X/\mathbf{F}_q$, the appropriately renormalised categorical trace of Frobenius on the automorphic sheaf category on $\mathrm{Bun}_G(\overline{\mathbf{F}}_q)$ identifies with unramified automorphic functions on $\mathrm{Bun}_G(\mathbf{F}_q)$. With component-balanced Hecke insertions, the same trace formalism identifies the trace object with the cohomology of the corresponding shtuka stacks. Under geometric Satake, the endomorphisms generated by creation, partial Frobenius, and annihilation are the excursion operators used in the function-field parametrisation.

Proof of Theorem 5.30.

This is the theorem proved in the series *Automorphic functions as the trace of Frobenius* by Arinkin, Gaitsgory, Kazhdan, Raskin, Rozenblyum, Varshavsky, and collaborators. The geometric mechanism has already been displayed here: the trace of Frobenius closes the graph against the diagonal, while insertion of a component-balanced Hecke kernel replaces the fixed-point stack by a shtuka stack. Their work supplies the substantial global inputs suppressed by the finite-type model: the renormalised sheaf category on the non-quasi-compact stack Bun_G, its dualisability, the nilpotent singular-support condition, and the comparison of the categorical trace with the sheaf–function space. Vincent Lafforgue's theorem identifies the resulting excursion algebra with semisimple global Langlands parameters (Lafforgue 2018). The present chapter uses this result as an established theorem and does not replace its proof by the formal kernel calculation.

□

5.15 Systematic function-field controls

The function-field comparison can now be applied to each principal assertion rather than only to the final Frobenius example. Let $X/\mathbf{F}_q$ be a smooth projective geometrically connected curve and let the sheaf theory be constructible $\overline{\mathbf{Q}}_\ell$-complexes, with the usual renormalisations on stacks.

5.15.1 Correspondences and six operations

For schemes of finite type over $\mathbf{F}_q$, the category of correspondences is formed with no conjectural geometry: fibre products exist, proper maps are stable under base change, and the graph of every morphism is a finite-type correspondence. The étale six operations were constructed in the Grothendieck school and organised in SGA 4 and SGA 5. For Artin stacks, the higher coherences and descent are supplied by the modern enhanced formalisms, including Liu–Zheng. Thus Proposition 5.7 and Proposition 5.8 are theorems in this control case. The vertical and horizontal pasting maps in §5.3.1 are not additional assumptions: they are part of the enhanced functoriality.

A direct test is furnished by finite morphisms. If $f : Y \to X$ is finite étale of degree n, then for the constant sheaf

$$f_!\overline{\mathbf{Q}}_\ell = f_*\overline{\mathbf{Q}}_\ell$$

is lisse of rank n. After a base change $g : X' \to X$, the fibre over a geometric point x' is the vector space on the finite set $Y \times_X x'$. Both sides of $g^*f_* \simeq f'_*g'^*$ are therefore the same vector space with the same Galois action. This stalk calculation is the elementary shadow of the full proper base-change theorem.

5.15.2 Kernels and convolution

On a proper scheme, constructible complexes with finite-dimensional stalks are dualisable for Verdier duality, external products satisfy Künneth, and proper pushforward preserves constructibility. Consequently the integral-transform and convolution formulas are theorems. For stacks such as Bun_G, compact generation and dualisability are subtler, but the geometric Langlands and automorphic-trace literature provides the needed renormalised categories in the cases used here.

The two-place calculation of §5.13 is the simplest factorisation check. Over the open subset

$$(X \times X) \setminus \Delta_X,$$

an x-modification and a y-modification have disjoint support. The two-leg Hecke stack is the fibre product of the one-leg stacks over Bun_G, independently of order. Its IC sheaf is the external product of the local IC sheaves, and the convolution comparison is the factorisation isomorphism. Beilinson–Drinfeld's Ran-space formalism upgrades this observation from two distinct points to every finite disjoint family of legs. When legs collide, the fusion product replaces the external product; that degeneration is precisely what constructs the tensor structure in geometric Satake.

5.15.3 Shadows, traces, and fixed points

For a proper $X/\mathbf{F}_q$, the shadow of an endokernel is

$$K \longmapsto R\Gamma_c(X, \Delta^* K).$$

Cyclicity is Theorem 5.15, additivity is the additivity of compactly supported cohomology, and Künneth is the ordinary Künneth theorem. For a graph kernel, SGA 5 identifies this shadow with a sum of Lefschetz local terms. Hence every trace rule in §5.8 has an established function-field realisation.

The non-invertible transfer rule can be seen without abstract bicategories. Let $j : U \hookrightarrow X$ be an open immersion stable under an endomorphism f. The natural transformation

$$j_! f_U^* \longrightarrow f_X^* j_!$$

induces a map from the trace on U to the trace on X. On fixed points it is extension by zero from U^f to X^f. This map is generally not an equivalence because fixed points on the boundary $X \setminus U$ contribute to the target. The failure of invertibility is therefore measured by an actual boundary trace. Applying additivity to $j_! \mathbb{1}_U \to \mathbb{1}_X \to i_* \mathbb{1}_{X \setminus U}$ gives

$$\mathrm{tr}(f_X) = \mathrm{tr}(f_U) + \mathrm{tr}(f_{X \setminus U}),$$

whenever the terms are defined. This is the geometric prototype of Eisenstein or boundary defects later in the book.

5.15.4 Hecke kernels and eigenvalues

For GL_1, the control is class field theory on the Picard stack: the Hecke kernel is translation by a divisor, and its eigenvalue on a character sheaf is the value of the corresponding rank-one local system at Frobenius. For spherical GL_2, geometric Satake identifies the minuscule IC kernel with the standard representation of the dual group, giving the explicit eigenvalue $\alpha_x + \beta_x$. At Iwahori level, the affine flag stratification gives the relation $(T_s + 1)(T_s - Q) = 0$. These are theorems, not predictions, and the direct counts in §5.10, §5.11, and §5.12 verify their normalisations.

For a global Hecke eigensheaf $\mathcal{F}_\rho$, the factorisation eigenvalue condition for two distinct places is

$$\mathsf{T}_{x,\mathrm{Std}} \mathsf{T}_{y,\mathrm{Std}}(\mathcal{F}_\rho) \simeq \mathcal{F}_\rho \otimes \rho_x^{\mathrm{Std}} \otimes \rho_y^{\mathrm{Std}}.$$

Taking Frobenius traces produces $(\alpha_x + \beta_x)(\alpha_y + \beta_y)$, exactly as in the direct double-coset calculation. Drinfeld proved the global correspondence for GL_2, Laurent Lafforgue for GL_n, and Vincent Lafforgue constructed the excursion-operator parametrisation for every reductive group (Drinfeld 1983; Lafforgue 2002; Lafforgue 2018).

5.15.5 Frobenius and the categorical automorphic trace

The graph of Frobenius exists globally on X and on Bun_G after base change to $\overline{\mathbf{F}}_q$. Its closure against the diagonal gives ordinary fixed points in the first case and G-bundles defined over $\mathbf{F}_q$ in the second. With balanced Hecke insertions the same closure gives shtuka stacks. The theorem of Arinkin, Gaitsgory, Kazhdan, Raskin, Rozenblyum, Varshavsky, and collaborators identifies the renormalised categorical trace with automorphic functions, while the shtuka cohomology carries the excursion operators. This establishes the complete chain

$$\text{correspondence} \longrightarrow \text{kernel} \longrightarrow \text{convolution} \longrightarrow \text{shadow} \longrightarrow \text{fixed-point integral}$$

over a function field. The arithmetic programme fails not in the arrows of this chain, whose formal proofs are given in this chapter, but at the existence of the global objects on which the arrows would act.

5.16 The exact arithmetic obstruction

Every step of Theorem 5.27 can be named. One needs an object X, an endomorphism Frob_q, its graph, the Cartesian intersection of that graph with the diagonal, a sheaf theory on all four corners, base change for the square, and a trace-class pushforward to the point. Over a curve over $\mathbf{F}_q$, all of these exist.

For the proposed arithmetic base, the missing step is exact:

$$\mathfrak{X}_{\mathbf{Z}}^{\Phi} := \mathfrak{X}_{\mathbf{Z}} \times_{\mathfrak{X}_{\mathbf{Z}} \times \mathfrak{X}_{\mathbf{Z}}, (\mathrm{id}, \Phi)} \mathfrak{X}_{\mathbf{Z}}$$

cannot presently be formed inside a constructed analytic geometry carrying the required six operations, because neither $\mathfrak{X}_{\mathbf{Z}}$ nor its absolute Frobenius endo-correspondence Φ has been constructed with those properties. Consequently the Beck–Chevalley step

$$\Delta^*(\mathrm{id}, \Phi)_! \simeq (i_\Phi)_! j_\Phi^*$$

has no arithmetic theorem behind it. This is the precise point at which the function-field computation stops. The absence is not merely a lack of a point count or a difficult convergence estimate. It is the absence of the Cartesian square to which base change would be applied.

The chôra axioms make the desired expression well-typed conditionally: KH.1 supplies the correspondence functor, KH.6 supplies the primitive trace, and KH.17 supplies Φ. They do not construct any of these data on $\mathfrak{X}_{\mathbf{Z}}$. The distinction is decisive. The formula

$$\mathrm{tr}(K_2 \star K_1) = \int K_1(x, y) K_2(y, x)$$

is a theorem once its operations exist; the existence of the arithmetic operations is a conjecture.

Status and Caveat. The weakest point of the chapter is not the correspondence or kernel calculus, which is established, but its intended arithmetic domain. No present theorem constructs the global arithmetic curve with a full six-functor formalism, a trace-class automorphic category, and an absolute Frobenius whose graph admits the required base change. In particular, the chapter has not computed an arithmetic Hecke trace on $\mathfrak{X}_{\mathbf{Z}}$. It has proved the unique formula such a trace must satisfy if the conjectural data exist, and has shown that the formula specialises to the established function-field theorem. If nuclearity fails, the global trace may not exist even after the base is constructed; if base change fails, kernel convolution itself ceases to be associative in the required sense.

5.17 What the formalism establishes, and what it does not

The established mathematical content has several sources. The higher category of correspondences and its duality are part of the Gaitsgory–Rozenblyum architecture (Gaitsgory and Rozenblyum 2017a). Liu–Zheng provide a construction and descent method for enhanced six operations on higher stacks, including the coherences needed for Lefschetz–Verdier arguments. Mann's work provides an axiomatic and geometric six-functor formulation in rigid analytic settings (Mann 2022), while Heyer–Mann connect such formalisms to smooth representations (Heyer and Mann 2024). The kernel equivalence for perfect stacks and its derived-geometric applications are due to Ben-Zvi, Francis, and Nadler (Ben-Zvi et al. 2010). The shadow formalism and its general bicategorical trace are due to Ponto and Shulman (Ponto and Shulman 2012). Additivity and higher traces are established in (May 2001; Hoyois et al. 2017).

The contribution of the present framework is organisational but not empty. It singles out the exact package required by later chapters: a correspondence functor, dualisable values, a shadow, and a localisation comparison. It proves from that package the fibre-product formula Theorem 5.15 and records the five identities in §5.8 as a stable calculus. It also identifies the Ponto–Shulman shadow as precisely the cyclic core of the primitive trace axiom. That last identification is a theorem about the definitions; the non-degenerate arithmetic localisation required by the book remains conjectural.

One should also distinguish the theorem

$$\mathrm{Fun}^{L}(\mathcal{C},\mathcal{D}) \simeq \mathcal{C}^{\vee} \otimes \mathcal{D}$$

from a geometric kernel theorem. The former follows formally from the dualisability of $\mathcal{C}$. To replace $\mathcal{C}^{\vee} \otimes \mathcal{D}$ by sheaves on $X \times Y$, one must prove self-duality of the sheaf category and a Künneth equivalence. These can fail for non-perfect, non-quasi-compact, or analytic stacks. Later references to a "kernel on the arithmetic product' ' therefore carry both geometric hypotheses, not only abstract dualisability.

Summary. Correspondences compose by pullback, and the universal property of their $(\infty,2)$-category turns Beck–Chevalley equivalences into functoriality. A six-functor formalism sends a span to pull–tensor–push; under dualisability, every continuous functor is a tensor, and under geometric Künneth hypotheses that tensor is a kernel. Kernels compose by convolution, while a Ponto–Shulman shadow closes an endokernel into a trace. The trace of $K_{21} \star K_{12}$ is the fibre-product integral of $K_{12}(x,y) \otimes K_{21}(y,x)$, making cyclicity and the Lefschetz formula two forms of one identity. The GL_2 Hecke kernel yields the explicit eigenvalue $\alpha_x + \beta_x$, and Frobenius yields the exact point-counting trace over $\mathbf{F}_q$. The arithmetic analogue stops at the construction of the graph of Φ and the base-change square for its fixed points.

5.18 Notes and references

The use of spans as morphisms is classical, but the higher coherence required for sheaf theories on derived stacks belongs to the modern $(\infty,2)$-categorical treatment. Gaitsgory and Rozenblyum's *A Study in Derived Algebraic Geometry*, especially the volume on correspondences and duality, is the principal reference for the architecture used here (Gaitsgory and Rozenblyum 2017a). Their conventions involve choices of vertical and horizontal classes and several opposites; readers comparing formulas should compare the induced pull–push functor rather than the drawn orientation of a span.

Liu and Zheng's work on enhanced six operations has a different emphasis. It constructs the étale formalism for higher Artin and Deligne–Mumford stacks by systematic descent and proves the coherences needed for base change and the Lefschetz–Verdier formula. It should not be cited as though it merely restated the universal property of spans. Mann, by contrast, deliberately uses a correspondence functor as the definition and interface of the formalism in p-adic geometry (Mann 2022). Heyer and Mann show how the same interface organises smooth representation categories (Heyer and Mann 2024). Gallauer's survey is a useful entry to the axiomatic literature (Gallauer 2021).

Integral transforms in derived algebraic geometry, including the tensor product description of sheaves on products and its relation to centres, are developed by Ben-Zvi, Francis, and Nadler (Ben-Zvi et al. 2010). The abstract equivalence $\mathrm{Fun}^{L}(\mathcal{C},\mathcal{D}) \simeq \mathcal{C}^{\vee}\otimes\mathcal{D}$ is part of the tensor calculus of presentable categories in *Higher Algebra* (Lurie 2017). The geometric identification of the right-hand side with a sheaf category is a separate theorem and should always be checked.

Ponto and Shulman's paper is the source for shadows and bicategorical traces (Ponto and Shulman 2012). Their framework clarifies why Hochschild homology is cyclic even when the ambient bicategory of algebras and bimodules is not symmetric. May proves additivity of traces in triangulated categories (May 2001); Hoyois, Scherotzke, and Sibilla place higher traces and the categorified Chern character in the setting of noncommutative motives (Hoyois et al. 2017). Toën and Vezzosi develop equivariant traces in derived algebraic geometry (Toën and Vezzosi 2015).

The fixed-point formula belongs historically to Grothendieck and Verdier's formalism in SGA 5 (Grothendieck and Illusie 1977). The point of the kernel proof is not to replace that theorem, whose local terms contain serious geometry, but to expose the formal composition that precedes localisation. Hoyois' quadratic refinement shows how much additional information a more structured coefficient category can retain (Hoyois 2014).

The local Hecke calculation uses Satake's spherical isomorphism (Satake 1963). Its geometric form is geometric Satake; in the present calculation only the standard representation of the dual GL_2 and the $Q+1$ points of $\mathbf{P}^1(k_x)$ are needed. The function-field trace theorem is the work of Arinkin, Gaitsgory, Kazhdan, Raskin, Rozenblyum, Varshavsky, and collaborators. Vincent Lafforgue's excursion formalism supplies the global semisimple parametrisation (Lafforgue 2018), with Drinfeld and Laurent Lafforgue as the decisive predecessors for GL_2 and GL_n (Drinfeld 1983; Lafforgue 2002).

A final comparison clarifies the lineage of the composition formula. Gaitsgory–Rozenblyum place correspondences before sheaf theory: the higher-categorical span is the geometric syntax. Liu–Zheng place descent before the final formalism: multisimplicial diagrams are the means by which operations on schemes are extended to higher stacks without losing coherence. Mann places the correspondence functor at the interface: once a theory has been shown to define that functor, proper base change and projection formula can be

consumed uniformly by later arguments. The present chapter follows all three choices in sequence rather than selecting one as a substitute for the others.

Ponto–Shulman's shadow is orthogonal to this construction. A correspondence category explains how to compose open strings; a shadow explains how to close them. The cyclicity isomorphism is additional structure on a bicategory, not a consequence of pullbacks. In the bicategory of algebras and bimodules it is Hochschild homology; in the geometric kernel bicategory it is diagonal restriction followed by integration. This is the precise reason the chôra axioms keep trace primitive even though dualisability constructs individual traces: the chosen shadow and, more strongly, its fixed-point localisation must be compatible across the entire geometry.

Ben-Zvi–Francis–Nadler and Toën–Vezzosi approach the same circle from derived geometry. The former identify broad classes of continuous functors with integral kernels and use loop spaces and centres to interpret Hochschild invariants (Ben-Zvi et al. 2010). The latter develop categorical and equivariant traces as derived-geometric functions on fixed and loop spaces (Toën and Vezzosi 2015). Their results explain why the middle term of Theorem 5.15 should be read simultaneously as a convolution integral, a Hochschild class, and a derived fixed-point intersection. The formulas coincide only after the duality and perfectness hypotheses have been verified.

The ramified calculation belongs to the Iwahori–Matsumoto side of Hecke theory, whereas the unramified calculation belongs to Satake. The contrast between Q and $Q+1$ is geometric: an Iwahori flag removes one point from the projective line of spherical modifications. The quadratic polynomial $(T_s+1)(T_s-Q)$ is therefore as concrete a trace calculation as the spherical polynomial $1-(\alpha_x+\beta_x)u+\alpha_x\beta_x u^2$. Later chapters use this contrast when ramification replaces dual-group representation rings by affine Hecke categories.

The word "kernel'' has several compatible histories. In classical analysis it is a function $K(x,y)$ defining an integral operator. In algebraic geometry it became a sheaf on a product defining a Fourier–Mukai transform. In category theory it is the tensor representing a continuous functor from a dualisable source. These meanings coincide only under hypotheses. The analytic kernel must be integrable, the geometric product must support the necessary pushforward, and the categorical source must be dualisable. The equivalence in Theorem 5.10 explains the common form but does not remove any of these finiteness conditions.

The same caution applies to "trace''. The trace of a linear map is a scalar; the trace of an endofunctor is a Hochschild-type object; the trace of a geometric correspondence is an integral over a derived fixed-point figure; and the Lefschetz number is a numerical realisation of that object. Toën and Vezzosi emphasise the derived fixed-point and equivariant character side (Toën and Vezzosi 2015). Ponto and Shulman emphasise the bicategorical cyclicity that makes the construction independent of the cut in a circular composite (Ponto and Shulman 2012). Hoyois, Scherotzke, and Sibilla emphasise higher traces as natural transformations on noncommutative motives (Hoyois et al. 2017). The primitive trace of a chôra asks for these compatible aspects together with localisation.

There is also a useful distinction between the correspondence calculus of this chapter and cohomological correspondences in the narrower sense of SGA 5. A cohomological correspondence includes a map between pullbacks of coefficient complexes on the middle space, and its local term depends on that map. Our weighted correspondence records the same information as a coefficient or a two-morphism of kernels. The fibre-product composition in Proposition 5.12 is therefore the higher categorical form of the classical composition of cohomological correspondences. What SGA 5 adds is the geometric construction and invariance of local terms, especially in non-transverse situations.

Gaitsgory–Rozenblyum's contribution is indispensable at a different point: they provide an ambient higher category in which these compositions and adjunctions coexist coherently for derived stacks (Gaitsgory and Rozenblyum 2017a). Without that ambient object, a list of base-change isomorphisms does not by itself say which higher compatibility must hold. Liu–Zheng show how those compatibilities can be constructed by descent for enhanced sheaf theories on higher stacks. Mann demonstrates that the resulting structure is an effective definition and comparison interface in non-archimedean analytic geometry (Mann 2022). The three works answer, respectively, where the coherence lives, how it is built in a major sheaf theory, and how it is packaged for reuse.

The Hecke examples display the same progression from geometry to algebra. The $Q + 1$ points of the spherical fibre produce the standard dual-group character. Choosing an Iwahori flag removes one point and replaces the representation ring by the affine-Hecke quadratic relation. Moving to two distinct places turns convolution into factorisation, while colliding the places turns factorisation into fusion. Geometric Satake is the theorem that organises all of these local calculations into the tensor category of representations of the dual group. The explicit matrices in §5.12 are therefore not separate from the categorical theory; they are its smallest ramified stalk computation.

Finally, the function-field theorem fixes the evidential standard for the arithmetic proposal. Over $X/\mathbf{F}_q$, Frobenius is an actual endomorphism, its graph is an actual correspondence, the fixed-point fibre product is an actual stack, and the trace can be compared with automorphic functions. The number-field chapter may reuse every formal identity proved here, but it may not cite the function-field theorem as construction of an absolute Frobenius over $\mathfrak{X}_{\mathbf{Z}}$. The missing graph and the missing nuclear trace remain missing. The value of the calculus is that this absence occurs at a named input rather than being dispersed through every later manipulation.

5.19 Exercises

Exercise A. Let $X \xleftarrow{p} Z \xrightarrow{q} Y$ and $Y \xleftarrow{r} W \xrightarrow{s} T$ be finite-set correspondences. Write their matrices with entries equal to fibre cardinalities and verify directly that pullback composition of spans is matrix multiplication.

Exercise B. Prove that the reverse of a map-span is its right adjoint in the bicategory of spans by writing the unit and counit as maps of spans and checking both triangle identities.

Exercise C. Starting only from Beck–Chevalley and the projection formula, rederive Proposition 5.11 for three kernels and identify the canonical associator.

Exercise D. Let A and B be associative algebras and let ${}_AM_B$, ${}_BN_A$ be perfect bimodules. Show that the Ponto–Shulman cyclicity map becomes the standard equivalence $HH(A; M \otimes_B N) \simeq HH(B; N \otimes_A M)$.

Exercise E. For a finite-dimensional vector space V and endomorphisms A, B, recover $\operatorname{tr}(BA) = \sum_{i,j} A_{ij}B_{ji}$ as the discrete form of Theorem 5.15.

Exercise F. Compute the lower modifications of the trivial bundle $\mathcal{O}_X^{\oplus 2}$ at a rational point x. Exhibit the $q+1$ quotient lines and verify the double-coset decomposition in §5.11.

Exercise G. With the normalisation of Proposition 5.26, compute T_x on an unramified principal series $\operatorname{Ind}(\chi_1 \otimes \chi_2)$ and obtain $\chi_1(\varpi_x) + \chi_2(\varpi_x)$ after normalising the Satake parameters.

Exercise H. Count the points of $y^2 = x^3 - x$ over $\mathbf{F}_7$ and use the Lefschetz formula to determine the trace and characteristic polynomial of Frobenius on H^1.

Exercise I. Show that for a smooth variety over $\mathbf{F}_q$ the graph of Frobenius meets the diagonal transversely by computing the derivative. Explain why the argument fails for a general endomorphism.

Exercise J. Assume conjecturally that $\mathfrak{X}_{\mathbf{Z}}$, Φ, and the required six operations exist. Write the derived fixed-point square for Φ, then list exactly which arrows must be proper, nuclear, or dualisable for Corollary 5.16 to define a global arithmetic trace.

Exercise K. For GL_1, take two points x, y of equal degree and verify directly that the graph of $L \mapsto L(x - y)$ has Frobenius fixed fibre a torsor under $\mathrm{Pic}^0(X)(\mathbf{F}_q)$. Compute the answer for an elliptic curve with $\#E(\mathbf{F}_q) = q + 1 - a_q$.

Exercise L. Multiply the matrix $\left(\begin{smallmatrix} 0 & Q \\ 1 & Q^{-1} \end{smallmatrix}\right)$ by itself, verify the Iwahori quadratic relation, and write explicit eigenvectors for the eigenvalues Q and -1. Interpret the latter as the Steinberg line.

Exercise M. Let $x \neq y$ be rational points. List the $(q+1)^2$ pairs of quotient lines for the composite Hecke correspondence and prove that the two orders of modification give the same subsheaf of E. Check the factorised eigenvalue on the trivial representation.

Chapter 6

The Language: Modal Directed Type Theory

Synopsis. This chapter specifies the formal language in which the axioms of the arithmetic topos are to be stated. The language is a dependent type theory with directed hom-types, a two-category of modes, adjoint modalities including $\flat \dashv \sharp$, and a primitive trace construction presented by explicit formation, introduction, elimination, and computation rules. Two derivations are carried out inside the calculus: cyclicity of trace and a proper base-change instance. A semantics in indexed categories is defined, and the assertion that its intended models are precisely chôrai is isolated as a conjecture. No implementation of the complete theory exists, and nothing in this chapter is machine-checked.

6.1 Why the language must be directed

The axioms assembled in the preceding chapters are not naturally statements of ordinary set theory. Their primitive operations are pullback and pushforward, composition of correspondences, adjunction, duality, localisation, and trace. These operations have sources and targets, and most of them are not invertible. If they are forced into a language in which every identity is a path and every path is reversible, the direction is not destroyed semantically, but it is hidden syntactically in structure carried by terms. That hiding is expensive. A base-change square becomes a proposition about four functors and a named natural transformation; a correspondence becomes a span together with an independently defined composition; a trace becomes a construction performed after dualizability has been unpacked. The formal language should expose these as grammatical operations.

Homotopy type theory gives the correct language for equivalence, higher coherence, and univalence (Program 2013; Kapulkin et al. 2021). It does not by itself distinguish a morphism $x \to y$ from a morphism $y \to x$, because identity types are groupoidal. Riehl and Shulman repaired precisely this defect by enriching type theory with a strict directed interval and extension types. Segal types then behave as synthetic $(\infty, 1)$-categories and Rezk types as complete ones (Riehl and Shulman 2017). Their construction is established mathematics; its adoption here is not a claim that the arithmetic theory has already been formalised.

Direction alone is insufficient. Arithmetic geometry continually changes ambient world: from a global object to its p-adic localisation, from a cohesive object to its discrete shadow, from a sheaf to its proper pushforward, and from an object with geometric variation to one on which only global sections remain. These changes are modalities. The work of Shulman on cohesive type theory, of Licata and Shulman on adjoint logic with a two-category of modes, and of Licata, Shulman, and Riley on fibrational modal logic provides the governing pattern (Licata and Shulman 2016; Shulman 2018). Mitchell Riley's work on commuting cohesions and Felix Wellen's modal treatment of Cartan geometry further demonstrate that several geometric structures can be carried simultaneously without collapsing them into one undifferentiated necessity operator.

The final ingredient is not available in those systems. A chôra contains a primitive trace theory, and the internal language therefore needs a trace former whose cyclicity is visible to the calculus. Making cyclicity a judgemental equality would be tempting, because rewriting $\mathrm{tr}(fg)$ to $\mathrm{tr}(gf)$ would then be immediate. It would also introduce an unoriented rewriting loop and threaten normalisation. The design adopted below is more conservative: cyclicity is a path constructor of a trace type, and the scalar trace is obtained by an eliminator. Thus cyclicity is derivable, but it is not built into definitional equality.

The governing mathematical relations are collected in Plate 6.1.

Notation 6.1 (The four primary judgements). [DESIGN]

For every mode m the theory has the following judgements:

$$\mathcal{M} \vdash m\ \mathsf{mode}, \qquad \Gamma\ \mathsf{ctx@}m, \qquad \Gamma \vdash A\ \mathsf{type@}m, \qquad \Gamma \vdash a : A@m.$$

It also has directed judgements

$$\Gamma \vdash x \longrightarrow_A y\ \mathsf{type@}m, \qquad \Gamma \vdash f : x \longrightarrow_A y@m,$$

and judgemental equality judgements for contexts, types, and terms. The symbol $\equiv$ is reserved for an internal equivalence; $\equiv$ is never used for judgemental equality, which is written $\equiv_{\mathrm{jud}}$ in prose and $\equiv$ only inside displayed inference rules when the subscript would obscure the rule.

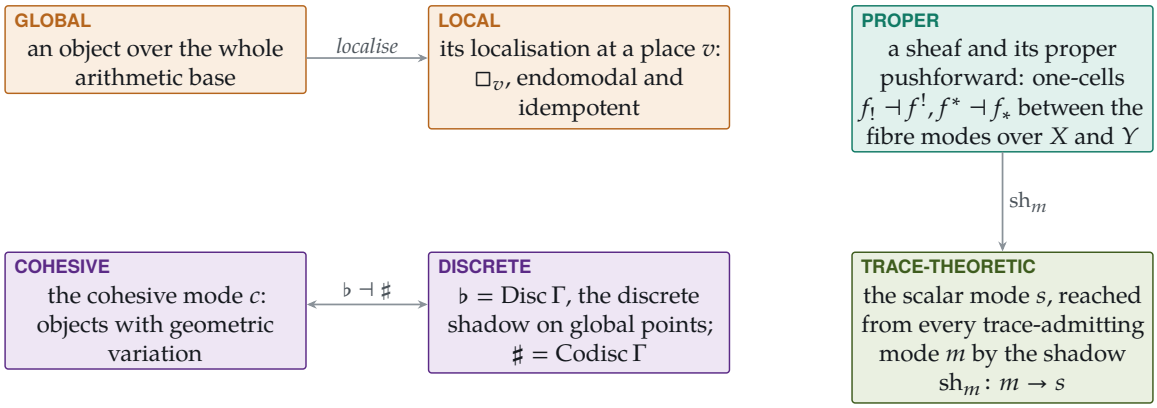


Plate 6.1. Modalities organise passage between global, local, discrete, cohesive, proper, and trace-theoretic viewpoints. The diagram should be read as a typing discipline: an arrow records a permitted change of mode, while an adjunction records the corresponding introduction and elimination principles.

What it does not show: a model: the diagram is the typing discipline of $\mathrm{TT}_{\mathrm{KH}}$*, and the claim that its models satisfying the KH signature are chôrai is conjectural.*

The last convention requires care. In ordinary mathematical prose the sign $\simeq$ denotes equivalence, while type-theoretic presentations often use $\equiv$ for definitional equality. Here displayed rules use $\equiv_{\mathrm{jud}}$ whenever there is any possibility of confusion. Identity types are written $\mathrm{Id}_A(x,y)$; directed hom-types are written $\mathrm{Hom}_A(x,y)$. The first are reversible, the second are not.

6.2 Modes, locks, and contexts

A modal calculus is only as precise as its account of contexts. A modality is not merely an endofunction on closed types. It changes which variables may be used and how they may be used. The lock notation records that change.

Definition 6.2 (Mode theory). [DESIGN]

A **mode theory** is a strict two-category $\mathcal{M}$ with:

1. objects $m, n, \ldots$, called modes;
2. one-cells $\mu : n \to m$, called mode morphisms;
3. two-cells $\alpha : \mu \Rightarrow \nu$ between parallel mode morphisms;
4. specified adjunctions $\mu \dashv \nu$ used by the type formers;
5. a symmetric monoidal product on the full sub-two-category of geometric modes;
6. a distinguished scalar mode s and, for every trace-admitting mode m, a shadow morphism $\mathrm{sh}_m : m \to s$.

Composition is written from right to left: $\mu\nu$ means first ν, then μ. Equality of one-cells is strict only for the chosen presentation of $\mathcal{M}$; geometric identifications are represented by invertible two-cells.

Strictness here is administrative. A bicategorical mode theory can be strictified for the purpose of syntax, while its semantics retains the relevant coherence. Nothing depends on equality between geometric functors being strict.

Definition 6.3 (Locked context). [DESIGN]

If Γ ctx@m and $\mu : n \to m$, then Γ/μ ctx@n is the context obtained by placing a **lock** μ at the right boundary of Γ. A variable declared to the left of the lock may be used to the right only through the action prescribed by μ. Locks satisfy

$$(\Gamma/\mu)/\nu \equiv_{\mathrm{jud}} \Gamma/(\mu\nu), \qquad \Gamma/1_m \equiv_{\mathrm{jud}} \Gamma.$$

For a two-cell $\alpha : \mu \Rightarrow \nu$ there is a context coercion $\alpha_\Gamma : \Gamma/\mu \to \Gamma/\nu$, functorial in vertical and horizontal composition.

The variable and context-extension rules are the ordinary dependent rules at a fixed mode:

$$\frac{\Gamma \text{ ctx@}m \qquad \Gamma \vdash A \text{ type@}m}{\Gamma, x : A \text{ ctx@}m} \qquad \frac{\Gamma, x : A, \Delta \text{ ctx@}m}{\Gamma, x : A, \Delta \vdash x : A@m}.$$

Substitution is admissible and mode-preserving. Modal substitution uses locks: if $\Gamma/\mu \vdash a : A@n$ and $\Delta \to \Gamma$ is a substitution at m, then $\Delta/\mu \vdash a[\Delta/\mu] : A[\Delta/\mu]@n$.

Structural discipline. Weakening, exchange, and contraction are admissible only within a region of a context bounded by the same locks. Moving a declaration across a lock is not a structural rule. It requires a term witnessing the relevant modal unit, counit, or comparison map.

This restriction is the point of the notation. For example, a global proof cannot silently use a p-local witness as though it were defined at every place. It must be introduced through the right adjoint to localisation or transported by an explicitly supplied descent term.

The theory includes dependent products, dependent sums, identity types, universes, finite inductive types, and extension types. Their rules are the standard Martin-Löf rules at each mode, with locks propagated through substitution. Universes are cumulative:

$$\Gamma \vdash \mathcal{U}_{m,i} \text{ type@}m, \qquad \Gamma \vdash A : \mathcal{U}_{m,i}@m \Longrightarrow \Gamma \vdash A : \mathcal{U}_{m,i+1}@m.$$

No resizing rule is included in the core. When a construction needs to move a large type into a smaller universe, it must state a resizing axiom or remain unformed.

6.2.1 Raw syntax and substitution

For metatheoretic purposes the preceding rules are generated from a raw grammar. A raw mode is an object expression in the chosen presentation of $\mathcal{M}$. A raw context is a finite list whose entries are either declarations $x : A$ or locks $/\mu$. A raw type is built from universe symbols, dependent products and sums, identity and extension types, modal types, directed hom-types, and trace types. A raw term is built from variables, the usual introduction and elimination forms, interval abstraction and application, modal introduction and elimination, Segal filler operations, and the two trace constructors. A derivation is a finite tree certifying that a raw expression is meaningful. This distinction between raw expression and derivable judgement is indispensable: the kernel never checks prose notation directly.

There are three forms of substitution. Ordinary substitution replaces a variable by a term in the same modal region. Mode substitution acts on an entire judgement by a two-functor between mode theories. Lock substitution reindexes declarations through a mode morphism. They satisfy an interchange law. If $\sigma : \Delta \to \Gamma$ is an ordinary substitution and $\mu : n \to m$, then

$$(A[\sigma])/\mu \equiv_{\mathrm{jud}} (A/\mu)[\sigma/\mu].$$

If $\alpha : \mu \Rightarrow \nu$ is a two-cell, coercion commutes with ordinary substitution:

$$\alpha_*(a[\sigma/\mu]) \equiv_{\mathrm{jud}} (\alpha_* a)[\sigma/\nu].$$

These are equations of the explicit syntax and must be oriented in any normalisation proof. They are not appeals to an informal principle that substitution is "evident".

Judgemental equality is the least congruence containing the beta rules for the ordinary type formers, the point beta rule for trace, identity and associativity for lock composition, and the chosen strict equations of the mode presentation. It does not contain univalence, cyclicity of trace, Beck–Chevalley isomorphisms, adjunction triangle identities not selected as strict, or equivalence of Morita presentations. Those are terms in identity or equivalence types. This boundary is a principal metatheoretic design choice. Every equation added to judgemental equality enlarges the kernel's conversion problem; every equation left propositional creates proof terms but preserves a more transparent reduction theory.

The theory is intensional. Function extensionality follows from univalence in the fragment where univalence is available, but it is not a judgemental rule. Proof irrelevance is not assumed for arbitrary identity types. Some status, support, and finite-certificate types may be declared propositions, in which case their inhabitants are propositionally equal. Directed hom-types are not propositions, even when their endpoints are equal, and no coercion converts a directed arrow into an identity proof without additional invertibility data.

The distinction can be seen in the simplest example. Let P be the directed type with two objects $0, 1$ and one non-identity arrow $u : 0 \to 1$. The term u inhabits $\mathrm{Hom}_P(0,1)$, while $\mathrm{Id}_P(0,1)$ is empty in the intended Rezk interpretation. Were directed homs identified with identity types, u would force 0 and 1 to be equivalent and the walking arrow would collapse to a groupoid. A correspondence category contains this walking-arrow phenomenon at every non-invertible pushforward.

Contextual dependency is equally important. If $x : A$ and $B(x)$ is a type, a directed dependent arrow over $f : x \to y$ is not merely a map $B(x) \to B(y)$. It is a section over the directed interval whose endpoints lie in the two fibres. Cocartesian and cartesian lifting structures recover covariant and contravariant transport. Thus a functor, fibration, and natural transformation can all be expressed in the same directed grammar. This is the mechanism by which a six-functor formalism can eventually be internalised rather than represented as a flat record of unrelated maps.

Universe levels are annotated on raw syntax even when omitted in print. The universe containing small figures is lower than the universe containing their presentable categories, and the universe containing a mode-indexed family of such categories is higher again. A proposed chôra-level univalence principle must therefore be stratified:

$$\mathrm{Id}_{\mathcal{U}_{\mathrm{Ch},i}}(A,B) \simeq \mathrm{Equiv}_{\mathrm{Ch},i}(A,B),$$

not asserted for a type containing itself. Large-cardinal assumptions in the semantic discussion below serve this stratification; they do not license unrestricted resizing in syntax.

6.2.2 A modal typing example

Consider a global type A and, for each place v, its localisation $\Box_v A$. Suppose a local predicate $P_v : \Box_v A \to \mathcal{U}_{\mathsf{ar}_v}$ is given, together with a local witness

$$\Gamma/\ell_v \vdash p_v : \prod_{x:\Box_v A} P_v(x)@\mathsf{ar}_v.$$

There is no rule permitting p_v to be used directly in the global context Γ. One may introduce its right-adjoint image:

$$\Gamma \vdash \mathrm{mod}_{(\ell_v)_*}(p_v) : \left\langle (\ell_v)_* \;\middle|\; \prod_{x:\Box_v A} P_v(x) \right\rangle @\mathsf{ar}.$$

This term says that the local witness has been packaged as global data with a v-locality marker. It does not say that P_v holds globally.

To descend a family $(p_v)_v$, one must supply overlap compatibilities. For places v, w, the commuting cell $\chi_{v,w} : \Box_v \Box_w \Rightarrow \Box_w \Box_v$ transports the two restrictions into a common type. A compatibility term has the form

$$c_{v,w} : \mathrm{transport}_{\chi_{v,w}}(\Box_w p_v) = \Box_v p_w.$$

For triples one needs a coherence identifying the two composites built from $c_{v,w}$, $c_{w,u}$, and $c_{v,u}$. The Yang–Baxter equation for χ ensures that the endpoints of this coherence are themselves definitionally well typed. Without it, the two triple-overlap terms would inhabit merely equivalent, rather than identical, target types and a further coercion would be required.

Assume now an adelic descent operator

$$\mathrm{desc}_A : \mathrm{DescentData}((\Box_v A)_v) \to A$$

with a beta law after every localisation. Applying it to the local terms and their compatibilities yields a global term only if the restrictedness condition is present. That condition can be represented by a finite set S of exceptional places and a proof that, for $v \notin S$, p_v lies in a chosen integral submodal type. Thus the expression $\bigotimes'_v \Box_v$ is not an untyped infinite product: its introduction rule asks for finite exceptional support and integral data elsewhere.

This example separates four claims that ordinary prose often merges. Local existence is the family (p_v). Pairwise compatibility is the family $(c_{v,w})$. Effective descent is the existence of desc_A. Adelic completeness is the assertion that the resulting global term localises back to the original family and is unique. Only the first two are data in an arbitrary mode diagram; the last two are substantive properties of the arithmetic chôra.

The $\flat/\sharp$ pair gives an analogous warning within one cohesive mode. For $a : A$, the unit gives $a \to \sharp A$, while the counit gives $\flat A \to A$. There is generally no map $A \to \flat A$ and no map $\sharp A \to A$. Consequently a term may be made codiscrete without choosing global points, and discrete data may be included into cohesive data, but cohesive variation cannot be discarded or recovered by an unlicensed coercion. This asymmetry is exactly what the adjunction records.

In a proof assistant, the surface notation might omit all four locks in a short calculation. Elaboration is allowed to reconstruct them only when the mode constraints have a unique solution or when a user-supplied annotation selects one. If two incomparable mode paths could type the same expression, the elaborator must report ambiguity rather than select a path whose mathematical normalisation is different. This is especially important for Frobenius, where geometric and arithmetic conventions differ by inversion, and for $f_!$ versus f_*, where equality is a properness theorem rather than notation.

6.3 Directed hom-types

Let $\mathbf{2}$ be a strict interval type with endpoints $0, 1 : \mathbf{2}$ and with no path identifying them. If A is a type and $x, y : A$, the directed hom-type is the extension type of maps $\mathbf{2} \to A$ with prescribed endpoints. This definition is internal and does not identify directed arrows with identity proofs.

Definition 6.4 (Directed hom-type). [DESIGN]
For $\Gamma \vdash A$ type@m and $\Gamma \vdash x, y : A@m$, define

$$\mathrm{Hom}_A(x,y) := \left\langle \prod_{t:\mathbf{2}} A \;\middle|\; f(0) = x,\, f(1) = y \right\rangle.$$

The constant extension gives $\mathrm{id}_x : \mathrm{Hom}_A(x,x)$. A type A is a **Segal type** if, for each composable pair $x \xrightarrow{f} y \xrightarrow{g} z$, the type of fillers of the corresponding inner horn is contractible. It is a **Rezk type** if it is Segal and the map from identity types to equivalences in its homotopy category is an equivalence.

The Segal condition produces composition without choosing a global composition operation in advance. Contractibility says that a composite exists and that all choices agree through a unique coherent tower. This is the synthetic replacement for writing associators, pentagons, and their higher analogues.

Theorem 6.5 (Riehl–Shulman synthetic infinity-categories). [ESTABLISHED]
In simplicial type theory, Segal types support coherently associative directed composition, and Rezk types support a synthetic theory of complete $(\infty,1)$-categories, including the Yoneda lemma and the usual relation between equivalences and identities (*Riehl and Shulman 2017*).

The present theory imports the Riehl–Shulman directed fragment as an established component. It does not import an implementation, and it does not infer that a combination with the trace rules below is normalising.

Proposition 6.6 (Composition from Segal filling). [THEOREM]
If A is a Segal type, then there are terms

$$\mathrm{comp}_{x,y,z} : \mathrm{Hom}_A(x,y) \to \mathrm{Hom}_A(y,z) \to \mathrm{Hom}_A(x,z)$$

whose left-unit, right-unit, and associativity laws hold as identity terms. The types of all choices of these data are contractible.

Proof of Proposition 6.6.
Given $f : x \to y$ and $g : y \to z$, form the inner horn $\Lambda^2_1 \to A$ whose two non-degenerate edges are f and g. By the Segal hypothesis, its filler type is contractible. Restriction of the centre edge of any filler gives $g \circ f : x \to z$. Taking one of f, g to be a constant edge gives the unit paths, because the relevant horn-filler types are again contractible. For three arrows, the two composites are restrictions of fillers of the same three-simplex spine. Contractibility of the iterated filler type gives associativity. Repeating the argument in one higher dimension gives coherence of the associativity path, and induction on dimension gives the remaining tower. Since every choice was made from a contractible type, the total type of composition data is contractible.

□

A functor between Segal types is simply a dependent function preserving the directed structure. For $F : A \to B$ and $f : x \to_A y$, its action is written $F(f) : F(x) \to_B F(y)$. Natural transformations are directed dependent terms $\eta_x : F(x) \to G(x)$ satisfying the usual naturality square as an identity term. Nothing in these definitions requires inverse arrows.

Correspondences need one further layer. A correspondence from X to Y is a type K with arrows $K \to X$ and $K \to Y$, the left arrow carrying a support class. Their composition is a dependent pullback. The type of correspondences is therefore itself directed, while equivalences between correspondences remain identity-like. This separation is what ordinary identity types alone cannot express economically.

6.4 Modal type formers and the adjoint pair

For every one-cell $\mu : n \to m$ the theory has a modal type former $\langle \mu \mid - \rangle$ from types at n in a locked context to types at m. Positive and negative modalities are distinguished by their introduction and elimination rules; adjunction is encoded in the mode theory, not asserted separately for every pair of types.

Definition 6.7 (Modal type formation). [DESIGN]

For $\mu : n \to m$, the basic rules are

$$\frac{\Gamma / \mu \vdash A \ \mathsf{type@}n}{\Gamma \vdash \langle \mu \mid A \rangle \ \mathsf{type@}m} \quad (\mathsf{Mod\text{-}F})$$

and

$$\frac{\Gamma / \mu \vdash a : A@n}{\Gamma \vdash \mathrm{mod}_\mu(a) : \langle \mu \mid A \rangle @m} \quad (\mathsf{Mod\text{-}I}).$$

Elimination is permitted into a type C whose dependency on a modal variable is stable under the μ-lock:

$$\frac{\Gamma, z : \langle \mu \mid A \rangle \vdash C(z) \ \mathsf{type@}m \qquad \Gamma / \mu, x : A \vdash c(x) : C(\mathrm{mod}_\mu\, x)@m}{\Gamma, z : \langle \mu \mid A \rangle \vdash \mathrm{let}_\mu \ \mathrm{mod}_\mu\, x = z \ \mathrm{in}\, c(x) : C(z)@m} \quad (\mathsf{Mod\text{-}E}),$$

with computation rule

$$\mathrm{let}_\mu \ \mathrm{mod}_\mu\, x = \mathrm{mod}_\mu\, a \ \mathrm{in}\, c(x) \equiv_{\mathrm{jud}} c(a). \quad (\mathsf{Mod}\text{-}\beta)$$

The stability side condition is represented formally by a lifting operation against the unit of the modality.

The side condition prevents an eliminator from observing hidden cohesive or local data. For a left-exact idempotent modality it is the usual requirement that the target family be modal. For an adjoint mode change it is expressed by placing the motive under the corresponding lock.

The cohesive fragment uses two modes, c for cohesive types and d for discrete types, with a triple of adjoints

$$\mathrm{Disc} : d \rightleftarrows c : \Gamma : c \rightleftarrows d : \mathrm{Codisc}, \qquad \mathrm{Disc} \dashv \Gamma \dashv \mathrm{Codisc}\,.$$

The endomodal composites are

$$\flat := \mathrm{Disc}\,\Gamma, \qquad \sharp := \mathrm{Codisc}\,\Gamma.$$

Thus $\flat$ forgets variation and returns the discrete object generated by global points, while $\sharp$ returns the codiscrete object on the same global points. These descriptions are semantic mnemonics, not extra reduction rules.

Proposition 6.8 (The flat–sharp adjunction). [THEOREM]
In any mode theory containing Disc $\dashv \Gamma \dashv$ Codisc, *the induced endomodal operators satisfy* $\flat \dashv \sharp$. *Internally there is an equivalence*

$$\mathrm{Hom}_c(\flat A, B) \simeq \mathrm{Hom}_c(A, \sharp B)$$

natural in cohesive types A and B.

Proof of Proposition 6.8.
By the first adjunction,

$$\mathrm{Hom}_c(\mathrm{Disc}\,\Gamma A, B) \simeq \mathrm{Hom}_d(\Gamma A, \Gamma B).$$

By the second adjunction,

$$\mathrm{Hom}_d(\Gamma A, \Gamma B) \simeq \mathrm{Hom}_c(A, \mathrm{Codisc}\,\Gamma B).$$

Composition gives the displayed equivalence. Naturality follows from naturality of the two adjunction equivalences. In the syntax, the forward map is a modal elimination through the Disc lock followed by modal introduction through the Codisc lock; the two triangle identities reduce the composites to the identity.

□

For the arithmetic theory, the mode category has more objects. There is a global mode ar, a local mode ar_v for each place v, a cohesive/analytic mode, a stable linear mode in which six operations act, and a scalar mode in which traces live. The place morphisms $\ell_v : \mathsf{ar}_v \to \mathsf{ar}$ induce idempotent endomodal operators $\square_v$. Their commutation is represented by invertible two-cells

$$\chi_{v,w} : \square_v \square_w \Rightarrow \square_w \square_v,$$

subject to the Yang–Baxter coherence for triples of places. Adelic completeness is not a judgemental identity of an infinite product. It is a term, on a restricted subuniverse $\mathcal{U}_{\mathrm{ad}}$, of type

$$\prod_{A:\mathcal{U}_{\mathrm{ad}}} \mathrm{Equiv}\left(\left\langle \bigotimes_v{}' \square_v \mid A \right\rangle, A\right).$$

The restricted tensor is part of the signature and carries finite-support data and an integrality witness outside that support.

The six operations are also mode morphisms. For a supported map $f : X \to Y$ one has $f^*, f_*, f_!, f^!$ as modal operators between fibre modes, with adjunction two-cells

$$f^* \dashv f_*, \qquad f_! \dashv f^!.$$

Tensor and internal hom live within a fibre mode. Projection formula and base change are two-cells generated by composition in the correspondence mode theory. This is more economical than adding them as unrelated constants.

6.4.1 Six operations as modal adjunctions

The modal notation is not intended to suggest that $f^*, f_*, f_!$, and $f^!$ are all idempotent endofunctors. A place localisation such as $\Box_v$ is endomodal and idempotent. A six-operation functor usually changes the fibre mode from X to Y or conversely. What unifies them is that both are one-cells in $\mathcal{M}$ and therefore both act on locked contexts.

For a map $f : X \to Y$, adopt the directions

$$f^* : Y \to X, \qquad f_* : X \to Y, \qquad f_! : X \to Y, \qquad f^! : Y \to X$$

in the mode category. A type A in the fibre over X gives modal types $\langle f_! \mid A\rangle$ and $\langle f_* \mid A\rangle$ over Y. A type B over Y gives $\langle f^* \mid B\rangle$ and $\langle f^! \mid B\rangle$ over X. The adjunction equivalence

$$\mathrm{Hom}_Y(\langle f_! \mid A\rangle, B) \simeq \mathrm{Hom}_X(A, \langle f^! \mid B\rangle)$$

is induced by the unit and counit two-cells of $f_! \dashv f^!$. The second adjunction gives

$$\mathrm{Hom}_X(\langle f^* \mid B\rangle, A) \simeq \mathrm{Hom}_Y(B, \langle f_* \mid A\rangle).$$

These equivalences are dependent when A and B vary in contexts.

Properness is then represented by a modal equivalence

$$\mathrm{proper}_f : \prod_A \mathrm{Equiv}(\langle f_! \mid A\rangle, \langle f_* \mid A\rangle).$$

Smoothness is represented by an invertible object ω_f and an equivalence

$$\mathrm{smooth}_f : \prod_B \mathrm{Equiv}\left(\langle f^! \mid B\rangle, \langle f^* \mid B\rangle \otimes \omega_f\right).$$

Étaleness adds a trivialisation $\omega_f \simeq \mathbb{1}$. These terms make the definitions of KH.3 literal in the internal language. They also show why geometric adjectives are propositions about modal operators rather than tags attached to raw maps.

Tensor and internal hom are not mode changes, but their compatibility with mode changes is typed by projection formula cells. For example,

$$\pi_{f,A,B} : \langle f_! \mid A \otimes \langle f^* \mid B\rangle\rangle \longrightarrow \langle f_! \mid A\rangle \otimes B.$$

When $\pi_{f,A,B}$ is an equivalence, the projection formula holds for those arguments. A blanket projection-formula theorem quantifies over the appropriate universes and records naturality. The formula is not admitted merely because the symbols can be written.

The same discipline governs Frobenius. The absolute Frobenius Φ is an endocorrespondence, hence an endomodal operator on the relevant fibre. Equivariance of a type A is a term

$$\rho_A : \mathrm{Equiv}(\langle \Phi \mid A\rangle, A)$$

together with coherence under iteration. A Φ-fixed point is a dependent pair (a, p) with $a : A$ and $p : \Phi(a) = a$ in the appropriate directed or identity sense. For a moduli Rezk type, the derived fixed-point object must retain the homotopy of the equation; truncating to a set of fixed points would discard automorphisms and local terms.

Place localisation and Frobenius interact through specified two-cells

$$\lambda_v : \Box_v \Phi \Rightarrow \Phi_v \Box_v.$$

Local–global compatibility is consequently a family of terms asserting that the localisation of a global Φ-equivariant object agrees with the local Φ_v-equivariant object. This is the type-theoretic content of the claim that a global parameter restricts to the prescribed local parameter. If a normalisation convention changes geometric to arithmetic Frobenius, the two-cell changes; the type system does not regard the two conventions as silently interchangeable.

The gauge modality is treated in the same fashion. It records the Rees direction and acts on filtered objects. Its special fibres at the Hodge and de Rham points are mode morphisms, and comparison maps between them are two-cells. At a finite place, Nygaard filtration and prismatic Frobenius form a commuting square only after the appropriate twist. At infinity, the split and compact factors of the Deligne torus act differently. The mode theory is therefore not a list of commuting unary boxes; it is a two-dimensional diagram whose nontrivial cells record exactly the twists and comparisons that arithmetic geometry cannot suppress.

The system-level dependencies of the construction are displayed in Plate 6.2.

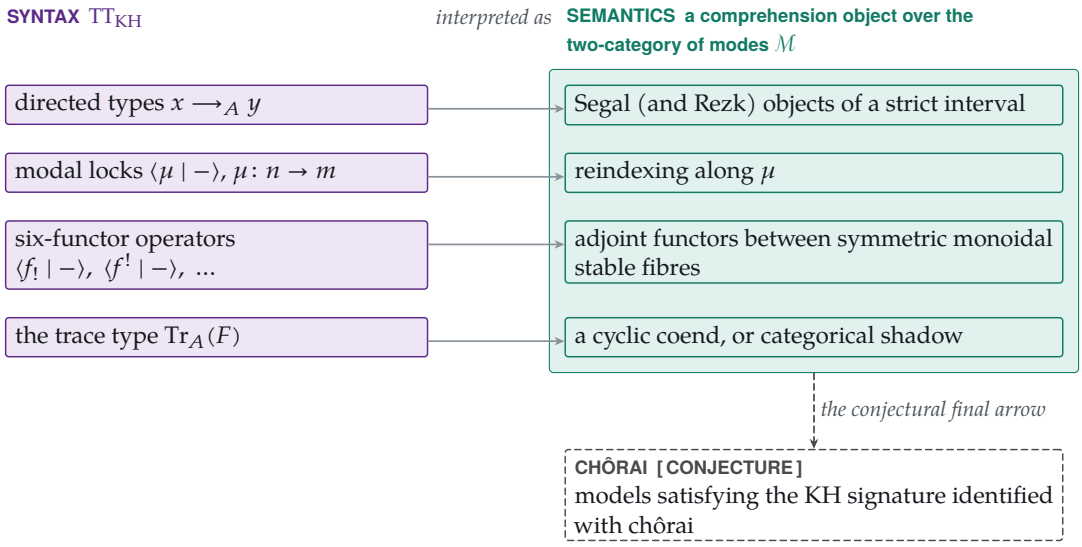


Plate 6.2. The syntax is interpreted by a comprehension object over the two-category of modes. Directed types map to Segal objects, modal locks map to reindexing, six-functor operators map to adjoint functors, and the trace type maps to a coend or categorical shadow. The conjectural final arrow identifies such models satisfying the KH signature with chôrai.

What it does not show: that the models are chôrai: the final identification of models satisfying the KH signature with chôrai is a conjecture, and no model of the full theory is constructed.

6.5 The two language axioms

The final two axioms of the KH system concern language and evidence. They are design axioms, not theorems about an already existing implementation.

Axiom 6.9 (KH.27: Internal language). [DESIGN]

There is a dependent type theory $\mathrm{TT}_{\mathrm{KH}}$ with cumulative universes, directed interval and extension types, Segal and Rezk completion principles, a two-category of modes containing the place, gauge, analytic, six-functor, Frobenius, and shadow modalities, a chôra-level univalence principle, and the trace rules of Construction 6.12. Every statement marked `[F]` *in the treatise is required to admit a faithful formal statement in this language. Expressibility does not entail that a proof term exists.*

Axiom 6.10 (KH.28: Verification adequacy and economy). [DESIGN]
The formal corpus obeys four rules. First, factual formal status is conferred only by a trusted proof kernel, by an independently checked computation certificate, or by an explicitly reviewed external theorem. Secondly, statistical or heuristic systems may propose terms but may not confer truth. Thirdly, there is no coercion from an approximation to an exact object without a convergence, stabilisation, or enclosure theorem. Fourthly, every formal claim records a finite dependency graph whose leaves have explicit epistemic status.

The first clause is **kernel sovereignty**. The second is **nonauthority of prediction**. The third is **no approximation erasure**. The fourth is **provenance closure**. These are part of the mathematical specification because they govern the meaning of `[F]`; they do not assert that the specification has been realised.

6.6 The trace modality

There are two related notions of trace. A directed category has an object-valued trace, the cyclic coend

$$\int^{x:A} \mathrm{Hom}_A(x, Fx),$$

for an endofunctor F. A dualizable object in a symmetric monoidal category has a scalar trace in $\mathrm{End}(\mathbb{1})$. The first is the primitive type former. The second is obtained by applying a shadow eliminator. This order keeps the cyclic relation explicit and covers both Hochschild-type traces and numerical traces.

Definition 6.11 (Trace-admissible directed type). [DESIGN]
A **trace-admissible directed type** at mode m is a Rezk type A equipped with:
1. small directed hom-types in a universe closed under the relevant coends;
2. an endofunctor action preserving Segal composition;
3. a shadow morphism $\mathrm{sh}_m : m \to s$;
4. a specified class of endofunctors for which the cyclic coend exists;
5. when scalar trace is required, a symmetric monoidal duality datum with coevaluation and evaluation satisfying the triangle identities.

The judgement is written $\Gamma \vdash A\ \mathsf{traceable@}m$.

The coend is presented syntactically as a higher-inductive quotient. Its points are cycles $u : x \to Fx$, and its path constructors impose dinaturality. No higher constructor is omitted: the phrase "higher-inductive quotient" means the free type with the displayed point and path constructors and all coherences generated by its universal property.

Construction 6.12 (Trace modality and rules). [DESIGN]
Let $\Gamma \vdash A\ \mathsf{traceable@}m$ and $\Gamma \vdash F : A \to A@m$. The trace modality is governed by the following rules.

Formation

$$\frac{\Gamma \vdash A\ \mathsf{traceable@}m \qquad \Gamma \vdash F : A \to A@m}{\Gamma \vdash \mathrm{Tr}_A(F)\ \mathsf{type@}s}\ (\mathsf{Tr\text{-}F}).$$

Point introduction

$$\frac{\Gamma \vdash x : A@m \qquad \Gamma \vdash u : \mathrm{Hom}_A(x, Fx)@m}{\Gamma \vdash \mathrm{cyc}_F(x,u) : \mathrm{Tr}_A(F)@s}\ (\mathsf{Tr\text{-}I}_0).$$

Cyclic path introduction

$$\frac{\Gamma \vdash a : \mathrm{Hom}_A(x,y)@m \qquad \Gamma \vdash b : \mathrm{Hom}_A(y,Fx)@m}{\Gamma \vdash \mathrm{cyc}_F(x, b \circ a) = \mathrm{cyc}_F(y, F(a) \circ b) : \mathrm{Tr}_A(F)@s} \quad (\mathsf{Tr\text{-}I}_1).$$

Dependent elimination

$$\frac{\begin{array}{c}\Gamma, z : \mathrm{Tr}_A(F) \vdash P(z)\ \mathsf{type}@s \\ \Gamma, x : A, u : x \to Fx \vdash c(x,u) : P(\mathrm{cyc}_F(x,u))@s \\ \Gamma, a : x \to y, b : y \to Fx \vdash q(a,b) : \\ \mathrm{transport}_P(\mathrm{cyc}_1(a,b), c(x, b \circ a)) = c(y, F(a) \circ b)\end{array}}{\Gamma, z : \mathrm{Tr}_A(F) \vdash \mathrm{rec}_{\mathrm{Tr}}(c, q; z) : P(z)@s} \quad (\mathsf{Tr\text{-}E}).$$

Point computation

$$\mathrm{rec}_{\mathrm{Tr}}(c, q; \mathrm{cyc}_F(x,u)) \equiv_{\mathrm{jud}} c(x,u) \quad (\mathsf{Tr}\text{-}\beta_0).$$

Path computation

$$\mathrm{ap}_{\mathrm{rec}_{\mathrm{Tr}}(c,q)}(\mathrm{cyc}_1(a,b)) = q(a,b) \quad (\mathsf{Tr}\text{-}\beta_1).$$

The path computation is propositional, not judgemental. The eliminator has a propositional uniqueness principle: two sections agreeing on point constructors and cyclic paths are equal.

The rule is the internal form of a coend. A non-dependent eliminator into a type B asks for a family

$$c_x : \mathrm{Hom}_A(x, Fx) \to B$$

such that

$$c_x(b \circ a) = c_y(F(a) \circ b).$$

This is precisely dinaturality. The correspondence with maps out of the trace is explicit. Given $h : \mathrm{Tr}_A(F) \to B$, define

$$c_x(u) := h(\mathrm{cyc}_F(x,u)).$$

Applying h to the cyclic path constructor proves $c_x(b \circ a) = c_y(F(a) \circ b)$. Conversely, given a dinatural family c, apply $\mathsf{Tr\text{-}E}$ with constant motive B; dinaturality supplies its path-coherence argument and yields $\bar{c} : \mathrm{Tr}_A(F) \to B$. Point computation gives

$$\bar{c}(\mathrm{cyc}_F(x,u)) = c_x(u).$$

Starting with h, constructing c, and eliminating again returns h by the uniqueness principle for Tr. Starting with c, evaluating the induced map on a point constructor returns c by $\mathsf{Tr}\text{-}\beta_0$. Hence maps $\mathrm{Tr}_A(F) \to B$ are equivalent to dinatural cocones from the twisted hom functor to B.

For a symmetric monoidal Rezk type $\mathcal{U}$ and a dualizable $X : \mathcal{U}$, the ordinary scalar trace is a term

$$\mathrm{tr}_X(f) : \mathrm{End}_{\mathcal{U}}(\mathbb{1})$$

defined from

$$\mathbb{1} \xrightarrow{\mathrm{coev}} X \otimes X^\vee \xrightarrow{f \otimes \mathrm{id}} X \otimes X^\vee \xrightarrow{\beta} X^\vee \otimes X \xrightarrow{\mathrm{ev}} \mathbb{1}.$$

This formula is not a custom reduction rule. It is the value of the shadow map on the cycle represented by f, and its invariance is obtained from the trace eliminator.

Theorem 6.13 (Scalar trace from the shadow). [ESTABLISHED]
Let $\mathcal{U}$ be a symmetric monoidal trace-admissible Rezk type and let X be dualizable. There is a map

$$\mathrm{shadow}_X : \mathrm{Tr}_{\mathcal{U}}(\mathrm{id}_{\mathcal{U}}) \longrightarrow \mathrm{End}_{\mathcal{U}}(\mathbb{1})$$

whose value on the cycle (X, f) is the evaluation–coevaluation composite displayed above. This is the standard shadow-to-scalar trace construction (Ponto and Shulman 2012).

Proof of Theorem 6.13.
For each object X and endomorphism $f : X \to X$, define $c(X, f)$ by the evaluation–coevaluation composite. Given $a : X \to Y$ and $b : Y \to X$, the two composites $c(X, b \circ a)$ and $c(Y, a \circ b)$ are equal by naturality of the symmetry and the triangle identities for the duals. Written as string diagrams, the equality slides a once around the closed loop; algebraically it inserts the identity $(\mathrm{id} \otimes \mathrm{ev})(\mathrm{coev} \otimes \mathrm{id}) = \mathrm{id}$ and uses naturality of β. Thus c is dinatural. Apply Tr-E with constant motive $\mathrm{End}(\mathbb{1})$. The point computation rule gives the claimed value. □

The same construction in presentable stable categories gives Hochschild homology and higher categorical traces (Ponto and Shulman 2012; Hoyois et al. 2017). The chapter does not identify every trace in a chôra with a scalar. The object-valued trace is primary; THH and TC retain homotopical data that a numerical trace discards.

6.6.1 Localisation, twisted traces, and cyclotomic structure

The notation $\mathrm{Tr}_A(F)$ is deliberately twisted by an endofunctor. For $F = \mathrm{id}$ it is the ordinary cyclic trace. For Frobenius it is the receptacle for Lefschetz-type fixed-point data. For a correspondence $X \leftarrow K \to X$, F is represented by the induced endofunctor on the sheaf fibre, and the trace type depends on the correspondence rather than only on an underlying map of spaces.

A localisation datum adds a fixed-point type $\mathrm{Fix}(F)$ and a local-term family

$$\mathrm{loc}_F : \mathrm{Fix}(F) \to \mathcal{U}_s,$$

together with a comparison

$$\mathrm{localise}_F : \mathrm{Tr}_A(F) \longrightarrow \int_{x:\mathrm{Fix}(F)} \mathrm{loc}_F(x).$$

Depending on the geometric hypotheses, this comparison may be an equivalence, a filtration, or merely a map. The language does not force a Lefschetz formula in circumstances where properness or transversality fails. Such hypotheses appear as arguments to the term establishing that $\mathrm{localise}_F$ is an equivalence.

The distinction matters for the arithmetic trace formula. On a proper smooth variety over a finite field, fixed-point localisation is an established theorem under the usual sheaf-theoretic hypotheses. On Bun_G, non-properness introduces Eisenstein and boundary terms. Type-theoretically, the naive localise map fails to be invertible on the whole category, while it may become invertible after restriction to a renormalised or nilpotent-singular-support modality. The cofibre of the map is itself a typed object rather than an omitted error term.

Cyclotomic structure requires more than one twisted trace. An object T at the scalar mode carries an S^1-action and, for each prime p, a Frobenius map

$$\varphi_p : T \longrightarrow T^{tC_p}.$$

The fixed-point and Tate constructions are modalities on an equivariant stable mode. In the intended semantics, THH is the object-valued trace of the identity, while TC is assembled as an equaliser involving the negative and periodic cyclic modalities. The type theory should not reduce this equaliser to a single primitive number. It should form the relevant limits in the scalar stable mode and preserve the maps naming each local Frobenius.

A schematic formation of the cyclotomic refinement is therefore

$$\frac{\Gamma \vdash T : \mathcal{U}_{S^1} \qquad \Gamma \vdash (\varphi_p : T \to T^{tC_p})_{p \text{ prime}}}{\Gamma \vdash \mathrm{Cyc}(T, \varphi) : \mathcal{U}_{\mathrm{cyc}}}.$$

This rule is not part of the small computational core because it quantifies over all primes and uses Tate constructions. A finite-prime truncation can be encoded, checked, and refined. The passage to all primes requires a universe and convergence discipline of its own.

The trace eliminator also explains local terms. To map a trace into a sum of local contributions, one supplies a dinatural family assigning a local term to every cycle and proves invariance under sliding a morphism around the cycle. In a geometric model the difficult theorem is that this family factors through the fixed-point object and that the resulting comparison is an equivalence. The syntax makes the theorem's input and output explicit but does not trivialise its proof.

There is a useful separation between additivity and localisation. Additivity says that a cofibre sequence of traceable endofunctors induces a sum or cofibre relation among traces. Localisation says that a trace decomposes over fixed-point strata. The first is a formal property of stable shadows in many settings; the second contains geometric content. A chôra's primitive trace theory includes both structures as named data only where they are available, and the type theory keeps their fields distinct.

6.6.2 Worked computation: the trace quotient of S_3

The trace type can be computed completely for a one-object groupoid. Let $A = BS_3$ have one object $*$ and endomorphism group

$$S_3 = \{1, (12), (13), (23), (123), (132)\}.$$

Take $F = \mathrm{id}$. The point constructors of $\mathrm{Tr}_A(F)$ are the six cycles $\mathrm{cyc}(*, \sigma)$, one for each permutation σ. The cyclic path rule gives

$$\mathrm{cyc}(*, ba) = \mathrm{cyc}(*, ab)$$

for every $a, b \in S_3$.

These relations identify precisely the conjugate elements. To see one direction, put $a = h$ and $b = h^{-1}g$. Then

$$ba = h^{-1}gh, \qquad ab = hh^{-1}g = g,$$

so

$$\mathrm{cyc}(*, h^{-1}gh) = \mathrm{cyc}(*, g).$$

Conversely, a generating cyclic relation $ba \sim ab$ is a conjugacy relation, because

$$ab = a(ba)a^{-1}.$$

Thus the set truncation of $\mathrm{Tr}_{BS_3}(\mathrm{id})$ is the set of conjugacy classes of S_3.

There are exactly three classes. The identity class is

$$C_1 = \{1\}.$$

The transposition class is

$$C_2 = \{(12), (13), (23)\}.$$

For example,

$$(123)^{-1}(12)(123) = (23), \qquad (132)^{-1}(12)(132) = (13).$$

The three-cycles form

$$C_3 = \{(123), (132)\},$$

since

$$(12)^{-1}(123)(12) = (132).$$

No element of one displayed class can be conjugate to an element of another, because conjugacy preserves cycle type.

The class sizes can be checked from centralisers. The centraliser of 1 has order 6, so $|C_1| = 6/6 = 1$. The centraliser of (12) is $\{1, (12)\}$, of order 2, so $|C_2| = 6/2 = 3$. The centraliser of (123) is $\{1, (123), (132)\}$, of order 3, so $|C_3| = 6/3 = 2$. The sum $1 + 3 + 2 = 6$ checks that no element has been omitted.

A map from the trace set to an abelian group R is consequently the same as a class function on S_3. The three irreducible characters give an explicit basis:

	C_1	C_2	C_3
χ_{triv}	1	1	1
χ_{sgn}	1	-1	1
χ_{std}	2	0	-1.

The inner products, with class-size weights, are

$$\langle \chi, \psi \rangle = \frac{1}{6} \left(\chi(C_1)\psi(C_1) + 3\chi(C_2)\psi(C_2) + 2\chi(C_3)\psi(C_3) \right).$$

For the standard character,

$$\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle = \frac{1}{6}(2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2) = 1.$$

Its inner products with the trivial and sign characters are respectively

$$\frac{1}{6}(2 + 0 - 2) = 0, \qquad \frac{1}{6}(2 + 0 - 2) = 0.$$

This calculation exhibits the passage from the object-valued cyclic quotient to scalar traces without suppressing the quotient relation. The relation is generated by $\mathsf{Tr}\text{-}\mathsf{I}_1$; the character is supplied by the eliminator as a dinatural function.

6.7 Worked derivation: cyclicity

The smallest nontrivial test of the rules is the familiar identity $\mathrm{tr}(gf) = \mathrm{tr}(fg)$. It must be obtained without declaring the two composites judgementally equal.

Theorem 6.14 (Cyclicity of scalar trace). [THEOREM]
Let $\mathcal{U}$ be a symmetric monoidal trace-admissible Rezk type. If $f : X \to Y$ and $g : Y \to X$ with X and Y dualizable, then

$$\mathrm{tr}_X(g \circ f) = \mathrm{tr}_Y(f \circ g) : \mathrm{End}_{\mathcal{U}}(\mathbb{1}).$$

Proof of Theorem 6.14.
The derivation has six explicit steps.

1. By the directed typing rules, $f : \mathrm{Hom}_{\mathcal{U}}(X, Y)$ and $g : \mathrm{Hom}_{\mathcal{U}}(Y, X)$.
2. Apply $\mathsf{Tr\text{-}I}_0$ to the composite $g \circ f : X \to X$:
$$\frac{X : \mathcal{U} \qquad g \circ f : X \to X}{\mathrm{cyc}(X, g \circ f) : \mathrm{Tr}_{\mathcal{U}}(\mathrm{id})}.$$
3. Apply $\mathsf{Tr\text{-}I}_0$ to $f \circ g : Y \to Y$:
$$\frac{Y : \mathcal{U} \qquad f \circ g : Y \to Y}{\mathrm{cyc}(Y, f \circ g) : \mathrm{Tr}_{\mathcal{U}}(\mathrm{id})}.$$
4. Instantiate $\mathsf{Tr\text{-}I}_1$ with $a = f : X \to Y$, $b = g : Y \to X$, and $F = \mathrm{id}_{\mathcal{U}}$. Since $F(f) \equiv_{\mathrm{jud}} f$, this gives
$$p : \mathrm{cyc}(X, g \circ f) = \mathrm{cyc}(Y, f \circ g).$$
5. Apply the function shadow of Theorem 6.13 to p:
$$\mathrm{ap}_{\mathrm{shadow}}(p) : \mathrm{shadow}(\mathrm{cyc}(X, gf)) = \mathrm{shadow}(\mathrm{cyc}(Y, fg)).$$
6. Reduce both endpoints by $\mathsf{Tr\text{-}}\beta_0$. They become $\mathrm{tr}_X(gf)$ and $\mathrm{tr}_Y(fg)$, respectively. The resulting identity term is the desired cyclicity proof.

No reversal of f or g was used. The equality arises from the cyclic quotient, not from treating the directed hom-types as identity types.

□

Proposition 6.15 (Multiplicativity). [ESTABLISHED]
For dualizable X, Y and endomorphisms $f : X \to X$, $g : Y \to Y$,

$$\mathrm{tr}_{X \otimes Y}(f \otimes g) = \mathrm{tr}_X(f) \otimes \mathrm{tr}_Y(g).$$

This is the standard multiplicativity of categorical trace (Ponto and Shulman 2012).

Proof of Proposition 6.15.
Choose the dual of $X \otimes Y$ to be $Y^\vee \otimes X^\vee$. Expand the left-hand scalar trace by Theorem 6.13. Naturality and the hexagon identity for the symmetry separate the closed diagram into the tensor product of the X-loop and the Y-loop. The triangle identities remove the two inserted unit–counit pairs. The remaining composites are exactly the two scalar traces on the right. This proof uses only the symmetric monoidal and duality rules, so it is stable under every modal reindexing preserving those structures.

□

6.8 Worked derivation: proper base change

The larger test is a base-change derivation. Consider a pullback square of figures

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y. \end{array}$$

Assume f is supported, so $f_!$ exists. In a six-functor formalism the comparison

$$g^* f_! \Longrightarrow f'_! g'^*$$

is induced by the equality of the two composite correspondences. For a cartesian square, that comparison is invertible. The type theory records this as an invertible two-cell in the mode theory and transports modal terms along it.

Theorem 6.16 (A typed proper base-change instance). [THEOREM]
Let the displayed square be cartesian, let f belong to the supported class, and let

$$\beta : g^* f_! \Rightarrow f'_! g'^*$$

be the invertible Beck–Chevalley two-cell generated by the pullback square. For every context Γ ctx@Y and every $\Gamma/(g^* f_!) \vdash A$ type@X, there is a term

$$\mathrm{bc}_A : \mathrm{Equiv}\,(\langle g^* f_! \mid A\rangle, \langle f'_! g'^* \mid \beta_* A\rangle)\,@Y',$$

natural in A. The inverse is induced by β^{-1}.

Proof of Theorem 6.16.

We write the derivation as numbered proof steps because each change of mode is load-bearing.

1. **Form the source modal type.** From $\Gamma/(g^* f_!) \vdash A$ type, apply Mod-F:

$$\frac{\Gamma/(g^* f_!) \vdash A\ \mathsf{type}}{\Gamma \vdash \langle g^* f_! \mid A\rangle\ \mathsf{type}}.$$

2. **Change the locked context along the two-cell.** The mode theory supplies $\beta : g^* f_! \Rightarrow f'_! g'^*$. By Definition 6.3, it induces

$$\beta_\Gamma : \Gamma/(g^* f_!) \longrightarrow \Gamma/(f'_! g'^*).$$

Reindexing A along this coercion gives

$$\Gamma/(f'_! g'^*) \vdash \beta_* A\ \mathsf{type}.$$

3. **Form the target modal type.**

$$\frac{\Gamma/(f'_! g'^*) \vdash \beta_* A\ \mathsf{type}}{\Gamma \vdash \langle f'_! g'^* \mid \beta_* A\rangle\ \mathsf{type}}.$$

4. **Construct the forward map.** Assume $z : \langle g^* f_! \mid A\rangle$. Eliminate z by Mod-E. In the locked branch we have $a : A$. Coerce it along β to $\beta_* a : \beta_* A$, and introduce it into the target modality:

$$\frac{a : A \qquad \beta_* a : \beta_* A}{\mathrm{mod}_{f'_! g'^*}(\beta_* a) : \langle f'_! g'^* \mid \beta_* A\rangle}.$$

Therefore

$$\overrightarrow{\mathrm{bc}_A}(z) := \mathrm{let}_{g^* f_!}\, \mathrm{mod}(a) = z\ \mathrm{in}\, \mathrm{mod}_{f'_! g'^*}(\beta_* a).$$

5. **Construct the inverse.** Since the square is cartesian, β is invertible. Repeating the previous step with β^{-1} gives
$$\mathrm{bc}_A^{\leftarrow} : \langle f'_! g'^* \mid \beta_* A \rangle \to \langle g^* f_! \mid A \rangle.$$
6. **Verify the first composite.** It suffices, by modal elimination, to evaluate on an introduced term $\mathrm{mod}(a)$. Two applications of Mod-β reduce the composite to
$$\mathrm{mod}((\beta^{-1})_*(\beta_* a)).$$
Functoriality of two-cell coercion and the identity $\beta^{-1}\beta = 1$ reduce this judgementally to $\mathrm{mod}(a)$.
7. **Verify the second composite.** The same calculation reduces it to $\mathrm{mod}(\beta_*((\beta^{-1})_* a'))$, hence to $\mathrm{mod}(a')$.
8. **Naturality.** For a modal map $h : A \to B$, the two composites around the naturality square reduce after modal elimination to $\beta_* h(\beta_* a)$ and $\beta_*(h(a))$. These are equal by functoriality of two-cell reindexing.

The forward map, inverse, two homotopies, and naturality term constitute bc_A. The proof does not postulate base change as an equation between arbitrary functors. It derives the instance from the invertible two-cell attached to the cartesian correspondence square.

□

This derivation displays the practical advantage of a mode theory. The comparison is typed before it is proved invertible. If the square is not cartesian, or if the supportedness hypothesis is absent, the relevant two-cell or modality is missing and the purported base-change formula does not typecheck.

6.8.1 Worked computation: base change for finite fibres

The abstract derivation can be checked on a square of finite sets where every functor is explicit linear algebra. Let
$$Y = \{*\}, \qquad Y' = \{u, v\}, \qquad X = \{1, 2, 3\},$$
and let $f : X \to Y$ and $g : Y' \to Y$ be the unique maps. The pullback is
$$X' = X \times_Y Y' = \{(1, u), (2, u), (3, u), (1, v), (2, v), (3, v)\}.$$
The maps are
$$g'(i, w) = i, \qquad f'(i, w) = w.$$

Work with finite-dimensional vector spaces over $\mathbf{Q}$. A sheaf on a finite discrete set is a family of vector spaces. Define A on X by
$$A_1 = \mathbf{Q}, \qquad A_2 = \mathbf{Q}^2, \qquad A_3 = \mathbf{Q}.$$
Choose bases
$$A_1 = \langle e_1 \rangle, \qquad A_2 = \langle e_2, e_3 \rangle, \qquad A_3 = \langle e_4 \rangle.$$
Since all fibres are finite, $f_! = f_*$ and
$$(f_! A)_* = A_1 \oplus A_2 \oplus A_3 \cong \mathbf{Q}^4.$$

Pullback along g repeats this vector space at both points of Y':

$$(g^* f_! A)_u \cong \mathbf{Q}^4, \qquad (g^* f_! A)_v \cong \mathbf{Q}^4.$$

In the ordered bases, a section is a pair

$$((a_1, a_2, a_3, a_4)_u, (b_1, b_2, b_3, b_4)_v).$$

On the other route, $g'^* A$ is the family

w	$(1,w)$	$(2,w)$	$(3,w)$
u	$\mathbf{Q}$	$\mathbf{Q}^2$	$\mathbf{Q}$
v	$\mathbf{Q}$	$\mathbf{Q}^2$	$\mathbf{Q}$.

Pushforward along f' sums the three vector spaces over each w:

$$(f'_! g'^* A)_w = (g'^* A)_{(1,w)} \oplus (g'^* A)_{(2,w)} \oplus (g'^* A)_{(3,w)} \cong \mathbf{Q}^4.$$

The Beck–Chevalley map at w sends

$$(c_1, c_2, c_3, c_4) \longmapsto (c_1, (c_2, c_3), c_4).$$

With the direct-sum basis flattened in the stated order, its matrix is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus the comparison is the identity matrix at both u and v.

Naturality can also be checked numerically. Let B be the family

$$B_1 = \mathbf{Q}, \qquad B_2 = \mathbf{Q}, \qquad B_3 = \mathbf{Q},$$

and define $h : A \to B$ by matrices

$$h_1 = (2), \qquad h_2 = \begin{pmatrix} 1 & -1 \end{pmatrix}, \qquad h_3 = (3).$$

Then $f_! h : \mathbf{Q}^4 \to \mathbf{Q}^3$ has matrix

$$H = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Both routes around the base-change naturality square produce H at u and H at v. For example,

$$H \begin{pmatrix} 4 \\ 5 \\ 7 \\ 11 \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 33 \end{pmatrix}.$$

The other route first regards the vector as $(4, (5,7), 11)$ and applies (h_1, h_2, h_3), giving $(8, 5-7, 33) = (8, -2, 33)$. This is the componentwise arithmetic shadow of step eight in Theorem 6.16.

The example is degenerate in a useful way: because finite maps are proper, $f_! = f_*$ and $f'_! = f'_*$. It therefore checks both proper base change and ordinary direct-image base change. For an infinite non-proper fibre, direct sum and direct product differ, and the corresponding comparison cannot be declared invertible without additional hypotheses.

6.9 Rendering the KH axioms as types

A formal language earns its place only if the axioms can actually be rendered in it. The following translations are schematic only where the objects they mention, such as the universe of presentable stable categories, have not yet been constructed internally. Their logical shape is explicit.

Axiom 6.17 (Type-theoretic KH.1). [FORMAL TARGET]

Let $\mathcal{C}$ be a Rezk type with finite limits, let $E : \prod_{X,Y:\mathcal{C}} \mathrm{Hom}_{\mathcal{C}}(X,Y) \to \mathcal{U}$ be closed under identities, composition, and pullback, and let $\mathrm{Corr}(\mathcal{C},E)$ be the resulting correspondence Rezk type. KH.1 is the type

$$\sum_{\mathcal{C}:\mathcal{U}_{\mathrm{Rezk}}} \sum_{E:\mathrm{Support}(\mathcal{C})} \mathrm{LaxSymMon}\left(\mathrm{Corr}(\mathcal{C},E), \mathrm{Pr}^{L}_{\mathrm{st}}\right).$$

A term of this type is the six-functor part of a chôra. No underlying point set occurs in the type.

Axiom 6.18 (Type-theoretic KH.2). [FORMAL TARGET]

For a term $\mathcal{D}$ of Axiom 6.17, KH.2 is the dependent type whose fields extract

$$f^{*},\ f_{*},\ f_{!},\ f^{!},\ \otimes,\ \underline{\mathrm{Hom}}$$

from elementary correspondences and whose remaining fields identify the base-change and projection-formula maps with the images under $\mathcal{D}$ of composition two-cells in $\mathrm{Corr}(\mathcal{C},E)$. No independent field named `baseChangeAxiom` is permitted.

In record notation the essential part is

```
structure SixOperations (D : Corr C E -> PrLSt) where
  pull       : ...
  push       : ...
  pushProper : ...
  pullExtra  : ...
  tensor     : ...
  internalHom : ...
  baseChange :
    image D corrComposition = beckChevalley
```

The ellipses are not proposed kernel syntax; they indicate interfaces that must be filled once the universes of categories and correspondences exist. The mathematical requirement is that the compatibility fields be images of functorial structure rather than unrelated proofs.

Axiom 6.19 (Type-theoretic KH.12). [FORMAL TARGET]

Let $\mathcal{P}$ be the type of places of $\mathbf{Q}$ and $\mathcal{U}_{\mathrm{ad}}$ the restricted adelic universe. KH.12 is rendered by terms

$$\square : \prod_{v:\mathcal{P}} \mathcal{U}_{\mathrm{ad}} \to \mathcal{U}_{\mathrm{ad}},$$

$$\mathrm{idem} : \prod_{v,A} \mathrm{Equiv}(\square_v \square_v A, \square_v A),$$

$$\mathrm{comm} : \prod_{v,w,A} \mathrm{Equiv}(\square_v \square_w A, \square_w \square_v A),$$

together with the triple coherence for comm *and an adelic descent term*

$$\mathrm{adelicComplete} : \prod_{A:\mathcal{U}_{\mathrm{ad}}} \mathrm{Equiv}\left(\left\langle \bigotimes_v' \square_v \mid A \right\rangle, A\right).$$

KH.27 and KH.28 have already been stated as Axiom 6.9 and Axiom 6.10. Their type-theoretic content is real: KH.27 asks for a signature and semantics, while KH.28 asks for an inductive status type and verified transition rules. A minimal status type is

```
inductive Status
  | kernelChecked
  | certifiedComputation
  | externalReviewed
  | conjecture
  | refuted
```

with no constructor converting `conjecture` or a numerical approximation into `kernelChecked`.

Example 6.20 (A provenance-closure derivation). [THEOREM]
Suppose a finite dependency graph is given by a well-founded relation $\prec$ on claims. If every leaf has a root-of-trust witness and every internal claim carries a checker taking witnesses for all predecessors to a witness for the claim, then every claim has a replay witness.

Proof of Example 6.20.
Use well-founded induction on $\prec$. For a leaf, return its stored root-of-trust witness. For an internal node c, the induction hypothesis produces witnesses for every $d \prec c$; apply the stored checker for c to that finite tuple. The resulting term is a replay witness for c. No claim about the mathematical truth of an external source is made beyond the status encoded by its leaf witness.

□

6.10 Semantics

Syntax without semantics would only rename the desired structures. A model must explain contexts, substitution, directed homs, modes, modalities, and trace in ordinary mathematical terms.

Definition 6.21 (Model of the core theory). [DESIGN]
A **model of the core of** $\mathrm{TT}_{\mathrm{KH}}$ consists of:

1. a split category with families $\mathcal{E}_m \to \mathcal{B}_m$ for every mode m, with cumulative universes, dependent products and sums, identity types, finite inductive types, and extension types;
2. for every $\mu : n \to m$, a substitution-preserving reindexing functor on contexts and a modal operation interpreting $\langle \mu \mid - \rangle$;
3. coherent interpretations of two-cells and specified adjunctions in $\mathcal{M}$;
4. a strict interval object whose Segal and Rezk objects interpret directed types;
5. symmetric monoidal stable fibres for the linear modes and adjoint functors interpreting the six operations;
6. for every trace-admissible A and $F : A \to A$, a cyclic coend interpreting $\mathrm{Tr}_A(F)$, stable under substitution and equipped with the stated dependent universal property;
7. univalent universes at the levels at which univalence is asserted.

A morphism of models preserves this structure up to the coherences named in the signature.

One may formulate the same definition as a comprehension object internal to a two-category of indexed Rezk categories. The split presentation is used because inference rules require literal substitution. Different splittings should yield equivalent internal languages; proving that invariance belongs to the coherence theorem, not to the definition.

Theorem 6.22 (Soundness for a specified model). [THEOREM]
Every derivable judgement of the rules stated in this chapter is valid in every model in the sense of Definition 6.21.

Proof of Theorem 6.22.
Proceed by induction on derivations. The structural, dependent type, and extension-type cases are the defining operations of the category with families. The modal formation, introduction, elimination, and computation cases are interpreted by the corresponding modal adjunction and its substitution stability. Directed composition is interpreted by Segal filling. For Tr-F use the chosen cyclic coend. The point constructor is its universal dinatural cocone; the cyclic path is dinaturality; the dependent eliminator is the dependent universal property; and the two computation rules are its point and path equations. Two-cell coercions are interpreted by the model's coherent action of $\mathcal{M}$. Since each rule is preserved by substitution, the induction also proves substitution soundness.

□

The theorem is deliberately conditional: it proves that a structure satisfying the semantic clauses validates the syntax. It does not construct such a structure for the full KH signature.

Conjecture 6.23 (Chôra adequacy). [CONJECTURE]
There is a two-category $\mathbf{Chôr}^{\mathrm{dual}}$ *of suitably small or universe-indexed chôrai such that:*

1. *every chôra determines a model of* $\mathrm{TT}_{\mathrm{KH}}$*;*
2. *every model satisfying geometricity, six-functor, duality, trace-nondegeneracy, and reconstruction conditions is equivalent to the internal-language model of a chôra;*
3. *these assignments induce an equivalence after localising both sides at Morita equivalences;*
4. *the arithmetic chôra, if it exists, interprets every* `[F]` *statement in the ledger.*

This is the exact form of the proposed correspondence between models and chôrai. It is not a theorem. Even the target $(\infty,3)$-category required by the field theory has not been constructed. The likely proof would combine an internal-language theorem for indexed higher categories, a coherence theorem for correspondences, and a reconstruction theorem identifying the semantic six-functor object with its geometric chôra. The reconstruction step is at least as hard as KH.5, and the trace-nondegeneracy step has no general theorem.

The forward direction is more plausible than the reverse. Given a chôra, its fibre categories, adjunctions, and trace should provide the modal semantics. Conversely, a model may have the formal operations of geometry without enough points, algebra objects, or dualizable generators to be reconstructed as a chôra. Adequacy therefore includes explicit hypotheses rather than asserting that every arbitrary model is geometric.

6.11 The function-field control

Over a smooth projective curve $X/\mathbf{F}_q$, the mode theory has a simpler model. The global mode is the étale or constructible sheaf theory on X; the place modalities are restrictions to formal discs or geometric points; the arithmetic endomorphism is Frobenius; and the trace is the categorical trace of Frobenius. The geometry exists, and the relevant six-functor formalisms are established. The global Langlands theorem for GL_n is due to Drinfeld and Laurent Lafforgue, while Vincent Lafforgue's excursion operators provide the general reductive-group parameterisation (Lafforgue 2002; Lafforgue 2018).

The finite discrete control model. Let C be a finite category, let $F : C \to C$ be an endofunctor, and interpret $\mathrm{Tr}_C(F)$ as the ordinary coend

$$\int^{x \in C} \mathrm{Hom}_C(x, Fx).$$

Then the constructors and eliminator of Construction 6.12 are interpreted by the quotient of $\coprod_x \mathrm{Hom}_C(x, Fx)$ under the relation $(x, b \circ a) \sim (y, F(a) \circ b)$.

This finite model verifies the logical shape of the trace rules without any higher-categorical assumptions. It also shows why cyclicity must be a relation rather than a definitional rewrite: the quotient may identify representatives in both directions, but a normal form for representatives is not canonical.

The genuine function-field model replaces finite categories by stable categories of sheaves and the set-valued coend by a homotopy coend. The trace-of-Frobenius formalism then supplies the bridge to functions on rational points. What is established in the function-field case is the geometric and categorical machinery in substantial settings, not the full syntax proposed here. No published proof identifies a completed $\mathrm{TT}_{\mathrm{KH}}$ kernel with the geometric Langlands machinery.

The principal assertions of this chapter have the following function-field controls.

For directed hom-types, the control is the category of correspondences of schemes or stacks over X. Pullback composition is non-invertible, while equivalences of correspondences are invertible higher cells. The abstract correspondence formalism is established in higher category theory; its internal presentation by Riehl–Shulman types remains a change of language rather than a new geometric theorem.

For place modalities, a closed point $x \in X$ gives restriction from X to a formal disc, punctured disc, or geometric fibre. These functors and their adjoints occur in the established étale and constructible six-functor formalisms. Distinct closed points are disjoint, so their localisations commute in the expected sense. The number-field claim is harder because the restricted adelic product and the archimedean place have no equally complete geometric construction.

For proper base change, the function-field assertion is a theorem in the usual constructible étale setting. The typed derivation of Theorem 6.16 isolates its formal component: once the six-functor functor carries the pullback correspondence cell to an invertible cell, the term is forced. The geometric theorem is the verification that the cell is invertible under the standard hypotheses.

For trace, the control is Frobenius acting on constructible complexes and on categories of sheaves. Grothendieck's trace formula and Deligne's work on weights establish the cohomological fixed-point formula and its arithmetic consequences for varieties over finite fields. Categorical trace constructions for Bun_G have been developed by Gaitsgory, Kazhdan, Rozenblyum, Varshavsky, Yom Din, and collaborators. These results support the semantics of twisted trace; they do not supply the proposed kernel rules.

For the global correspondence, Drinfeld proved the rank-two function-field case, Laurent Lafforgue proved the correspondence for GL_n, and Vincent Lafforgue constructed excursion operators and global parameters for general reductive groups. The theorem there depends on the actual curve, its Frobenius, and shtuka geometry. The arithmetic theory must reproduce this structure after replacing X by $\mathfrak{X}_{\mathbf{Z}}$; it cannot cite the function-field theorem as a formal proof of the number-field statement.

6.11.1 Worked computation: $\mathbf{P}^1$ over $\mathbf{F}_2$

The trace formula can be checked without any unproved input for the projective line. For every $n \geq 1$,

$$\#\mathbf{P}^1(\mathbf{F}_{2^n}) = 2^n + 1.$$

Indeed, the affine chart has 2^n points and the point at infinity contributes one more. Hence the zeta function is

$$Z(\mathbf{P}^1/\mathbf{F}_2, t) = \exp\left(\sum_{n\geq 1} \frac{(2^n+1)t^n}{n}\right).$$

Using

$$\sum_{n\geq 1} \frac{t^n}{n} = -\log(1-t)$$

and

$$\sum_{n\geq 1} \frac{2^n t^n}{n} = -\log(1-2t),$$

we obtain

$$Z(\mathbf{P}^1/\mathbf{F}_2, t) = \frac{1}{(1-t)(1-2t)}.$$

The étale cohomology groups over $\overline{\mathbf{F}}_2$ are

$$H^0_{\text{ét}}(\mathbf{P}^1, \mathbf{Q}_\ell) \cong \mathbf{Q}_\ell, \qquad H^1_{\text{ét}}(\mathbf{P}^1, \mathbf{Q}_\ell) = 0,$$

$$H^2_{\text{ét}}(\mathbf{P}^1, \mathbf{Q}_\ell) \cong \mathbf{Q}_\ell(-1), \qquad \ell \neq 2.$$

Geometric Frobenius has trace 1 on H^0 and 2^n on H^2 after the n-th iterate. Therefore the alternating trace is

$$\sum_{i=0}^{2} (-1)^i \operatorname{tr}\left(\mathrm{Frob}_2^n \mid H^i_{\text{ét}}\right) = 1 + 2^n,$$

which agrees with the point count.

The determinant form gives the same polynomial:

$$Z(\mathbf{P}^1/\mathbf{F}_2, t) = \prod_{i=0}^{2} \det\left(1 - t\,\mathrm{Frob}_2 \mid H^i_{\text{ét}}\right)^{(-1)^{i+1}} = \frac{1}{(1-t)(1-2t)}.$$

The degree-zero factor records the fixed connected component; the degree-two factor records the Tate class. There is no degree-one factor.

At the level of the trace rules, the endofunctor is Frobenius and the scalar shadow is the alternating supertrace on cohomology. Fixed-point localisation maps the global trace to the sum of local contributions of the $2^n + 1$ rational points, each equal to 1 because the fixed points are transverse. This is a complete function-field instance of the passage

$$\text{categorical trace} \longrightarrow \text{cohomological trace} \longrightarrow \text{fixed-point count}.$$

The analogous number-field statement is not obtained by replacing 2 with a prime in this formula. It requires the global arithmetic base, cyclotomic localisation, and the archimedean contribution.

Small characteristic introduces no anomaly in this example. At $p = 2$ the condition $\ell \neq p$ merely requires an auxiliary prime such as $\ell = 3$. For $\mathbf{P}^1/\mathbf{F}_3$ the same calculation gives

$$\#\mathbf{P}^1(\mathbf{F}_{3^n}) = 3^n + 1, \qquad Z(\mathbf{P}^1/\mathbf{F}_3, t) = \frac{1}{(1-t)(1-3t)}.$$

Thus the trace grammar is uniform at the two smallest primes even though specific integral or prismatic models may have additional low-prime complications.

6.11.2 Worked computation: an elliptic curve over $\mathbf{F}_2$

Consider

$$E: \quad y^2 + y = x^3 + x$$

over $\mathbf{F}_2$. For $x = 0$, the right-hand side is 0, and both $y = 0$ and $y = 1$ solve $y^2 + y = 0$. For $x = 1$,

$$x^3 + x = 1 + 1 = 0,$$

so again there are two values of y. Adding the point at infinity gives

$$\#E(\mathbf{F}_2) = 2 + 2 + 1 = 5.$$

The Frobenius trace is therefore

$$a_2 = 2 + 1 - \#E(\mathbf{F}_2) = -2.$$

The local characteristic polynomial on H^1 is

$$P_2(T) = T^2 - a_2 T + 2 = T^2 + 2T + 2,$$

and the numerator of the zeta function is

$$1 - a_2 t + 2t^2 = 1 + 2t + 2t^2.$$

The count over $\mathbf{F}_4$ can be checked directly. Write

$$\mathbf{F}_4 = \{0, 1, \omega, \omega^2\}, \qquad \omega^2 + \omega + 1 = 0.$$

Then $\omega^2 = \omega + 1$ and $\omega^3 = 1$. The values of $x^3 + x$ are

x	0	1	ω	ω^2
$x^3 + x$	0	0	ω^2	ω.

The values of $y^2 + y$ are

y	0	1	ω	ω^2
$y^2 + y$	0	0	1	$1.$

Thus $x = 0$ and $x = 1$ each contribute two affine points, while $x = \omega$ and $x = \omega^2$ contribute none. Including infinity,

$$\#E(\mathbf{F}_4) = 5.$$

Let α, β be the roots of $P_2(T)$. They satisfy

$$\alpha + \beta = -2, \qquad \alpha\beta = 2.$$

Hence

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 4 - 4 = 0,$$

and the cohomological trace formula predicts

$$\#E(\mathbf{F}_4) = 2^2 + 1 - (\alpha^2 + \beta^2) = 5,$$

agreeing with the enumeration.

For the third iterate,

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (-2)^3 - 3 \cdot 2 \cdot (-2) = 4.$$

Therefore

$$\#E(\mathbf{F}_8) = 2^3 + 1 - 4 = 5.$$

The first three point counts are consequently $5, 5, 5$, although the field sizes are $2, 4, 8$. The equality is not periodic behaviour of the curve; it results from the particular Frobenius eigenvalues and does not persist for all n.

The zeta function is

$$Z(E/\mathbf{F}_2, t) = \frac{1 + 2t + 2t^2}{(1 - t)(1 - 2t)}.$$

Its numerator satisfies the genus-one functional equation. Indeed,

$$2t^2 \left(1 + 2(2t)^{-1} + 2(2t)^{-2}\right) = 1 + 2t + 2t^2.$$

The roots of $P_2(T)$ have complex absolute value $\sqrt{2}$, since

$$T^2 + 2T + 2 = (T + 1)^2 + 1$$

has roots $-1 \pm i$. This is the purity input in its smallest explicit form.

In the type-theoretic interpretation, the directed endomorphism is geometric Frobenius, the trace object retains the action on the whole cohomology complex, and the scalar shadow gives

$$1 - \operatorname{tr}(\mathrm{Frob}_2^n \mid H^1) + 2^n.$$

The calculation shows what "weights replace the heart" means in a case that can be checked by hand: only the graded cohomology groups, Frobenius traces, and purity of the two degree-one eigenvalues enter the zeta function.

The arithmetic case differs in three decisive respects. The global base $\mathfrak{X}_{\mathbf{Z}}$ is conjectural; the archimedean modality has no completed six-functor model; and cyclotomic trace introduces an interaction among all primes and the S^1-direction. The control case therefore tests the directed and trace fragments, but it does not settle the adelic mode theory.

6.12 Metatheory: what is and is not claimed

Four metatheoretic questions must be separated: canonicity, normalisation, decidability of type-checking, and consistency. They are often conflated because, for a small intensional type theory, one proof may establish several at once. The present calculus combines directed extension types, modal locks, adjoint mode change, univalence, and a higher-inductive trace quotient. No such combined proof exists.

Canonicity asks whether a closed term of a data type reduces to a canonical constructor. The relevant form here is homotopy canonicity for ordinary inductive types and a separate statement about modal normal forms. Trace types should not be expected to have unique constructor representatives, because they are quotients by cyclic paths. A sensible canonicity theorem would concern closed natural numbers and booleans, not canonical representatives of traces.

Normalisation asks whether every term admits a normal form and whether definitional equality is detected by comparison of normal forms. The directed and modal fragments each have proof-theoretic precedents, but their combination with the trace rules has not been analysed. In particular, the path computation rule and the stability conditions on modal elimination must be incorporated into a gluing or normalisation-by-evaluation model.

Conjecture 6.24 (Normalisation for the computational fragment). [CONJECTURE]
Let $\mathrm{TT}_{\mathrm{KH}}^{\mathrm{comp}}$ *omit resizing, chôra-level univalence, unrestricted higher inductive types, and infinite adelic tensors, while retaining a finite decidable mode theory, directed extension types, and the trace type with propositional cyclicity. Then every well-typed term has a normal form, substitution preserves normalisation, and judgemental equality is decidable.*

No proof is claimed. A proof would require at least the following. First, one needs a category of normal forms stable under modal locks and two-cell coercions. Secondly, a logical-relations or gluing model must cover extension types and Segal operations. Thirdly, the trace type must be handled as a higher-inductive quotient whose point computation is strict and whose path computation is propositional. Fourthly, critical pairs among modal beta-reduction, substitution, and two-cell coercion must be shown confluent. Finally, universe cumulativity must be proved compatible with the normalisation construction. Making cyclicity judgemental would add the bidirectional rewrite $\mathrm{tr}(fg) \leftrightarrow \mathrm{tr}(gf)$ and would destroy the evident termination measure; this is why Construction 6.12 does not adopt that design.

Criterion 6.25 (Conditional decidability of type-checking). [CONJECTURE]
Type-checking for $\mathrm{TT}_{\mathrm{KH}}^{\mathrm{comp}}$ *is decidable if Conjecture 6.24 holds, the mode two-category has decidable equality of generating cells and a terminating coherence procedure, and all side conditions on modal elimination are represented by decidable typing data. Type-checking for the full theory is not claimed decidable.*

The status is conjectural because its principal premise is conjectural. Elaboration is a further problem: even if checking an explicit term is decidable, inferring omitted modes, universe levels, and coherence cells may be undecidable or practically intractable. The kernel should check fully elaborated terms; automation may search for the omitted data without entering the trust base.

Theorem 6.26 (Model-relative consistency). [THEOREM]
If the rules of $\mathrm{TT}_{\mathrm{KH}}$ *have a model in the sense of Definition 6.21 in which the empty type is interpreted by an object with no global section, then there is no closed term of the empty type in the syntax.*

Proof of Theorem 6.26.
Assume a closed term $t : \mathbf{0}$. By Theorem 6.22, its interpretation is a global section of the interpretation of $\mathbf{0}$. This contradicts the hypothesis that the latter has no global section. Therefore no such term exists.

□

This theorem is the ordinary semantic consistency argument. It does not produce the model.

Conjecture 6.27 (Relative consistency of the full KH theory). [CONJECTURE]
Assume ZFC together with a finite tower of strongly inaccessible cardinals sufficient to form the required universes of presentable stable $(\infty,1)$*-categories and indexed higher categories. Assume additionally the existence of a chôra satisfying KH.1–KH.28 and of cyclic coends stable under its modal reindexing. Then* $\mathrm{TT}_{\mathrm{KH}}$ *has a model and is consistent relative to those assumptions.*

The qualification "assume additionally" is essential. This chapter does not claim consistency relative to a bare inaccessible cardinal. Large cardinals manage size; they do not construct the arithmetic chôra, prove trace-nondegeneracy, or supply the archimedean six operations. Removing the additional existence assumption is precisely part of Conjecture 6.23.

Canonicity is similarly open for the combined theory. Homotopy canonicity is known in important univalent settings, including inverse-diagram models, but those results do not automatically cover directed extension types and the trace quotient (Shulman 2015). The only claim made here is the conditional one: a successful normalisation proof for the stated computational fragment should imply canonicity for its ordinary inductive data types. No theorem number is attached because the implication has not been formalised for this syntax.

There are several possible strengths of canonicity, and the distinction should be retained in any future result. **Numeral canonicity** says that every closed natural-number term is propositionally equal to a numeral. **Definitional canonicity** says that it reduces to a numeral by kernel conversion. **Homotopy canonicity** allows the equality to be produced in a semantic or propositional sense even when conversion is not complete. For a theory with univalence and higher constructors, the first or third may be attainable when the second is not.

Modal canonicity asks a different question. A closed term of $\langle \mu \mid A \rangle$ should be represented by modal introduction only when μ has the appropriate positive character and the term contains no neutral modal eliminations. Right-adjoint modalities may have canonical forms governed by their counits rather than by a simple constructor theorem. The $\flat/\sharp$ adjunction therefore requires a polarity-sensitive statement, not one uniform claim for all modalities.

Trace types are intentionally non-canonical at the level of representatives. The cycles $(x, b \circ a)$ and $(y, F(a) \circ b)$ are distinct point constructors joined by a path. A theorem demanding a unique representative would contradict the intended coend semantics. The appropriate result is elimination canonicity: every map out of a trace is determined by a dinatural family. That property is already part of the inference rules, but its computational realisation still depends on a coherent treatment of the path constructor.

Consequently a future metatheory should be modular. It should prove ordinary data canonicity for the dependent core, modal normal forms for a classified family of mode

morphisms, and a universal-property theorem for traces. Combining these into one slogan would conceal which part fails if the trace quotient or an adjoint modality resists normalisation.

> **Status and Caveat.** The metatheoretic weakness is not peripheral. $\mathrm{TT}_{\mathrm{KH}}$ has no implementation; the combined calculus has no proved normalisation, canonicity, or decidable type-checking theorem; and the intended chôra model is conjectural. Nothing in this chapter is machine-checked. The inference rules are a mathematical specification of a research language, not evidence that a kernel accepting them is sound. Any later chapter using `[F]` means "formalisation target", never "verified result".

6.13 The bridge to Lean 4 and mathlib

Lean 4 provides an intensional dependent type theory, an elaborator, metaprogramming facilities, and a small kernel (Moura and Ullrich 2021). Mathlib supplies a broad library of ordinary category theory, algebra, topology, analysis, number theory, sheaves, sites, and substantial algebraic geometry (Community 2020). It is therefore the practical host for the ledger and for large parts of the one-categorical and arithmetic library. It is not an implementation of $\mathrm{TT}_{\mathrm{KH}}$.

The fragment expressible today without changing Lean's kernel includes the following. A finite mode category can be represented as data. Modalities can be represented as monads, comonads, reflective subcategories, or adjoint functors. Ordinary categories, functors, natural transformations, adjunctions, monoidal categories, dualizable objects, and one-categorical traces can be defined. The syntax and derivation relation of a finitary fragment can be encoded inductively, and a verified checker for that encoded syntax can be written. Finite dependency graphs, status labels, hashes, and certificate checkers are well within Lean's present scope. Parts of condensed mathematics and the Liquid Tensor Experiment demonstrate that very large categorical proofs can be organised in Lean (Commelin and Scholze 2023; Buzzard et al. 2020).

The missing fragment is precisely the one on which the intended semantics depends. Lean 4 does not natively provide univalence, higher inductive types, Riehl–Shulman extension types, Rezk types as a built-in language of $(\infty,1)$-categories, a general $(\infty,2)$-category of correspondences, or a chôra-level universe. These may be encoded by explicit simplicial or categorical models, but then coherence is proved rather than supplied by the type theory. Such an encoding is valuable and may be the first implementation, but it is not the same as making the directed syntax native.

There are three viable implementation strategies. The first is an **external object theory**: define the syntax of $\mathrm{TT}_{\mathrm{KH}}^{\mathrm{comp}}$ as an inductive datatype in Lean, write an elaborator outside the kernel, and prove in Lean that the checker is sound for an explicit model. The second is a **semantic encoding**: represent directed categories by simplicial objects or complete Segal objects in mathlib, and formulate modalities and trace by ordinary category theory. The third is a **new kernel or frontend** inspired by cubical and simplicial type theories, with certificates translated to a smaller core. Only the first two can presently claim to use the stock Lean kernel without extension.

Principle 6.28 (The Lean trust boundary). [DESIGN]

In a Lean 4 development, a theorem is verified only relative to the Lean kernel and every axiom appearing in its transitive dependency list. If univalence, a trace quotient, chôra adequacy, six-functor coherence, or an archimedean comparison is introduced by `axiom`*, then any theorem using it is not a verification of that mathematical assertion. It is a checked derivation conditional on the axiom.*

This point must be stated without euphemism: anything axiomatised is not verified. Kernel checking guarantees that a conclusion follows from the declared assumptions; it does not establish the assumptions. An axiom audit is therefore part of every `[F]` artifact.

The corresponding algorithmic and verification flow is summarised in Plate 6.3.

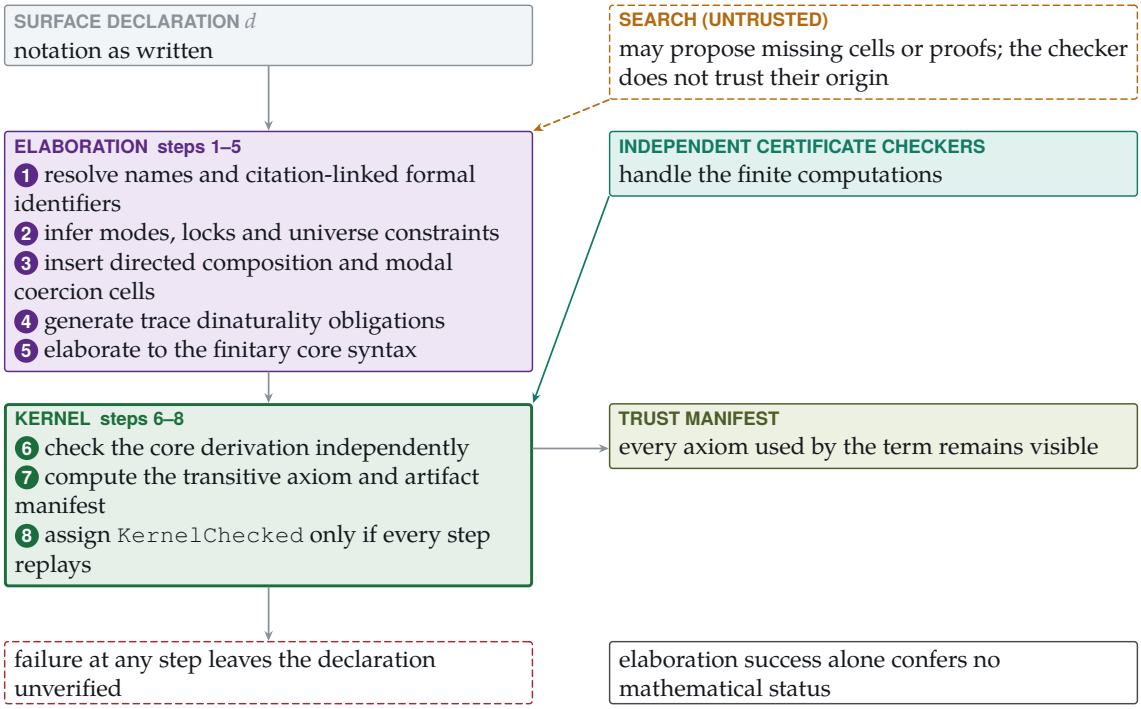


Plate 6.3. Elaboration translates surface notation into explicit modes, locks, universe levels, directed compositions, and coherence cells. A small checker then validates the elaborated term, while independent certificate checkers handle finite computations. Every axiom used by the term remains visible in the trust manifest; elaboration success alone confers no mathematical status.

What it does not show: a working implementation: the pipeline is proposed, and the formal fragment it would check is a statement skeleton whose proofs are not supplied.

Elaboration and kernel checking. The proposed elaboration pipeline is:

```
input: surface declaration d
1. resolve names and citation-linked formal identifiers
2. infer modes, locks, and universe constraints
3. insert directed composition and modal coercion cells
4. generate trace dinaturality obligations
5. elaborate to the finitary core syntax
6. check the core derivation independently
7. compute the transitive axiom and artifact manifest
8. assign KernelChecked only if every step replays
```

Failure at any step leaves the declaration unverified. Search procedures may propose missing cells or proofs, but the checker does not trust their origin.

6.14 Formalisation ledger for the [F] statements

The statement ledger contains nine statements carrying [F]: GC.7 and CV.1–CV.8. The twelve falsification tests are also computational targets, but they are protocols rather than ledger theorems. The table records the difficulty of formalising the nine statements, not the difficulty of proving their mathematical content.

Ledger item	Required formal fragment	Difficulty	Present status
GC.7, analytic EGA	Analytic stacks, stable categories, six operations, proper base change, Verdier duality	Extreme	Not expressible end to end in mathlib; major components absent
CV.1, kernel sovereignty	Kernel replay, axiom audit, immutable statement identity	Moderate	Implementable now as infrastructure
CV.2, provenance closure	Finite DAGs, well-founded induction, typed statuses	Low	The abstract theorem is routine; no treatise ledger implementation exists
CV.3, artifact identity and replay	Canonical serialisation, hashes, deterministic build semantics	Moderate	Engineering target; collision assumptions external
CV.4, enclosure soundness	Exact semantics of interval operations and outward rounding	Moderate	Feasible for chosen numerical kernels
CV.5, certified algebraic kernels	Hecke, Satake, orbital-integral specifications and certificates	High to extreme	Feasible only in bounded, explicitly specified cases
CV.6, fair search and counterexamples	Enumerated tactic system, witness checking, statistical protocol hypotheses	High	Logical core feasible; arithmetic and statistical assumptions remain domain-specific
CV.7, nonauthority and no approximation erasure	Status type, absence of coercions, semantic review records	Moderate	Enforceable by interface design, not yet built
CV.8, $\mathrm{TT}_{\mathrm{KH}}$ adequacy	Full syntax, chôra semantics, higher correspondences, trace modality	Unknown	Conjectural; no implementation

The table prevents a recurrent ambiguity. CV.2 is mathematically simple once the graph is finite, while GC.7 and CV.8 require mathematics not currently available. All carry [F] because each has a formal target, not because the targets are equally near.

The falsification tests divide into three formal strata. Finite exact tests, such as checking a bounded point count or a matrix identity, can produce certificates now. Analytic tests, such as a regularised determinant giving a gamma factor, require verified analytic estimates and normalisation data. Geometric tests, such as constructing the archimedean chôra, cannot be reduced to computation until the objects themselves are defined. A test protocol may be fully formalised while the conjecture it tests remains open.

6.14.1 A staged implementation without false certification

The first implementation should not begin by adding axioms until the examples compile. It should begin with an external syntax and a finite mode theory. The walking arrow, walking adjunction, walking pullback square, and finite coend model are sufficient to test variable locking, two-cell transport, cyclicity, and the base-change derivation. These examples have set-valued or ordinary categorical semantics and therefore do not require an $(\infty, 3)$-category.

The next stage should add explicit simplicial models. A Segal object can be represented as a simplicial object satisfying horn conditions stated in ordinary Lean mathematics. Rezk

completeness can be a predicate. Extension types can be interpreted as limits of diagrams. This loses the concision of native simplicial type theory, but it creates a semantic laboratory in which the directed claims can be checked by the stock kernel. The price is that every coherence theorem must be proved for the explicit model.

A third stage should formalise stable shadows and finite trace coends. The target is not immediately THH of arbitrary presentable $(\infty, 1)$-categories. A realistic target is a dg category with finitely many compact generators, an explicitly bounded cyclic bar construction, and a theorem identifying the checked complex with the desired trace under stated finiteness hypotheses. This stage would support genuine computations while preserving the distinction between a truncated bar complex and the full trace.

Only after those stages should one formalise the abstract six-functor interface. The interface can be meaningful before every geometric instance exists: a correspondence category, a lax monoidal functor, adjunction records, and images of composition cells can be defined in a restricted two-categorical setting. The base-change derivation in Theorem 6.16 then becomes an early regression test. The theorem should fail to elaborate when the square is not marked cartesian or when $f_!$ is unavailable.

The final stage, chôra adequacy, cannot be scheduled as engineering. It requires new mathematics. A repository may contain a declaration `axiom choraAdequacy`; doing so is useful for exploring downstream statement shapes, but every dependent theorem must display that axiom in its trust manifest. Removing the declaration and replacing it with a proof is the milestone. The number of downstream declarations that compile under the axiom is not evidence for the axiom.

This staged plan gives `[F]` a disciplined interpretation. At an early stage, an `[F]` theorem may have only a faithful formal statement. Later it may have a proof conditional on named axioms. Later still some axioms may be discharged by semantic constructions or imported established theorems. The status should record all three states rather than compressing them into a single badge.

6.14.2 What a complete formal artifact must contain

A printed theorem and a kernel declaration are not by themselves a complete formal artifact for this project. The declaration must be accompanied by its semantic profile. At minimum that profile records the mode theory revision, universe levels, normalisation conventions, imported axioms, external theorems, finite computation certificates, and the exact prose statement to which the declaration is linked. These fields prevent a proof of a nearby statement from being presented as a proof of the intended one.

Consider the base-change theorem. Its surface statement suppresses the direction conventions of the four mode morphisms, the proof that the square is cartesian, the supportedness witness for f, the universe containing A, and the chosen Beck–Chevalley cell. The elaborated declaration must include all of them. A possible core telescope is schematically:

```
(X Y X' Y' : Figure i)
(f : Hom X Y) (g : Hom Y' Y)
(f' : Hom X' Y') (g' : Hom X' X)
(sq : PullbackSquare f g f' g')
(supp : Supported f)
(A : Fibre X j)
```

The conclusion then names the two modal composites and the equivalence constructed from the cell attached to `sq`. If `supp` is removed, the term $f_!A$ is unformed. If `sq` is weakened

to a merely commutative square, the comparison map may still be formed, but the inverse field of the equivalence cannot be filled.

The semantic profile must also state which theorem turns the syntactic `PullbackSquare` into an invertible correspondence cell. In a finite category of sets, that theorem is elementary. In an $(\infty, 2)$-category of correspondences, it includes coherence for homotopy pullbacks. In an analytic six-functor formalism, it further includes the theorem that the functor $\mathcal{D}$ carries the cell to the required natural equivalence. The same surface derivation may therefore have different dependency depths in different models.

For trace cyclicity, the artifact records the trace constructor revision. If a later kernel changes cyclicity from a path constructor to judgemental equality, existing proofs may still elaborate but their metatheoretic meaning has changed. Content addressing must include the rule set, not only the final term. The artifact also records the shadow map used to pass from the object-valued trace to the scalar trace. Two shadow maps can agree on decategorification and differ on higher homotopy; a numerical equality alone does not identify them.

Normalisation conventions are especially important in Langlands theory. The local reciprocity map may send a uniformiser to geometric Frobenius or to arithmetic Frobenius. The Satake isomorphism may be normalised or unnormalised. Haar measures may give a hyperspecial compact subgroup volume one or use a motivic normalisation. Tate twists may be homological or cohomological. Each choice changes the types of compatibility maps. A formal artifact should represent the convention as a parameter or a structure, never as an explanatory comment.

External theorems require two links. The first is bibliographic and identifies the source theorem. The second is semantic and states the formal proposition being imported. A reference to an article does not make every plausible formalisation of its main theorem valid. Until the source theorem itself is formalised, the imported proposition remains an external reviewed assumption with a human entailment record. Kernel checking downstream proves only a conditional statement.

Finite computations require a different interface. A certificate contains the input specification, output, checker version, and proof that the checker's acceptance implies the abstract proposition needed by the theorem. For a matrix trace, the checker verifies entries and arithmetic. A refinement lemma connects the matrix representation to a linear endomorphism. A second lemma connects linear trace to the categorical shadow. Omitting either lemma leaves a gap even when the arithmetic is exact.

Approximate computations add an error model. Suppose a complex number is represented by a ball $B(c, r)$. A certificate that the exact value lies in the ball is not a certificate that it equals c. To prove nonvanishing the ball must exclude zero. To prove equality of two exact values, their enclosures must be combined with a theorem forcing equality, or refined until an exact algebraic certificate is available. The type system should make the absence of a coercion from `Ball` to `Complex` visible.

The dependency graph for a release theorem is then a mathematical object. Its leaves fall into three classes: kernel primitives and declared axioms, certified finite checkers, and external reviewed sources. Internal nodes are proof terms or verified transformations. A replay traverses the graph in topological order. If a leaf changes status, every dependent release is recomputed. This is KH.28 expressed operationally.

There is no requirement that every theorem be formalised before it can be discussed in the book. There is a requirement that the status be visible. A theorem from the literature may remain `[E]` with a precise citation and no Lean term. A conjecture may have a formal statement and no proof. A computation may be certified only within a stated bound. What is

prohibited is a status transition caused by the fluency of an explanation or the size of a test suite.

The prose-to-formal link also has to be checked. A useful procedure is dual control. One reviewer formalises the prose statement. A second reviewer, starting from the formal term rather than the prose, writes a back-translation including every hypothesis and convention. The two texts are compared. Disagreement does not show that the theorem is false; it shows that the formalisation is not yet an adequate representation of the intended claim.

This procedure is particularly necessary for statements involving "canonical" maps. In higher category theory a map may be canonical only up to a contractible choice. The formal statement should expose the contractibility or work in a universe where the choice is identified by univalence. Selecting one representative and hiding the selection may make later naturality impossible to state. The trace and base-change constructions in this chapter were designed so that their choices arise from universal properties rather than arbitrary constants.

A complete artifact should finally include negative information: failed elaboration attempts, counterexamples, bounds searched, and axioms that were considered but rejected. This information does not enter the proof term, but it prevents repeated exploration of known dead ends and records the scope of computational evidence. If a conjecture was tested only for split groups of rank at most two, the ledger must not describe it as tested for reductive groups.

The formal language thus has two outputs. Its first output is a proof term or an explicit conjecture marker. Its second is a trust manifest explaining why that term has the status displayed in the book. The kernel governs the first; the evidence architecture governs the second. Neither can replace the other.

Formal difficulty is not epistemic status. The difficulty of formalising a statement and its mathematical status are independent. An established theorem may be extremely difficult to encode; a conjecture may be easy to state; and an incorrect assertion may typecheck perfectly. `[F]` records the existence of a precise verification target, not truth, importance, or proximity to completion.

6.15 Design alternatives and failure modes

The theory could have been designed differently. Three alternatives deserve an explicit rejection.

The first is to use ordinary HoTT and encode every directed category as a type together with a hom-family. This is logically possible and may be the best route inside Lean. It does not give synthetic higher coherence for free: composition, associativity, units, naturality, and every higher compatibility must be carried as structure. For a small category this cost is acceptable. For the iterated correspondence categories of a chôra it is the very coherence burden the language is meant to control.

The second is to make all six operations endomodal operators on one universe. That presentation suppresses their changing bases. The expression $f_!A$ does not live where A lives, and a base-change formula is well typed only because four base modes fit into a square. A single-mode syntax can recover this with indexed types, but then the indexing is a mode theory under another name.

The third is to postulate a scalar trace constant with cyclicity as a rewrite rule. It would be adequate for finite-dimensional linear algebra and inadequate for the categorical trace used by the treatise. The trace of Frobenius on a sheaf category is an object with endomorphisms and local terms, not merely a number. Moreover a cyclic rewrite has no preferred orientation. The coend presentation retains the object and records cyclicity by a path.

There are also failure modes internal to the adopted design. The mode theory may be too large to admit decidable coherence. Chôra-level univalence may be inconsistent with universe stratification unless stated only for a suitably small universe. The trace coend may fail to be stable under an analytic pushforward. Adelic completion may not be a modality on any useful subuniverse. The directed interval and the cubical interval used for univalence may interact badly. Each is a concrete proof obligation.

Problem 6.29 (Coherence for arithmetic modes). [CONJECTURE]
Construct a finite presentation of the mode two-category needed for $\mathrm{TT}^{\mathrm{comp}}_{\mathrm{KH}}$ and prove a coherence theorem deciding equality of generated two-cells, while retaining the Beck–Chevalley, projection-formula, place-commutation, Frobenius, and $\flat \dashv \sharp$ structures used in this chapter.

A positive solution would not construct the semantic functors, but it would make their formal composites manageable. A negative solution would force a proof-relevant representation of coherence cells in the kernel and would weaken the decidability prospect.

Problem 6.30 (Modal stability of trace). [CONJECTURE]
Determine hypotheses on a modal functor $\langle \mu \mid - \rangle$ under which the canonical comparison

$$\langle \mu \mid \mathrm{Tr}_A(F) \rangle \longrightarrow \mathrm{Tr}_{\langle \mu | A \rangle}(\langle \mu \mid F \rangle)$$

is an equivalence. Establish the result for pullback, proper pushforward, and place localisation in the function-field control case before asserting it for $\mathfrak{X}_{\mathbf{Z}}$.

This problem is the type-theoretic form of commuting trace with base change. Dualizability and preservation of the relevant coend are natural sufficient conditions, but the analytic categories of the arithmetic theory may not satisfy them without nuclearity or compactness.

6.16 Comparison with neighbouring type theories

The proposed calculus should be located precisely among existing systems. It is not a replacement for Martin-Löf type theory, homotopy type theory, cubical type theory, simplicial type theory, cohesive type theory, or multimodal type theory. It uses each for a different layer and adds one operation, trace, whose intended semantics is categorical rather than merely logical.

Ordinary intensional Martin-Löf type theory supplies dependent products, dependent sums, identity types, universes, and inductive families. Those constructors are enough to encode categories as records and to represent a finite mode theory. They do not make equivalence invariant by construction. The univalent layer repairs that deficiency: equivalences can be transported as identities in a universe. For the treatise this matters because a chôra is defined only up to equivalence and because replacing a presentation by a Morita-equivalent one must not alter a theorem.

Cubical type theory offers computational univalence through interval and composition operations. Its interval is path-like: reversing a path is available. The Riehl–Shulman directed interval has a different purpose and must not be identified with the cubical interval. A combined implementation therefore needs two interval structures or a more general shape theory. One controls equality and univalence; the other controls directed morphisms. Confusing them would either make every directed arrow invertible or deprive identity types of symmetry.

Simplicial type theory contributes extension types and shape variables. Its Segal conditions turn horn filling into composition, and its Rezk condition relates equivalences to identities without groupoidifying every arrow. This is the closest existing language to the directed core

used here. The arithmetic additions are the explicit two-category of modes, the six-functor signature, and the trace coend. None follows merely from declaring that a type is Segal.

Cohesive type theory contributes the chain of adjoints separating shape, discrete, cohesive, and codiscrete information. The pair $\flat \dashv \sharp$ used here is a portion of that chain. Arithmetic localisation is analogous to cohesion only formally. Localising at p is not the same semantic operation as taking a discrete reflection, and the theory therefore gives them distinct generators. Relations between them, such as a comparison of global points after p-adic completion, must be represented by two-cells proved in a model.

Adjoint logic and multimodal type theory contribute the lock discipline. Their central insight is that a modality acts on contexts, not only on closed types. The arithmetic calculus adopts that insight literally. It differs in the size and geometry of its mode theory: a place for every prime, an archimedean mode, fibre modes for figures, and shadow modes for traces. Because an infinite mode signature threatens decidability, the computational fragment works with finite generated subtheories and treats passage to all places as a separate limit.

Linear and bunched type theories may also be relevant because trace requires dualizability and tensor products. The present chapter does not introduce a linear usage discipline for variables. Instead, linearity resides in the semantic fibre $\mathrm{Pr}^{L}_{\mathrm{st}}$ and in the symmetric monoidal structure of trace-admissible modes. A future kernel may profit from making dual resources linear in syntax, but that refinement is not required for the inference rules stated here.

The trace modality is closest to a higher-inductive coend. It differs from a modal necessity operator: it is indexed by a directed type and an endofunctor, and its eliminator asks for dinaturality rather than modal stability. Calling it a modality emphasises that it changes from the geometric mode to the shadow mode and forgets the chosen base point of a cycle. Its algebra is the algebra of shadows and traces, not the idempotence algebra of a reflective subuniverse.

This distinction explains why Tr is not assumed idempotent. In general $\mathrm{Tr}_{\mathrm{Tr}_A(F)}(\mathrm{id})$ is a new trace, not canonically the original one. Nor does Tr preserve all dependent products. What one expects are Morita invariance, cyclicity, additivity under stable cofibre sequences, multiplicativity under tensor products, and compatibility with selected modalities. Each property has its own semantic hypotheses.

There is also a distinction between univalence and directed completeness. Univalence says that equivalent objects in a universe can be identified. Rezk completeness says that equivalences inside a directed type correspond to identities of objects. Chôra-level univalence applies one universe higher: equivalent chôrai should be identified. None of these principles says that an arbitrary correspondence is invertible. The calculus uses three different notions because the geometry requires all three.

The resulting architecture is conservative in one sense and ambitious in another. It is conservative because no known theorem of ordinary type theory is restated as an arithmetic novelty; the components are attributed to their sources. It is ambitious because no existing proof assistant combines them with the required semantics, and because the trace and chôra adequacy claims are new. The novelty lies in the interface and in the mathematical objects that the interface is meant to govern.

6.16.1 A finite linear control calculation

Finite-dimensional vector spaces provide a second control model, intermediate between the finite-category coend and the desired stable higher categories. Let k be a field, $V = k^2$, and

$$f = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \qquad g = \begin{pmatrix} 4 & 0 \\ 5 & -1 \end{pmatrix}.$$

Then

$$gf = \begin{pmatrix} 4 & 8 \\ 5 & 7 \end{pmatrix}, \qquad fg = \begin{pmatrix} 14 & -2 \\ 15 & -3 \end{pmatrix}.$$

Both traces are 11. In the formal system this equality is not justified by the arithmetic calculation alone. The calculation certifies the endpoints, while Theorem 6.14 supplies the general identity term. A reflection theorem may then identify the matrix trace computed by a finite certificate with the scalar categorical trace of the corresponding linear endomorphism.

Now take $W = k$,

$$a : W \to V, \quad a(t) = (t, 0), \qquad b : V \to W, \quad b(x, y) = x + 2y.$$

The composite $ba : W \to W$ is multiplication by 1, so its trace is 1. The composite $ab : V \to V$ has matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

and also has trace 1. This is the two-object form of the cyclic derivation: a and b are not inverse, and V and W do not even have the same dimension, yet $\mathrm{tr}_W(ba) = \mathrm{tr}_V(ab)$. The trace path constructor is designed to express exactly this phenomenon.

The calculation also clarifies what a certificate proves. A matrix checker can verify multiplication and diagonal sums. It does not prove that the abstract trace on all dualizable objects agrees with matrix trace. That agreement is a separate refinement theorem. Without it, the certificate is about arrays over k, not about the semantic trace former.

6.16.2 Worked computation: supertrace on a two-term complex

Trace additivity can be seen in a two-term cochain complex over $\mathbf{Q}$. Let

$$C^0 = \mathbf{Q}^2, \qquad C^1 = \mathbf{Q}^2,$$

with bases e_1, e_2 and u_1, u_2, and differential

$$d = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad d(e_1) = u_1, \quad d(e_2) = 0.$$

Define a cochain endomorphism F by

$$F^0 = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad F^1 = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

It is a cochain map because

$$dF^0 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} = F^1 d.$$

The chain-level supertrace is

$$\mathrm{str}(F \mid C^\bullet) = \mathrm{tr}(F^0) - \mathrm{tr}(F^1) = (2 + 3) - (2 + 5) = -2.$$

Now compute cohomology. Since

$$\ker d = \langle e_2 \rangle,$$

we have

$$H^0(C^\bullet) \cong \mathbf{Q}\langle [e_2] \rangle.$$

The image of d is $\langle u_1 \rangle$, so

$$H^1(C^\bullet) = C^1 / \operatorname{im} d \cong \mathbf{Q}\langle [u_2] \rangle.$$

The induced endomorphism on H^0 is multiplication by 3, and the induced endomorphism on H^1 is multiplication by 5. Hence

$$\operatorname{str}(F \mid H^\bullet(C)) = 3 - 5 = -2,$$

equal to the chain-level supertrace.

The cancellation can be displayed without invoking a general theorem. The acyclic summand

$$\mathbf{Q}\langle e_1 \rangle \xrightarrow{1} \mathbf{Q}\langle u_1 \rangle$$

has trace contribution $2 - 2 = 0$. The zero-differential summand

$$\mathbf{Q}\langle e_2 \rangle \xrightarrow{0} \mathbf{Q}\langle u_2 \rangle$$

has contribution $3 - 5 = -2$ and is its own cohomology. Thus the equality of supertraces results from additivity and the vanishing of trace on the contractible summand.

This example is the finite algebraic shadow of localisation for stable categories. A cofibre sequence splits the trace into contributions, and an equivalence identifies the trace of an acyclic object with zero. To lift the calculation into the formal system, one needs a stable mode, a trace on endomorphisms of dualizable complexes, and a theorem that the scalar shadow is additive over cofibre sequences. The matrices certify the numerical result; the stable trace theorem supplies its invariant meaning.

6.16.3 From a semantic model towards a chôra

Suppose a model of Definition 6.21 is given. To recover a candidate chôra, one first extracts its Rezk type of figures $\mathcal{C}$ and the family of fibre categories $\mathcal{D}(X)$. Supported maps are those for which the $f_!$ mode is present and stable under pullback. Composition two-cells yield a functor from correspondences. Tensor structure in the fibres and its projection cells yield lax symmetric monoidality.

The duality datum must then be reconstructed. A mere right adjoint $f^!$ does not supply Verdier duality. One needs dualising objects ω_X, an internal-hom operation, and an involutive equivalence on a constructible subtype. The assertion that double dualisation is the identity is a term, and its compatibility with $f_!, f_*, f^*, f^!$ is a family of higher cells. These are fields of the chôra candidate, not consequences of the core modal syntax.

Next one extracts the trace. The type former $\operatorname{Tr}_A(F)$ gives a shadow for directed categories internal to the model. A chôra requires the trace on the particular stable categories $\mathcal{D}(X)$ and their endofunctors, together with symmetric monoidality, additivity, localisation, and nondegeneracy. The basic inference rules provide cyclicity and the universal property of the coend. They do not automatically prove that the pairing on π_0 has trivial radical or that fixed-point localisation is an equivalence.

Finally, reconstruction asks whether the extracted operations determine the figures. Two non-equivalent geometric presentations may have Morita-equivalent categories and identical

traces. The intended equivalence with chôrai must therefore localise at a carefully selected Morita relation, while still retaining enough commutative algebra objects to recover geometry when KH.5 applies. This tension is the principal obstruction to the reverse direction of Conjecture 6.23.

The forward construction, from a chôra to a model, has its own coherence problem. A six-functor formalism is naturally an $(\infty, 2)$-functor, while the inference rules use split substitution. One must choose a strict presentation and prove that different choices induce equivalent syntactic models. Similarly, a homotopy coend must be selected functorially enough to interpret substitution. Large-cardinal universes make the choices small relative to a higher universe, but do not make them canonical.

Thus the adequacy conjecture has four separable components: an internal-language theorem, a strictification theorem, a trace-representation theorem, and a reconstruction theorem. Failure of one component need not invalidate the others. Even if full reconstruction fails, a sound internal language for six-functor formalisms with trace would remain useful. Conversely, a reconstruction theorem without normalisation would not produce a practical proof assistant. The ledger should record these dependencies separately.

6.17 Summary

Summary. The language $\mathrm{TT}_{\mathrm{KH}}$ has four layers. Its dependent core provides contexts, types, terms, universes, and identity types. Its directed layer follows Riehl and Shulman: hom-types are extension types, and Segal filling supplies coherent composition without making arrows invertible. Its modal layer is governed by a two-category of modes, with locks controlling variable use, adjunctions controlling introduction and elimination, and the $\flat \dashv \sharp$ pair arising from discrete, global-section, and codiscrete mode changes. Its trace layer is a cyclic coend presented by explicit inference rules; scalar trace, cyclicity, multiplicativity, and a base-change instance were derived from those rules and the mode coherences.

KH.27 specifies this language and KH.28 specifies its evidence discipline. Five KH axioms were rendered as actual types or records, showing that the translation is not merely verbal. A model is an indexed comprehension structure with directed, modal, stable, and trace operations. The claim that the intended models correspond to chôrai remains conjectural. Normalisation, canonicity, and decidability of type-checking for the combined calculus are open; consistency is proved only relative to a model, and existence of the required model needs universe assumptions plus substantive geometric hypotheses. Lean 4 can host an external syntax, a checker, ordinary categorical fragments, and the verification ledger today, but every missing higher or modal principle introduced as an axiom remains unverified.

6.18 Notes and references

The directed core is taken from Riehl and Shulman's simplicial type theory (Riehl and Shulman 2017). Its importance here is not a new theorem about synthetic categories, but the observation that correspondences and non-invertible functors should inhabit the grammar of the foundational language. The terminology "Segal type" and "Rezk type" follows their work. The underlying homotopical interpretation of identity types and univalence belongs to the development initiated by Voevodsky and presented systematically in the HoTT Book (Program 2013). The simplicial-set model of univalent foundations is due to Kapulkin, Lumsdaine, and Voevodsky (Kapulkin et al. 2021).

The modal organisation descends from several related lines. Licata and Shulman's adjoint logic makes a two-category of modes part of the syntax (Licata and Shulman 2016). Shulman's real-cohesive homotopy type theory demonstrates that adjoint modalities can support serious synthetic mathematics (Shulman 2018). Riley's work on multiple commuting cohesions and Wellen's modal Cartan geometry are especially close in spirit to the simultaneous place, analytic, and differential modalities required here. The precise bibliography for those two later developments is not yet included in the treatise database, so they have been attributed in prose rather than by invented keys.

The categorical trace and its cyclicity have a long history. The formulation through shadows in bicategories is developed by Ponto and Shulman (Ponto and Shulman 2012); higher traces and their relation to noncommutative motives are treated by Hoyois, Scherotzke, and Sibilla (Hoyois et al. 2017). The present trace modality is not claimed to occur in those sources. Its coend semantics is standard, while its dependent inference rules and their combination with arithmetic modes are a design of this book.

Lean's kernel and programming architecture are described by de Moura and Ullrich (Moura and Ullrich 2021), and the scale and organisation of mathlib by the mathlib community (Community 2020). The perfectoid-spaces and Liquid Tensor projects are the relevant precedents for large formal developments in modern arithmetic and condensed mathematics (Buzzard et al. 2020; Commelin and Scholze 2023). They are precedents, not implementations of higher chôrai.

The function-field control rests on the work of Drinfeld, Laurent Lafforgue, and Vincent Lafforgue. The cited sources here are Laurent Lafforgue's global correspondence for GL_n and Vincent Lafforgue's shtuka and excursion operator construction (Lafforgue 2002; Lafforgue 2018). The chapter uses that theory only as a semantic control: it does not claim that the proofs have been transported into the proposed type theory.

The separation between identity types and directed hom-types should also be read historically. Homotopical models of identity types explain why equality has groupoidal structure; they do not turn categories into groupoids (Awodey and Warren 2009). Classical dependent type theory can represent a category by a type of objects and a family of morphism types, but then every coherence datum is external to the identity eliminator. Riehl–Shulman syntax internalises the simplicial shapes responsible for that coherence. The gain is greatest for the present project because a correspondence is already a directed object and the six operations form a network of adjunctions rather than a single algebraic structure.

There are two distinct uses of univalence in the chapter. The first is the ordinary univalence of a universe of homotopy types, whose established models justify transport along equivalence. The second is the proposed univalence of a universe of chôrai. The latter requires a notion of equivalence preserving six operations, duality, and trace, and it lives at a higher universe level. No result cited for ordinary univalence establishes this chôra-level principle. The chapter

therefore treats it as part of KH.27's design and of the adequacy conjecture, not as an imported theorem.

The modal notation has ancestors in categorical logic as well as in homotopy type theory. An adjunction gives introduction and elimination rules because maps out of a left adjoint are maps into the right adjoint. A two-category of modes organises several such adjunctions and their transformations. What is specific here is the interpretation of mode objects as geometric fibres and of mode morphisms as six-functor operations or place changes. This interpretation must be validated separately in each semantic example. A formal mode theory can be perfectly coherent while having no arithmetic model.

The $\flat/\sharp$ notation belongs to cohesive mathematics. Its use here is restricted to the adjunction obtained from discrete, global-section, and codiscrete functors. It should not be confused with the place modalities $\Box_v$, which are localisations rather than discrete reflections. Nor should either be confused with truncation modalities. A future library should use distinct namespaces or mode indices even if surface typography uses familiar box-like symbols.

The trace construction draws on two mathematical traditions. In algebra, the quotient by commutators gives degree-zero Hochschild homology and explains why matrix trace is cyclic. In higher category theory, the trace of an endomorphism of a dualizable object is a shadow, and the trace of an endofunctor of a category is Hochschild homology. Ponto–Shulman and Hoyois–Scherotzke–Sibilla provide the relevant formal structures (Ponto and Shulman 2012; Hoyois et al. 2017). The S_3 calculation above is the finite groupoid instance of the first tradition; the coend rules are written so that the same syntax can receive the higher interpretation.

Cyclotomic trace adds a further layer. Nikolaus and Scholze's formulation of topological cyclic homology isolates the circle action, fixed-point constructions, Tate constructions, and Frobenius maps (Nikolaus and Scholze 2018). The chapter does not attempt to place those constructions in the judgemental core. It treats them as structures formed in a stable equivariant mode. This decision keeps the primitive trace rules small and leaves the infinitary all-primes construction to the library and semantics.

The base-change derivation belongs to the modern packaging of six operations by correspondence categories. Gaitsgory and Rozenblyum develop the higher correspondence machinery (Gaitsgory and Rozenblyum 2017a), while modern six-functor formalisms use functoriality on correspondences to generate Beck–Chevalley and projection-formula maps. The derivation in this chapter does not reprove geometric proper base change. It shows that, once the semantic functor sends the pullback cell to an equivalence, the internal proof has no remaining geometric choice.

The finite-fibre calculation makes this distinction transparent. In finite sets, the pullback square can be enumerated, pushforward is a finite direct sum, and the Beck–Chevalley map is a permutation of summands. For constructible sheaves on schemes, the analogous equivalence is a theorem with cohomological hypotheses. For analytic chôrai, it is part of the conjectural analytic EGA package. The type-theoretic term has the same shape in all three cases, while its semantic dependency changes.

The supertrace calculation is an elementary instance of additivity. The contractible summand contributes zero because the same eigenvalue occurs in adjacent degrees with opposite signs. In a stable category this cancellation is expressed by a cofibre sequence and the additivity of trace. Formalising the matrix calculation is straightforward; formalising the theorem that categorical trace is additive requires the stable and higher-categorical library. The example was included to prevent those two tasks from being confused.

The projective-line and elliptic-curve computations serve a parallel purpose. Their point counts, cohomology groups, and Frobenius polynomials are classical and explicit. They demonstrate the formal sequence from an endomorphism to a cohomological trace and then to a fixed-point count. Deligne's theory of weights supplies the general purity theorem behind the absolute values of Frobenius eigenvalues (Deligne 1974a). The arithmetic theory seeks an analogue in which all places, including infinity, participate in one trace. Nothing in the finite-field calculation proves that such an analogue exists.

The metatheoretic caution follows experience with univalent systems. Model existence establishes relative consistency, while normalisation and canonicity require separate syntactic or semantic arguments. Shulman's work on homotopy canonicity for inverse diagrams shows one route (Shulman 2015), but the directed interval, modal locks, and trace quotient add proof obligations not present there. The decision to make cyclicity propositional is intended to preserve a possible normalisation argument; it is not itself such an argument.

The practical Lean discussion should be read against two complementary facts. Lean and mathlib have demonstrated that large developments in category theory, algebraic geometry, and condensed mathematics are possible. They have not made Lean's kernel into a univalent or simplicial type theory. An external syntax checked in Lean can therefore provide trustworthy experiments with the proposed rules, but its soundness depends on the semantic theorem proved about that encoding. Declaring the missing principles as axioms is useful for interface design and useless as evidence for the principles themselves.

Finally, the evidence discipline of KH.28 has precedents in proof-assistant practice but is elevated here because the mathematical programme is unusually heterogeneous. A single headline claim may depend on a kernel proof, a finite point count, an interval enclosure, a theorem imported from the literature, and a conjectural geometric comparison. The dependency graph must preserve those distinctions. A proof term can show that the conclusion follows from the leaves; only the status manifest tells the reader what the leaves are.

6.19 Exercises

6.1. Write the complete non-dependent eliminator for $\mathrm{Tr}_A(F)$ and prove directly that maps $\mathrm{Tr}_A(F) \to B$ are equivalent to dinatural families $\prod_x \mathrm{Hom}_A(x, Fx) \to B$.

6.2. In a finite category with two objects and arrows $x \overset{a}{\to} y \overset{b}{\to} x$, compute $\mathrm{Tr}_C(\mathrm{id})$ as a quotient set and identify the classes of ba and ab.

6.3. Expand the algebraic proof of Theorem 6.14 using evaluation, coevaluation, symmetry, and the two triangle identities, without string diagrams.

6.4. Formulate a left-exactness condition on a modality $\Box$ and show that it preserves identity types of $\Box$-modal objects.

6.5. Starting from Disc $\dashv \Gamma \dashv$ Codisc, derive the unit and counit of $\flat \dashv \sharp$ and verify both triangle identities as modal terms.

6.6. Repeat Theorem 6.16 for the comparison $g^* f_* \to f'_* g'^*$, listing exactly which hypotheses make the comparison invertible.

6.7. Give a finite presentation of the mode theory generated by one adjunction $\mu \dashv \nu$ and prove a normal-form theorem for its two-cells.

6.8. Encode the syntax of the point and cyclic-path constructors of Tr as an inductive family in Lean 4. State precisely which quotient or higher-inductive principle is missing for the intended eliminator.

6.9. Audit a small Lean theorem that uses one declared axiom. Produce its transitive axiom list and explain why kernel acceptance is conditional rather than an independent verification of the axiom.

6.10. In the finite function-field control model, show that an equivalence of categories $C \simeq D$ induces an equivalence $\mathrm{Tr}_C(F) \simeq \mathrm{Tr}_D(EFE^{-1})$.

6.11. Compute the set truncation of $\mathrm{Tr}_{BD_8}(\mathrm{id})$ for the dihedral group of order 8. List every conjugacy class, compute its size from a centraliser, and check that the class sizes sum to 8.

6.12. In the finite-fibre base-change calculation above, replace the dimensions $(1, 2, 1)$ by $(2, 1, 3)$ and choose explicit matrices $h_1 : \mathbf{Q}^2 \to \mathbf{Q}$, $h_2 : \mathbf{Q} \to \mathbf{Q}^2$, and $h_3 : \mathbf{Q}^3 \to \mathbf{Q}^2$. Compute both routes around the naturality square on a numerical vector.

6.13. Test Problem 6.30 in the category of finite-dimensional representations of S_3 with the invariants modality. Determine whether forming coinvariant trace classes commutes with taking invariants for the trivial, sign, and standard representations.

6.14. Test Conjecture 6.24 on the fragment generated by one adjunction and the trace of a finite monoid. Orient all beta rules, enumerate terms through a fixed size, and report either a reduction loop or the observed normal forms. Explain why the bounded experiment proves neither termination nor confluence in general.

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